A Course in Differential Privacy

One topic per chapter, with every proof

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Preface

This course develops differential privacy from motivating failures to formal guarantees and algorithms. Later parts cover query answering, private statistics and machine learning, lower bounds, and deployments.

Each chapter covers one topic. Numbered definitions, theorems, and proofs make the mathematics explicit. Proofs are complete or cite a standard source and state the needed prerequisites. The prose explains each result and its implications.

The course assumes discrete probability, including expectation, variance, independence, and Markov's, Chebyshev's, and Hoeffding's inequalities. It also assumes basic linear algebra and calculus. The course states and proves any additional probability results where it first uses them.

Notation. A dataset is $D = (x_1, \dots, x_n) \in \mathcal{X}^n$, a list of n records from a universe $\mathcal{X}$. Two datasets are neighbours, $D \sim D'$, when they differ in one record, which is the replace-one relation. A mechanism M maps a dataset to a random output in a set $\mathcal{Y}$. The privacy parameters are $\varepsilon \geq 0$ and $\delta \in [0, 1]$. The natural logarithm is written ln, and the logarithm to base 2 is written $\log_2$.

Part I.

Privacy failures

This part examines privacy failures before defining privacy. Its first chapter shows how attackers combine releases protected by pseudonyms, identifier removal, summary statistics, or model training with information from other sources. Its second proves that accurate answers to counting queries can reveal almost every secret bit. These failures motivate a definition that allows arbitrary side information, covers every output, degrades gracefully under repeated releases, and permits enough error to prevent reconstruction.

Table 1.: The chapters of this part.

Chapter	Question it answers
Attempts at Data Privacy	Why do pseudonyms, removing identifiers, releasing models or statistics, and k-anonymity fail to protect people?
Reconstruction Attacks	How much error must answers to counting queries have before an analyst cannot rebuild the data?

1. Attempts at data privacy

This chapter examines five releases of sensitive data that their owners considered safe. Attackers defeated each protection by joining the release with information from another source. The protections lacked a formal guarantee against a defined class of attackers. The final section shows that k-anonymity, an early formal response, fails for the same reason.

1.1. Pseudonyms: the taxi trips

In 2014, a public-records request obtained records of every New York City taxi trip in 2013. Each record included pickup and drop-off times and locations, the fare, the cab's medallion number, and the driver's licence number. Because the identifiers revealed drivers' work locations and earnings, the regulator replaced each with a string of 32 hexadecimal digits. This pseudonymization preserves one key property: the same identifier always receives the same code.

Definition 1.1 (Pseudonymization)**.** Let $\mathcal{U}$ be a set of identifiers, such as licence numbers. A **pseudonymization** replaces each identifier $u \in \mathcal{U}$ in a table by $h(u)$, where h is a fixed injective function, so that the same person gets the same pseudonym in every row.

An analyst found that one licence pseudonym earned more per day than any driver could. It was the MD5 hash of `0`, the placeholder for a missing licence; thus h was unkeyed MD5. A hash can be hard to invert when its input space is large. Here the identifier formats allowed only about 22 million strings: about 3 million licence numbers and 19 million medallion numbers. Anyone could hash every candidate and look up each pseudonym.

Proposition 1.1 (Dictionary attack)**.** *Let the set of identifiers $\mathcal{U}$ be finite, and let the pseudonymization function h be injective and computable by anyone. Then $|\mathcal{U}|$ evaluations of h recover the identifier behind every pseudonym in the table.*

Proof.

1. Compute $h(u)$ for every $u \in \mathcal{U}$ and store the pairs $(h(u), u)$ in a table sorted by $h(u)$. This takes $|\mathcal{U}|$ evaluations of h.
2. For each pseudonym v in the released table, look up v. Since h is injective, exactly one stored pair has first entry v, and its second entry is the identifier.

□

A licence number does not identify a name by itself. The attacker joined the decoded table with a public list of licensed drivers, which mapped each licence number to a name. This revealed every driver's trips and income. The same kind of join with a named auxiliary table underlies the attacks in this chapter.

Definition 1.2 (Linkage attack)**.** Let T be a released table and T' an auxiliary table that names people, and let Q be a set of attributes that both tables contain. A **linkage attack** joins T and T' on Q. A row of T is **re-identified** when its values on Q match exactly one row of T' and no other row of T.

Random pseudonyms would stop the dictionary attack but not linkage. A passenger knows the time and place of their pickup and can read the driver's licence number in the cab. The matching row reveals the driver's pseudonym, which links to every other trip by that driver. Pickup time and location also identify a passenger's trip; the drop-off may be their home.

1.2. Rich records: the Netflix Prize

In 2006, Netflix released about 100 million film ratings from about 480,000 subscribers for a contest. Each rating included a user pseudonym, film, score, and date. Such histories can reveal sensitive traits, including a person's politics or sexuality.

Narayanan and Shmatikov showed in 2008 that the pseudonyms protected little, because many subscribers also post reviews on the Internet Movie Database (IMDb) under their name or a stable handle. A linkage attack in the sense of Definition 1.2, with Q the films, scores and dates, matches the two sources. The match need not be exact. The authors scored each candidate pair of users by how many films both rated with similar scores on nearby dates, and they gave rarely rated films more weight. They reported that 8 ratings, 2 of which may be wrong, with dates known only to within 14 days, single out 99% of the records.

In a sparse table with many columns, records are often unique. A small set of ratings can identify a person.

1.3. Releasing a model: memorization

A trained model may seem safer to release than a table. In 2019, Carlini and coauthors tested this assumption on language models. A language model predicts each word from the preceding words, using parameters θ learned from a text corpus. The authors measured how surprising a string was to the model, in bits; strings the model considered likely received lower scores.

Definition 1.3 (Log-perplexity)**.** Write $\Pr_\theta$ for the probability that the model with parameters θ assigns to a sequence of words, or to the next word given the words before it. The **log-perplexity** of a sequence $x_1, \dots, x_\ell$ of ℓ words under this model is

$$P_\theta(x_1, \dots, x_\ell) = -\log_2 \Pr_\theta[x_1, \dots, x_\ell] = \sum_{i=1}^{\ell} -\log_2 \Pr_\theta [x_i \mid x_1, \dots, x_{i-1}] \,. \tag{1.1}$$

The second form follows from the chain rule of probability. An English model assigns a low score to a common sentence and a high score to an unlikely sequence. The test asks whether a rare training string, such as a password or identity number, receives a much lower score than comparable strings. A low score indicates memorization and may let anyone querying the model recover the string.

To measure this, the authors put a **canary** into the training text: a sentence of a fixed format with a random secret, such as "my identity number is 314-15-9265". They then compared its score with those of all other sentences of that format. Exposure is large when the model ranks the true canary above most of them.

Definition 1.4 (Exposure)**.** Let $\mathcal{R}$ be the finite set of candidate secrets, all sentences of the canary's format, and let $N = |\mathcal{R}|$. Sort $\mathcal{R}$ by increasing log-perplexity under the model, and let $\operatorname{rank}(r) \in \{1, \dots, N\}$ be the position of r, so that rank 1 is the most likely candidate. The **exposure** of r is

$$\operatorname{exposure}(r) = \log_2 N - \log_2 \operatorname{rank}(r). \tag{1.2}$$

Exposure ranges from 0 for the least likely candidate to $\log_2 N$ for the most likely. An attacker can test candidates in rank order to recover a high-ranked canary. The next result bounds exposure when the model did not see the canary.

Proposition 1.2 (Exposure without memorization)**.** *Suppose the canary r is drawn uniformly from $\mathcal{R}$, independently of the model, and the ranking breaks ties in a fixed way. Then for every $t \geq 0$,*

$$\Pr[\operatorname{exposure}(r) \geq t] \leq 2^{-t}, \qquad \mathbb{E}[\operatorname{exposure}(r)] < \frac{1}{\ln 2} \approx 1.443. \tag{1.3}$$

Proof.

1. The ranking depends only on the model, and r is uniform and independent of it. So $\operatorname{rank}(r)$ is uniform on $\{1, \dots, N\}$.
2. The exposure is at least t exactly when $\operatorname{rank}(r) \leq N2^{-t}$. By step 1 this has probability $\lfloor N2^{-t} \rfloor / N \leq 2^{-t}$.
3. By step 1, the average exposure is $\log_2 N - \frac{1}{N}\sum_{k=1}^{N} \log_2 k = \frac{1}{N \ln 2}\,(N \ln N - \ln N!)$.
4. Since ln is increasing, $\ln k$ is at least the integral of $\ln x$ over $[k-1, k]$ for $k \geq 2$. Summing over k gives

$$\ln N! \geq \int_1^N \ln x \, \mathrm{d}x = N \ln N - N + 1. \tag{1.4}$$

5. Steps 3 and 4 give $\mathbb{E}[\operatorname{exposure}(r)] \leq \frac{N-1}{N \ln 2} < \frac{1}{\ln 2}$.

□

So if the model never saw the canary, an exposure of t bits is as surprising as an event of probability 2^{-t}. With nine-digit secrets, $N = 10^9$, and the largest possible exposure is $\log_2 10^9 \approx 29.9$.

The experiments showed different levels of memorization. A translation model trained on about 100,000 sentence pairs assigned maximum exposure to a canary containing a nine-digit identity number after four insertions. Gmail Smart Compose, trained on billions of word sequences, used a candidate space of order 10^{12}; after 10,000 insertions, canary exposure reached about 10 out of 39.9. Dropout and regularization did not reduce exposure. The authors found that differential privacy, the subject of the next part, did.

1.4. Releasing statistics: genomic studies

Medical studies must publish findings, and aggregate statistics can appear safer than individual records. But if a study enrolls only people with a stigmatized disease, participation reveals a diagnosis. A test that infers membership from published statistics is then an attack.

Homer and coauthors showed in 2008 that such a test can use genetic data. It compares a person's genotype with published allele frequencies for the study group and a reference population. Given sufficient genetic markers and a suitable group size, the test can infer whether the person's DNA was included. In practice, the test has more false positives than the analysis predicts because its threshold depends on how closely the reference population matches the study group. The United States National Institutes of Health nevertheless placed aggregate genomic statistics behind an approval process.

1.5. Removing identifiers: k-anonymity

In the mid-1990s, the Massachusetts Group Insurance Commission gave researchers hospital records for state employees after removing names and Social Security numbers. Each record retained the patient's ZIP code, birth date, sex, and diagnosis. Latanya Sweeney bought the Cambridge voter roll, which listed each voter's name, ZIP code, birth date, and sex. A linkage attack (Definition 1.2) on the first three attributes identified the governor's hospital records. The same attributes identify many people: Sweeney later estimated that 87% of the United States population was unique by five-digit ZIP code, sex, and birth date.

Sweeney and Samarati classified table attributes into three types. A release removes **identifiers**, such as names, and retains **sensitive attributes**, such as diagnoses. **Quasi-identifiers**, such as ZIP code, age, and sex, can identify people in combination. The release therefore generalizes them, for example by shortening ZIP codes, until each person shares their quasi-identifier values with at least k rows.

Definition 1.5 (k-anonymity)**.** Let $k \geq 1$ be an integer. A table is **k-anonymous** with respect to a set of quasi-identifiers if every combination of quasi-identifier values that appears in the table appears in at least k rows. The rows that share a combination form an **equivalence class**.

A k-anonymous table stops the attack on the governor, because a linkage on the quasi-identifiers matches every person to at least k rows. But the definition says nothing about the sensitive values in a class, or about other releases. Two examples show what goes wrong, with tables from two hospitals in the same town.

Table 1.1.: Hospital A's release, 3-anonymous with respect to ZIP code and age.

ZIP code	Age	Diagnosis
021**	20–29	influenza
021**	20–29	asthma
021**	20–29	diabetes
021**	30–39	hepatitis
021**	30–39	hepatitis
021**	30–39	hepatitis
024**	40–59	influenza
024**	40–59	migraine
024**	40–59	asthma

Table 1.2.: Hospital B's release, 3-anonymous with respect to ZIP code and age.

ZIP code	Age	Diagnosis
02***	18–24	influenza
02***	18–24	asthma
02***	18–24	fracture
02***	25–34	diabetes
02***	25–34	migraine
02***	25–34	fracture

The first example shows that a class whose rows all share one sensitive value hides nothing.

Example 1.1 (Homogeneity)**.** A neighbour aged 35, living in ZIP code 02139, was a patient of hospital A. In Table 1.1, the neighbour must be in the class with ZIP code 021** and age 30–39. All three of its rows say hepatitis, so the attacker learns the diagnosis without knowing which row is the neighbour's.

The second example has varied diagnoses in each class, but combines the two releases.

Example 1.2 (Intersection)**.** A colleague aged 27, living in ZIP code 02139, was treated at both hospitals for the same condition. In Table 1.1 the colleague's class allows influenza, asthma or diabetes. In Table 1.2 the class with age 25–34 allows diabetes, migraine or fracture. Only diabetes is in both lists, so the two releases together reveal the diagnosis, although each is 3-anonymous.

The second example is a failure of composition: each hospital checked its own table and could not know what the other released. Ganta, Kasiviswanathan and Smith showed in 2008 that such attacks succeed in practice, also against refinements of k-anonymity.

These cases motivate three requirements for a privacy definition. It must allow arbitrary side information, apply to any output, and degrade gracefully under composition. Differential privacy, defined in the chapter on randomized response, meets all three.

1.6. Summary

Table 1.3.: The chapter's results.

Result	Statement
Definition 1.1	replace each identifier u by $h(u)$ for a fixed injective h
Proposition 1.1	a public h on a set of N identifiers is inverted by N evaluations
Definition 1.2	join a release with named auxiliary data on shared attributes
Definition 1.4	$\log_2 N - \log_2 \operatorname{rank}(r)$ for a canary among N candidates
Proposition 1.2	without memorization, exposure $\geq t$ has probability at most 2^{-t} and mean below $1/\ln 2$
Definition 1.5	every quasi-identifier combination appears in at least k rows
Example 1.1	a class with one sensitive value reveals it
Example 1.2	two k-anonymous releases together can reveal a value

1.7. References

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2. Reconstruction attacks

This chapter studies when slightly inaccurate answers to counting queries let an attacker reconstruct almost all of a dataset. It defines this failure without yet defining privacy, then proves that many query-answering algorithms permit it. To prevent reconstruction, a curator answering every query needs error of order n, the number of people. For about n random queries, error of order $\sqrt{n}$ is necessary. ## Reconstruction from exact statistics {#sec-rec-census}

Exact statistics can reveal the underlying records. A census collects **microdata**, one record per person with fields such as age, sex, and race. It publishes counts, medians, and means for small areas such as city blocks. Suppressing cells with fewer than three people is insufficient because tables overlap. For example, if three of four residents are women, a published median age for the men reveals the age of the only man.

Every published number is a constraint on the unknown microdata, and enough constraints leave only one possibility. The next example counts the age combinations that one line of a census table leaves for three people.

Example 2.1 (One line of a census table)**.** A block's table says that it has three men, with median age 35 and mean age 47. Call the youngest man's age u_1, the middle one's u_2 and the oldest one's u_3, so that $u_1 \le u_2 \le u_3$. Ages are whole numbers between 0 and 125.

1. Without the statistics, the number of such triples is the number of multisets of size 3 from 126 values, $\binom{128}{3} = 341{,}376$.
2. The median of three numbers is the middle one, so $u_2 = 35$. Then u_1 has 36 values and u_3 has 91, which leaves $36 \times 91 = 3{,}276$ triples.
3. The mean gives $u_1 + u_2 + u_3 = 141$, so $u_3 = 106 - u_1$. Each u_1 from 0 to 35 gives a u_3 between 71 and 106, and every such u_3 lies between u_2 and 125. So exactly 36 triples remain, from $(0, 35, 106)$ to $(35, 35, 71)$.

One table row reduces the possible triples by nearly a factor of 10,000. Other rows group the same people differently and reduce the possibilities further. A reconstruction attack finds microdata satisfying all published constraints. This is an integer program, which is hard in the worst case but often tractable for practical instances. For a fictional block of seven people, Garfinkel, Abowd, and Martindale found that the full table's constraints had a unique solution; a solver found it in a fraction of a second.

The attack also worked at scale. Applied to the 2010 tables, it reconstructed records that matched the confidential records exactly for 46% of the population and matched within one year of age for 71%. Linking the records with commercial databases identified more than 50 million people. The Bureau adopted differential privacy for the 2020 census.

2.1. Subset queries and blatant non-privacy

To analyse these attacks, consider n people with public identifying attributes and one secret bit each. Their bits form $d = (d_1, \dots, d_n) \in \{0,1\}^n$. An analyst asks how many people selected by a condition on the identifiers have bit 1. Since such a condition can select any set of people, represent a query by a subset $S \subseteq [n] = \{1, \dots, n\}$ and its indicator vector, also denoted $S \in \{0,1\}^n$, with entry i equal to 1 exactly when person i is selected.

A curator holds the secret bits and answers the queries. The definition assumes only a bound E on how far each answer may be from the truth.

Definition 2.1 (Subset queries and a curator with error E)**.** Let $d \in \{0,1\}^n$ be the vector of secret bits. The true answer to a **subset query** $S \in \{0,1\}^n$ is the number of people in S whose bit is 1,

$$a(S) = \sum_{i \in S} d_i = \langle S, d \rangle, \tag{2.1}$$

where $\langle S, d \rangle = \sum_i S_i d_i$ is the inner product of the indicator vector S with d.

A **curator with error** E answers each query S with a response $r(S)$, a number within E of the truth, $|r(S) - a(S)| \leq E$. Within that bound, the curator may choose its answers in any way, depending on d and on all the queries.

The error need not be random: the curator may choose the most misleading allowed answer. Therefore, the attacks also apply to random noise. Some error is necessary because the exact answer to $S = \{i\}$ is person i's secret bit.

Measure reconstruction error by the number of incorrect bits. The attacker outputs $c \in \{0,1\}^n$ as a guess for d. Their **Hamming distance**, $\mathrm{dist}(c, d) = |\{i : c_i \neq d_i\}|$, counts the errors. The next definition formalizes a severe privacy failure: for every secret vector, an attacker recovers all but a vanishing fraction of its bits with probability approaching one.

Definition 2.2 (Blatant non-privacy)**.** Consider a family of curators, one for each number of people n. The family is **blatantly non-private** if three things exist. The first is an attacker that interacts with the curator and outputs a guess $c \in \{0,1\}^n$. The second is a bound $g(n)$ on the number of wrong bits, with $g(n)/n \to 0$. The third is a failure probability $\beta(n)$, with $\beta(n) \to 0$. They must satisfy, for every n and every secret vector $d \in \{0,1\}^n$,

$$\Pr\left[\mathrm{dist}(c, d) \leq g(n)\right] \geq 1 - \beta(n). \tag{2.2}$$

An attacker with $o(n)$ errors recovers the secrets of all but a vanishing fraction of people. This is a privacy failure under any useful definition.

2.2. Asking every query

The first attack ignores cost. It asks all 2^n subset queries and outputs any candidate that agrees with every answer.

Theorem 2.1 (Reconstruction from all subset queries)**.** *Let $E \geq 0$, and let a curator with error E answer all 2^n subset queries, giving the answer $r(S)$ to the query S. An attacker that outputs any $c \in \{0,1\}^n$ satisfying*

$$|\langle S, c\rangle - r(S)| \leq E \quad \text{for every } S \subseteq [n] \tag{2.3}$$

finds such a c, and every such c differs from the secret vector d in at most $4E$ positions, $\operatorname{dist}(c, d) \leq 4E$.

Proof.

1. The true vector d satisfies Equation 2.3 by Definition 2.1, so a candidate exists.
2. Let $I_0 = \{i : d_i = 0\}$. The attacker asked the query I_0, so Equation 2.3 gives $\langle I_0, c\rangle \leq r(I_0) + E$. The curator's bound gives $r(I_0) \leq \langle I_0, d\rangle + E = E$. Hence $\langle I_0, c\rangle \leq 2E$. This sum counts the positions of I_0 where $c_i = 1 \neq d_i$, so c and d differ in at most $2E$ positions of I_0.
3. Let $I_1 = \{i : d_i = 1\}$. In the same way, $\langle I_1, c\rangle \geq r(I_1) - E \geq \langle I_1, d\rangle - 2E = |I_1| - 2E$. The number of positions of I_1 where $c_i = 0$ is $|I_1| - \langle I_1, c\rangle \leq 2E$.
4. The sets I_0 and I_1 partition $[n]$, so $\operatorname{dist}(c, d) \leq 4E$.

□

The attacker does not know I_0 or I_1, so it must ask every query. If $E = o(n)$, the error bound meets Definition 2.2.

Corollary 2.1. *If $E = n/401$, the attacker of Theorem 2.1 recovers more than 99% of the bits. If $E = E(n)$ with $E(n)/n \to 0$, the curator is blatantly non-private, with $g(n) = 4E(n)$ and $\beta(n) = 0$.*

Proof. For the first claim, $4n/401 < n/100$, since $4 \times 100 < 401$. The second claim is Theorem 2.1 with Definition 2.2, because the attacker never fails. □

Counting-query answers lie in $[0, n]$, so error of order n is maximal. The next result shows that error E can still hide about $E/2$ bits, regardless of the attacker's queries.

Proposition 2.1 (Error E hides $E/2$ bits)**.** *Let $E \geq 1$ and $k = \min(\lfloor E \rfloor, n)$. For every attacker, which may ask any queries in any adaptive order, there are a vector $d \in \{0,1\}^n$ and a curator with error E for which* $\Pr[\operatorname{dist}(c, d) \geq k/2] \geq 1/2$.

Proof.

1. Let d be all zeros, and let d' have ones in its first k positions and zeros elsewhere, so $\operatorname{dist}(d, d') = k$.

2. On the data d', let the curator answer every query exactly, $r(S) = \langle S, d' \rangle$. On the data d, let it give the same answers. Their error is $|\langle S, d' - d \rangle| \le k \le E$, so both are curators with error E.
3. The answers to every query are the same on d and on d'. So the attacker's output c has the same distribution in the two cases.
4. For every c, the triangle inequality gives $\mathrm{dist}(c, d) + \mathrm{dist}(c, d') \ge k$, so at least one of the two distances is at least $k/2$. Hence $\Pr[\mathrm{dist}(c, d) \ge k/2] + \Pr[\mathrm{dist}(c, d') \ge k/2] \ge 1$, and one of the two probabilities is at least $1/2$.

□

Thus, preventing reconstruction from all 2^n queries requires error of order n, and Theorem 2.1 is tight up to a factor of 8. However, this attack is impractical once n exceeds a few dozen.

2.3. Random queries

The second attack asks about n uniformly random subsets of people. It reconstructs data despite errors of order $\sqrt{n}$, below the first attack's tolerance but still large relative to the error of a useful statistic.

Fix a wrong candidate c. If it differs from d in k positions, a random subset query separates their answers by a sum of about $k/2$ random signs, typically of magnitude $\sqrt{k}$. A curator with error E can hide a difference of at most $2E$. Thus, when $\sqrt{k} \gg E$, each query rules out c with constant probability. The proof formalizes this using a lower bound on the magnitude of a random sum, rather than a concentration bound.

Lemma 2.1 (A random subset sees a fixed vector)**.** *Let $w \in \mathbb{R}^n$, with Euclidean length $\|w\|_2 = (\sum_i w_i^2)^{1/2}$. Let S be a uniformly random subset of $[n]$, so that each i is in S independently with probability $1/2$. Then the sum $\langle S, w \rangle$ of the entries of w in S satisfies*

$$\Pr\left[\, |\langle S, w \rangle| \ge \tfrac{1}{4}\|w\|_2 \,\right] \ge \frac{3}{32}. \tag{2.4}$$

Proof.

1. If $w = 0$ the event always holds, so let $w \ne 0$. Write $S_i = (1 + \xi_i)/2$, where the signs $\xi_i \in \{-1, +1\}$ are independent and uniform. Then $\langle S, w \rangle = \frac{1}{2}(a + Z)$, with the constant $a = \sum_i w_i$ and the random sum $Z = \sum_i \xi_i w_i$.
2. *Moments of Z.* Expand the powers and use independence. A product of the ξ_i has mean 0 unless each index appears an even number of times. So $\mathbb{E}[Z^2] = \sum_i w_i^2 = \|w\|_2^2$, and

$$\mathbb{E}[Z^4] = \sum_i w_i^4 + 3\sum_{i \ne j} w_i^2 w_j^2 = 3\|w\|_2^4 - 2\sum_i w_i^4 \le 3\|w\|_2^4. \tag{2.5}$$

3. *Paley–Zygmund inequality.* Let $Y \ge 0$ have $\mathbb{E}[Y] > 0$, and let $s \in [0, 1]$. Split $\mathbb{E}[Y]$ according to whether $Y > s\,\mathbb{E}[Y]$, and apply the Cauchy–Schwarz inequality to the second part:

$$\mathbb{E}[Y] \le s\,\mathbb{E}[Y] + \mathbb{E}\left[Y\mathbf{1}[Y > s\,\mathbb{E}[Y]]\right] \le s\,\mathbb{E}[Y] + \sqrt{\mathbb{E}[Y^2]\,\Pr[Y > s\,\mathbb{E}[Y]]}. \tag{2.6}$$

Rearranging gives $\Pr[Y > s\,\mathbb{E}[Y]] \ge (1-s)^2\,\mathbb{E}[Y]^2/\mathbb{E}[Y^2]$.

4. Apply step 3 to $Y = Z^2$ with $s = 1/4$, using step 2. The event $|Z| > \|w\|_2/2$ has probability at least $(3/4)^2/3 = 3/16$.
5. *Removing the constant a.* For every z, $|a+z|+|a-z| \ge 2|z|$, so $|a+Z| \ge |Z|$ or $|a-Z| \ge |Z|$. Flipping all signs maps Z to $-Z$ and preserves the distribution and the event of step 4. Hence the events $\{|Z| > \|w\|_2/2,\ |a+Z| \ge |Z|\}$ and $\{|Z| > \|w\|_2/2,\ |a-Z| \ge |Z|\}$ have the same probability. Together they cover the event of step 4, so each has probability at least 3/32.
6. On the first of these events, $|\langle S, w\rangle| = \frac{1}{2}|a+Z| \ge \frac{1}{2}|Z| > \frac{1}{4}\|w\|_2$.

□

The lemma rules out any fixed wrong candidate with constant probability per query. A union bound over the 2^n candidates then shows that about n queries rule out every wrong candidate.

Theorem 2.2 (Reconstruction from random queries)**.** *Let $E \ge 0$ and $\beta \in (0,1)$, and let*

$$m \ge \frac{32}{3}\left(n\ln 2 + \ln\frac{1}{\beta}\right). \tag{2.7}$$

The attacker draws m independent, uniformly random subsets $S_1, \dots, S_m$ of $[n]$ before seeing any answer. A curator with error E answers them with $r_1, \dots, r_m$. The attacker outputs any $c \in \{0,1\}^n$ with $|\langle S_j, c\rangle - r_j| \le E$ for all j. Then such a c exists and, with probability at least $1-\beta$ over the queries, every such c has

$$\operatorname{dist}(c,d) \le 64E^2. \tag{2.8}$$

Proof.

1. The true vector d satisfies all m constraints by Definition 2.1, so a candidate exists.
2. *One far candidate.* Fix c with $k = \operatorname{dist}(c,d) > 64E^2$, and let $w = c - d$. Its entries are in $\{-1,0,1\}$ and exactly k are nonzero, so $\|w\|_2 = \sqrt{k} > 8E$. By Lemma 2.1, each query has $|\langle S_j, w\rangle| \ge \sqrt{k}/4 > 2E$ with probability at least 3/32. The queries are independent, so the probability that $|\langle S_j, w\rangle| \le 2E$ for every j is at most $(1-3/32)^m \le e^{-3m/32}$.
3. *All far candidates.* There are at most 2^n candidates. By the union bound, the event B that some candidate with $\operatorname{dist}(c,d) > 64E^2$ has $|\langle S_j, c-d\rangle| \le 2E$ for every j has probability at most $2^n e^{-3m/32}$, which is at most β by the choice of m.
4. *Consistent candidates are close.* Suppose B does not occur. If c is consistent with the answers, the triangle inequality gives $|\langle S_j, c-d\rangle| \le |\langle S_j, c\rangle - r_j| + |r_j - \langle S_j, d\rangle| \le 2E$ for every j. So c is not far, which is Equation 2.8. The event B depends only on the queries, so this holds whatever answers the curator chose.

□

The number of queries is about $7.4n$: for $n = 1{,}000$ and $\beta = 0.01$ the theorem asks for $m = 7{,}443$. Setting $\beta = 2^{-n}$ gives the asymptotic statement.

Corollary 2.2. *If $E = E(n)$ with $E(n)/\sqrt{n} \to 0$, the curator is blatantly non-private against $15n$ random queries, with $g(n) = 64E(n)^2$ and $\beta(n) = 2^{-n}$. If $E = \alpha\sqrt{n}$, the same attacker gets at most $64\alpha^2 n$ bits wrong with probability at least $1 - 2^{-n}$.*

Proof. With $\beta = 2^{-n}$, the bound on m in Theorem 2.2 is $\frac{32}{3} \cdot 2n \ln 2 \approx 14.79\, n \le 15n$. Then $g(n) = 64E(n)^2 = 64n\,(E(n)/\sqrt{n})^2$, and $g(n)/n \to 0$. □

2.3.1. An efficient attack

The attacker of Theorem 2.2 asks few queries but searches 2^n candidates. A search over fractional vectors is a linear program, so this attack is called linear programming (LP) decoding. It looks for $x \in [0,1]^n$ with $|\langle S_j, x\rangle - r_j| \le E$ for all j, which takes time polynomial in n and m, and rounds x to the nearest bit vector. The analysis is the same as before with one extra step: a fractional candidate is not one of 2^n points, so a fine grid replaces the set of candidates.

Theorem 2.3 (Reconstruction by linear programming)**.** *Let $E > 0$, $\beta \in (0,1)$, $K = \lceil n/(2E) \rceil$, and*

$$m \ge \frac{32}{3}\left(n \ln(2K+1) + \ln\frac{1}{\beta}\right). \tag{2.9}$$

The attacker draws m random subsets and receives answers from a curator with error E, as in Theorem 2.2. It finds any $x \in [0,1]^n$ with $|\langle S_j, x\rangle - r_j| \le E$ for all j, and it outputs $c_i = 1$ if $x_i \ge 1/2$ and $c_i = 0$ otherwise. With probability at least $1 - \beta$ over the queries,

$$\operatorname{dist}(c, d) \le 676E^2. \tag{2.10}$$

Proof.

1. The vector d is feasible, so the linear program has a solution x. Let $z = x - d \in [-1,1]^n$. As in step 4 of Theorem 2.2, $|\langle S_j, z\rangle| \le 2E$ for every j.
2. *Rounding errors.* If $c_i \ne d_i$, then either $d_i = 0$ and $x_i \ge 1/2$, or $d_i = 1$ and $x_i < 1/2$. In both cases $|z_i| \ge 1/2$, so $(2z_i)^2 \ge 1$. Summing over i gives $\operatorname{dist}(c,d) \le 4\|z\|_2^2$.
3. *The grid.* Let $h = 1/K \le 2E/n$, and let G be the set of vectors whose entries are multiples of h in $[-1,1]$. It has $(2K+1)^n$ points. Round each entry of z to the nearest multiple of h to get $z' \in G$, with $|z_i - z'_i| \le h/2 \le E/n$. Then $|\langle S_j, z - z'\rangle| \le \sum_i |z_i - z'_i| \le E$, so $|\langle S_j, z'\rangle| \le 3E$ for every j. Also $\|z - z'\|_2 \le \sqrt{n} \cdot E/n \le E$.
4. *Far grid points are ruled out.* Fix $g \in G$ with $\|g\|_2 > 12E$. By Lemma 2.1, each query has $|\langle S_j, g\rangle| \ge \|g\|_2/4 > 3E$ with probability at least $3/32$. As in Theorem 2.2, the probability that $|\langle S_j, g\rangle| \le 3E$ for every j is at most $e^{-3m/32}$. A union bound over the $(2K+1)^n$ grid points bounds the probability that this happens for some far grid point by $(2K+1)^n e^{-3m/32} \le \beta$.

5. Outside that event, step 3 shows that z' is not far, so $\|z'\|_2 \le 12E$ and $\|z\|_2 \le 13E$. Step 2 gives $\mathrm{dist}(c, d) \le 4 \cdot 169E^2 = 676E^2$.

□

For $1 \le E \le n$, the factor $\ln(2K+1)$ is at most $\ln(n+3)$, so the attack runs in polynomial time with $O(n \log n)$ queries and gets at most $676E^2$ bits wrong. So error $o(\sqrt{n})$ again lets an attacker recover all but a vanishing fraction of the bits. The constants 64 and 676 are far from the truth, as Figure 2.1 shows: on simulated data the attack recovers almost every bit at errors the theorem cannot handle.

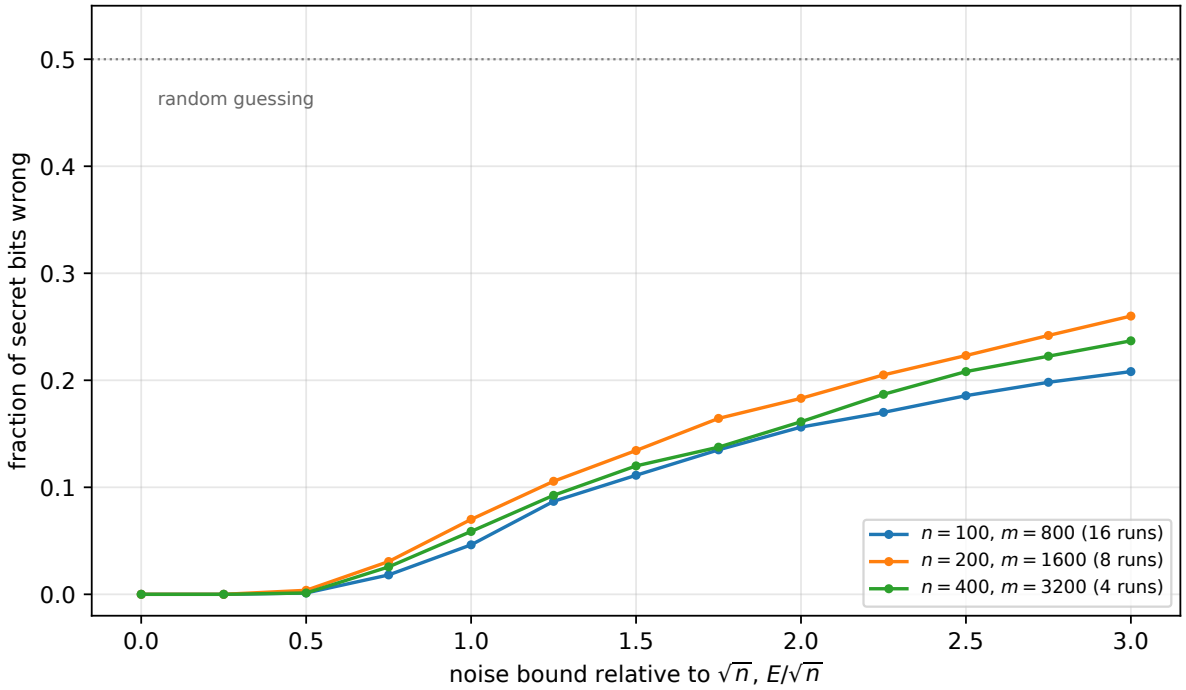


Figure 2.1.: The linear-programming attack on simulated data, with $m = 8n$ random subset queries and an error drawn uniformly from $[-E, E]$ for each answer. The attacker finds a point of the feasible set of Theorem 2.3, up to a violation of at most 0.01, by projected gradient descent and rounds it. Each point averages $1600/n$ runs. The curves for different n nearly coincide, so the error the attack tolerates grows like $\sqrt{n}$. For $n = 400$, a single run gets none of the bits wrong at $E = \sqrt{n}/2$ and 17% of them at $E = 2\sqrt{n}$.

Bounds on random matrices, which this course does not develop, show that $O(n)$ queries suffice for the linear program (Dwork and Roth, 2014, Section 8.1). They also show that a linear program that minimizes the total error tolerates a constant fraction of arbitrary answers (Dwork, McSherry and Talwar, 2007). The argument above gives an inefficient version of that tolerance.

Proposition 2.2 (Tolerating some arbitrary answers)**.** *Let $E \ge 0$, let $\beta \in (0, 1)$ be a failure probability, let $\rho \in [0, 3/128)$ be the fraction of answers that may be arbitrary, and let*

$$m \geq \frac{2048}{9}\left(n \ln 2 + \ln \frac{1}{\beta}\right). \tag{2.11}$$

The attacker draws m random subsets as in Theorem 2.2. The curator answers at least $(1-\rho)m$ of them with error at most E and the others arbitrarily, choosing which after seeing the queries. The attacker outputs any $c \in \{0,1\}^n$ with $|\langle S_j, c\rangle - r_j| \leq E$ for at least $(1-\rho)m$ of the queries. With probability at least $1-\beta$, every such c has $\operatorname{dist}(c,d) \leq 64E^2$.

Proof.

1. The true vector d meets the condition, so a candidate exists.
2. Fix c with $\operatorname{dist}(c,d) > 64E^2$ and let $V_j = \mathbf{1}[|\langle S_j, c-d\rangle| > 2E]$ indicate that query j exposes c. As in step 2 of Theorem 2.2, the V_j are independent with mean at least $3/32$. Hoeffding's inequality gives $\Pr\left[\sum_j V_j < 3m/64\right] \leq \exp(-2m(3/64)^2) = e^{-9m/2048}$.
3. A union bound over at most 2^n candidates shows that, with probability at least $1 - 2^n e^{-9m/2048} \geq 1-\beta$, every far candidate has $\sum_j V_j \geq 3m/64$.
4. Suppose this holds, and let c be an output. The queries where c is consistent and those answered with error at most E each number at least $(1-\rho)m$, so at least $(1-2\rho)m$ queries are of both kinds. On each of them, the triangle inequality gives $|\langle S_j, c-d\rangle| \leq 2E$, so $V_j = 0$. Hence $\sum_j V_j \leq 2\rho m < 3m/64$, and c is not far.

□

These attacks are nearly tight. The chapters on approximate differential privacy and advanced composition show that a curator can answer m subset queries with error of order $\sqrt{m}$, up to logarithmic factors and privacy parameters, while retaining differential privacy. For n queries, the threshold between blatant reconstruction and a formal guarantee is therefore of order $\sqrt{n}$.

2.4. Reconstruction in practice

The random-query attack may seem theoretical because it uses arbitrary subsets, but it broke a deployed system. Aircloak's Diffix interface allowed unlimited counting queries on sensitive data and promised less error than differential privacy.

Diffix added independent Gaussian noise with standard deviation 1 for each query condition, so error grew with the number of conditions. It also rejected OR conditions, suppressed small counts, and altered extreme values. Describing a random subset of n people requires about $n/2$ OR conditions, so these defenses blocked the direct attack.

Cohen and Nissim changed the attack order. They sampled a single condition from a family of conditions whose selected sets appear random. Each condition tests a digit of a power of each person's numeric identifier. One query retained a row when

$$\lfloor 100\,(2 \cdot \text{identifier})^{0.7} + 0.5 \rfloor = \lfloor 100\,(2 \cdot \text{identifier})^{0.7} \rfloor, \tag{2.12}$$

which holds exactly when the third decimal digit of $(2 \cdot \text{identifier})^{0.7}$ is at most 4. Among the 1,001 identifiers from 2000 to 3000, it selects 501, about half.

Each query used few conditions, so its error did not grow with the number of rows. Diffix added Gaussian noise with standard deviation 4. Cohen and Nissim targeted four identifier ranges, each containing 73 to 142 rows, and asked 3,500 queries per range. Because Gaussian noise is unbounded, they used a linear program minimizing total error; like Proposition 2.2, it tolerates a few large errors. It recovered the secret attribute for every row in all four ranges.

The theorems do not directly apply because these queries are not random subsets. Their proofs use query randomness only through Lemma 2.1, which requires each query to expose a fixed wrong candidate with constant probability. Experiments show that the digit tests have this property on real identifiers; this is empirical evidence, not a theorem.

2.5. Summary

Table 2.1.: The chapter's results.

Result	Statement
Example 2.1	a median and a mean cut 341,376 possible ages to 36
Definition 2.2	an attacker recovers all but $o(n)$ secret bits with probability tending to 1
Theorem 2.1	all 2^n queries with error E reveal all but $4E$ bits
Proposition 2.1	error E can hide $E/2$ bits from any attacker
Lemma 2.1	a random subset has $\lvert\langle S, w\rangle\rvert \geq \lVert w\rVert_2/4$ with probability at least 3/32
Theorem 2.2	$\frac{32}{3}(n \ln 2 + \ln \frac{1}{\beta})$ random queries with error E reveal all but $64E^2$ bits
Theorem 2.3	a linear program does the same in polynomial time with $O(n \log n)$ queries, all but $676E^2$ bits
Proposition 2.2	a fraction below 3/128 of arbitrary answers does not stop the attack

2.6. References

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- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014. Presents both attacks in Section 8.1.
- Raymond Paley and Antoni Zygmund, "A Note on Analytic Functions in the Unit Circle", 1932. Gives the inequality in Lemma 2.1.
- Aloni Cohen and Kobbi Nissim, "Linear Program Reconstruction in Practice", 2020. Breaks Diffix with digit-test queries.

Part II.

The basics of differential privacy

The first part showed that statistics can expose individuals. This part defines differential privacy (DP), the course's central guarantee. Under DP, replacing one record changes the probability of any event by at most a factor of e^{ε}, where ε is the privacy parameter. The chapters then prove basic mechanisms and composition rules with exact constants.

Table 2.2.: The chapters of this part and the questions they answer.

Chapter	Question it answers
Randomized response and the definition	How can one collect a sensitive bit from each person, and what does it mean for an algorithm to be private?
The Laplace mechanism	How much noise makes a numeric statistic private, and how do private algorithms combine?
Approximate differential privacy	What does a small additive slack δ allow, how small must it be, and what does the Gaussian mechanism gain?
Advanced composition	Why does the privacy cost of k analyses grow only like $\sqrt{k}$?
The exponential mechanism	How can one privately choose the best object from a set?

3. Randomized response and the definition

Each person holds one sensitive bit, and an analyst wants the fraction of ones without learning any individual's bit. Randomized response uses a biased coin to estimate that fraction. Its plausible-deniability guarantee also motivates differential privacy (DP), the subject of the rest of the course.

3.1. Randomized response

Write n for the number of people. Person number i holds a bit $x_i \in \{0, 1\}$, for instance whether they cheated on an exam, and the analyst wants the fraction of ones,

$$p = \frac{1}{n} \sum_{i=1}^{n} x_i. \tag{3.1}$$

Each person sends one message Y_i, determined by their bit x_i and private randomness. Truthful reports reveal p exactly but expose every bit. Fair-coin reports hide every bit but have the same distribution, $\frac{1}{n}\mathrm{Bin}(n, \frac{1}{2})$, for every dataset and therefore reveal nothing about p. Randomized response trades accuracy for privacy between these extremes.

Definition 3.1 (Randomized response)**.** For a bias $\kappa \in [0, 1/2]$, which sets the probability $\frac{1}{2} + \kappa$ of reporting the truth, person i reports

$$Y_i = \begin{cases} x_i & \text{with probability } \frac{1}{2} + \kappa, \\ 1 - x_i & \text{with probability } \frac{1}{2} - \kappa, \end{cases} \tag{3.2}$$

with fresh, independent coins for each person.

At $\kappa = 1/2$, reports are truthful; at $\kappa = 0$, they are fair coins. For $\kappa = 1/4$, a report of 1 is consistent with a true bit of 0, because the coin flips the bit with probability $1/4$. This ambiguity increases as κ decreases.

The reports are biased toward the true bits, so their average does not estimate p directly. The analyst can correct for the bias because the expected report is linear in x_i:

$$\mathbb{E}[Y_i] = \left(\tfrac{1}{2} + \kappa\right) x_i + \left(\tfrac{1}{2} - \kappa\right)(1 - x_i) = 2\kappa x_i + \tfrac{1}{2} - \kappa, \tag{3.3}$$

Solving for x_i gives an unbiased estimate of each bit; averaging these estimates gives an unbiased estimate of p.

Definition 3.2 (Debiased estimator)**.** For $\kappa > 0$, the analyst outputs

$$\tilde{p} = \frac{1}{n} \sum_{i=1}^{n} \frac{Y_i - \frac{1}{2} + \kappa}{2\kappa}. \tag{3.4}$$

The next theorem establishes unbiasedness and computes the estimator's variance.

Theorem 3.1. *For every $\kappa \in (0, 1/2]$ and every input $D \in \{0,1\}^n$, the estimator is unbiased, $\mathbb{E}[\tilde{p}] = p$, and its variance is*

$$\mathrm{Var}(\tilde{p}) = \frac{1 - 4\kappa^2}{16\kappa^2 n} \leq \frac{1}{16\kappa^2 n}. \tag{3.5}$$

Proof.

1. Each term $Z_i = (Y_i - \frac{1}{2} + \kappa)/(2\kappa)$ has mean x_i, by the formula for $\mathbb{E}[Y_i]$ above. Averaging over i and using linearity of expectation gives $\mathbb{E}[\tilde{p}] = p$.
2. The constants $\frac{1}{2} - \kappa$ and 2κ do not change the variance except through the scaling, so $\mathrm{Var}(Z_i) = \mathrm{Var}(Y_i)/(4\kappa^2)$.
3. The report Y_i is a Bernoulli variable that equals 1 with probability $\frac{1}{2} + \kappa$ or $\frac{1}{2} - \kappa$, depending on x_i. In both cases its variance is $(\frac{1}{2} + \kappa)(\frac{1}{2} - \kappa) = \frac{1}{4} - \kappa^2$.
4. The reports are independent because each person uses their own coins. So the variance of the sum is the sum of the variances, and

$$\mathrm{Var}(\tilde{p}) = \frac{1}{n^2} \sum_{i=1}^{n} \frac{\mathrm{Var}(Y_i)}{4\kappa^2} = \frac{\frac{1}{4} - \kappa^2}{4\kappa^2 n} = \frac{1 - 4\kappa^2}{16\kappa^2 n}. \tag{3.6}$$

□

Chebyshev's inequality gives $|\tilde{p} - p| \leq 1/(2\kappa\sqrt{n})$ with probability at least $3/4$. Hoeffding's inequality gives a sharper bound for failure probability β, because each Z_i has a bounded range.

Theorem 3.2. *For every $\kappa \in (0, 1/2]$ and $\beta \in (0, 1)$, with probability at least $1 - \beta$ over the coins,*

$$|\tilde{p} - p| \leq \frac{1}{2\kappa} \sqrt{\frac{\ln(2/\beta)}{2n}}. \tag{3.7}$$

Proof.

1. The term Z_i takes the values $(\frac{1}{2} + \kappa)/(2\kappa)$ and $-(\frac{1}{2} - \kappa)/(2\kappa)$, so it lies in an interval of length $1/(2\kappa)$.
2. The Z_i are independent with mean x_i. Hoeffding's inequality for the average of n independent variables, each in an interval of length ℓ, gives

$$\Pr\left[\, |\tilde{p} - p| \geq t \,\right] \leq 2\exp\left(-\frac{2nt^2}{\ell^2}\right) = 2\exp(-8\kappa^2 n t^2)\,. \tag{3.8}$$

3. Setting the right-hand side equal to β and solving for t gives Equation 3.7.

□

Thus the error decreases as $1/(\kappa\sqrt{n})$. Equivalently, accuracy α with failure probability at most β requires

$$n \geq \frac{\ln(2/\beta)}{8\kappa^2\alpha^2} \tag{3.9}$$

people. A smaller κ hides each bit better and needs more people for the same accuracy. To say how much better it hides a bit, the next section gives a definition.

3.2. Differential privacy

In the **central model**, each person sends their record to a **trusted curator**. The curator runs an algorithm M on $D = (x_1, \ldots, x_n) \in \mathcal{X}^n$ and publishes its output, which an adversary may observe. The algorithm, called a **mechanism**, must ensure that changing one person's record has a limited effect on the output distribution. This requires comparing datasets that differ in one record.

Definition 3.3 (Neighbouring datasets)**.** Two datasets $D, D' \in \mathcal{X}^n$ are **neighbours**, written $D \sim D'$, if they differ in exactly one record. This is the **replace-one** relation.

The definition bounds how much the output distribution can change between neighbours. For any event, its probability can increase by at most a factor of e^{ε}.

Definition 3.4 (Differential privacy)**.** Let $\varepsilon \geq 0$. A randomized algorithm $M \colon \mathcal{X}^n \to \mathcal{Y}$, which maps a dataset to an output in a set $\mathcal{Y}$, is **ε-differentially private** (ε-DP) if for all neighbours $D \sim D'$ and all events $S \subseteq \mathcal{Y}$,

$$\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S]. \tag{3.10}$$

The probability is over the coins of M only. The datasets are fixed.

This guarantee is also called **pure** differential privacy, distinguishing it from the relaxation in a later chapter. Four properties are useful.

First, the guarantee is quantitative. At $\varepsilon = 0$, the output distribution is identical on every dataset; this gives perfect privacy but no data-dependent accuracy. A deterministic algorithm is ∞-DP and provides no finite guarantee. Typical ε values range from about 0.1 to 5; smaller values give stronger privacy. Since the bound is e^{ε}, it weakens rapidly when $\varepsilon > 1$.

Second, it is a worst-case guarantee over every neighbouring pair, including unlikely datasets. It assumes no probability distribution over datasets.

Third, it is symmetric. Swapping D and D' in Equation 3.10 gives the lower bound $\Pr[M(D) \in S] \geq e^{-\varepsilon} \Pr[M(D') \in S]$.

Fourth, it bounds a probability ratio rather than a difference. For small ε, e^{ε} is close to $1+\varepsilon$. Ratios multiply across steps, so $e^{\varepsilon_1}e^{\varepsilon_2} = e^{\varepsilon_1+\varepsilon_2}$; the next chapter uses this fact for group privacy. A small additive distance, such as total variation, does not provide the same guarantee.

This replace-one relation defines **bounded** DP. The **add/remove**, or **unbounded**, relation instead adds or removes one record. A replacement takes two such changes, so group privacy from the next chapter gives 2ε-DP under replace-one neighbours from an ε-DP add/remove guarantee. The converse may fail because the dataset size is private under add/remove adjacency. This course uses replace-one adjacency. Compare privacy guarantees only when they use the same relation.

For discrete outputs it is enough to check single outcomes, because the probability of an event is the sum of the probabilities of its outcomes.

Lemma 3.1. *Let $\mathcal{Y}$ be countable. If $\Pr[M(D) = y] \le e^{\varepsilon}\Pr[M(D') = y]$ for all neighbours $D \sim D'$ and all outputs $y \in \mathcal{Y}$, then M is ε-DP.*

Proof. For an event S, sum the pointwise inequality over the outputs $y \in S$:

$$\Pr[M(D) \in S] = \sum_{y\in S}\Pr[M(D) = y] \le e^{\varepsilon}\sum_{y\in S}\Pr[M(D') = y] = e^{\varepsilon}\Pr[M(D') \in S]. \tag{3.11}$$

□

A finite privacy guarantee requires randomization.

Proposition 3.1. *For $\varepsilon < \infty$, a deterministic ε-DP algorithm is constant.*

Proof.

1. Let M be deterministic and $D \sim D'$. Take the event $S = \{M(D)\}$. The definition gives $1 = \Pr[M(D) \in S] \le e^{\varepsilon}\Pr[M(D') \in S]$, so $\Pr[M(D') = M(D)] > 0$. Since $M(D')$ is a single value, $M(D') = M(D)$.
2. Any two datasets in $\mathcal{X}^n$ are joined by a path of at most n neighbours, changing one record at a time. By step 1, M takes the same value at every step of the path, so it is constant.

□

3.2.1. Hypothesis testing

The definition also bounds an adversary's ability to distinguish neighbouring datasets. Suppose the adversary knows that the dataset is either D or D' and observes the output of M. It tests the **null hypothesis** against the **alternative**:

$$H_0 : \text{the dataset is } D, \qquad H_1 : \text{the dataset is } D'. \tag{3.12}$$

A test is a function $\phi\colon \mathcal{Y} \to [0,1]$ giving the probability of choosing H_1 after observing an output. Its **false positive rate** FPR is the probability of choosing H_1 when H_0 is true; its **false negative rate** FNR is the probability of choosing H_0 when H_1 is true:

$$\mathrm{FPR} = \mathbb{E}[\phi(M(D))], \qquad \mathrm{FNR} = 1 - \mathbb{E}[\phi(M(D'))]. \tag{3.13}$$

The definition bounds event probabilities, but a test can take any value in $[0, 1]$. The following lemma extends the bound to these functions.

Lemma 3.2. *Let M be ε-DP and let $h \colon \mathcal{Y} \to [0, 1]$. For all neighbours $D \sim D'$,*

$$\mathbb{E}[h(M(D))] \leq e^{\varepsilon}\, \mathbb{E}[h(M(D'))]. \tag{3.14}$$

Proof.

1. A variable with values in $[0, 1]$ has mean equal to the integral of its tail, so $\mathbb{E}[h(M(D))] = \int_0^1 \Pr[h(M(D)) > t]\, \mathrm{d}t$.
2. Each tail is the probability of the event $\{y : h(y) > t\}$, which Equation 3.10 bounds by $e^{\varepsilon} \Pr[h(M(D')) > t]$.
3. Integrating over $t \in [0, 1]$ gives $e^{\varepsilon}\, \mathbb{E}[h(M(D'))]$.

□

An adversary wants both error rates to be small. The next theorem characterizes differential privacy by a lower bound on their joint values for every test.

Theorem 3.3. *A mechanism M is ε-DP if and only if, for every pair of neighbours and every test, the false positive rate* FPR *and the false negative rate* FNR *satisfy*

$$\mathrm{FPR} + e^{\varepsilon}\mathrm{FNR} \geq 1 \qquad \textit{and} \qquad e^{\varepsilon}\mathrm{FPR} + \mathrm{FNR} \geq 1. \tag{3.15}$$

Proof.

1. *DP extends from events to tests.* Let M be ε-DP and let ϕ take values in $[0, 1]$. Then $\mathbb{E}[\phi(M(D))] \leq e^{\varepsilon}\, \mathbb{E}[\phi(M(D'))]$, and the same holds with D and D' swapped. This is Lemma 3.2, applied to ϕ in each direction.
2. *DP implies the inequalities.* Apply step 1 to ϕ, from D' to D: $1 - \mathrm{FNR} = \mathbb{E}[\phi(M(D'))] \leq e^{\varepsilon}\, \mathbb{E}[\phi(M(D))] = e^{\varepsilon}\mathrm{FPR}$. This is the second inequality. Apply step 1 to $1 - \phi$, from D to D': $1 - \mathrm{FPR} = \mathbb{E}[1 - \phi(M(D))] \leq e^{\varepsilon}\, \mathbb{E}[1 - \phi(M(D'))] = e^{\varepsilon}\mathrm{FNR}$. This is the first.
3. *The inequalities imply DP.* Take an event S and the test $\phi(y) = \mathbf{1}[y \in S]$, which answers H_1 exactly when the output lands in S. Its error rates are $\mathrm{FPR} = \Pr[M(D) \in S]$ and $\mathrm{FNR} = 1 - \Pr[M(D') \in S]$. The second inequality says $e^{\varepsilon} \Pr[M(D) \in S] + 1 - \Pr[M(D') \in S] \geq 1$, that is, $\Pr[M(D') \in S] \leq e^{\varepsilon} \Pr[M(D) \in S]$. The pair of neighbours was arbitrary, so the same holds with the roles of D and D' swapped, which is Equation 3.10.

□

At $\varepsilon = 0$, the inequalities require $\mathrm{FPR} + \mathrm{FNR} \geq 1$, achieved by a test that ignores the output, such as a fair random guess. As ε increases, lower error pairs become possible and the adversary can distinguish the datasets more accurately.

3.2.2. What the definition promises

The probability of any event changes by at most a factor e^{ε} when one person's record is replaced. Thus participation cannot substantially increase the probability of an output-caused harm. For example, a linkage attack succeeds at nearly the same rate when that person's record is absent, regardless of auxiliary information. At $\varepsilon = 0.1$, the probability can increase by at most $e^{0.1} \approx 1.105$. This information-theoretic guarantee holds even against an adversary with unlimited computation and side information.

The definition does not prevent inference from population patterns. If a private study finds an association between smoking and cancer, an insurer may raise a participant's premium. The study would reach the same conclusion without that person's record; the inference concerns the population, not participation. Statistical analysis is intended to reveal such population-level facts.

3.3. Randomized response revisited

The next theorem gives randomized response a privacy guarantee. It treats the report vector as the output; the estimator Equation 3.4 inherits the same guarantee by post-processing, proved in the next chapter.

Theorem 3.4. *Randomized response with bias $\kappa \in [0, 1/2)$, releasing $(Y_1, \ldots, Y_n)$, is ε-DP for*

$$\varepsilon = \ln \frac{\frac{1}{2} + \kappa}{\frac{1}{2} - \kappa}. \tag{3.16}$$

Equivalently, to obtain ε-DP, report the truth with probability $\frac{1}{2} + \kappa = e^{\varepsilon}/(1 + e^{\varepsilon})$.

Proof.

1. Fix an output $y \in \{0, 1\}^n$. The coins are independent across people, so $\Pr[M(D) = y] = \prod_{i=1}^{n} \Pr[Y_i = y_i]$, where each factor depends only on x_i.
2. Let D and D' differ only in record j, and write Y_j' for the report of person j on D'. All factors with $i \neq j$ are equal for the two datasets and cancel in the ratio:

$$\frac{\Pr[M(D) = y]}{\Pr[M(D') = y]} = \frac{\Pr[Y_j = y_j]}{\Pr[Y_j' = y_j]}. \tag{3.17}$$

3. Each probability on the right is $\frac{1}{2} + \kappa$ or $\frac{1}{2} - \kappa$, so the ratio is at most $(\frac{1}{2} + \kappa)/(\frac{1}{2} - \kappa) = e^{\varepsilon}$.
4. Lemma 3.1 turns this pointwise bound into the definition.
5. Solving $e^{\varepsilon} = (\frac{1}{2} + \kappa)/(\frac{1}{2} - \kappa)$ for $\frac{1}{2} + \kappa$ gives $e^{\varepsilon}/(1 + e^{\varepsilon})$.

□

For small κ the privacy parameter is $\varepsilon = 4\kappa + O(\kappa^3)$, so a bias of order ε gives ε-DP. Exactly, the bias that gives ε-DP is

$$2\kappa = \frac{e^{\varepsilon} - 1}{e^{\varepsilon} + 1} = \tanh \frac{\varepsilon}{2}, \tag{3.18}$$

so Theorem 3.2 and Theorem 3.1 become, with probability at least $1 - \beta$,

$$|\tilde{p} - p| \le \coth\!\left(\frac{\varepsilon}{2}\right) \sqrt{\frac{\ln(2/\beta)}{2n}}, \qquad \operatorname{Var}(\tilde{p}) = \frac{e^{\varepsilon}}{(e^{\varepsilon} - 1)^2\, n}. \tag{3.19}$$

Here p is again the fraction of ones and $\tilde{p}$ its estimate. The factor $\coth(\varepsilon/2)$ lies between $2/\varepsilon$ and $2/\varepsilon + \varepsilon/6$ for every $\varepsilon > 0$, so the error is $O(1/(\varepsilon\sqrt{n}))$ for $\varepsilon \le 1$. Figure 3.1 compares this bound with simulation, and Example 3.1 gives the sizes at a typical privacy level.

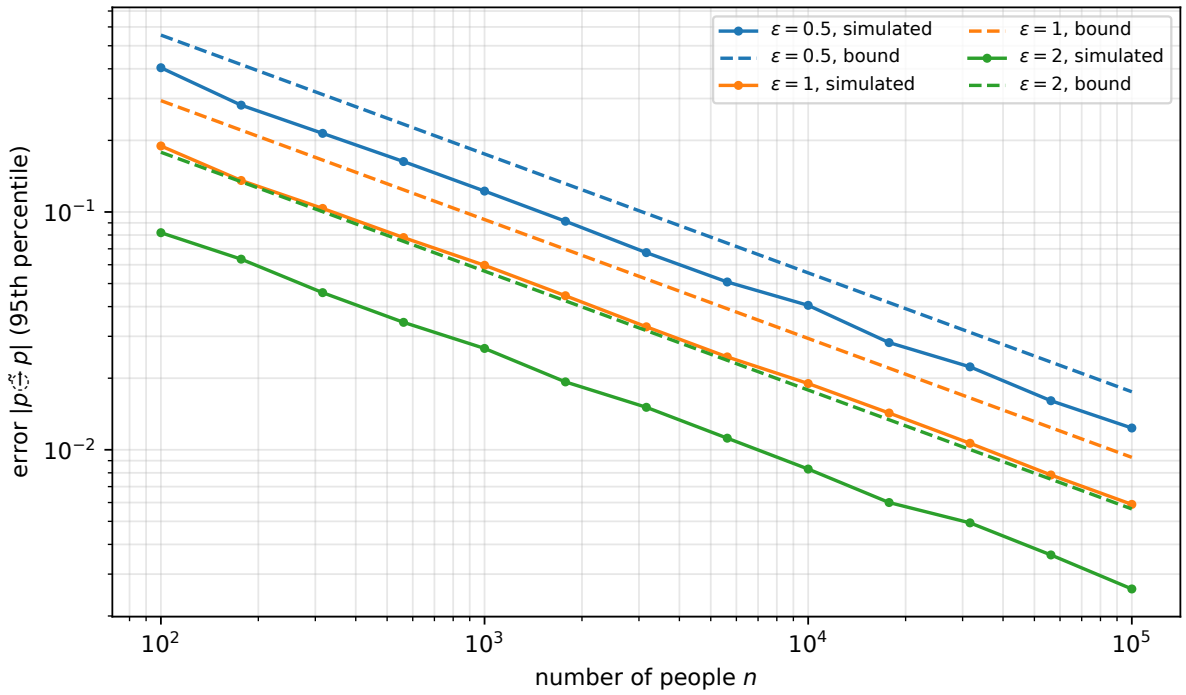


Figure 3.1.: Error of randomized response. For each ε, the solid line is the 95th percentile of $|\tilde{p} - p|$ over 2,000 simulated runs with $p = 0.3$, and the dashed line is the bound of Equation 3.19 with $\beta = 0.05$. Both fall as $1/\sqrt{n}$. The bound is 1.4 to 2.2 times the simulated error, and it is loosest for large ε, because Hoeffding's inequality uses only the range of each term and not its smaller variance.

Example 3.1. With $\varepsilon = 1$, each person tells the truth with probability $e/(1 + e) = 0.7311$. With $n = 10{,}000$ and $\beta = 0.05$, the bound Equation 3.19 is $\coth(0.5)\sqrt{\ln 40/20{,}000} = 2.1640 \times 0.01358 = 0.0294$. The standard deviation from Equation 3.19 is $\sqrt{e/((e-1)^2 \cdot 10{,}000)} = 0.0096$.

Randomized response gives more than the definition asks for. Each person randomizes their own bit before sending it, so no one ever sees a true bit and the curator needs no trust. This is **local differential privacy**, the subject of a later chapter, and it implies the central guarantee proved here. The price is accuracy, because the $\sqrt{n}$ in the error is not necessary under central privacy: the next chapter's Laplace mechanism estimates p with error $O(1/(\varepsilon n))$.

3.4. Summary

Table 3.1.: The chapter's results.

Result	Statement
Theorem 3.2	randomized response with bias κ has error $\frac{1}{2\kappa}\sqrt{\ln(2/\beta)/(2n)}$
Definition 3.4	$\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S]$ for all neighbours and events
Lemma 3.1	for discrete outputs, single outcomes suffice
Proposition 3.1	a deterministic private algorithm is constant
Lemma 3.2	the bound extends from events to functions with values in $[0, 1]$
Theorem 3.3	ε-DP holds exactly when every test has $\mathrm{FPR} + e^{\varepsilon}\mathrm{FNR} \ge 1$ and $e^{\varepsilon}\mathrm{FPR} + \mathrm{FNR} \ge 1$
Theorem 3.4	truth with probability $e^{\varepsilon}/(1 + e^{\varepsilon})$ gives ε-DP
Equation 3.19	error $\coth(\varepsilon/2)\sqrt{\ln(2/\beta)/(2n)} = O(1/(\varepsilon\sqrt{n}))$

3.5. References

- Stanley L. Warner, "Randomized Response: A Survey Technique for Eliminating Evasive Answer Bias", 1965. Gives the original protocol, for surveys on sensitive questions.
- Cynthia Dwork, Frank McSherry, Kobbi Nissim and Adam Smith, "Calibrating Noise to Sensitivity in Private Data Analysis", 2006. Defines differential privacy.
- Larry Wasserman and Shuheng Zhou, "A Statistical Framework for Differential Privacy", 2010. Gives the hypothesis-testing reading of the definition.
- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014. Explains the definition and its meaning, with the smoking example.
- Salil Vadhan, "The Complexity of Differential Privacy", 2017. Shows in Section 1.6 why total variation distance gives no meaningful guarantee.
- Thomas Steinke and Jonathan Ullman, "The Pitfalls of Average-Case Differential Privacy", 2020, a post on the DifferentialPrivacy.org blog. Explains why the worst case over datasets matters.

4. The Laplace mechanism

Randomized response privatizes one bit per person. Many analyses instead compute a statistic of the full dataset, such as a count or histogram. The Laplace mechanism adds noise proportional to the maximum change one record can cause. For the fraction of ones, its error is of order $1/(\varepsilon n)$, compared with $1/(\varepsilon\sqrt{n})$ for randomized response. This chapter also proves three properties of differential privacy (DP): post-processing, group privacy, and basic composition.

4.1. Sensitivity

The output must hide the contribution of any one person, so the noise must be at least as large as the change that one person can cause in the answer. The sensitivity measures the largest such change. Recall from the randomized response chapter that two datasets are neighbours, written $D \sim D'$, when they differ in exactly one record.

Definition 4.1 ($\ell 1$ sensitivity)**.** Let $f\colon \mathcal{X}^n \to \mathbb{R}^k$ map a dataset of n records to a vector of d numbers. The ℓ_1 **sensitivity** of f is the largest change in f, measured by the sum of the absolute changes of its coordinates, between neighbours $D \sim D'$:

$$\Delta_1 f = \sup_{D \sim D'} \|f(D) - f(D')\|_1, \tag{4.1}$$

where $\|v\|_1 = \sum_{j=1}^{d} |v_j|$ for a vector $v \in \mathbb{R}^d$.

The ℓ_1 norm determines the privacy proof of Theorem 4.1. Other mechanisms use other norms; the Gaussian mechanism in the next chapter uses ℓ_2. In one dimension, both norms are absolute value. In d dimensions, the sensitivities can differ by up to a factor of $\sqrt{d}$.

The running example is the fraction of ones from the randomized response chapter. Replacing one bit changes the fraction by at most $1/n$.

Example 4.1. Let $\mathcal{X} = \{0, 1\}$ and let $f(D) = \frac{1}{n}\sum_{i=1}^{n} x_i$ be the fraction of ones. Changing one bit changes the sum by at most 1, so $\Delta_1 f = 1/n$.

4.2. The Laplace distribution

The noise must mask shifts of the true answer up to its sensitivity. The Laplace density decays exponentially with distance from zero, so a bounded shift changes its value by at most a bounded factor.

Definition 4.2 (Laplace distribution)**.** The **Laplace distribution** $\mathrm{Lap}(b)$ with scale $b > 0$ has density

$$\frac{1}{2b}\exp\left(-\frac{|x|}{b}\right), \qquad x \in \mathbb{R}. \tag{4.2}$$

The Laplace distribution is a two-sided exponential distribution with tails proportional to $e^{-|x|/b}$. These tails decay more slowly than Gaussian tails. Its mean, variance, and tail probability determine the mechanism's accuracy.

Lemma 4.1. *Let $Z \sim \mathrm{Lap}(b)$. Then $\mathbb{E}[Z] = 0$, $\mathrm{Var}(Z) = 2b^2$, and for every $t \geq 0$,*

$$\Pr[|Z| \geq tb] = e^{-t}. \tag{4.3}$$

Proof.

1. The density is symmetric about 0, so the mean is 0.
2. The absolute value $|Z|$ has density $\frac{1}{b}e^{-u/b}$ on $[0, \infty)$, the exponential distribution with mean b. Its second moment is $\int_0^\infty u^2 \frac{1}{b}e^{-u/b}\,\mathrm{d}u = 2b^2$, by integrating by parts twice. Since the mean of Z is 0, $\mathrm{Var}(Z) = \mathbb{E}[Z^2] = 2b^2$.
3. The tail of $|Z|$ is $\int_{tb}^\infty \frac{1}{b}e^{-u/b}\,\mathrm{d}u = e^{-t}$.

□

4.3. The Laplace mechanism

The mechanism adds independent Laplace noise to each coordinate, with scale $\Delta_1 f/\varepsilon$.

Definition 4.3 (Laplace mechanism)**.** For $f\colon \mathcal{X}^n \to \mathbb{R}^d$ with $0 < \Delta_1 f < \infty$ and for $\varepsilon > 0$, the Laplace mechanism outputs

$$M(D) = f(D) + (Z_1, \ldots, Z_d), \tag{4.4}$$

where $Z_1, \ldots, Z_d$ are independent with distribution $\mathrm{Lap}(\Delta_1 f/\varepsilon)$.

On the running example, the mechanism is unbiased. Lemma 4.1 gives its variance and tail bound.

Example 4.2. For $f(D) = p$, the fraction of ones, Example 4.1 gives $\Delta_1 f = 1/n$. The mechanism outputs $\tilde{p} = p + Z$ with $Z \sim \mathrm{Lap}(1/(\varepsilon n))$. So $\mathbb{E}[\tilde{p}] = p$, and $\mathrm{Var}(\tilde{p}) = 2/(\varepsilon n)^2$. By Equation 4.3 with $t = \ln(1/\beta)$, with probability at least $1 - \beta$,

$$|\tilde{p} - p| \leq \frac{\ln(1/\beta)}{\varepsilon n}. \tag{4.5}$$

With $\varepsilon = 1$, $n = 10{,}000$ and $\beta = 0.05$, this is $\ln(20)/10{,}000 = 3.0 \times 10^{-4}$. Randomized response with the same ε, n and β has the bound 0.0294, which is 98 times larger.

The accuracy gap grows with n: randomized response has error of order $1/(\varepsilon\sqrt{n})$, while the Laplace mechanism has error of order $1/(\varepsilon n)$. The latter requires a trusted curator who sees the data. The next theorem proves its privacy guarantee.

Theorem 4.1 (Privacy of the Laplace mechanism)**.** *For every $f\colon \mathcal{X}^n \to \mathbb{R}^d$ with $0 < \Delta_1 f < \infty$ and every $\varepsilon > 0$, the Laplace mechanism is ε-DP.*

Proof.

1. Fix neighbours $D \sim D'$, and write p and q for the densities of $M(D)$ and $M(D')$ on $\mathbb{R}^d$. In this proof p is a density, not the fraction of ones. Let $b = \Delta_1 f/\varepsilon$. The coordinates are independent, so at a point $y \in \mathbb{R}^d$,

$$\frac{p(y)}{q(y)} = \prod_{j=1}^{d} \frac{\exp(-|f(D)_j - y_j|/b)}{\exp(-|f(D')_j - y_j|/b)} = \exp\left(\frac{\varepsilon}{\Delta_1 f} \sum_{j=1}^{d} \left(|f(D')_j - y_j| - |f(D)_j - y_j| \right) \right). \tag{4.6}$$

2. By the triangle inequality, $|f(D')_j - y_j| - |f(D)_j - y_j| \leq |f(D')_j - f(D)_j|$ for each j. Summing over j gives at most $\|f(D) - f(D')\|_1$, which is at most $\Delta_1 f$ by Definition 4.1. So $p(y) \leq e^{\varepsilon} q(y)$ for every y.
3. For an event $S \subseteq \mathbb{R}^d$, integrate the pointwise bound over S: $\Pr[M(D) \in S] = \int_S p \leq e^{\varepsilon} \int_S q = e^{\varepsilon} \Pr[M(D') \in S]$.

□

The proof uses the matching exponential forms of the Laplace density and the privacy ratio. Shifting the density centre by at most the sensitivity changes its value by a factor of at most e^{ε}. Figure 4.1 shows the densities for a one-dimensional statistic.

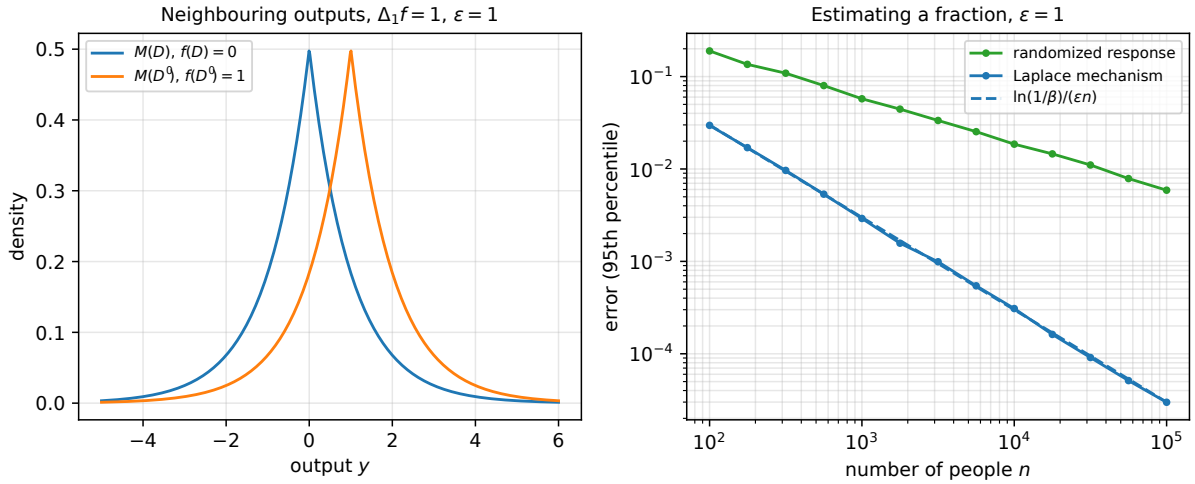


Figure 4.1.: Left, the densities of the Laplace mechanism at $\varepsilon = 1$ on two neighbours whose values differ by the sensitivity $\Delta_1 f = 1$. Their ratio stays between e^{-1} and e. Right, the 95th percentile of the error in estimating a fraction of ones, over 2,000 simulated runs with $p = 0.3$. Randomized response falls as $1/\sqrt{n}$ and the Laplace mechanism as $1/n$. The dashed line is the exact 95th percentile $\ln(20)/(\varepsilon n)$ from Equation 4.3.

4.4. Counting queries

A **counting query** asks how many records satisfy a predicate $\phi\colon \mathcal{X} \to \{0,1\}$. Its value is $f(D) = \sum_{i=1}^{n} \phi(x_i)$. Replacing one record changes the count by at most 1, so $\Delta_1 f = 1$ and the mechanism adds $\mathrm{Lap}(1/\varepsilon)$. By Equation 4.3, the error is at most $\ln(1/\beta)/\varepsilon$ with probability $1-\beta$, independent of n.

Often an analyst asks many counting queries $f = (f_1, \dots, f_k)$, with predicates $\phi_1, \dots, \phi_k$, on the same data, such as whether a person smokes and whether their income is above a threshold. One person can then change all of the answers at once.

Proposition 4.1. *The vector of k counting queries has ℓ_1 sensitivity at most k, with equality when some record satisfies every predicate and another satisfies none. For every $\beta \in (0,1)$, with probability at least $1-\beta$, the Laplace mechanism answers all k queries with error at most*

$$\frac{k \ln(k/\beta)}{\varepsilon}. \tag{4.7}$$

Proof.

1. Let D and D' differ in one record, x in D and x' in D'. In each query the other terms cancel, so $\|f(D) - f(D')\|_1 = \sum_{j=1}^{k} |\phi_j(x) - \phi_j(x')| \le k$. The sum equals k when $\phi_j(x) = 1$ and $\phi_j(x') = 0$ for all j.
2. The noise in each answer is $\mathrm{Lap}(k/\varepsilon)$. By Equation 4.3 with $t = \ln(k/\beta)$, each noise is at least $k \ln(k/\beta)/\varepsilon$ in absolute value with probability β/k.
3. The union bound over the k answers gives a failure probability of at most β.

□

This analysis needs all k queries to be fixed in advance. The composition theorem of Section 4.6 gives the same privacy when each query is chosen after seeing the earlier answers.

Compare this error of order k/ε per query with the reconstruction chapter, where about n counting queries answered with error $o(\sqrt{n})$ let an attacker recover almost the whole dataset. Here, n counting queries get error of order n/ε, up to a logarithm. The wide gap between $\sqrt{n}$ and n is real slack: advanced composition, under a relaxation of the definition, lowers the error to about $\sqrt{k}/\varepsilon$ per query, up to logarithmic factors.

4.5. Histograms

When the queries have structure, a sharper sensitivity bound is possible. In a **histogram**, each record falls into exactly one of k bins, such as an age in whole years, and the query counts the records in each bin: $f_j(D) = |\{i : x_i \text{ is in bin } j\}|$.

Proposition 4.2. *A histogram with $k \geq 2$ bins has ℓ_1 sensitivity 2. For every $\beta \in (0, 1)$, with probability at least $1 - \beta$, the Laplace mechanism, which adds $\mathrm{Lap}(2/\varepsilon)$ to each bin, answers every bin with error at most*

$$\frac{2\ln(k/\beta)}{\varepsilon}. \tag{4.8}$$

Proof.

1. Let D and D' differ in one record, which lies in bin a in D and in bin a' in D'. If $a = a'$ the histograms are equal. Otherwise count a drops by 1 and count a' rises by 1, so $f(D) - f(D') = e_a - e_{a'}$, where e_j is the j-th unit vector, and $\|e_a - e_{a'}\|_1 = 2$.
2. By Equation 4.3 with $t = \ln(k/\beta)$, each of the k noises is at least $2\ln(k/\beta)/\varepsilon$ in absolute value with probability β/k. The union bound over the bins finishes the proof.

□

The sensitivity is 2 because a replacement moves a record from one bin to another. A single bin has error at most $2\ln(1/\beta)/\varepsilon$ with probability $1 - \beta$, and the proposition pays only $\ln k$ more to control all bins at once. So private histograms over very large domains are practical. For $k = 100$ bins, $\varepsilon = 1$ and $\beta = 0.05$, the bound is $2\ln(2000) = 15.20$, and in a simulation of 20,000 runs the 95th percentile of the largest error is 15.17. The same 100 bins treated as arbitrary counting queries would get the bound $100\ln(2000) = 760.1$ from Proposition 4.1.

4.6. Properties of differential privacy

Three properties support analysis of private algorithms built from multiple steps.

4.6.1. Post-processing

Post-processing cannot weaken privacy if it uses only the private output and independent randomness. The randomized response chapter used this property to estimate the fraction of ones.

Theorem 4.2 (Post-processing)**.** *Let $M\colon \mathcal{X}^n \to \mathcal{Y}$ be ε-DP, and let $F\colon \mathcal{Y} \to \mathcal{Z}$ be a randomized map whose coins are independent of those of M. Then $F \circ M$ is ε-DP.*

Proof.

1. A randomized map is a random choice of a deterministic one. Write $F(y) = g_R(y)$, where R is the coins of F and each $g_r\colon \mathcal{Y} \to \mathcal{Z}$ is deterministic.
2. Fix neighbours $D \sim D'$ and an event $S \subseteq \mathcal{Z}$. For fixed coins, $g_R(M(D)) \in S$ exactly when $M(D) \in g_R^{-1}(S)$, so

$$\Pr[F(M(D)) \in S] = \mathbb{E}_R\Big[\Pr\left[M(D) \in g_R^{-1}(S)\right]\Big]. \tag{4.9}$$

3. Inside the expectation, apply the definition of ε-DP to the event $g_R^{-1}(S)$, then undo step 2 for D':

$$\mathbb{E}_R\Big[\Pr\left[M(D) \in g_R^{-1}(S)\right]\Big] \le e^{\varepsilon}\,\mathbb{E}_R\Big[\Pr\left[M(D') \in g_R^{-1}(S)\right]\Big] = e^{\varepsilon}\Pr[F(M(D')) \in S]. \tag{4.10}$$

□

4.6.2. Group privacy

The definition protects one record. Group privacy extends the guarantee to datasets that differ in several records, with a cost proportional to their number.

Theorem 4.3 (Group privacy)**.** *Let $M\colon \mathcal{X}^n \to \mathcal{Y}$ be ε-DP, and let $D, D' \in \mathcal{X}^n$ differ in k records. For every event $S \subseteq \mathcal{Y}$,*

$$\Pr[M(D) \in S] \le e^{k\varepsilon}\Pr[M(D') \in S]. \tag{4.11}$$

Proof.

1. The proof is a **hybrid argument**. Change the k differing records one at a time, from their values in D to their values in D'. This gives datasets $D = D^{(0)}, D^{(1)}, \ldots, D^{(k)} = D'$ in which each consecutive pair are neighbours.
2. Apply the definition to each consecutive pair, from the first to the last:

$$\Pr[M(D^{(0)}) \in S] \le e^{\varepsilon}\Pr[M(D^{(1)}) \in S] \le e^{2\varepsilon}\Pr[M(D^{(2)}) \in S] \le \ldots \le e^{k\varepsilon}\Pr[M(D^{(k)}) \in S]. \tag{4.12}$$

□

Thus, a group of k records has a $k\varepsilon$ guarantee, which is informative only when $k\varepsilon$ is small. The theorem also relates the adjacency conventions from the randomized response chapter: replacement is one removal and one addition, so ε-DP under add/remove adjacency implies 2ε-DP under replace-one adjacency.

4.6.3. Basic composition

Analysts often run several mechanisms on the same dataset, choosing later mechanisms from earlier outputs.

Definition 4.4 (Adaptive composition)**.** Let $M_1, \dots, M_k$ be mechanisms. For each i and each sequence of earlier outputs $y_{<i} = (y_1, \dots, y_{i-1})$, the mechanism $M_i(\cdot\,; y_{<i})$ takes the dataset D and returns y_i. Their **adaptive composition** M runs them in order on the same D, each with fresh coins, and releases $(y_1, \dots, y_k)$.

Basic composition adds the privacy parameters, even when each mechanism depends on earlier outputs.

Theorem 4.4 (Basic composition)**.** *If each $M_i(\cdot\,; y_{<i})$ is ε_i-DP for every $y_{<i}$, then the adaptive composition of $M_1, \dots, M_k$ is $(\sum_{i=1}^{k} \varepsilon_i)$-DP. In particular, k mechanisms that are each ε-DP compose to a $k\varepsilon$-DP mechanism.*

For countable output spaces, the probability of an output sequence factors into conditional probabilities.

Proof. Countable outputs.

1. Fix neighbours $D \sim D'$ and outputs $y = (y_1, \dots, y_k)$. Each M_i uses fresh coins, so the chain rule gives $\Pr[M(D) = y] = \prod_{i=1}^{k} \Pr[M_i(D; y_{<i}) = y_i]$, and likewise for D'.
2. Each factor for D is at most e^{ε_i} times the matching factor for D', by the privacy of $M_i(\cdot\,; y_{<i})$ applied to the event $\{y_i\}$. Multiplying the k bounds gives

$$\Pr[M(D) = y] = \prod_{i=1}^{k} \Pr[M_i(D; y_{<i}) = y_i] \le \prod_{i=1}^{k} e^{\varepsilon_i} \Pr[M_i(D'; y_{<i}) = y_i] = \exp\Big(\sum_{i=1}^{k} \varepsilon_i \Big) \Pr[M(D') = y]. \tag{4.13}$$

3. The pointwise lemma of the randomized response chapter turns this bound on single outcomes into the definition.

□

For uncountable outputs, such as real-valued Laplace releases, the proof uses conditional expectations instead of products. It relies on the following extension to functions h with values in $[0, 1]$: if M is ε-DP, then for all neighbours

$$\mathbb{E}[h(M(D))] \le e^{\varepsilon}\, \mathbb{E}[h(M(D'))]. \tag{4.14}$$

This is Lemma 3.2 of the randomized response chapter.

Proof. General outputs.

1. The composition of the first $k-1$ mechanisms is itself a mechanism, so by induction on k it suffices to treat $k = 2$.
2. Fix neighbours $D \sim D'$ and an event S of pairs. For an output y_1, write $S_{y_1} = \{y_2 : (y_1, y_2) \in S\}$ and $g_D(y_1) = \Pr[M_2(D; y_1) \in S_{y_1}]$, and define $g_{D'}$ in the same way. Conditioning on the first output gives $\Pr[M(D) \in S] = \mathbb{E}_{y_1 \sim M_1(D)}[g_D(y_1)]$, and likewise for D'.
3. By the privacy of $M_2(\cdot\,; y_1)$, $g_D(y_1) \le e^{\varepsilon_2} g_{D'}(y_1)$ for every y_1.
4. The function $g_{D'}$ takes values in $[0, 1]$, so the fact above, applied to M_1, gives

$$\Pr[M(D) \in S] \le e^{\varepsilon_2}\, \mathbb{E}_{y_1 \sim M_1(D)}[g_{D'}(y_1)] \le e^{\varepsilon_1+\varepsilon_2}\, \mathbb{E}_{y_1 \sim M_1(D')}[g_{D'}(y_1)] = e^{\varepsilon_1+\varepsilon_2} \Pr[M(D') \in S]. \tag{4.15}$$

□

Basic composition recovers the analysis of Proposition 4.1. Each counting query with $\mathrm{Lap}(k/\varepsilon)$ noise is (ε/k)-DP, and k of them compose to ε-DP, now even when the analyst chooses the queries adaptively. So the cost of k analyses grows linearly in k. It can grow like $\sqrt{k}$ instead, but only under the relaxation of the definition that the next chapter introduces.

4.7. Summary

Table 4.1.: The chapter's results.

Result	Statement
Definition 4.1	$\Delta_1 f$ is the largest ℓ_1 change in f between neighbours
Lemma 4.1	$\mathrm{Lap}(b)$ has variance $2b^2$ and $\Pr[\lvert Z\rvert \ge tb] = e^{-t}$
Theorem 4.1	adding $\mathrm{Lap}(\Delta_1 f/\varepsilon)$ to each coordinate is ε-DP
Example 4.2	a fraction of ones with error $\ln(1/\beta)/(\varepsilon n)$
Proposition 4.1	k counting queries: sensitivity k, error $k \ln(k/\beta)/\varepsilon$
Proposition 4.2	a histogram: sensitivity 2, error $2\ln(k/\beta)/\varepsilon$ over all bins

Result	Statement
Theorem 4.2	post-processing keeps ε-DP
Theorem 4.3	datasets differing in k records: factor $e^{k\varepsilon}$
Theorem 4.4	adaptive composition adds the ε_i

4.8. References

- Cynthia Dwork, Frank McSherry, Kobbi Nissim and Adam Smith, "Calibrating Noise to Sensitivity in Private Data Analysis", 2006. Defines sensitivity and the Laplace mechanism.
- Irit Dinur and Kobbi Nissim, "Revealing Information while Preserving Privacy", 2003. Gives the reconstruction attacks that set the scale for the error.
- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014. Covers the mechanism and its uses in Section 3.3, and the properties in Sections 2.3 and 3.5.
- Cynthia Dwork, Guy Rothblum and Salil Vadhan, "Boosting and Differential Privacy", 2010. Proves advanced composition, which improves the linear growth of basic composition.

5. Approximate differential privacy

Pure differential privacy (DP) bounds the ratio of output probabilities by e^{ε} for every event. This chapter adds a slack δ, defining (ε, δ)-DP. The examples show why δ must be much smaller than $1/n$. This relaxation permits the Gaussian mechanism, which can add less noise than the Laplace mechanism to high-dimensional statistics, and enables the advanced composition theorem of the next chapter.

5.1. The definition

An additive bound alone does not provide a useful privacy guarantee. Suppose it required only that, for all neighbours and events,

$$\Pr[M(D) \in S] \leq \Pr[M(D') \in S] + \delta. \tag{5.1}$$

When δ is small, this bound prevents useful accuracy; when δ is large, it permits harmful mechanisms (Vadhan, 2017, Section 1.6), as Section 5.3 shows. Combining it with the multiplicative bound of pure DP gives the required definition. Neighbours $D \sim D'$ differ in one record under the replace-one relation.

Definition 5.1 (Approximate differential privacy)**.** Let $\varepsilon \geq 0$ and $\delta \in [0, 1]$. A randomized algorithm $M \colon \mathcal{X}^n \to \mathcal{Y}$ is (ε, δ)**-differentially private** if for all neighbours $D \sim D'$ and all events $S \subseteq \mathcal{Y}$,

$$\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S] + \delta. \tag{5.2}$$

At $\delta = 0$, this is ε-DP from the randomized response chapter, also called **pure DP**. For $\delta > 0$, it is **approximate DP**.

5.2. The privacy loss random variable

Definition 5.1 bounds event probabilities. To analyse individual outcomes, fix neighbours $D \sim D'$ and compare the likelihood of an output y under each dataset. Write P and Q for the distributions of $M(D)$ and $M(D')$, and $p(y)$ and $q(y)$ for their probabilities or densities. In general, these densities exist with respect to the measure $P + Q$.

Privacy loss is the log-likelihood ratio $\ln(p(y)/q(y))$. If $p(y) > 0$ but $q(y) = 0$, then observing y rules out D', and the loss is $+\infty$.

Definition 5.2 (Privacy loss)**.** Let $p(y)$ and $q(y)$ be the probabilities, or densities, of an outcome y under $M(D)$ and under $M(D')$. The **privacy loss** of an outcome y with $p(y) > 0$ is

$$L_{D,D'}(y) = \ln \frac{p(y)}{q(y)}, \tag{5.3}$$

where $L_{D,D'}(y) = +\infty$ if $q(y) = 0$. Once the pair is fixed we write it $L(y)$. The **privacy loss random variable** is $L = L(Y)$ with $Y = M(D)$: the privacy loss of the outcome the mechanism actually produces on D.

Under $M(D)$, $p(M(D)) > 0$ with probability 1, so L is defined. Positive loss favours D over D'; negative loss favours D'. Infinite loss means the outcome is impossible under D'.

The Laplace mechanism of the previous chapter gives a first example. For a function $f\colon \mathcal{X}^n \to \mathbb{R}$ whose value changes by at most $\Delta_1 f$ between neighbours, it releases $f(D) + Z$, where Z has the Laplace density $\frac{1}{2b}e^{-|z|/b}$ with scale $b = \Delta_1 f/\varepsilon$, and it is ε-DP.

Example 5.1. Take neighbours with $f(D) = a$ and $f(D') = a - \Delta_1 f$, the largest gap allowed. The densities give

$$L(y) = \frac{\varepsilon}{\Delta_1 f}(|y - a + \Delta_1 f| - |y - a|), \tag{5.4}$$

which equals ε for $y \geq a$, equals $-\varepsilon$ for $y \leq a - \Delta_1 f$, and is linear in between.

1. The event $y \geq a$ has probability $\frac{1}{2}$ under P, because the noise is symmetric. So L has an atom of mass $\frac{1}{2}$ at ε.
2. The event $y \leq a - \Delta_1 f$ needs noise below $-\Delta_1 f = -\varepsilon b$, which has probability $\frac{1}{2}e^{-\varepsilon}$. So L has an atom of mass $\frac{1}{2}e^{-\varepsilon}$ at $-\varepsilon$.
3. In between, write $y = a - s$ with $s \in (0, \Delta_1 f)$, so that $L = \varepsilon - 2\varepsilon s/(\Delta_1 f)$. The variable s has density $\frac{\varepsilon}{2\Delta_1 f}e^{-\varepsilon s/(\Delta_1 f)}$, and the change of variables gives L the density $u \mapsto \frac{1}{4}e^{(u-\varepsilon)/2}$ on $(-\varepsilon, \varepsilon)$, of total mass $\frac{1}{2}(1 - e^{-\varepsilon})$.

The three masses add up to 1. The privacy loss never leaves $[-\varepsilon, \varepsilon]$, as Figure 5.1 shows on the left.

More generally, pure DP bounds privacy loss by ε with probability 1, while approximate DP allows it to exceed ε with probability about δ. The next identity expresses the smallest valid δ for a pair as the largest event-wise violation of the multiplicative bound, averaged through the positive part $(z)_+ = \max(z, 0)$.

Lemma 5.1. *For neighbours $D \sim D'$ and $\varepsilon \geq 0$,*

$$\sup_{S \subseteq \mathcal{Y}} \Big(\Pr[M(D) \in S] - e^{\varepsilon} \Pr[M(D') \in S] \Big) = \mathbb{E}\Big[\big(1 - e^{\varepsilon - L}\big)_+ \Big], \tag{5.5}$$

where L is the privacy loss random variable of Definition 5.2 for this pair, $(z)_+ = \max(z, 0)$ is the positive part of z, and $e^{\varepsilon - L}$ is read as 0 when $L = +\infty$. So M is (ε, δ)-DP if and only if the right-hand side is at most δ for every ordered pair of neighbours.

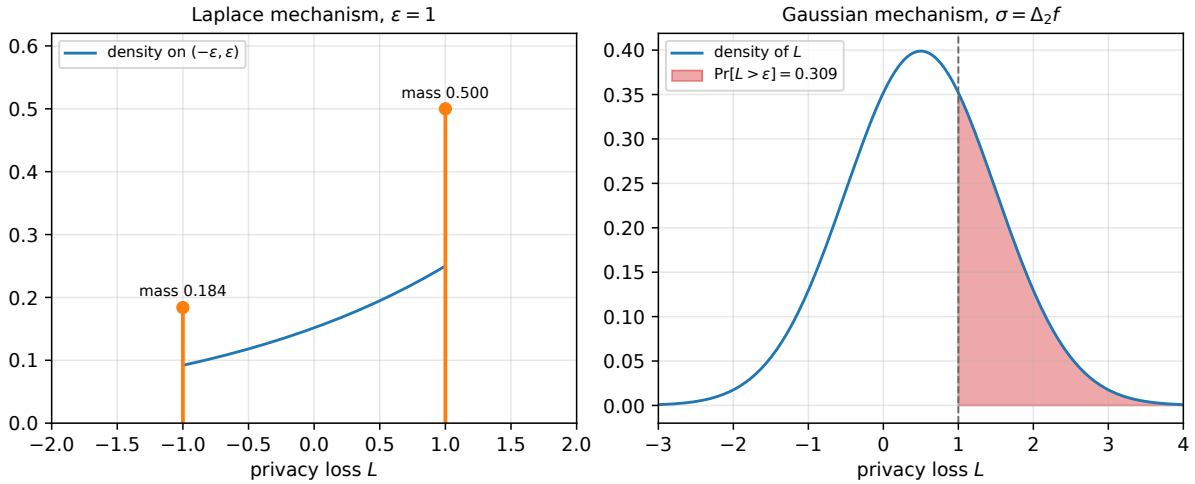


Figure 5.1.: Privacy loss for two mechanisms on neighbours whose values differ by the full sensitivity. Left, the Laplace mechanism with $\varepsilon = 1$, from Example 5.1: atoms of mass 0.5 at ε and 0.184 at $-\varepsilon$, and a density in between. Right, the Gaussian mechanism with noise standard deviation $\sigma = \Delta_2 f$, from Lemma 5.4: the loss is Gaussian with mean $1/2$ and variance 1, so it exceeds $\varepsilon = 1$ with probability 0.309.

Proof.

1. For an event S, write the difference as an integral, $\Pr[M(D) \in S] - e^{\varepsilon} \Pr[M(D') \in S] = \int_S (p - e^{\varepsilon} q)$. The integrand is positive exactly on $B = \{y : p(y) > e^{\varepsilon} q(y)\}$, so the integral is largest for $S = B$, where it equals $\int (p - e^{\varepsilon} q)_+$.
2. Where $p(y) = 0$, the integrand $(p - e^{\varepsilon} q)_+$ is 0. Where $p(y) > 0$, it equals $p(y)(1 - e^{\varepsilon} q(y)/p(y))_+ = p(y)(1 - e^{\varepsilon - L(y)})_+$, which is also right when $q(y) = 0$ and $L(y) = \infty$.
3. Integrating $p(y)$ times a function of y is taking the expectation under P, so $\int (p - e^{\varepsilon} q)_+ = \mathbb{E}[(1 - e^{\varepsilon - L})_+]$.
4. Definition 5.1 asks that the left-hand side of Equation 5.5 be at most δ for every ordered pair of neighbours, which proves the last sentence.

□

The right-hand side of Equation 5.5 is the **hockey-stick divergence** of P from Q, named for the shape of $z \mapsto (1 - e^{\varepsilon - z})_+$. Its integrand is zero when $L \le \varepsilon$ and at most one otherwise. The next theorem relates this divergence to the probability that L exceeds ε.

Theorem 5.1. *Let $M : \mathcal{X}^n \to \mathcal{Y}$ be a randomized algorithm, and let L be its privacy loss random variable for an ordered pair of neighbours $D \sim D'$.*

1. *M is ε-DP if and only if $L \le \varepsilon$ with probability 1 for every ordered pair of neighbours. In that case $|L| \le \varepsilon$ with probability 1.*
2. *If $\Pr[L > \varepsilon] \le \delta$ for every ordered pair of neighbours, then M is (ε, δ)-DP.*
3. *If M is (ε, δ)-DP, then for every $t > 0$ and every ordered pair of neighbours,*

$$\Pr[L > \varepsilon + t] \le \frac{\delta}{1 - e^{-t}}. \tag{5.6}$$

Proof.

1. With $\delta = 0$, Lemma 5.1 says that M is ε-DP exactly when $\mathbb{E}[(1 - e^{\varepsilon - L})_+] = 0$ for every pair. The integrand is nonnegative, so this holds exactly when it is 0 with probability 1, that is, when $L \le \varepsilon$ with probability 1. For the lower bound, apply the same fact to the pair (D', D). It gives $q \le e^{\varepsilon} p$ except on a set of Q-probability 0, and that set also has P-probability 0, because pure DP gives $\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S]$. So $L \ge -\varepsilon$ with probability 1 under P.
2. The integrand $(1 - e^{\varepsilon - L})_+$ is 0 when $L \le \varepsilon$ and at most 1 when $L > \varepsilon$. So its expectation is at most $\Pr[L > \varepsilon] \le \delta$, and Lemma 5.1 gives (ε, δ)-DP.
3. On the event $L > \varepsilon + t$, the integrand is at least $1 - e^{-t}$. So $(1 - e^{-t}) \Pr[L > \varepsilon + t] \le \mathbb{E}[(1 - e^{\varepsilon - L})_+] \le \delta$, by Lemma 5.1.

□

The converse of part 2 fails with the same parameters, which is why part 3 pays in both ε and δ. For example, the Gaussian mechanism on the right of Figure 5.1 has privacy loss above 1 with probability 0.309, yet Proposition 5.3 below shows that it is $(1, 0.127)$-DP. With $t = \varepsilon$, part 3 says that an (ε, δ)-DP mechanism has privacy loss above 2ε with probability at most $\delta/(1 - e^{-\varepsilon})$. So, up to these constants, (ε, δ)-DP says that the privacy loss is at most ε except with probability about δ, and it says nothing about what happens on that rare event.

5.3. How small δ must be

On the event of probability δ, the privacy loss may be infinite. The first example shows this in its crudest form.

Example 5.2. Let $\perp$ be a fixed symbol that carries no information about the data. The mechanism $M(D)$ outputs $\perp$ with probability $1 - \delta$ and outputs the whole dataset D with probability δ. For neighbours $D \ne D'$ and an event S,

$$\Pr[M(D) \in S] \le (1 - \delta)\, \mathbf{1}[\perp \in S] + \delta \le \Pr[M(D') \in S] + \delta, \tag{5.7}$$

so M is $(0, \delta)$-DP. With probability $1 - \delta$ the privacy loss is $\ln \frac{1-\delta}{1-\delta} = 0$, and with probability δ it is infinite, because the output D is impossible under D'.

This mechanism publishes everyone's data with probability δ, so δ must be small. The next example shows how small, because it publishes each record on its own.

Definition 5.3 (Name and shame)**.** For $\delta \in [0, 1]$, the mechanism NS_δ draws independent coins $c_1, \dots, c_n$, each equal to 1 with probability δ, and outputs the set of pairs $\{(i, x_i) : c_i = 1\}$.

Each record, which might be a person's medical file, is published in the clear with probability δ. Even so, the mechanism meets the definition with $\varepsilon = 0$, because a person's record affects the output only when that person's coin comes up 1, which has probability δ.

Proposition 5.1. *For every $\delta \in [0, 1]$, NS_δ is $(0, \delta)$-DP.*

Proof.

1. Let D and D' differ only in record i, and run the mechanism on both datasets with the same coins. Let E be the event $c_i = 1$, so $\Pr[E] = \delta$.
2. On the complement $\bar{E}$, record i is not published, and the output is a function of the coins and of the records $j \neq i$, which the two datasets share. So on $\bar{E}$ the two outputs are equal.
3. For an event S, split on E and use step 2:

$$\Pr[\mathrm{NS}_\delta(D) \in S] = \Pr[\mathrm{NS}_\delta(D) \in S,\ \bar{E}] + \Pr[\mathrm{NS}_\delta(D) \in S,\ E] \leq \Pr[\mathrm{NS}_\delta(D') \in S,\ \bar{E}] + \Pr[E]. \tag{5.8}$$

4. The first term is at most $\Pr[\mathrm{NS}_\delta(D') \in S]$ and the second is δ, which is Equation 5.2 with $\varepsilon = 0$.

□

So a mechanism that plainly harms people meets $(0, \delta)$-DP. Its harm depends on the chance that it publishes at least one record, which the next proposition computes.

Proposition 5.2. *The probability that NS_δ publishes at least one record is $1 - (1 - \delta)^n$, and*

$$n\delta - \frac{(n\delta)^2}{2} \leq 1 - (1 - \delta)^n \leq n\delta. \tag{5.9}$$

Proof.

1. The coins are independent, so no record is published with probability $(1 - \delta)^n$, and the complement gives the formula.
2. The upper bound is the union bound over the n records, each published with probability δ.
3. For the lower bound, $1 - \delta \leq e^{-\delta}$ gives $(1 - \delta)^n \leq e^{-n\delta}$. The inequality $e^{-x} \leq 1 - x + x^2/2$ for $x \geq 0$ then gives $1 - (1 - \delta)^n \geq n\delta - (n\delta)^2/2$.

□

So the chance that someone is named is about $n\delta$ when $n\delta$ is small. At $\delta = 1/n$, step 3 of the proof shows that some record is in the clear with probability at least $1 - e^{-1} \approx 0.632$. A guarantee that allows this is not privacy, so δ must be well below $1/n$. Even $\delta = n^{-1.1}$ allows a chance of up to $n\delta = n^{-0.1}$, which is 0.25 for a million records. Common choices are $\delta \leq 1/n^2$, or a **cryptographically small** δ, smaller than every inverse polynomial in n.

In both examples the privacy loss is 0 or infinite, so the rare event is a catastrophe. Without more information about a mechanism, you must assume this worst case. But many mechanisms degrade gracefully. Their privacy loss exceeds ε with probability about δ but rarely exceeds it by much, so they meet a family of guarantees at once, such as $(1, 10^{-3})$-DP and $(2, 10^{-6})$-DP. The Gaussian mechanism of the next section is the main example, as Example 5.5 shows.

The definition must be stated for events, not single outcomes. For pure DP the two are equivalent on countable outputs, by the pointwise lemma of the randomized response chapter, but for approximate DP the pointwise version is much weaker.

Example 5.3. Let $m = \lceil 1/\delta \rceil$ for some $\delta \in (0, 1)$. The mechanism $M(D) = (D, U)$ outputs the whole dataset together with a number U drawn uniformly from $\{1, \dots, m\}$. Every single outcome has probability at most $1/m \le \delta$, so for all neighbours and all outcomes y,

$$\Pr[M(D) = y] \le e^{\varepsilon} \Pr[M(D') = y] + \delta. \tag{5.10}$$

Yet the event $S = \{(D, u) : 1 \le u \le m\}$ has probability 1 under D and 0 under D', so M is not (ε, δ_0)-DP for any ε and any $\delta_0 < 1$.

The privacy loss of this mechanism is infinite with probability 1. So Theorem 5.1, which looks at the privacy loss, catches the failure that the pointwise inequality misses.

5.4. The Gaussian mechanism

The Laplace mechanism uses ℓ_1 sensitivity. The Gaussian mechanism uses ℓ_2 sensitivity, which can be smaller.

Definition 5.4 ($\ell 2$ sensitivity)**.** The ℓ_2 **sensitivity** of a function $f \colon \mathcal{X}^n \to \mathbb{R}^d$ is

$$\Delta_2 f = \sup_{D \sim D'} \|f(D) - f(D')\|_2. \tag{5.11}$$

The ℓ_1 sensitivity $\Delta_1 f$ of the previous chapter is the same supremum with the ℓ_1 norm. The two sensitivities are within a factor $\sqrt{d}$ of each other, because the two norms are.

Lemma 5.2. *For every $v \in \mathbb{R}^d$, $\|v\|_2 \le \|v\|_1 \le \sqrt{d}\,\|v\|_2$. Hence $\Delta_2 f \le \Delta_1 f \le \sqrt{d}\,\Delta_2 f$.*

Proof.

1. Expanding the square, $\|v\|_1^2 = \sum_i v_i^2 + \sum_{i \ne j} |v_i||v_j| \ge \|v\|_2^2$.
2. The Cauchy–Schwarz inequality applied to $(|v_1|, \dots, |v_d|)$ and the all-ones vector gives $\|v\|_1 = \sum_i |v_i| \cdot 1 \le \|v\|_2 \sqrt{d}$.
3. Applying both inequalities to $f(D) - f(D')$ and taking the supremum over neighbours gives the bounds on the sensitivities.

□

The Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$ has mean μ, variance σ^2, and density

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{5.12}$$

Its tails decay as $e^{-x^2/(2\sigma^2)}$, faster than Laplace tails. Write $\mathcal{N}(0, \sigma^2 I_d)$ for a vector of d independent $\mathcal{N}(0, \sigma^2)$ variables. The Gaussian mechanism adds this vector to the true answer.

Definition 5.5 (Gaussian mechanism)**.** For $f\colon \mathcal{X}^n \to \mathbb{R}^d$ and a noise level $\sigma > 0$, the Gaussian mechanism outputs

$$M(D) = f(D) + Z, \qquad Z \sim \mathcal{N}(0, \sigma^2 I_d). \tag{5.13}$$

The next theorem specifies a sufficient noise level σ for given ε and δ.

Theorem 5.2 (Privacy of the Gaussian mechanism)**.** *Let f have $0 < \Delta_2 f < \infty$, and let $\varepsilon > 0$ and $\delta \in (0, 1)$. The Gaussian mechanism with*

$$\sigma = \frac{\Delta_2 f}{\varepsilon} \sqrt{2\ln(1/\delta) + \varepsilon} \tag{5.14}$$

is (ε, δ)-DP.

The proof is in Section 5.4.1. As $\delta \to 0$, $\sigma = \frac{\Delta_2 f}{\varepsilon}\sqrt{2\ln(1/\delta)}(1 + o(1))$. Compared with the Laplace scale $\Delta_1 f/\varepsilon$ per coordinate, the Gaussian mechanism pays a factor $\sqrt{2\ln(1/\delta)}$ for approximate DP but may save up to $\sqrt{d}$ through the ℓ_2 sensitivity. The next example quantifies this trade-off.

Example 5.4. Each of n people answers d yes-or-no questions, so the dataset is $D \in \{0,1\}^{n \times d}$ with rows $x_i \in \{0,1\}^d$. The analyst wants the vector of averages $f(D) = \frac{1}{n}\sum_i x_i$, with small error in ℓ_2 norm.

1. Replacing row x by row x' changes f by $(x - x')/n$. The largest change is from the all-ones row to the all-zeros row, so $\Delta_1 f = d/n$ and $\Delta_2 f = \sqrt{d}/n$.
2. The Laplace mechanism adds $\mathrm{Lap}(d/(\varepsilon n))$ to each coordinate, and each coordinate has variance $2d^2/(\varepsilon n)^2$. The root mean squared ℓ_2 error is

$$\sqrt{\mathbb{E}\|Z\|_2^2} = \sqrt{d \cdot \frac{2d^2}{\varepsilon^2 n^2}} = \frac{\sqrt{2}\, d^{3/2}}{\varepsilon n}. \tag{5.15}$$

3. The Gaussian mechanism with Equation 5.14 adds $\mathcal{N}(0, \sigma^2)$ to each coordinate with $\sigma = \frac{\sqrt{d}}{\varepsilon n}\sqrt{2\ln(1/\delta) + \varepsilon}$, so its root mean squared error is

$$\sqrt{d}\,\sigma = \frac{d\sqrt{2\ln(1/\delta) + \varepsilon}}{\varepsilon n}. \tag{5.16}$$

4. The ratio of the two errors is $\sqrt{2d/(2\ln(1/\delta)+\varepsilon)}$. So the Gaussian mechanism is more accurate as soon as $d > \ln(1/\delta) + \varepsilon/2$, and its advantage grows like $\sqrt{d}$.

With $d = 100$ questions, $n = 10{,}000$ people, $\varepsilon = 1$ and $\delta = 10^{-6}$, the Laplace error is $\sqrt{2} \cdot 1000/10{,}000 = 0.1414$ and the Gaussian error is $100\sqrt{27.63+1}/10{,}000 = 0.0535$, smaller by a factor 2.64. A simulation of 20,000 runs of each mechanism gives 0.1413 and 0.0535.

The Laplace mechanism provides pure DP. The Gaussian mechanism provides approximate DP and can use less noise in high dimensions.

5.4.1. Proof of privacy

The proof first derives the Gaussian distribution of privacy loss, then bounds its probability of exceeding ε by δ. Part 2 of Theorem 5.1 then gives privacy. The following lemma supplies the needed Gaussian distribution and tail facts.

Lemma 5.3. *Let $W \sim \mathcal{N}(0,1)$.*

1. *For constants μ and $a \neq 0$, $\mu + aW \sim \mathcal{N}(\mu, a^2)$.*
2. *If $V_1 \sim \mathcal{N}(0, a_1^2)$ and $V_2 \sim \mathcal{N}(0, a_2^2)$ are independent, then $V_1 + V_2 \sim \mathcal{N}(0, a_1^2 + a_2^2)$.*
3. *For every $u \geq 0$, $\Pr[W > u] \leq e^{-u^2/2}$.*
4. *For every $u > 0$, $\Pr[W > u] \leq e^{-u^2/2}/(u\sqrt{2\pi})$.*

Proof.

1. Completing the square in the density gives the moment generating function $\mathbb{E}[e^{\lambda W}] = e^{\lambda^2/2}$. A distribution is determined by its moment generating function when that function is finite near 0, as it is for all the variables here.
2. For part 1, $\mathbb{E}[e^{\lambda(\mu+aW)}] = e^{\lambda\mu+\lambda^2 a^2/2}$, which is the moment generating function of $\mathcal{N}(\mu, a^2)$.
3. For part 2, independence makes the moment generating function of the sum the product $e^{\lambda^2 a_1^2/2} e^{\lambda^2 a_2^2/2} = e^{\lambda^2(a_1^2+a_2^2)/2}$.
4. For part 3, Markov's inequality applied to e^{uW} gives $\Pr[W > u] \leq e^{-u^2}\,\mathbb{E}[e^{uW}] = e^{-u^2/2}$.
5. For part 4, write $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$ for the density, whose derivative is $-z\varphi(z)$. On $z \geq u$ the ratio z/u is at least 1, so $\Pr[W > u] \leq \int_u^\infty \frac{z}{u}\varphi(z)\,\mathrm{d}z = \frac{\varphi(u)}{u}$.

□

The first step computes the privacy loss distribution; its mean and variance depend on the distance between $f(D)$ and $f(D')$.

Lemma 5.4. *Let $D \sim D'$, let M be the Gaussian mechanism with noise level σ, and write $v = f(D) - f(D')$. The privacy loss random variable of $M(D)$ against $M(D')$ is distributed as*

$$L \sim \mathcal{N}(\rho,\ 2\rho)\,, \qquad \rho = \frac{\|v\|_2^2}{2\sigma^2}. \tag{5.17}$$

Proof.

1. The output is $f(D) + z$ with $z \sim \mathcal{N}(0, \sigma^2 I_d)$. Its density under D is proportional to $\exp(-\|z\|_2^2/(2\sigma^2))$. Under D', the same point is at distance $z + v$ from the mean $f(D')$, so its density is proportional to $\exp(-\|z + v\|_2^2/(2\sigma^2))$, with the same constant.
2. The constants cancel in the ratio, and the logarithm gives

$$L = \frac{\|z + v\|_2^2 - \|z\|_2^2}{2\sigma^2} = \frac{1}{2\sigma^2} \sum_{j=1}^{d} \left((z_j + v_j)^2 - z_j^2 \right) = \frac{1}{2\sigma^2} \sum_{j=1}^{d} \left(2 z_j v_j + v_j^2 \right). \tag{5.18}$$

3. The constant part is $\sum_j v_j^2/(2\sigma^2) = \rho$, the claimed mean.
4. The random part is $\frac{1}{\sigma^2} \sum_j v_j z_j$, and the terms with $v_j = 0$ vanish. By part 1 of Lemma 5.3 each remaining term $v_j z_j$ is $\mathcal{N}(0, \sigma^2 v_j^2)$, and the terms are independent. By part 2 their sum is $\mathcal{N}(0, \sigma^2 \|v\|_2^2)$. Dividing by σ^2 gives $\mathcal{N}(0, \|v\|_2^2/\sigma^2) = \mathcal{N}(0, 2\rho)$.

□

The privacy-loss variance is twice its mean. In one dimension, $L = vz/\sigma^2 + v^2/(2\sigma^2)$. This Gaussian loss is specific to the Gaussian mechanism; the Laplace mechanism's loss has atoms (Example 5.1).

Proof. Proof of Theorem 5.2.

1. By part 2 of Theorem 5.1, it suffices to show $\Pr[L > \varepsilon] \le \delta$ for every ordered pair of neighbours. If $v = 0$ then $L = 0$, so assume $v \neq 0$.
2. By Lemma 5.4 and part 1 of Lemma 5.3, $L = \rho + \sqrt{2\rho}\, W$ with $W \sim \mathcal{N}(0, 1)$. So

$$\Pr[L > \varepsilon] = \Pr\left[W > g(\rho) \right], \qquad g(\rho) = \frac{\varepsilon - \rho}{\sqrt{2\rho}} = \frac{\varepsilon}{\sqrt{2\rho}} - \sqrt{\frac{\rho}{2}}. \tag{5.19}$$

3. The function g decreases in ρ, and $\rho \le (\Delta_2 f)^2/(2\sigma^2)$ by the definition of $\Delta_2 f$. Write $t = \sqrt{2\ln(1/\delta) + \varepsilon}$, so that Equation 5.14 reads $\sigma = t\, \Delta_2 f/\varepsilon$ and the largest value of ρ is $\varepsilon^2/(2t^2)$. Then

$$g(\rho) \ge g\left(\frac{\varepsilon^2}{2t^2} \right) = t - \frac{\varepsilon}{2t} =: u. \tag{5.20}$$

4. The value u is positive, because $t^2 \ge \varepsilon > \varepsilon/2$. Its square is $u^2 = t^2 - \varepsilon + \varepsilon^2/(4t^2) \ge t^2 - \varepsilon = 2\ln(1/\delta)$.
5. By part 3 of Lemma 5.3, $\Pr[L > \varepsilon] \le \Pr[W > u] \le e^{-u^2/2} \le e^{-\ln(1/\delta)} = \delta$.

□

Only the upper tail of L matters in this proof, because part 2 of Theorem 5.1 bounds $\Pr[L > \varepsilon]$, not $\Pr[|L| > \varepsilon]$. For $\varepsilon \le 1$ the noise level Equation 5.14 is at most $\frac{\Delta_2 f}{\varepsilon} \sqrt{2\ln(1.65/\delta)}$. The sharper tail bound of part 4 of Lemma 5.3 lowers the constant 1.65 to 1.25, the form that later chapters use.

Corollary 5.1 (Gaussian mechanism for $\varepsilon \le 1$)**.** *Let f have $0 < \Delta_2 f < \infty$, and let $\varepsilon \in (0, 1]$ and $\delta \in (0, 1)$. The Gaussian mechanism with*

$$\sigma = \frac{\Delta_2 f}{\varepsilon} \sqrt{2 \ln(1.25/\delta)} \tag{5.21}$$

is (ε, δ)-DP.

Proof.

1. Write $t = \sqrt{2 \ln(1.25/\delta)}$, so that $\sigma = t\,\Delta_2 f/\varepsilon$ and $e^{-t^2/2} = \delta/1.25$. Steps 1 to 3 of the proof of Theorem 5.2 used only $\sigma = t\,\Delta_2 f/\varepsilon$, so they give $\Pr[L > \varepsilon] \le \Pr[W > u]$ with $u = t - \varepsilon/(2t)$.
2. Since $\varepsilon \le 1$, the value u is at least $t - 1/(2t)$, which increases in t. Also $u^2 = t^2 - \varepsilon + \varepsilon^2/(4t^2) \ge t^2 - 1$.
3. *Case $\delta \le 0.74$.* Then $t \ge \sqrt{2 \ln(1.25/0.74)} \ge 1.0239$, and by step 2, $u \ge 0.5356$. By part 4 of Lemma 5.3 and step 2,

$$\Pr[W > u] \le \frac{e^{-u^2/2}}{u\sqrt{2\pi}} \le \frac{e^{1/2}\, e^{-t^2/2}}{0.5356\sqrt{2\pi}} = \frac{e^{1/2}}{1.25 \cdot 0.5356\sqrt{2\pi}}\, \delta \le 0.983\, \delta. \tag{5.22}$$

4. *Case $\delta > 0.74$.* Then $t > \sqrt{2 \ln 1.25} \ge 0.6680$, and by step 2, $u > -0.0805$. The density of W is at most $1/\sqrt{2\pi}$, so $\Pr[W > u] \le \frac{1}{2} + 0.0805/\sqrt{2\pi} \le 0.533 < \delta$.
5. In both cases $\Pr[L > \varepsilon] \le \delta$, and part 2 of Theorem 5.1 gives (ε, δ)-DP.

□

This is the bound of Dwork and Roth (2014, Theorem A.1). On a grid of ε from 0.01 to 1 and δ from 0.89 down to 10^{-20}, the exact δ of Proposition 5.3 at this noise level is at most 0.233 times the target. Both constants are loose, because the tail bounds of Lemma 5.3 are loose. The exact δ comes from the identity of Lemma 5.1 for the Gaussian privacy loss, evaluated instead of bounded.

Proposition 5.3. *Let Φ be the distribution function of $\mathcal{N}(0, 1)$, let f have $0 < \Delta_2 f < \infty$, and let $s = \Delta_2 f/\sigma$. For every $\varepsilon \ge 0$, the Gaussian mechanism with noise level σ is $(\varepsilon, \delta(\varepsilon))$-DP for*

$$\delta(\varepsilon) = \Phi\Big(\frac{s}{2} - \frac{\varepsilon}{s}\Big) - e^{\varepsilon}\, \Phi\Big(-\frac{s}{2} - \frac{\varepsilon}{s}\Big), \tag{5.23}$$

and for no smaller δ when some pair of neighbours has $\|f(D) - f(D')\|_2 = \Delta_2 f$.

Proof.

1. Fix a pair of neighbours and let $r = \|v\|_2/\sigma \in (0, s]$, so that $\rho = r^2/2$ and $L = r^2/2 + rW$. By Lemma 5.1, the smallest δ for this pair is $\mathbb{E}[(1 - e^{\varepsilon - L})\mathbf{1}[L > \varepsilon]]$.
2. The event $L > \varepsilon$ is $W > c$ with $c = \varepsilon/r - r/2$, so it has probability $\Phi(r/2 - \varepsilon/r)$.

3. Write φ for the standard normal density. Completing the square gives $e^{-rz}\varphi(z) = e^{r^2/2}\varphi(z+r)$, so $\mathbb{E}[e^{-rW}\mathbf{1}[W > c]] = e^{r^2/2}\,\Phi(-c-r)$. Since $e^{\varepsilon-L} = e^{\varepsilon-r^2/2}e^{-rW}$ and $-c-r = -r/2-\varepsilon/r$,

$$\mathbb{E}\big[e^{\varepsilon-L}\mathbf{1}[L>\varepsilon]\big] = e^{\varepsilon-r^2/2}\,e^{r^2/2}\,\Phi\Big(-\frac{r}{2}-\frac{\varepsilon}{r}\Big) = e^{\varepsilon}\,\Phi\Big(-\frac{r}{2}-\frac{\varepsilon}{r}\Big). \tag{5.24}$$

4. Subtracting step 3 from step 2 gives Equation 5.23 with r in place of s. Call that function $h(r)$.
5. Write $c_1 = r/2-\varepsilon/r$ and $c_2 = -r/2-\varepsilon/r$. Then $(c_1^2-c_2^2)/2 = (c_1-c_2)(c_1+c_2)/2 = -\varepsilon$, so $e^{\varepsilon}\varphi(c_2) = \varphi(c_1)$. Differentiating, $h'(r) = \varphi(c_1)(\frac{1}{2}+\varepsilon/r^2) + e^{\varepsilon}\varphi(c_2)(\frac{1}{2}-\varepsilon/r^2) = \varphi(c_1) > 0$. 6. So h increases in r, the worst pair is the one with the largest r, and $h(r) \le h(s) = \delta(\varepsilon)$, with equality when some pair attains $\Delta_2 f$. □

The exact formula quantifies the graceful degradation discussed in Section 5.3: one noise level gives a family of privacy guarantees.

Example 5.5. For $\varepsilon = 1$ and $\delta = 10^{-5}$, Equation 5.14 gives $\sigma = \sqrt{2\ln(10^5)+1}\,\Delta_2 f = 4.9016\,\Delta_2 f$. At this noise level, Equation 5.23 gives the following guarantees, all at once:

- $(0.5,\ 6.1\times10^{-4})$-DP,
- $(1,\ 3.0\times10^{-8})$-DP,
- $(2,\ 3.0\times10^{-24})$-DP.

The guarantee at $\varepsilon = 1$ is far better than the $\delta = 10^{-5}$ that was asked for, and the smallest noise level that gives $(1, 10^{-5})$-DP is $\sigma = 3.7306\,\Delta_2 f$. As ε grows, δ falls faster than exponentially, so the privacy loss of this mechanism degrades gracefully.

5.5. Properties

The pure-DP properties proved in the Laplace mechanism chapter extend to approximate DP. Post-processing and composition add the corresponding δ terms; group privacy incurs a larger δ.

Theorem 5.3 (Post-processing)**.** *Let $M\colon \mathcal{X}^n \to \mathcal{Y}$ be (ε,δ)-DP, and let $F\colon \mathcal{Y} \to \mathcal{Z}$ be a randomized map whose coins are independent of those of M. Then $F\circ M$ is (ε,δ)-DP.*

Proof.

1. The proof of Theorem 4.2 writes F as a random choice g_R of deterministic functions and conditions on the coins R. Those steps do not use the definition of privacy, so they are unchanged.
2. Only its last step changes. For neighbours $D \sim D'$ and an event $S \subseteq \mathcal{Z}$, apply Equation 5.2, instead of pure DP, to the event $g_R^{-1}(S)$, so δ joins the bound inside the expectation:

$$\Pr[F(M(D)) \in S] = \mathbb{E}_R\Big[\Pr[M(D) \in g_R^{-1}(S)]\Big] \le \mathbb{E}_R\Big[e^\varepsilon \Pr[M(D') \in g_R^{-1}(S)] + \delta\Big] = e^\varepsilon \Pr[F(M(D')) \in S] + \delta. \tag{5.25}$$

□

For group privacy, recall that an ε-DP mechanism meets the bound with $e^{k\varepsilon}$ on datasets that differ in k records (Theorem 4.3). With approximate DP, each step of the chain also multiplies the earlier δ terms by e^ε.

Theorem 5.4 (Group privacy)**.** *Let M be (ε, δ)-DP, and let $D, D' \in \mathcal{X}^n$ differ in $k \ge 1$ records. For every event S,*

$$\Pr[M(D) \in S] \le e^{k\varepsilon} \Pr[M(D') \in S] + \delta \sum_{j=0}^{k-1} e^{j\varepsilon} \le e^{k\varepsilon} \Pr[M(D') \in S] + k e^{(k-1)\varepsilon}\delta. \tag{5.26}$$

Proof.

1. Change the differing records one at a time to get a chain $D = D^{(0)} \sim D^{(1)} \sim \cdots \sim D^{(k)} = D'$ of neighbours. Write $P_j = \Pr[M(D^{(j)}) \in S]$.
2. We show $P_0 \le e^{j\varepsilon} P_j + \delta \sum_{i=0}^{j-1} e^{i\varepsilon}$ by induction on j. The case $j = 1$ is Equation 5.2.
3. If the bound holds for j, apply Equation 5.2 to $D^{(j)} \sim D^{(j+1)}$, which gives $P_j \le e^\varepsilon P_{j+1} + \delta$. Substituting,

$$P_0 \le e^{j\varepsilon}(e^\varepsilon P_{j+1} + \delta) + \delta \sum_{i=0}^{j-1} e^{i\varepsilon} = e^{(j+1)\varepsilon} P_{j+1} + \delta \sum_{i=0}^{j} e^{i\varepsilon}. \tag{5.27}$$

4. At $j = k$ this is the first inequality in Equation 5.26. Each of the k terms of the sum is at most $e^{(k-1)\varepsilon}$, which gives the second.

□

The additive term grows exponentially with group size. The bound becomes uninformative once k exceeds roughly $\ln(1/\delta)/\varepsilon$.

Composition requires a bound for expectations of functions valued in $[0, 1]$, not only event probabilities. Lemma 3.2 gives this bound for pure DP; the following lemma extends it to approximate DP.

Lemma 5.5. *Let M be (ε, δ)-DP and let $h \colon \mathcal{Y} \to [0, 1]$. For all neighbours $D \sim D'$,*

$$\mathbb{E}[h(M(D))] \le e^\varepsilon \, \mathbb{E}[h(M(D'))] + \delta. \tag{5.28}$$

Proof.

1. The proof is that of Lemma 3.2 with $+\delta$. Its first step, which writes $\mathbb{E}[h(M(D))] = \int_0^1 \Pr[h(M(D)) > t]\,\mathrm{d}t$, does not use privacy and is unchanged.
2. Only the bound on each tail changes. Equation 5.2 bounds the probability of the event $\{y : h(y) > t\}$ by $e^{\varepsilon}\Pr[h(M(D')) > t] + \delta$, and integrating over $t \in [0,1]$ gives $e^{\varepsilon}\,\mathbb{E}[h(M(D'))] + \delta$.

□

As in Definition 4.4, each mechanism may depend on earlier outputs. Here each conditional mechanism $M_i(\cdot\,; y_{<i})$ is $(\varepsilon_i, \delta_i)$-DP.

Theorem 5.5 (Basic composition)**.** *If each $M_i(\cdot\,; y_{<i})$ is $(\varepsilon_i, \delta_i)$-DP for every $y_{<i}$, where the parameters ε_i and δ_i are fixed in advance, then the adaptive composition of $M_1, \ldots, M_k$ is $(\sum_i \varepsilon_i, \sum_i \delta_i)$-DP.*

Proof. The proof follows the general-output proof of Theorem 4.4, with an additive δ term. Steps 1 and 2 are unchanged, and induction reduces the argument to $k = 2$. For an event S of output pairs, let $S_{y_1} = \{y_2 : (y_1, y_2) \in S\}$ and $g_D(y_1) = \Pr[M_2(D; y_1) \in S_{y_1}]$. Conditioning on the first output gives $\Pr[M(D) \in S] = \mathbb{E}_{y_1 \sim M_1(D)}[g_D(y_1)]$. The additive term changes the remaining steps.

3. Since $M_2(\cdot\,; y_1)$ is $(\varepsilon_2, \delta_2)$-DP and $g_D \le 1$, we have $g_D(y_1) \le \min\{1, e^{\varepsilon_2} g_{D'}(y_1) + \delta_2\} \le h(y_1) + \delta_2$, where $h = \min\{1, e^{\varepsilon_2} g_{D'}\}$ takes values in $[0,1]$.
4. By Lemma 5.5 applied to M_1 and h,

$$\Pr[M(D) \in S] \le \mathbb{E}_{M_1(D)}[h] + \delta_2 \le e^{\varepsilon_1}\,\mathbb{E}_{M_1(D')}[h] + \delta_1 + \delta_2. \tag{5.29}$$

5. Finally $h \le e^{\varepsilon_2} g_{D'}$, and $\mathbb{E}_{y_1 \sim M_1(D')}[g_{D'}(y_1)] = \Pr[M(D') \in S]$ by the conditioning identity above, applied to D'. So $\Pr[M(D) \in S] \le e^{\varepsilon_1 + \varepsilon_2}\Pr[M(D') \in S] + \delta_1 + \delta_2$.

□

Fix the parameters in advance. If the analyst chooses ε_i from earlier outputs, the sum is no longer a valid bound without further work. With k mechanisms that are each (ε, δ)-DP, basic composition gives $(k\varepsilon, k\delta)$-DP. The next chapter proves that, for every $\delta' > 0$, the same composition is also $(\varepsilon', k\delta + \delta')$-DP with

$$\varepsilon' = \varepsilon\sqrt{2k\ln(1/\delta')} + k\varepsilon\,(e^{\varepsilon} - 1). \tag{5.30}$$

When ε is of order $1/\sqrt{k}$ or smaller, the second term is no larger than the first, so ε' grows like $\sqrt{k}$ rather than k. That allows quadratically more analyses for the same privacy, at the price of a $\delta' > 0$ even when every mechanism is pure.

5.6. Summary

Table 5.1.: The chapter's results.

Result	Statement
Definition 5.1	$\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S] + \delta$ for all neighbours and events
Lemma 5.1	the largest violation over events is $\mathbb{E}[(1 - e^{\varepsilon - L})_+]$
Theorem 5.1	pure DP is $L \le \varepsilon$ almost surely; $\Pr[L > \varepsilon] \le \delta$ implies (ε, δ)-DP; the converse holds with $\varepsilon + t$ and $\delta/(1 - e^{-t})$
Proposition 5.1	publishing each record with probability δ is $(0, \delta)$-DP, so δ must be well below $1/n$
Example 5.3	for approximate DP, single outcomes do not suffice
Lemma 5.2	$\Delta_2 f \le \Delta_1 f \le \sqrt{d}\, \Delta_2 f$
Lemma 5.4	the Gaussian mechanism's privacy loss is $\mathcal{N}(\rho, 2\rho)$ with $\rho = \lVert f(D) - f(D') \rVert_2^2 / (2\sigma^2)$
Theorem 5.2	$\sigma = \Delta_2 f \sqrt{2 \ln(1/\delta) + \varepsilon}/\varepsilon$ gives (ε, δ)-DP
Corollary 5.1	for $\varepsilon \le 1$, $\sigma = \Delta_2 f \sqrt{2 \ln(1.25/\delta)}/\varepsilon$ suffices
Proposition 5.3	exact $\delta(\varepsilon) = \Phi(s/2 - \varepsilon/s) - e^{\varepsilon} \Phi(-s/2 - \varepsilon/s)$ with $s = \Delta_2 f / \sigma$
Theorem 5.3	post-processing keeps (ε, δ)
Theorem 5.4	groups of k: $(k\varepsilon,\ k e^{(k-1)\varepsilon} \delta)$
Theorem 5.5	adaptive composition adds the ε_i and the δ_i

5.7. References

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6. Advanced composition

Basic composition adds the privacy costs of k analyses on the same data. This chapter proves a tighter bound, of order $\sqrt{k}$, at the cost of an additional failure probability. The proof represents each step by a pair of distributions on four outcomes, then applies Hoeffding's inequality to their composition.

6.1. The theorem

Write $D \sim D'$ for neighbouring datasets, which differ in one record (the replace-one relation). Recall from the approximate differential privacy (DP) chapter that a mechanism M is (ε, δ)**-DP** if, for all neighbours $D \sim D'$ and all events S,

$$\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S] + \delta. \tag{6.1}$$

The proofs below compare two output distributions at a time, so the next definition names this condition for a pair of random variables.

Definition 6.1 (Indistinguishable pair)**.** Let $\varepsilon, \delta \geq 0$. Two random variables A and B with values in the same space are (ε, δ)**-indistinguishable** if, for every event S,

$$\Pr[A \in S] \leq e^{\varepsilon} \Pr[B \in S] + \delta \qquad \text{and} \qquad \Pr[B \in S] \leq e^{\varepsilon} \Pr[A \in S] + \delta. \tag{6.2}$$

Thus, M is (ε, δ)-DP exactly when its outputs on every neighbouring pair are (ε, δ)-indistinguishable.

The theorem concerns adaptive composition as defined in Definition 4.4. Mechanisms $M_1, \ldots, M_k$ run on the same dataset in sequence; $M_j(D; y_{<j})$ may depend on earlier outputs, and the composition releases the transcript $(y_1, \ldots, y_k)$.

Basic composition from the previous chapters gives $(k\varepsilon, k\delta)$-DP when each conditional mechanism is (ε, δ)-DP. This chapter improves the first parameter. All output spaces here are countable, so probabilities are sums over outcomes.

Theorem 6.1 (Advanced composition)**.** *Let $\varepsilon \geq 0$, $\delta \in [0, 1)$ and $\delta' \in (0, 1)$. Let M be the adaptive composition of k mechanisms such that $M_j(\,\cdot\,; y_{<j})$ is (ε, δ)-DP for every j and every $y_{<j}$. Then M is $(\tilde{\varepsilon}, \tilde{\delta})$-DP with*

$$\tilde{\varepsilon} = \varepsilon\sqrt{2k \ln(1/\delta')} + k\varepsilon \, \frac{e^{\varepsilon} - 1}{e^{\varepsilon} + 1}, \qquad \tilde{\delta} = k\delta + \delta'. \tag{6.3}$$

The first term grows as $\varepsilon\sqrt{k}$ rather than $k\varepsilon$. Since $(e^\varepsilon - 1)/(e^\varepsilon + 1) = \tanh(\varepsilon/2) \le \varepsilon/2$, the second term is at most $k\varepsilon^2/2$. It is also at most $k\varepsilon(e^\varepsilon - 1)$, so the theorem implies the weaker bound quoted in the previous chapter. The parameter δ' controls a trade-off: increasing it lowers $\tilde{\varepsilon}$ and increases the total $\tilde{\delta}$.

Example 6.1 and Figure 6.1 compare the theorem with basic composition before the proof in Section 6.3.

Example 6.1. Take $\varepsilon = 0.1$ and $\delta' = 10^{-6}$, so that $\ln(1/\delta') = 13.8155$ and $\tanh(0.05) = 0.04996$. For $k = 100$ mechanisms, basic composition gives 10. Advanced composition gives $0.1\sqrt{200 \times 13.8155} + 100 \times 0.1 \times 0.04996 = 5.257 + 0.500 = 5.756$. For $k = 1{,}000$, basic composition gives 100 and advanced composition gives $16.623 + 4.996 = 21.618$. Advanced composition is the smaller of the two from $k = 31$ on.

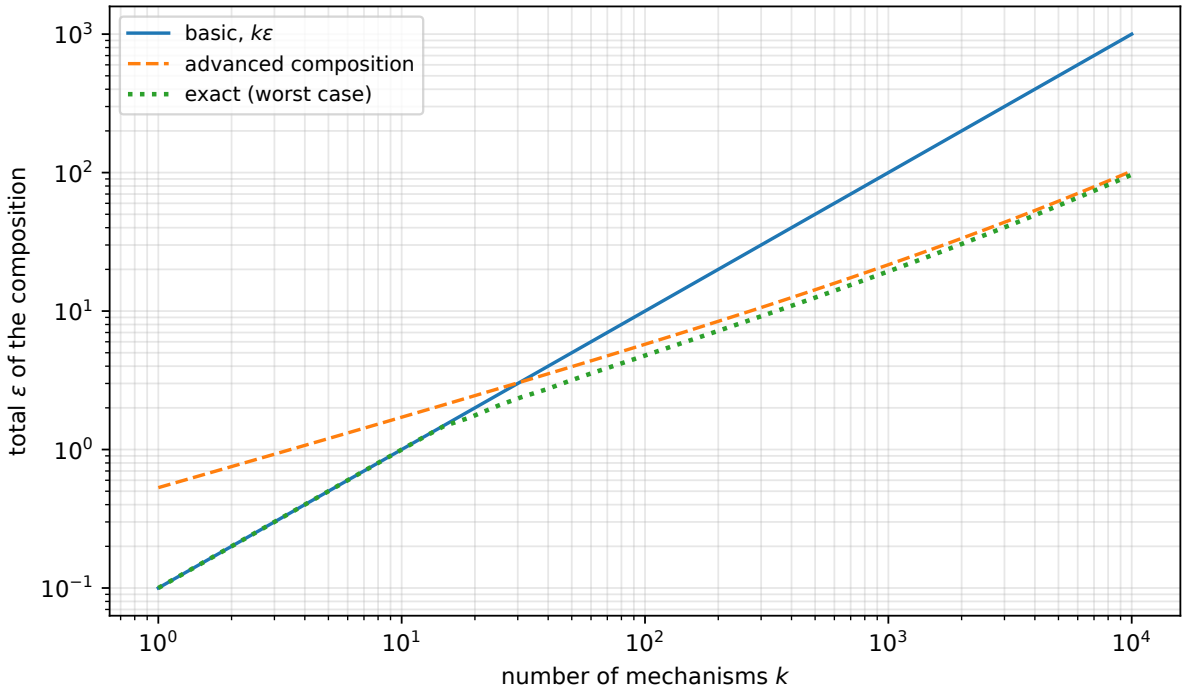


Figure 6.1.: The total privacy parameter of k mechanisms that are each 0.1-DP, at $\delta' = 10^{-6}$. The solid line is basic composition, the dashed line is Equation 6.3, and the dotted line is the exact smallest $\tilde{\varepsilon}$ for k independent randomized responses, which is the worst case for pure DP by Section 6.2. Advanced composition overtakes basic composition at $k = 31$ and is within 21% of the exact value for every k from 100 to 10,000.

In Figure 6.1, Theorem 6.1 is within 7% of the exact bound at $k = 10{,}000$. The observed growth, $\varepsilon\sqrt{k\ln(1/\delta')}$, is therefore close to optimal, not merely a proof artifact. Kairouz, Oh, and Viswanath (2015) derived the exact, tighter bound.

Given a total budget $\varepsilon_{\text{total}}$ for k analyses, Equation 6.3 can be inverted. The next corollary gives a sufficient per-mechanism budget.

Corollary 6.1. *Let $\varepsilon_{\text{total}} \in (0, 1]$ and $\delta' \in (0, 1/2]$. If each of k adaptively composed mechanisms is (ε, δ)-DP with*

$$\varepsilon = \frac{\varepsilon_{\text{total}}}{2\sqrt{2k \ln(1/\delta')}}, \tag{6.4}$$

then the composition is $(\varepsilon_{\text{total}}, k\delta + \delta')$-DP.

Proof.

1. By Theorem 6.1, the composition is $(\tilde{\varepsilon}, k\delta + \delta')$-DP. Its first term is $\varepsilon\sqrt{2k \ln(1/\delta')} = \varepsilon_{\text{total}}/2$.
2. The second term is $k\varepsilon \tanh(\varepsilon/2) \leq k\varepsilon^2/2$, because $\tanh x \leq x$. Substituting ε gives $k\varepsilon^2/2 = \varepsilon_{\text{total}}^2/(16 \ln(1/\delta'))$.
3. Since $\delta' \leq 1/2$, we have $\ln(1/\delta') \geq \ln 2 > 1/8$, and since $\varepsilon_{\text{total}} \leq 1$, the second term is at most $\varepsilon_{\text{total}}^2/2 \leq \varepsilon_{\text{total}}/2$.
4. Adding the two terms gives $\tilde{\varepsilon} \leq \varepsilon_{\text{total}}$. A mechanism that is $(\tilde{\varepsilon}, \tilde{\delta})$-DP is also $(\varepsilon_{\text{total}}, \tilde{\delta})$-DP for every $\varepsilon_{\text{total}} \geq \tilde{\varepsilon}$.

□

The corollary is conservative by about a factor of 2. For $\varepsilon_{\text{total}} = 1$, $k = 1{,}000$, and $\delta' = 10^{-6}$, it permits $\varepsilon = 0.00301$ per mechanism, yielding a total of 0.5045 under Equation 6.3. Numerical inversion permits $\varepsilon = 0.00591$, nearly twice as much; basic composition at that value gives 5.911.

The gain matters when the number of analyses grows with the problem, as in the following example.

Example 6.2 (Mean estimation with the Laplace mechanism)**.** Each of n people holds a vector $x_i \in \{0, 1\}^d$, and the analyst wants the mean $f(D) = \frac{1}{n}\sum_i x_i$. Recall the Laplace mechanism: if f has ℓ_1 sensitivity $\Delta_1 f$, meaning that $\|f(D) - f(D')\|_1 \leq \Delta_1 f$ for all neighbours, then adding independent noise from the Laplace distribution with scale $\Delta_1 f/\varepsilon$ to each coordinate is ε-DP. A Laplace variable with scale b has mean 0 and variance $2b^2$, so the expected squared ℓ_2 error of the noise vector is $2db^2$.

All coordinates at once. Replacing one person changes each coordinate of the mean by at most $1/n$, so $\Delta_1 f = d/n$, and the noise has scale $d/(n\varepsilon)$. The root mean squared ℓ_2 error is $\sqrt{2d}\, d/(n\varepsilon) = \sqrt{2}\, d^{3/2}/(n\varepsilon)$.

One coordinate at a time. Each coordinate is a mechanism with sensitivity $1/n$, released with Laplace noise of scale $1/(n\varepsilon_1)$, which makes it ε_1-DP. With ε_1 from Corollary 6.1 for $k = d$ and $\delta = 0$, the d releases together are (ε, δ')-DP for $\varepsilon \leq 1$. The root mean squared error is

$$\frac{\sqrt{2d}}{n\varepsilon_1} = \frac{\sqrt{2d} \cdot 2\sqrt{2d \ln(1/\delta')}}{n\varepsilon} = \frac{4d\sqrt{\ln(1/\delta')}}{n\varepsilon}. \tag{6.5}$$

The error now grows as d instead of $d^{3/2}$, which is the order the Gaussian mechanism achieves. Take $n = 100{,}000$, $\varepsilon = 1$ and $\delta' = 10^{-6}$. At $d = 1{,}000$, the two errors are 0.4472 and 0.1487. At $d = 10{,}000$, they are 14.14 and 1.487, nearly ten times apart. The price is the move from pure to approximate DP.

6.2. Reduction to four outcomes

A pair of output distributions $M_j(D)$ and $M_j(D')$ can be complicated. The first step of the proof replaces every such pair by the simplest (ε, δ)-indistinguishable pair. It has two ordinary outcomes, 0 and 1, whose probabilities differ by the factor e^{ε}, and two catastrophic outcomes, each with probability δ, which reveal which distribution the sample came from.

Definition 6.2 (The four-outcome pair)**.** For $\varepsilon \geq 0$ and $\delta \in [0, 1)$, let U and V take values in $\{0, 1, \text{I am } U, \text{I am } V\}$ with

$$\begin{aligned} &\Pr[U=0] = \frac{e^{\varepsilon}(1-\delta)}{1+e^{\varepsilon}}, \quad \Pr[U=1] = \frac{1-\delta}{1+e^{\varepsilon}}, \qquad \Pr[U=\text{I am } U] = \delta, \quad \Pr[U=\text{I am } V] = 0, \\ &\Pr[V=0] = \frac{1-\delta}{1+e^{\varepsilon}}, \qquad \Pr[V=1] = \frac{e^{\varepsilon}(1-\delta)}{1+e^{\varepsilon}}, \quad \Pr[V=\text{I am } U] = 0, \quad \Pr[V=\text{I am } V] = \delta. \end{aligned} \tag{6.6}$$

For $\delta = 0$ this is randomized response with truth probability $e^{\varepsilon}/(1+e^{\varepsilon})$, from the first chapter of this part. Here U is the report of a person whose bit is 0 and V that of a person whose bit is 1. The first check is that the pair meets the condition it stands for.

Lemma 6.1. *The pair U, V is (ε, δ)-indistinguishable.*

Proof.

1. On the outcomes 0 and 1, the ratio $\Pr[U = t]/\Pr[V = t]$ is e^{ε} or $e^{-\varepsilon}$, so $\Pr[U = t] \leq e^{\varepsilon} \Pr[V = t]$.
2. Take an event S. Sum step 1 over the outcomes of S in $\{0, 1\}$. The outcome "I am U" adds at most δ to $\Pr[U \in S]$, and "I am V" adds nothing. So $\Pr[U \in S] \leq e^{\varepsilon} \Pr[V \in S] + \delta$.
3. Swapping the labels $0 \leftrightarrow 1$ and "I am U" $\leftrightarrow$ "I am V" turns the distribution of U into that of V and back, so the same argument gives the other inequality.

□

Indistinguishability survives any further processing, as privacy does. The proof is that of post-processing in the approximate DP chapter, written for a pair of random variables.

Lemma 6.2 (Post-processing)**.** *Let A, B be (ε, δ)-indistinguishable, and let F be a randomized map whose coins are independent of A and B. Then $F(A)$ and $F(B)$ are (ε, δ)-indistinguishable.*

Proof.

1. *Functions with values in* $[0, 1]$. Let g take values in $[0, 1]$. The proof of Lemma 5.5 uses only the bound of the definition on events, so it applies to A and B unchanged and gives $\mathbb{E}[g(A)] \leq e^{\varepsilon} \mathbb{E}[g(B)] + \delta$.
2. *Post-processing.* Fix an event S and let $g(a) = \Pr[F(a) \in S]$, a probability over the coins of F. Because those coins are independent of A, $\Pr[F(A) \in S] = \mathbb{E}[g(A)]$, and likewise for B.

3. Step 1 applied to g gives $\Pr[F(A) \in S] \le e^{\varepsilon} \Pr[F(B) \in S] + \delta$. The other direction is the same with A and B swapped.

□

The four-outcome pair is the hardest (ε, δ)-indistinguishable pair to tell apart, in a strong sense. One randomized map produces any other such pair A, B from U, V.

Theorem 6.2 (Simulation by the four-outcome pair)**.** *Let $\varepsilon \ge 0$ and $\delta \in [0, 1)$, and let A, B be (ε, δ)-indistinguishable random variables on a countable space. Then there is a randomized map F such that $F(U)$ has the distribution of A and $F(V)$ has the distribution of B.*

The map sends each of the four outcomes to a distribution on the space. Some outcomes y are more than e^{ε} times as likely under A as under B. The definition bounds the total excess of A on these outcomes by δ, so "I am U" can carry it, and "I am V" carries the reverse excess. What is left is a pair whose probabilities are within a factor e^{ε} at every outcome. The outcome 0 puts more weight where A is the more likely of the two, and 1 where B is. Kairouz, Oh and Viswanath (2015) first proved the theorem, with hypothesis tests and Blackwell's theorem on the comparison of statistical experiments. The proof below builds the map directly.

Proof. Write $p(y) = \Pr[A = y]$ and $q(y) = \Pr[B = y]$.

1. *The excess.* Let $a(y) = \max(p(y) - e^{\varepsilon} q(y), 0)$ and $b(y) = \max(q(y) - e^{\varepsilon} p(y), 0)$, with totals $\alpha = \sum_y a(y)$ and $\beta = \sum_y b(y)$. Take S, the set of outcomes with $p(y) > e^{\varepsilon} q(y)$. Then $\alpha = \Pr[A \in S] - e^{\varepsilon} \Pr[B \in S] \le \delta$ by the definition. In the same way, $\beta \le \delta$.

2. *The rest.* Let $p_1(y) = p(y) - a(y) = \min(p(y), e^{\varepsilon} q(y))$ and $q_1(y) = q(y) - b(y) = \min(q(y), e^{\varepsilon} p(y))$. Then $p_1 \le e^{\varepsilon} q$ and $p_1 \le p \le e^{2\varepsilon} p$, so $p_1 \le e^{\varepsilon} q_1$. In the same way, $q_1 \le e^{\varepsilon} p_1$. The totals of p_1 and q_1 are $1 - \alpha$ and $1 - \beta$.

3. *Equal totals.* Suppose $\alpha \le \beta$; otherwise swap the roles of p and q. The totals of step 2 differ by $\sum_y (p_1(y) - q_1(y)) = \beta - \alpha$. So the outcomes with $p_1(y) > q_1(y)$ carry an excess $c = \sum_y \max(p_1(y) - q_1(y), 0) \ge \beta - \alpha$. If $\beta > \alpha$, remove the fraction $\theta = (\beta - \alpha)/c \le 1$ of that excess:
$$p_2(y) = p_1(y) - \theta \max(p_1(y) - q_1(y), 0), \tag{6.7}$$
and if $\beta = \alpha$, let $p_2 = p_1$. Let $q_2 = q_1$. Where p_2 differs from p_1, it lies between q_1 and p_1, so the bounds of step 2 hold for p_2 and q_2. Both now have the total $1 - \beta$.

4. *Scale.* Let $p_0 = t\, p_2$ and $q_0 = t\, q_2$ with $t = (1 - \delta)/(1 - \beta)$, which is in $(0, 1]$ because $\beta \le \delta < 1$. Then p_0 and q_0 have the total $1 - \delta$, satisfy $p_0 \le e^{\varepsilon} q_0$ and $q_0 \le e^{\varepsilon} p_0$, and $p_0 \le p$, $q_0 \le q$.

5. *The two catastrophic outcomes.* If $\delta > 0$, let F map "I am U" to a sample from $R_U = (p - p_0)/\delta$ and "I am V" to a sample from $R_V = (q - q_0)/\delta$. By step 4 both are non-negative, and each sums to $(1 - (1 - \delta))/\delta = 1$. If $\delta = 0$, these two outcomes never occur.

6. *The two ordinary outcomes.* If $\varepsilon = 0$, step 4 gives $p_0 = q_0$; let F map 0 and 1 to a sample from $p_0/(1-\delta)$. If $\varepsilon > 0$, define

$$P_0(y) = \frac{e^\varepsilon p_0(y) - q_0(y)}{(e^\varepsilon - 1)(1-\delta)}, \qquad P_1(y) = \frac{e^\varepsilon q_0(y) - p_0(y)}{(e^\varepsilon - 1)(1-\delta)}. \tag{6.8}$$

   Both are non-negative by step 4, and each sums to $(e^\varepsilon - 1)(1-\delta)/((e^\varepsilon - 1)(1-\delta)) = 1$. Let F map 0 to a sample from P_0 and 1 to a sample from P_1.

7. *Check.* U is 0 with probability $e^\varepsilon(1-\delta)/(1+e^\varepsilon)$, 1 with probability $(1-\delta)/(1+e^\varepsilon)$ and "I am U" with probability δ. So, for $\varepsilon > 0$, $F(U)$ equals y with probability

$$\frac{e^\varepsilon(e^\varepsilon p_0(y) - q_0(y)) + e^\varepsilon q_0(y) - p_0(y)}{(1+e^\varepsilon)(e^\varepsilon - 1)} + \delta R_U(y) = p_0(y) + p(y) - p_0(y) = p(y). \tag{6.9}$$

   For $\varepsilon = 0$, the probability is $(1-\delta)\,p_0(y)/(1-\delta) + p(y) - p_0(y) = p(y)$. In the same way, $F(V)$ equals y with probability $q(y)$.

For $\delta = 0$, steps 1 to 4 change nothing: $\alpha = \beta = 0$, $p_0 = p$ and $q_0 = q$. □

Applying Theorem 6.2 once per step of an adaptive composition reduces the whole transcript to a map of k independent copies of the four-outcome pair.

Theorem 6.3. *Let M be the adaptive composition of $M_1, \dots, M_k$ as in Theorem 6.1, and fix neighbours $D \sim D'$. Let $U_1, \dots, U_k$ be independent copies of U and $V_1, \dots, V_k$ independent copies of V. Then there is a randomized map F^* such that $F^*(U_1, \dots, U_k)$ has the distribution of $M(D)$ and $F^*(V_1, \dots, V_k)$ has the distribution of $M(D')$.*

Proof.

1. Fix a step j and an earlier transcript $y_{<j}$. The mechanism $M_j(\,\cdot\,; y_{<j})$ is (ε, δ)-DP, so its outputs $M_j(D; y_{<j})$ and $M_j(D'; y_{<j})$ are (ε, δ)-indistinguishable. Theorem 6.2 gives a randomized map $F_j^{y_{<j}}$ that turns U into $M_j(D; y_{<j})$ and V into $M_j(D'; y_{<j})$.
2. Define F^* on an input $(z_1, \dots, z_k)$ step by step. Set $y_1 = F_1(z_1)$, then $y_2 = F_2^{y_1}(z_2)$, and in general $y_j = F_j^{y_{<j}}(z_j)$, with fresh coins at each step. Output $(y_1, \dots, y_k)$.
3. By induction on j, the prefix $(y_1, \dots, y_j)$ computed from $(U_1, \dots, U_k)$ has the distribution of the first j outputs of $M(D)$. For $j = 1$ this is step 1. Given the prefix $y_{<j}$, the input U_j is still a fresh copy of U, because it is independent of $U_1, \dots, U_{j-1}$ and of the coins used so far. So $y_j = F_j^{y_{<j}}(U_j)$ has the distribution of $M_j(D; y_{<j})$, which is how M draws its j-th output.
4. The same induction with $V_1, \dots, V_k$ gives the distribution of $M(D')$.

□

The map F^* depends on D and D', but that is harmless, because privacy is a statement about one pair of neighbours at a time and Lemma 6.2 applies to any map. So it remains to show that $(U_1, \dots, U_k)$ and $(V_1, \dots, V_k)$ are $(\tilde\varepsilon, \tilde\delta)$-indistinguishable.

6.3. Composing the four-outcome pair

Write $\mathbf{U} = (U_1, \dots, U_k)$ and $\mathbf{V} = (V_1, \dots, V_k)$. As in the approximate DP chapter, the privacy loss of an outcome compares how likely it is under $\mathbf{U}$ and under $\mathbf{V}$. An outcome that is possible under $\mathbf{U}$ but impossible under $\mathbf{V}$ has infinite loss, and for an outcome z that is possible under $\mathbf{V}$, the **privacy loss** is

$$L(z) = \ln \frac{\Pr[\mathbf{U} = z]}{\Pr[\mathbf{V} = z]}. \tag{6.10}$$

The plan is to show that, for z drawn from $\mathbf{U}$, the loss rarely exceeds $\tilde{\varepsilon}$. Two kinds of outcome are bad. An outcome that contains "I am U" is impossible under $\mathbf{V}$, so its loss is infinite, which happens with probability about $k\delta$. On the other outcomes, the loss is a sum of k independent terms, and Hoeffding's inequality bounds the chance that it strays far above its mean by δ'. The first step computes the loss on the outcomes with no catastrophe, in which every coordinate is 0 or 1.

Lemma 6.3. *For every $z \in \{0,1\}^k$, the privacy loss is $L(z) = \varepsilon \sum_{j=1}^{k} (1 - 2z_j)$.*

Proof.

1. The coordinates are independent, so both probabilities factor and the log of the ratio is a sum: $L(z) = \sum_{j=1}^{k} \ln\left(\Pr[U = z_j] / \Pr[V = z_j]\right)$.
2. By Definition 6.2, the ratio is e^{ε} when $z_j = 0$ and $e^{-\varepsilon}$ when $z_j = 1$. In both cases its logarithm is $\varepsilon(1 - 2z_j)$.

□

Given no catastrophic outcome, the coordinates of $\mathbf{U}$ are independent bits, so the loss is a sum of independent terms, each ε or $-\varepsilon$. Hoeffding's inequality bounds its tail.

Take k independent random variables $W_1, \dots, W_k$, each with values in an interval $[l, u]$ of length $u - l$. **Hoeffding's inequality** bounds the chance that their sum exceeds its mean by a margin $\tau \geq 0$ or more,

$$\Pr\left[\sum_{j=1}^{k} W_j \geq \sum_{j=1}^{k} \mathbb{E}[W_j] + \tau\right] \leq \exp\left(-\frac{2\tau^2}{k(u-l)^2}\right). \tag{6.11}$$

With the loss and its tail bound, the main estimate follows: k copies of the four-outcome pair are as hard to tell apart as Theorem 6.1 claims.

Proposition 6.1. *Let $\varepsilon \geq 0$, $\delta \in [0, 1)$ and $\delta' \in (0, 1)$. Then $\mathbf{U}$ and $\mathbf{V}$ are $(\tilde{\varepsilon}, \tilde{\delta})$-indistinguishable, with $\tilde{\varepsilon}$ and $\tilde{\delta}$ as in Equation 6.3.*

Proof.

1. *The events.* Let E_1 be the event that some coordinate of $\mathbf{U}$ is "I am U". Let G be the set of good outcomes, the $z \in \{0,1\}^k$ with $L(z) \leq \tilde{\varepsilon}$. Every outcome z with $\Pr[\mathbf{U} = z] > 0$ that is not in G either lies in E_1 or lies in $\{0,1\}^k$ with $L(z) > \tilde{\varepsilon}$. Outcomes containing "I am V" have probability 0 under $\mathbf{U}$.

2. *The first bad event.* Each coordinate avoids "I am U" with probability $1-\delta$, independently, so $\Pr[E_1] = 1-(1-\delta)^k$. Bernoulli's inequality $(1-\delta)^k \geq 1-k\delta$ gives $\Pr[E_1] \leq k\delta$.

3. *The coordinates given no catastrophe.* Given that U_j is not "I am U", it is 0 with probability $\frac{e^\varepsilon(1-\delta)/(1+e^\varepsilon)}{1-\delta} = \frac{e^\varepsilon}{1+e^\varepsilon}$ and 1 otherwise. Conditioned on the complement $\bar{E}_1$, the coordinates stay independent, because $\bar{E}_1$ is an intersection of independent events, one per coordinate.

4. *The mean of the loss.* By Lemma 6.3, given $\bar{E}_1$ the loss is $\sum_j W_j$ with $W_j = \varepsilon(1-2U_j)$. Each W_j is ε with probability $e^\varepsilon/(1+e^\varepsilon)$ and $-\varepsilon$ otherwise, so

$$\mathbb{E}\Big[\sum_{j=1}^{k} W_j \,\Big|\, \bar{E}_1\Big] = k\varepsilon\,\frac{e^\varepsilon - 1}{e^\varepsilon + 1}, \tag{6.12}$$

   which is the second term of $\tilde{\varepsilon}$.

5. *The second bad event.* If $\varepsilon = 0$, the loss is $0 \leq \tilde{\varepsilon}$ and there is nothing to bound. Otherwise apply Equation 6.11, conditioned on $\bar{E}_1$, with $[l,u] = [-\varepsilon,\varepsilon]$ and $\tau = \varepsilon\sqrt{2k\ln(1/\delta')}$, the first term of $\tilde{\varepsilon}$. The exponent is $2\tau^2/(4k\varepsilon^2) = \ln(1/\delta')$, so

$$\Pr\left[L(\mathbf{U}) > \tilde{\varepsilon} \,\middle|\, \bar{E}_1\right] \leq \delta'. \tag{6.13}$$

   Multiplying by $\Pr[\bar{E}_1] \leq 1$ bounds the probability that $\mathbf{U} \in \{0,1\}^k$ and $L(\mathbf{U}) > \tilde{\varepsilon}$ by δ'.

6. *Assembly.* Take any event S. Split it into $S \cap G$ and the rest. On G, each outcome has $\Pr[\mathbf{U} = z] \leq e^{\tilde{\varepsilon}} \Pr[\mathbf{V} = z]$ by the definition of L. By step 1, the rest has probability at most that of the two bad events. Steps 2 and 5 then give

$$\Pr[\mathbf{U} \in S] \leq e^{\tilde{\varepsilon}} \Pr[\mathbf{V} \in S \cap G] + k\delta + \delta' \leq e^{\tilde{\varepsilon}} \Pr[\mathbf{V} \in S] + \tilde{\delta}. \tag{6.14}$$

7. *The other direction.* Swapping the labels $0 \leftrightarrow 1$ and "I am U" $\leftrightarrow$ "I am V" in every coordinate maps $\mathbf{U}$ to $\mathbf{V}$ and back. The same argument then gives $\Pr[\mathbf{V} \in S] \leq e^{\tilde{\varepsilon}} \Pr[\mathbf{U} \in S] + \tilde{\delta}$.

□

So the term $k\delta$ of $\tilde{\delta}$ pays for a catastrophic outcome in some step, and δ' pays for an unusually long run of outcomes that favour $\mathbf{U}$. The proof of the theorem puts the pieces together.

Proof. Proof of Theorem 6.1. Fix neighbours $D \sim D'$.

1. By Theorem 6.3, $M(D)$ and $M(D')$ have the distributions of $F^*(\mathbf{U})$ and $F^*(\mathbf{V})$ for one randomized map F^*.
2. By Proposition 6.1, $\mathbf{U}$ and $\mathbf{V}$ are $(\tilde{\varepsilon},\tilde{\delta})$-indistinguishable.
3. By Lemma 6.2, so are $F^*(\mathbf{U})$ and $F^*(\mathbf{V})$. The neighbours were arbitrary, so M is $(\tilde{\varepsilon},\tilde{\delta})$-DP.

□

6.4. Summary

Table 6.1.: The chapter's results.

Result	Statement
Theorem 6.1	k adaptive (ε, δ)-DP steps are $(\varepsilon\sqrt{2k\ln(1/\delta')} + k\varepsilon\tanh(\varepsilon/2),\ k\delta + \delta')$-DP
Corollary 6.1	per-step $\varepsilon = \varepsilon_{\text{total}}/(2\sqrt{2k\ln(1/\delta')})$ gives a total of $\varepsilon_{\text{total}} \le 1$
Example 6.2	Laplace noise per coordinate estimates a mean in $\{0,1\}^d$ with error $4d\sqrt{\ln(1/\delta')}/(n\varepsilon)$
Lemma 6.1	the four-outcome pair is (ε, δ)-indistinguishable
Lemma 6.2	indistinguishability survives post-processing
Theorem 6.2	every (ε, δ)-indistinguishable pair is a randomized map of the four-outcome pair
Theorem 6.3	an adaptive composition is a map of k independent copies of the four-outcome pair
Lemma 6.3	on $\{0,1\}^k$, the privacy loss is $\varepsilon\sum_j(1-2z_j)$
Proposition 6.1	k copies of the four-outcome pair are $(\tilde{\varepsilon}, \tilde{\delta})$-indistinguishable

6.5. References

- Cynthia Dwork, Guy N. Rothblum and Salil Vadhan, "Boosting and Differential Privacy", 2010. First proves the advanced composition theorem.
- Peter Kairouz, Sewoong Oh and Pramod Viswanath, "The Composition Theorem for Differential Privacy", 2015. Proves the simulation by the four-outcome pair and finds the exact optimal composition.
- Wassily Hoeffding, "Probability Inequalities for Sums of Bounded Random Variables", 1963. Gives the tail bound used for the privacy loss.
- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014. Proves advanced composition with the privacy loss in Section 3.5, with $k\varepsilon(e^{\varepsilon}-1)$ as the second term.
- Salil Vadhan, "The Complexity of Differential Privacy", 2017. Presents basic and advanced composition in Section 2.
- Jack Murtagh and Salil Vadhan, "The Complexity of Computing the Optimal Composition of Differential Privacy", 2016. Shows that the exact optimal composition is hard to compute for mixed parameters.

7. The exponential mechanism

The Laplace and Gaussian mechanisms release numeric values. Other tasks select an object, such as a price or classifier, and small numeric perturbations may sharply reduce its quality. The exponential mechanism samples objects with probability proportional to an exponential of their score. High-scoring objects are more likely, while the selection satisfies differential privacy (DP). This chapter proves its privacy and utility guarantees and applies it to three problems.

7.1. Selecting an object

A seller offers a digital good, such as an e-book, that costs nothing to copy. Each of n buyers values it at v_i and buys if the price p is at most v_i. Revenue at price p is the price multiplied by the number of buyers,

$$p \cdot \big|\{i : v_i \geq p\}\big|. \tag{7.1}$$

The revenue-maximizing price can reveal an individual valuation. If one buyer values the good much more than others, that valuation may determine the optimal price.

Adding noise to the optimal price can sharply reduce revenue. Suppose three buyers value the good at \$1 and one at \$2.50. A price of \$1 earns \$4; a price of \$1.01 earns \$1.01 because only one buyer purchases. Thus one cent of noise reduces revenue by about three quarters. Instead of perturbing the price, select among candidate prices according to their revenue.

In general, a selection problem specifies:

- a dataset $D \in \mathcal{X}^n$,
- a set $\mathcal{R}$ of objects,
- a **score function** $u \colon \mathcal{X}^n \times \mathcal{R} \to \mathbb{R}$, where $u(D, r)$ says how good the object r is for the dataset D.

For pricing, D is the list of valuations, $\mathcal{R}$ is the set of prices, and $u(D, p)$ is revenue. The candidate objects and score function are public; only D is private. Score sensitivity measures the maximum change in a fixed candidate's score when one record changes.

Definition 7.1 (Sensitivity of a score function)**.** The **sensitivity** of a score function u is

$$\Delta u = \sup_{r \in \mathcal{R}} \sup_{D \sim D'} \big|u(D, r) - u(D', r)\big|, \tag{7.2}$$

where the inner supremum is over neighbouring datasets, which differ in one record (the replace-one relation).

The score is scalar, so no norm is needed. The mechanism assigns weight $\exp(\varepsilon u(D, r)/(2\Delta))$ to each object r, where Δ upper-bounds the score sensitivity. Dividing by the sum of weights gives probabilities. For an infinite object set, such as an interval of prices, the weights define a density and the normalizer is an integral with respect to a fixed measure μ.

Definition 7.2 (Exponential mechanism)**.** Let $\varepsilon > 0$, and let $\Delta > 0$ satisfy $\Delta \geq \Delta u$. Let μ be a measure on $\mathcal{R}$, taken to be the counting measure, which gives each object weight 1, when $\mathcal{R}$ is finite. Assume that the normalizer

$$Z(D) = \int_{\mathcal{R}} \exp\left(\frac{\varepsilon\, u(D, r')}{2\Delta}\right) \mu(\mathrm{d}r') \tag{7.3}$$

is positive and finite for every D. The **exponential mechanism** M_E outputs an object $r \in \mathcal{R}$ drawn with density, with respect to μ,

$$p_D(r) = \frac{1}{Z(D)} \exp\left(\frac{\varepsilon\, u(D, r)}{2\Delta}\right). \tag{7.4}$$

For finite $\mathcal{R}$, Equation 7.4 assigns probability proportional to $\exp(\varepsilon u(D, r)/(2\Delta))$. Higher scores are more likely. Smaller ε or larger Δ makes the distribution more uniform, increasing privacy and reducing utility.

The definition does not specify an efficient sampling method. For finite $\mathcal{R}$, direct sampling takes time proportional to $|\mathcal{R}|$, which is infeasible when the set is exponentially large. For infinite sets, discretization is one option, as in Section 7.4.2. The mechanism is well-defined but not always computationally efficient.

7.2. Privacy

Between neighbours, each score changes by at most Δ, so its weight changes by at most $e^{\varepsilon/2}$. The normalizer changes by the same factor. Their ratio therefore changes by at most e^{ε}.

Theorem 7.1. *The exponential mechanism is ε-DP.*

Proof.

1. Fix neighbours $D \sim D'$ and an object r. By Equation 7.4, the ratio of the densities is

$$\frac{p_D(r)}{p_{D'}(r)} = \exp\left(\frac{\varepsilon\,(u(D, r) - u(D', r))}{2\Delta}\right) \cdot \frac{Z(D')}{Z(D)}. \tag{7.5}$$

2. By Definition 7.1 and $\Delta \geq \Delta u$, the difference $u(D, r) - u(D', r)$ is at most Δ, so the first factor is at most $e^{\varepsilon/2}$.

3. For every object r', the same bound gives $u(D', r') \leq u(D, r') + \Delta$, so $\exp(\varepsilon u(D', r')/(2\Delta)) \leq e^{\varepsilon/2} \exp(\varepsilon u(D, r')/(2\Delta))$. Integrating over r' gives $Z(D') \leq e^{\varepsilon/2} Z(D)$, so the second factor is at most $e^{\varepsilon/2}$.

4. So $p_D(r) \le e^{\varepsilon} p_{D'}(r)$ for every r. For an event $S \subseteq \mathcal{R}$, integrating over S gives $\Pr[M_E(D) \in S] \le e^{\varepsilon} \Pr[M_E(D') \in S]$.

□

The factor of 2 in the denominator allocates half the privacy budget to the score and half to the normalizer. If the normalizer is data-independent, the mechanism does not need this allocation.

Corollary 7.1. *Suppose that $Z(D)$, computed with Δ in place of 2Δ, is the same for every dataset D. Then the mechanism with density proportional to $\exp(\varepsilon u(D,r)/\Delta)$ is ε-DP.*

Proof. In step 1 of the proof of Theorem 7.1, with Δ in place of 2Δ, the factor $Z(D')/Z(D)$ equals 1, and the first factor is at most $\exp(\varepsilon\Delta/\Delta) = e^{\varepsilon}$. Step 4 then applies unchanged. □

7.3. Utility

Utility requires an output whose score is near the maximum. Define the maximum score and its set of maximizers by

$$\mathrm{OPT}(D) = \max_{r \in \mathcal{R}} u(D,r), \qquad \mathcal{R}^*(D) = \{r \in \mathcal{R} : u(D,r) = \mathrm{OPT}(D)\}. \tag{7.6}$$

The theorem bounds the score loss by approximately $(2\Delta/\varepsilon)\ln|\mathcal{R}|$. The loss is linear in sensitivity and $1/\varepsilon$, but logarithmic in the number of candidates.

Theorem 7.2. *Let $\mathcal{R}$ be finite. For every dataset D and every $t \ge 0$,*

$$\Pr\left[u(D, M_E(D)) \le \mathrm{OPT}(D) - \frac{2\Delta}{\varepsilon}\left(\ln\frac{|\mathcal{R}|}{|\mathcal{R}^*(D)|} + t\right)\right] \le e^{-t}. \tag{7.7}$$

Proof.

1. Fix a threshold τ. The output has score at most τ exactly when it is one of the objects with $u(D,r) \le \tau$, so

$$\Pr[u(D, M_E(D)) \le \tau] = \frac{\sum_{r : u(D,r) \le \tau} \exp(\varepsilon u(D,r)/(2\Delta))}{\sum_{r' \in \mathcal{R}} \exp(\varepsilon u(D,r')/(2\Delta))}. \tag{7.8}$$

2. *Numerator.* There are at most $|\mathcal{R}|$ terms, and each is at most $\exp(\varepsilon\tau/(2\Delta))$ because its score is at most τ.

3. *Denominator.* Every term is positive, so keeping only the $|\mathcal{R}^*(D)|$ terms with score $\mathrm{OPT}(D)$ gives a lower bound of $|\mathcal{R}^*(D)|\exp(\varepsilon\,\mathrm{OPT}(D)/(2\Delta))$.

4. Together, steps 2 and 3 give

$$\Pr[u(D, M_E(D)) \le \tau] \le \frac{|\mathcal{R}|}{|\mathcal{R}^*(D)|} \exp\left(\frac{\varepsilon\,(\tau - \mathrm{OPT}(D))}{2\Delta}\right). \tag{7.9}$$

5. Set $\tau = \mathrm{OPT}(D) - (2\Delta/\varepsilon)(\ln(|\mathcal{R}|/|\mathcal{R}^*(D)|) + t)$. The exponent becomes $-\ln(|\mathcal{R}|/|\mathcal{R}^*(D)|) - t$, and the right-hand side becomes e^{-t}.

□

Since $|\mathcal{R}^*(D)| \ge 1$, the following corollary gives the commonly used, simpler bound.

Corollary 7.2. *Let $\mathcal{R}$ be finite. For every dataset D and every $t \ge 0$,*

$$\Pr\left[u(D, M_E(D)) \le \mathrm{OPT}(D) - \frac{2\Delta}{\varepsilon}(\ln|\mathcal{R}| + t)\right] \le e^{-t}. \tag{7.10}$$

Proof.

1. At least one object attains the best score, so $|\mathcal{R}^*(D)| \ge 1$ and $\ln(|\mathcal{R}|/|\mathcal{R}^*(D)|) \le \ln|\mathcal{R}|$. The threshold in Equation 7.10 is therefore at most the threshold of Theorem 7.2 for the same t, so the event here is contained in the event there, and its probability is at most e^{-t}.

□

7.4. Applications

Each application specifies the candidate set and score, bounds sensitivity, and applies Theorem 7.1 and Corollary 7.2.

7.4.1. The Laplace mechanism

Let $f\colon \mathcal{X}^n \to \mathbb{R}$ have sensitivity Δf, which means that $|f(D) - f(D')| \le \Delta f$ for all neighbours. Take the real line as the set of objects, with Lebesgue measure, and score each candidate answer by its distance from the true one,

$$u(D, r) = -|f(D) - r|. \tag{7.11}$$

By the triangle inequality, $\big||f(D) - r| - |f(D') - r|\big| \le |f(D) - f(D')| \le \Delta f$, so $\Delta u \le \Delta f$. With $\Delta = \Delta f$, the density Equation 7.4 is proportional to $\exp(-\varepsilon|f(D) - r|/(2\Delta f))$, which is the density of $f(D)$ plus Laplace noise of scale $2\Delta f/\varepsilon$. So the exponential mechanism contains a Laplace mechanism, but with twice the noise scale $\Delta f/\varepsilon$ of the Laplace mechanism. Corollary 7.1 removes the extra factor.

Proposition 7.1. *Let $f : \mathcal{X}^n \to \mathbb{R}$ have sensitivity $\Delta f > 0$. The mechanism that outputs $f(D)$ plus noise from the Laplace distribution with scale $\Delta f/\varepsilon$ is ε-DP.*

Proof.

1. The output has density proportional to $\exp(-\varepsilon|f(D) - r|/(\Delta f)) = \exp(\varepsilon u(D, r)/(\Delta f))$, with the score above, whose sensitivity is at most Δf.
2. The normalizer is $\int_{\mathbb{R}} \exp(-\varepsilon|f(D)-r|/(\Delta f))\,\mathrm{d}r = 2\Delta f/\varepsilon$, after the substitution $u = r-f(D)$. It does not depend on D.
3. Corollary 7.1, with $\Delta = \Delta f$, applies and gives ε-DP.

□

7.4.2. Pricing a digital good

Return to the seller of Section 7.1, now with a dataset of valuations $D = (v_1, \dots, v_n) \in [0, 1]^n$ and prices in $[0, 1]$. Write γ for the best revenue over all prices,

$$\gamma = \max_{p \in [0,1]} p \cdot |\{i : v_i \geq p\}|. \tag{7.12}$$

The maximum exists: revenue increases linearly between consecutive valuations and drops immediately above each valuation, so an optimum occurs at a valuation or at 0.

Because the continuous price set is difficult to sample from, use a grid. For an integer $m \geq 1$, set $\alpha = 1/m$ and use the m prices

$$\mathcal{R} = \{\alpha, 2\alpha, \dots, 1\}. \tag{7.13}$$

Write $\mathrm{OPT}(D)$ for the maximum grid revenue. The next lemma bounds the loss from rounding the optimal price down to the grid.

Lemma 7.1. *The best revenue on the grid satisfies* $\mathrm{OPT}(D) \geq \gamma - \alpha n$.

Proof.

1. Let p^* attain γ. If $p^* < \alpha$, then $\gamma \leq p^* n < \alpha n$, so $\gamma - \alpha n < 0$. Revenue is never negative, so $\mathrm{OPT}(D) \geq 0 > \gamma - \alpha n$.
2. Otherwise, round p^* down to the grid, $p' = \alpha \lfloor p^*/\alpha \rfloor \in \mathcal{R}$, so that $p^* - \alpha < p' \leq p^*$.
3. Every buyer with $v_i \geq p^*$ also has $v_i \geq p'$. So the revenue at p' is at least $p' \cdot |\{i : v_i \geq p^*\}| \geq (p^* - \alpha)|\{i : v_i \geq p^*\}| \geq \gamma - \alpha n$.

□

The private price incurs rounding loss and selection loss. The theorem bounds their sum using a grid of about εn prices.

Theorem 7.3. *Let $\varepsilon > 0$, $m = \lceil \varepsilon n \rceil$ and $\alpha = 1/m$. The exponential mechanism with the grid $\mathcal{R}$, the revenue $u(D, p) = p \cdot |\{i : v_i \geq p\}|$ as score and $\Delta = 1$ is ε-DP. For every $D \in [0,1]^n$ and $t \geq 0$, with probability at least $1 - e^{-t}$, the revenue at its output price is at least*

$$\gamma - \frac{1 + 2\ln\lceil \varepsilon n \rceil + 2t}{\varepsilon}. \tag{7.14}$$

Proof.

1. *Sensitivity.* Changing one valuation changes the number of buyers at any price p by at most 1, and $p \leq 1$. So $\Delta u \leq 1 = \Delta$, and Theorem 7.1 gives ε-DP.
2. *Selection.* By Corollary 7.2 with $|\mathcal{R}| = m$, with probability at least $1 - e^{-t}$ the output has revenue at least $\mathrm{OPT}(D) - (2/\varepsilon)(\ln m + t)$.
3. *Discretization.* By Lemma 7.1, $\mathrm{OPT}(D) \geq \gamma - n/m$, and $n/m \leq n/(\varepsilon n) = 1/\varepsilon$ because $m \geq \varepsilon n$.
4. Adding the two losses gives Equation 7.14.

□

The loss is $O(\log(\varepsilon n)/\varepsilon)$. If a constant fraction of buyers values the good above a constant price, then γ grows linearly with n, and the loss is a vanishing fraction of optimal revenue. Grid size balances rounding loss, αn, against selection loss, $(2/\varepsilon)\ln(1/\alpha)$. The next example compares the bound with observed utility.

Example 7.1. Draw $n = 1{,}000$ valuations uniformly from $[0, 1]$ and take $\varepsilon = 0.5$, so that the grid has $m = 500$ prices. The best revenue is $\gamma = 267.40$, and the best grid price earns 266.88. With $t = \ln 20$, Equation 7.14 guarantees a revenue of at least $267.40 - 38.84 = 228.56$ with probability 0.95. The exact output distribution does much better. Its mean revenue is 263.86, and its 5th percentile is 256.52, a loss of 10.88, so the guarantee is about 3.6 times too pessimistic. Figure 7.1 shows the output distribution on a smaller example.

7.4.3. Private PAC learning

In the probably approximately correct (PAC) model, a learner receives labelled records and outputs a classifier accurate on new samples from the same distribution. Here, records are points in $\{0,1\}^d$ with labels in $\{0,1\}$. Accuracy is specified by error α and failure probability β; here α is not the pricing grid step.

Definition 7.3 (PAC learning)**.** A **concept class** is a finite set $\mathcal{C}$ of functions $c \colon \{0,1\}^d \to \{0,1\}$. An unknown **target** $c^* \in \mathcal{C}$ and an unknown distribution $\mathcal{D}$ on $\{0,1\}^d$ generate the dataset. Each record is a point x_i, drawn independently from $\mathcal{D}$, together with its label $y_i = c^*(x_i)$. So the dataset is $D = (z_1, \ldots, z_n)$ with labelled records $z_i = (x_i, y_i)$. The **error** of a function h is

$$\mathrm{err}(h) = \Pr_{x \sim \mathcal{D}}[h(x) \neq c^*(x)]. \tag{7.15}$$

A learner is (α, β)**-accurate** if, for every target and every $\mathcal{D}$, it outputs a function $\hat{h}$ with $\mathrm{err}(\hat{h}) \leq \alpha$ with probability at least $1 - \beta$.

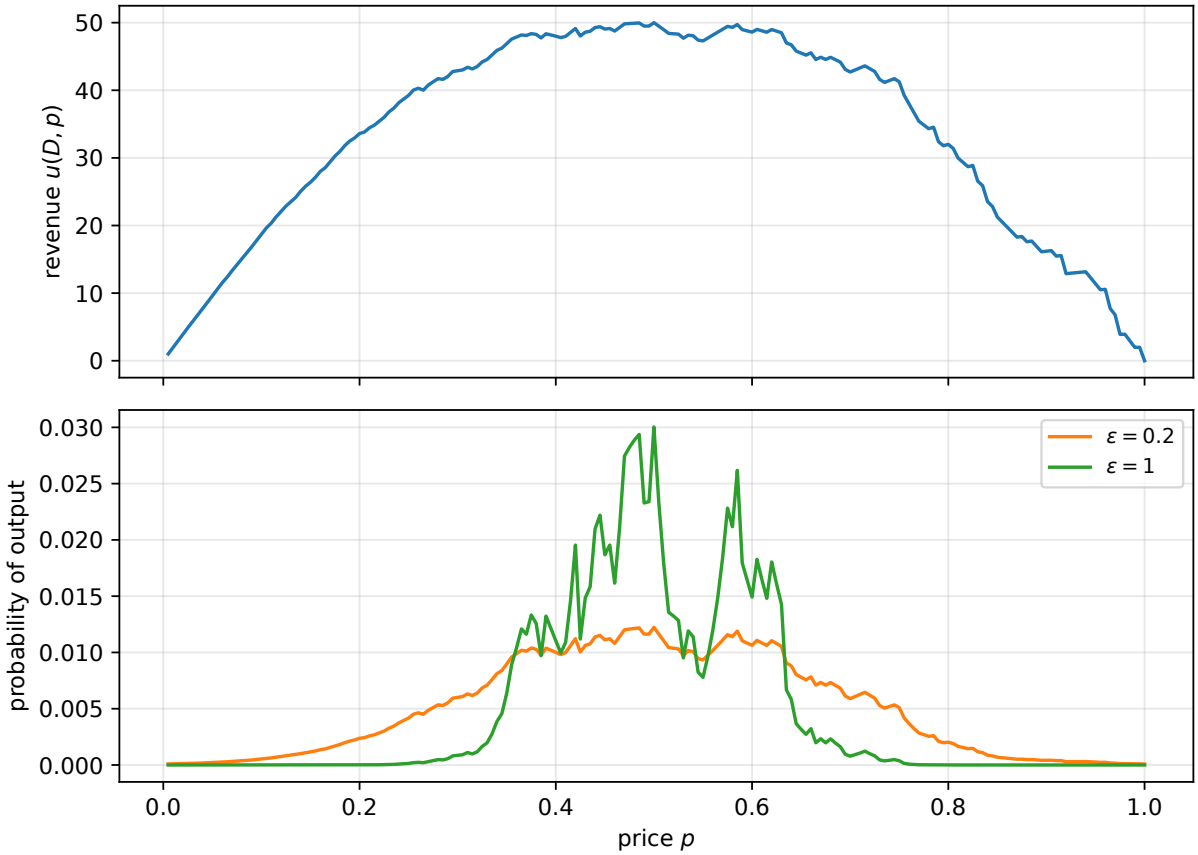


Figure 7.1.: The exponential mechanism for pricing, with 200 buyers whose valuations are uniform on $[0, 1]$ and a grid of 200 prices. Top, the revenue. Bottom, the probability of each output price for $\varepsilon = 0.2$ and $\varepsilon = 1$. A larger ε puts more weight on prices near the best revenue.

For example, $\mathcal{C}$ may be a finite set of half-spaces that label points on one side of a hyperplane as 1.

Without privacy, a consistent learner outputs a function in $\mathcal{C}$ that labels every record correctly; the target guarantees such a function exists. Uniform convergence ensures that every function's sample error is close to its true error. Define the sample error by

$$\widehat{\mathrm{err}}(h) = \frac{1}{n}\big|\{i : h(x_i) \neq y_i\}\big|. \tag{7.16}$$

Hoeffding's inequality bounds the probability that an average of n independent $[0,1]$ variables falls below its mean by at least s by $\exp(-2ns^2)$. Apply it to each function's error indicators and take a union bound over $\mathcal{C}$.

Lemma 7.2. *Let $\alpha, \beta \in (0,1)$ and $n \geq 2\ln(|\mathcal{C}|/\beta)/\alpha^2$. With probability at least $1-\beta$ over the sample, every $h \in \mathcal{C}$ satisfies $\mathrm{err}(h) \leq \widehat{\mathrm{err}}(h) + \alpha/2$. In particular, every function in $\mathcal{C}$ that labels the whole sample correctly has error at most $\alpha/2$.*

Proof.

1. Fix $h \in \mathcal{C}$. The indicators $W_i = \mathbf{1}[h(x_i) \neq c^*(x_i)]$ are independent, lie in $[0,1]$, and have mean $\mathrm{err}(h)$, because the x_i are independent draws from $\mathcal{D}$. Their average is $\widehat{\mathrm{err}}(h)$.
2. Hoeffding's inequality with $s = \alpha/2$ gives $\Pr[\widehat{\mathrm{err}}(h) \leq \mathrm{err}(h) - \alpha/2] \leq \exp(-n\alpha^2/2) \leq \beta/|\mathcal{C}|$, by the bound on n.
3. A union bound over the $|\mathcal{C}|$ functions shows that some h fails with probability at most β.
4. A function that labels the sample correctly has $\widehat{\mathrm{err}}(h) = 0$, so its error is at most $\alpha/2$.

□

Thus, $O(\ln|\mathcal{C}|/\alpha^2)$ records suffice.

For privacy, neighbours differ in one labelled record, and the replacement (x_i', y_i') need not follow $\mathcal{D}$ or satisfy $y_i' = c^*(x_i')$. Accuracy assumes data from the learning model; privacy must hold for every dataset.

The private learner selects a function using the exponential mechanism with score equal to negative sample error. Functions with fewer mistakes receive higher probability.

Theorem 7.4 (Private PAC learning)**.** *Let $\varepsilon > 0$ and $\alpha, \beta \in (0,1)$. Run the exponential mechanism with $\mathcal{R} = \mathcal{C}$, the score $u(D,h) = -\widehat{\mathrm{err}}(h)$ and $\Delta = 1/n$. It is ε-DP for every dataset. If*

$$n \geq \frac{2\ln(2|\mathcal{C}|/\beta)}{\alpha^2} \qquad \text{and} \qquad n \geq \frac{4\ln(2|\mathcal{C}|/\beta)}{\alpha\varepsilon}, \tag{7.17}$$

then it is an (α,β)-accurate learner.

Proof.

1. *Privacy.* Replacing one labelled record changes the number of mistakes of each h by at most 1, so $\Delta u \le 1/n$. Theorem 7.1 gives ε-DP for all datasets, whatever their labels.
2. *Selection.* The target makes no mistakes, so $\mathrm{OPT} = 0$. By Corollary 7.2 with $t = \ln(2/\beta)$, with probability at least $1 - \beta/2$ over the mechanism's coins, the output $\hat{h}$ has

$$\widehat{\mathrm{err}}(\hat{h}) \le \frac{2}{\varepsilon n}\left(\ln|\mathcal{C}| + \ln(2/\beta)\right) = \frac{2\ln(2|\mathcal{C}|/\beta)}{\varepsilon n} \le \frac{\alpha}{2}, \tag{7.18}$$

   where the last step uses the second bound on n.
3. *Generalization.* The first bound on n is the condition of Lemma 7.2 with $\beta/2$ in place of β. So with probability at least $1 - \beta/2$ over the sample, every $h \in \mathcal{C}$, including $\hat{h}$, has $\mathrm{err}(h) \le \widehat{\mathrm{err}}(h) + \alpha/2$.
4. By a union bound, both events hold with probability at least $1 - \beta$, and then $\mathrm{err}(\hat{h}) \le \alpha/2 + \alpha/2 = \alpha$.

□

The first bound on n is what learning needs without privacy, and the second is the cost of privacy. Both are logarithmic in $|\mathcal{C}|$, and for ε of constant size and small α, the cost of privacy is a lower-order term. For instance, with $|\mathcal{C}| = 2^{20}$, $\alpha = \beta = 0.05$ and $\varepsilon = 1$, we have $\ln(2|\mathcal{C}|/\beta) = 17.552$, and the two bounds are 14,041 and 1,404. The price is computation, because the mechanism scores every function in $\mathcal{C}$.

7.5. Summary

Table 7.1.: The chapter's results.

Result	Statement
Definition 7.2	output r with probability proportional to $\exp(\varepsilon u(D, r)/(2\Delta))$
Theorem 7.1	the exponential mechanism is ε-DP
Corollary 7.1	with a normalizer independent of D, $\exp(\varepsilon u/\Delta)$ is ε-DP
Theorem 7.2	the score falls below $\mathrm{OPT} - \frac{2\Delta}{\varepsilon}(\ln\frac{\lvert\mathcal{R}\rvert}{\lvert\mathcal{R}^*\rvert} + t)$ with probability at most e^{-t}
Corollary 7.2	the same with $\ln\lvert\mathcal{R}\rvert$
Proposition 7.1	the Laplace mechanism is the exponential mechanism with score $-\lvert f(D) - r\rvert$
Theorem 7.3	a private price earns $\gamma - O(\log(\varepsilon n)/\varepsilon)$
Lemma 7.2	$2\ln(\lvert\mathcal{C}\rvert/\beta)/\alpha^2$ records make every sample error close to the true error
Theorem 7.4	private PAC learning with $O(\ln(\lvert\mathcal{C}\rvert/\beta)(1/\alpha^2 + 1/(\alpha\varepsilon)))$ records

7.6. References

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Part III.

Query release and its limits

The previous part introduced mechanisms for answering a few questions under differential privacy (DP). This part studies how to answer many queries on one dataset. The first two chapters avoid assigning a separate privacy-budget share to every query. The third calibrates noise to the dataset instead of the worst case. The last proves that pure DP limits improvements over simple methods for marginals and histograms.

Table 7.2.: The chapters of this part.

Chapter	Question it answers
Private multiplicative weights	How many records are needed to answer a large set of linear queries at once?
Sparse vector	How can a stream of queries be answered when the cost should depend only on the few large ones?
Beyond global sensitivity	When can the noise follow the sensitivity of the actual dataset instead of the worst case?
Packing lower bounds	How much error must any pure DP mechanism incur on marginals and histograms?

8. Private multiplicative weights

An analyst may need answers to many statistical queries, such as the fraction of records in each table cell. Independent noise on each answer can require a number of records polynomial in the number of queries. Under approximate differential privacy (DP), private multiplicative weights needs a sample size logarithmic in the query-set and universe sizes. The algorithm adapts online learning: it constructs a distribution that matches the data on each query while accessing the data only through a small number of private measurements.

8.1. Linear queries

Records lie in the universe $\mathcal{X}$. A linear query returns the dataset average of a bounded function.

Definition 8.1 (Linear query)**.** A **linear query** is a function $q\colon \mathcal{X} \to [0,1]$ on the universe. On a dataset $D \in \mathcal{X}^n$ with records $x_1, \dots, x_n$ it takes the value

$$q(D) = \frac{1}{n} \sum_{i=1}^{n} q(x_i). \tag{8.1}$$

A counting query has a binary-valued q. Two common families are **histograms**, with one query $q_s(x) = \mathbf{1}[x = s]$ per universe element, and **one-way marginals**, with universe $\{0,1\}^d$ and query $q_j(x) = x_j$ for each coordinate. The analysis applies to any finite query set.

The mechanism must answer every query at once. Its error is the largest absolute error across the set, and it succeeds if this is at most α with probability at least $1 - \beta$.

Definition 8.2 (Query release)**.** Let $\mathcal{Q}$ be a finite set of linear queries, $\alpha \in (0,1]$ a target error and $\beta \in (0,1)$ a failure probability. A mechanism $M\colon \mathcal{X}^n \to [0,1]^{\mathcal{Q}}$ outputs one answer $M(D)_q$ for each query $q \in \mathcal{Q}$. It answers $\mathcal{Q}$ **to accuracy** α with probability $1 - \beta$ if, for every dataset D,

$$\Pr\left[\max_{q \in \mathcal{Q}} |M(D)_q - q(D)| \le \alpha\right] \ge 1 - \beta. \tag{8.2}$$

The following sufficient sample-size bounds quantify the data required.

Changing one record changes a linear query by at most $1/n$, so its **sensitivity** is at most $1/n$. The Laplace mechanism adds noise from $\mathrm{Lap}(b)$, whose density is $e^{-|z|/b}/(2b)$. At scale equal to ℓ_1 sensitivity divided by ε, it is ε-DP, and $\Pr[|\mathrm{Lap}(b)| \ge tb] = e^{-t}$. Applying it independently to each query gives a baseline for comparison.

Proposition 8.1 (Laplace baseline)**.** *Let $\varepsilon > 0$ and $\beta \in (0,1)$. Release $M(D)_q = q(D) + Z_q$ for every $q \in \mathcal{Q}$, with independent $Z_q \sim \mathrm{Lap}(|\mathcal{Q}|/(\varepsilon n))$. This mechanism is ε-DP, and it answers $\mathcal{Q}$ to accuracy α with probability $1-\beta$ whenever*

$$n \geq \frac{|\mathcal{Q}| \ln(|\mathcal{Q}|/\beta)}{\alpha\varepsilon}. \tag{8.3}$$

Proof.

1. The vector $(q(D))_{q\in\mathcal{Q}}$ changes by at most $1/n$ in each of its $|\mathcal{Q}|$ coordinates when one record changes, so its ℓ_1 sensitivity is at most $|\mathcal{Q}|/n$. The Laplace mechanism with scale $|\mathcal{Q}|/(\varepsilon n)$ is therefore ε-DP.
2. Write $b = |\mathcal{Q}|/(\varepsilon n)$ and $t = \ln(|\mathcal{Q}|/\beta)$. By the tail formula and a union bound over the $|\mathcal{Q}|$ coordinates, $\Pr[\max_q |Z_q| \geq tb] \leq |\mathcal{Q}| e^{-t} = \beta$.
3. The condition on n says exactly that $tb \leq \alpha$. So with probability at least $1-\beta$, every error is below α.

□

The advanced composition theorem improves the bound under approximate DP. It implies that k adaptive mechanisms, each ε_0-DP, are (ε', δ)-DP for every $\delta \in (0,1)$, where

$$\varepsilon' = \varepsilon_0 \sqrt{2k \ln(1/\delta)} + k\varepsilon_0 \left(e^{\varepsilon_0} - 1\right). \tag{8.4}$$

For a target budget ε, the next lemma bounds the per-mechanism budget ε_0 that keeps $\varepsilon' \leq \varepsilon$. It restates the budget-splitting corollary for Equation 8.4.

Lemma 8.1 (Splitting a budget)**.** *Let $\varepsilon \in (0,1]$, $\delta \in (0, 1/2]$ and $k \geq 1$. The adaptive composition of k mechanisms, each ε_0-DP with*

$$\varepsilon_0 = \frac{\varepsilon}{2\sqrt{2k \ln(1/\delta)}}, \tag{8.5}$$

is (ε, δ)-DP.

Proof.

1. The first term of Equation 8.4 is $\varepsilon_0 \sqrt{2k \ln(1/\delta)} = \varepsilon/2$.
2. Since $\ln(1/\delta) \geq \ln 2$, $\varepsilon \leq 1$ and $k \geq 1$, we have $\varepsilon_0 \leq 1$. The function e^x is convex, so $e^x - 1$ lies below the chord $(e-1)x \leq 2x$ on $[0,1]$. Hence the second term is at most $2k\varepsilon_0^2 = \varepsilon^2/(4\ln(1/\delta))$.
3. That is at most $\varepsilon/2$ because $\varepsilon \leq 1 \leq 2\ln 2 \leq 2\ln(1/\delta)$. Adding the two terms gives $\varepsilon' \leq \varepsilon$.

□

With one Laplace mechanism per query, the lemma permits a budget of order $\varepsilon/\sqrt{|\mathcal{Q}|}$ per query instead of $\varepsilon/|\mathcal{Q}|$. This reduces the baseline's factor $|\mathcal{Q}|$ to $\sqrt{|\mathcal{Q}|}$.

Proposition 8.2 (Laplace with advanced composition)**.** *Let $\varepsilon \in (0,1]$, $\delta \in (0,1/2]$ and $\beta \in (0,1)$. Release $M(D)_q = q(D) + Z_q$ with independent $Z_q \sim \mathrm{Lap}(1/(\varepsilon_0 n))$, where $\varepsilon_0 = \varepsilon/(2\sqrt{2|\mathcal{Q}|\ln(1/\delta)})$. This mechanism is (ε,δ)-DP, and it answers $\mathcal{Q}$ to accuracy α with probability $1-\beta$ whenever*

$$n \geq \frac{2\sqrt{2|\mathcal{Q}|\ln(1/\delta)}\,\ln(|\mathcal{Q}|/\beta)}{\alpha\varepsilon}. \tag{8.6}$$

Proof.

1. Each coordinate is the Laplace mechanism for a query of sensitivity $1/n$, so it is ε_0-DP. Lemma 8.1 with $k = |\mathcal{Q}|$ makes the whole release (ε,δ)-DP.
2. As in Proposition 8.1, $\Pr[\max_q |Z_q| \geq \ln(|\mathcal{Q}|/\beta)/(\varepsilon_0 n)] \leq \beta$, and the condition on n says that $\ln(|\mathcal{Q}|/\beta)/(\varepsilon_0 n) \leq \alpha$.

□

The Gaussian mechanism with advanced composition improves the rate to $\sqrt{|\mathcal{Q}|\ln|\mathcal{Q}|\ln(1/\delta)}/(\alpha\varepsilon)$ because its tails reduce the union-bound factor from $\ln|\mathcal{Q}|$ to $\sqrt{\ln|\mathcal{Q}|}$. These bounds remain polynomial in $|\mathcal{Q}|$; in the worst case, this dependence cannot be improved. A smaller universe can help, as Table 8.1 shows.

Table 8.1.: Sufficient sample sizes for answering every query in $\mathcal{Q}$ to accuracy α.

Method	Privacy	Records needed, up to constants
Laplace, basic composition	ε-DP	$\lvert\mathcal{Q}\rvert\ln\lvert\mathcal{Q}\rvert/(\alpha\varepsilon)$, improvable to $\lvert\mathcal{Q}\rvert/(\alpha\varepsilon)$
Gaussian, advanced composition	(ε,δ)-DP	$\sqrt{\lvert\mathcal{Q}\rvert\ln\lvert\mathcal{Q}\rvert\ln(1/\delta)}/(\alpha\varepsilon)$
Noisy histogram	ε-DP	$\sqrt{\lvert\mathcal{X}\rvert\ln\lvert\mathcal{Q}\rvert}/(\alpha\varepsilon)$
Exponential mechanism over small datasets	ε-DP	$\ln\lvert\mathcal{Q}\rvert\ln\lvert\mathcal{X}\rvert/(\alpha^3\varepsilon)$, up to logarithmic factors
Private multiplicative weights	(ε,δ)-DP	$\ln\lvert\mathcal{Q}\rvert\sqrt{\ln\lvert\mathcal{X}\rvert\ln(1/\delta)}/(\alpha^2\varepsilon)$, up to logarithmic factors

The noisy histogram releases one noisy count per element of the universe and answers every query from it. So it pays polynomially in $|\mathcal{X}|$, which is 2^d for d binary features. The fourth row is logarithmic in both $|\mathcal{Q}|$ and $|\mathcal{X}|$, but its running time is $|\mathcal{X}|^{O(\ln|\mathcal{Q}|/\alpha^2)}$. Private multiplicative weights keeps both logarithms and runs in time polynomial in $|\mathcal{Q}|$, $|\mathcal{X}|$ and n. This one outputs synthetic data, and for such algorithms that is close to the best possible under standard cryptographic assumptions.

8.2. Learning from experts

Multiplicative weights is an algorithm for a game of repeated predictions, which has nothing to do with privacy. This section develops it from the simplest case.

A group of N experts makes predictions over T days. Each day, every expert predicts whether a stock goes up or down, and the algorithm sees these predictions before it makes its own. Then the true outcome is revealed, and a wrong prediction counts as a **mistake**. Suppose first that one expert, whose identity is unknown, is always right. Then the algorithm can stop listening to any expert as soon as it errs.

Proposition 8.3 (Halving)**.** *Predict with the majority of the experts that have never erred, breaking ties towards "down", and discard every expert that errs. If some expert never errs, this algorithm makes at most* $\log_2 N$ *mistakes.*

Proof.

1. Let S^t be the set of experts still kept on day t, so $|S^1| = N$.
2. On a day with a mistake, the experts who predicted like the algorithm are wrong. They are at least half of S^t, because the algorithm followed the majority. All of them are discarded, so $|S^{t+1}| \le |S^t|/2$.
3. The perfect expert is never discarded, so $|S^t| \ge 1$ on every day. After M mistakes, $1 \le N/2^M$, so $M \le \log_2 N$.

□

The algorithm is measured against the best expert in hindsight. Its **regret** is its total loss minus the total loss of the best single expert. Here the best expert makes no mistakes, so the regret is at most $\log_2 N$.

Now suppose instead that the best expert makes OPT mistakes. Then halving fails, because the best expert may err on the first day and be discarded, after which the rest may all be wrong forever. The repair is to lower trust instead of removing it.

Proposition 8.4 (Weighted majority)**.** *Give each expert weight* 1*. Each day, predict with the side whose experts have the larger total weight, breaking ties towards "down", and then halve the weight of every expert that erred. If the best expert makes* OPT *mistakes, the algorithm makes* M *mistakes with*

$$M \le \frac{\mathrm{OPT} + \log_2 N}{\log_2(4/3)} \le 2.41\left(\mathrm{OPT} + \log_2 N\right). \tag{8.7}$$

Proof.

1. Let W^t be the total weight at the start of day t, so $W^1 = N$. Weights never increase.
2. *Upper bound.* On a day with a mistake, the side the algorithm followed carries at least half of W^t, and all of its weight is halved. So $W^{t+1} \le W^t - W^t/4 = \frac{3}{4}W^t$. Over T days, $W^{T+1} \le N(3/4)^M$.

3. *Lower bound.* The best expert's weight is halved exactly OPT times, so $W^{T+1} \geq (1/2)^{\mathrm{OPT}}$.
4. Combining the bounds gives $(4/3)^M \leq N\, 2^{\mathrm{OPT}}$. Taking $\log_2$ of both sides gives the first inequality, and $1/\log_2(4/3) = 2.409\ldots$ gives the second.

□

The guarantee is competitive with the best expert, but its regret, $1.41\,\mathrm{OPT} + 2.41 \log_2 N$, grows with OPT. A randomized algorithm with gentler updates removes the factor in front of OPT.

The general game has losses instead of mistakes. In each of T rounds, every expert i suffers a loss $\ell_i^t \in [-1, 1]$, chosen by an adversary, and a negative loss is a gain. A randomized algorithm picks one expert and suffers the loss of that expert. It is cleaner to let the algorithm pick a distribution p^t over the experts and suffer the average loss $\ell^t \cdot p^t = \sum_i p_i^t \ell_i^t$. Multiplicative weights keeps one weight per expert and plays the distribution proportional to the weights. After each round it multiplies each weight by a factor that falls as the loss of that expert rises.

Definition 8.3 (Multiplicative weights)**.** Fix a learning rate $\eta \in (0, 1/2]$. Start with weights $w_i^1 = 1$ for $i = 1, \ldots, N$. In round $t = 1, 2, \ldots$:

1. Play the distribution $p_i^t = w_i^t / W^t$, where $W^t = \sum_{i=1}^N w_i^t$.
2. Observe the loss vector $\ell^t \in [-1, 1]^N$, which may depend on $p^1, \ldots, p^t$, and suffer $\ell^t \cdot p^t$.
3. Update $w_i^{t+1} = w_i^t\,(1 - \eta \ell_i^t)$ for every i.

The factor $1 - \eta \ell_i^t$ lies in $[1/2, 3/2]$, so the weights stay positive, and an expert with a positive loss loses weight while one with a negative loss gains it. The analysis takes the logarithm of the total weight, which turns the products of factors $1 - \eta \ell_i^t$ into sums of $\ln(1 - \eta \ell_i^t)$. So it needs two bounds on that logarithm, one from above and one from below.

Lemma 8.2. *For every $x < 1$, $\ln(1 - x) \leq -x$. For every $x \in [-1/2, 1/2]$, $\ln(1 - x) \geq -x - x^2$.*

Proof.

1. The function $\ln(1 - x)$ is concave, so it lies below its tangent line at 0, which is $-x$.
2. Let $h(x) = \ln(1 - x) + x + x^2$. Then $h(0) = 0$ and $h'(x) = -\frac{1}{1-x} + 1 + 2x = \frac{x(1-2x)}{1-x}$.
3. On $[0, 1/2]$, $h' \geq 0$, so $h(x) \geq h(0) = 0$. On $[-1/2, 0]$, $h' \leq 0$, so h decreases to $h(0) = 0$ and is non-negative there too.

□

The main result compares the total loss of multiplicative weights with the total loss of any fixed distribution p chosen in hindsight. The difference is the regret, and the theorem bounds it by a term that grows with η and a term that shrinks with it.

Theorem 8.1 (Regret of multiplicative weights)**.** *Let $p^1, p^2, \dots$ be the distributions that multiplicative weights plays. For every learning rate $\eta \in (0, 1/2]$, every number of rounds $T \geq 1$, every sequence of loss vectors $\ell^1, \dots, \ell^T$ in $[-1, 1]^N$ and every fixed distribution p over the experts,*

$$\sum_{t=1}^{T} \ell^t \cdot p^t \leq \sum_{t=1}^{T} \ell^t \cdot p + \eta T + \frac{\ln N}{\eta}. \tag{8.8}$$

If $N \geq 2$, $T \geq 4 \ln N$ and $\eta = \sqrt{\ln N / T}$, the last two terms add up to $2\sqrt{T \ln N}$.

Proof.

1. *Upper bound on the total weight.* By the update rule and $p_i^t = w_i^t / W^t$,

$$W^{t+1} = \sum_{i=1}^{N} w_i^t (1 - \eta \ell_i^t) = W^t \left(1 - \eta\, \ell^t \cdot p^t\right). \tag{8.9}$$

Unrolling from $W^1 = N$ and applying the first bound of Lemma 8.2 to each factor,

$$\ln W^{T+1} = \ln N + \sum_{t=1}^{T} \ln\left(1 - \eta\, \ell^t \cdot p^t\right) \leq \ln N - \eta \sum_{t=1}^{T} \ell^t \cdot p^t. \tag{8.10}$$

2. *Lower bound through one expert.* The total weight is at least the weight of any expert i. Each $\eta \ell_i^t$ lies in $[-1/2, 1/2]$, so the second bound of Lemma 8.2 applies, and $(\ell_i^t)^2 \leq 1$:

$$\ln W^{T+1} \geq \ln w_i^{T+1} = \sum_{t=1}^{T} \ln(1 - \eta \ell_i^t) \geq -\eta \sum_{t=1}^{T} \ell_i^t - \eta^2 T. \tag{8.11}$$

3. *Lower bound through a distribution.* Multiply the inequality of step 2 by p_i and sum over i. The weights p_i sum to 1, so $\ln W^{T+1} \geq -\eta \sum_t \ell^t \cdot p - \eta^2 T$.
4. Combining steps 1 and 3 gives $-\eta \sum_t \ell^t \cdot p - \eta^2 T \leq \ln N - \eta \sum_t \ell^t \cdot p^t$. Dividing by η and rearranging gives Equation 8.8.
5. With $\eta = \sqrt{\ln N / T}$, both ηT and $\ln N / \eta$ equal $\sqrt{T \ln N}$. The condition $N \geq 2$ makes $\eta > 0$, and $T \geq 4 \ln N$ makes $\eta \leq 1/2$.

□

The regret is of order $\sqrt{T}$, compared with cumulative losses of order T. Thus, average loss approaches that of the best fixed distribution, with an error term of order $\sqrt{\ln N / T}$. A point-mass distribution recovers the guarantee for an algorithm that selects one expert at each round.

Corollary 8.1. *Let $N \geq 2$. Fix losses $\ell^1, \dots, \ell^T \in [-1, 1]^N$ in advance, with $T \geq 4 \ln N$, and let $L_i = \sum_t \ell_i^t$. Pick expert i_t with probability $p_{i_t}^t$, where p^t comes from Definition 8.3 with $\eta = \sqrt{\ln N / T}$. Then*

$$\mathbb{E}\Big[\sum_{t=1}^{T} \ell_{i_t}^t\Big] \leq \min_i L_i + 2\sqrt{T \ln N}. \tag{8.12}$$

Proof. The losses are fixed, so each p^t is fixed too, and $\mathbb{E}[\ell^t_{i_t}] = \ell^t \cdot p^t$. Sum over t and apply Theorem 8.1 with p the point mass on the best expert. □

8.3. Multiplicative weights for queries

For query release, each universe element is an expert, and each p^t is a candidate dataset distribution. Represent the true dataset by its histogram distribution for comparison.

Definition 8.4 (Histogram of a dataset)**.** The **histogram** of $D \in \mathcal{X}^n$ is the distribution p_D on $\mathcal{X}$ with $p_D(x) = |\{i : x_i = x\}|/n$, the fraction of records equal to x. For a linear query q and a distribution p on $\mathcal{X}$, write $\langle q, p\rangle = \sum_{x\in\mathcal{X}} q(x)p(x)$.

Grouping records by value gives $q(D) = \langle q, p_D\rangle$. The goal is to find $\hat{p}$ such that $\langle q, \hat{p}\rangle$ approximates $\langle q, p_D\rangle$ for every $q \in \mathcal{Q}$.

Consider a game in which an adversary chooses query q^t after the algorithm selects distribution p^t. The loss is $|\langle q^t, p^t\rangle - \langle q^t, p_D\rangle|$. Although p_D would answer every query exactly, computing it reads the full dataset. Multiplicative weights accesses the data through one query and its error sign per round; both can be privatized. A bound on cumulative error then limits the number of rounds with large error.

Define the candidate's error by $f(p') = |\langle q^t, p'\rangle - \langle q^t, p_D\rangle|$. This is convex and equals zero at p_D. Away from zero error, its gradient at p^t is $s_t q^t$, where s_t is the sign of $\langle q^t, p^t - p_D\rangle$; take $s_t = +1$ at zero. The next lemma shows that the loss vector $\ell^t = s_t q^t$ equals the query error in the regret comparison.

Lemma 8.3. *Let q be a linear query, p' and p distributions on $\mathcal{X}$, and s the sign of $\langle q, p' - p\rangle$, with $s = +1$ if it is 0. The vector $\ell = sq$ lies in $[-1, 1]^{\mathcal{X}}$, and*

$$\ell \cdot (p' - p) = |\langle q, p'\rangle - \langle q, p\rangle|. \tag{8.13}$$

Proof. The entries of q lie in $[0, 1]$, so those of sq lie in $[-1, 1]$. By linearity, $\ell\cdot(p'-p) = s\langle q, p'-p\rangle$, which is the absolute value of $\langle q, p' - p\rangle$ by the choice of s. □

For these losses, regret against the fixed distribution p_D equals the candidates' cumulative query error. The regret bound therefore limits that total.

Theorem 8.2. *Run multiplicative weights over the experts $\mathcal{X}$ with learning rate $\eta \in (0, 1/2]$, and let p^t be the distribution it plays in round t. Feed it the losses $\ell^t = s_t q^t$, where $q^1, q^2, \ldots$ are any queries in $\mathcal{Q}$ and s_t is the sign of $\langle q^t, p^t - p_D\rangle$. After any number U of rounds,*

$$\sum_{t=1}^{U} |\langle q^t, p^t\rangle - q^t(D)| \le \eta U + \frac{\ln|\mathcal{X}|}{\eta}. \tag{8.14}$$

With $|\mathcal{X}| \ge 2$, $U \ge 4\ln|\mathcal{X}|$ and $\eta = \sqrt{\ln|\mathcal{X}|/U}$, the right-hand side is $2\sqrt{U\ln|\mathcal{X}|}$.

Proof. By Lemma 8.3, the left-hand side equals $\sum_t \ell^t \cdot p^t - \sum_t \ell^t \cdot p_D$. Apply Theorem 8.1 with $N = |\mathcal{X}|$ and the fixed distribution $p = p_D$. □

The regret bound controls cumulative error, but the goal is a single distribution accurate on every query. Call a round a **mistake** when its error exceeds α. If each round selects a query with such an error, cumulative error grows by more than α per round, while the regret bound grows by only η per round. Thus the number of mistakes is bounded. The algorithm implements this strategy by selecting the query with largest error.

Definition 8.5 (Multiplicative weights for queries)**.** Given D, $\mathcal{Q}$ and $\alpha \in (0, 1]$, set $\eta = \alpha/2$ and let p^1 be uniform on $\mathcal{X}$. In round $t = 1, 2, ...$:

1. If $|\langle q, p^t\rangle - q(D)| \le \alpha$ for every $q \in \mathcal{Q}$, output p^t and stop.
2. Otherwise pick $q^t \in \mathcal{Q}$ with $|\langle q^t, p^t\rangle - q^t(D)| > \alpha$, and let s_t be the sign of $\langle q^t, p^t\rangle - q^t(D)$.
3. Update $p^{t+1}(x) \propto p^t(x)\,(1 - \eta s_t q^t(x))$.

In step 3, $\propto$ means normalize the right-hand side to sum to one. Each update adds more than α to cumulative error, while its upper bound grows by only $\eta = \alpha/2$ per round. Therefore the algorithm must terminate.

Corollary 8.2 (Mistake bound)**.** *Let $|\mathcal{X}| \ge 2$. The algorithm of Definition 8.5 stops after fewer than $4 \ln |\mathcal{X}|/\alpha^2$ updates, and its output $\hat{p}$ satisfies $|\langle q, \hat{p}\rangle - q(D)| \le \alpha$ for every $q \in \mathcal{Q}$.*

Proof.

1. The algorithm is multiplicative weights with $\eta = \alpha/2 \le 1/2$ and the losses of Theorem 8.2, since normalizing the weights gives the update in step 3.
2. Suppose it makes U updates. Each of them has error above α, so by Theorem 8.2

$$\alpha U < \frac{\alpha}{2} U + \frac{2 \ln |\mathcal{X}|}{\alpha}, \qquad \text{that is,} \qquad U < \frac{4 \ln |\mathcal{X}|}{\alpha^2}. \tag{8.15}$$

3. So the algorithm stops, which happens only at step 1, where every query is within α.

□

The bound is worst-case. Figure 8.1 shows an example with substantially fewer updates.

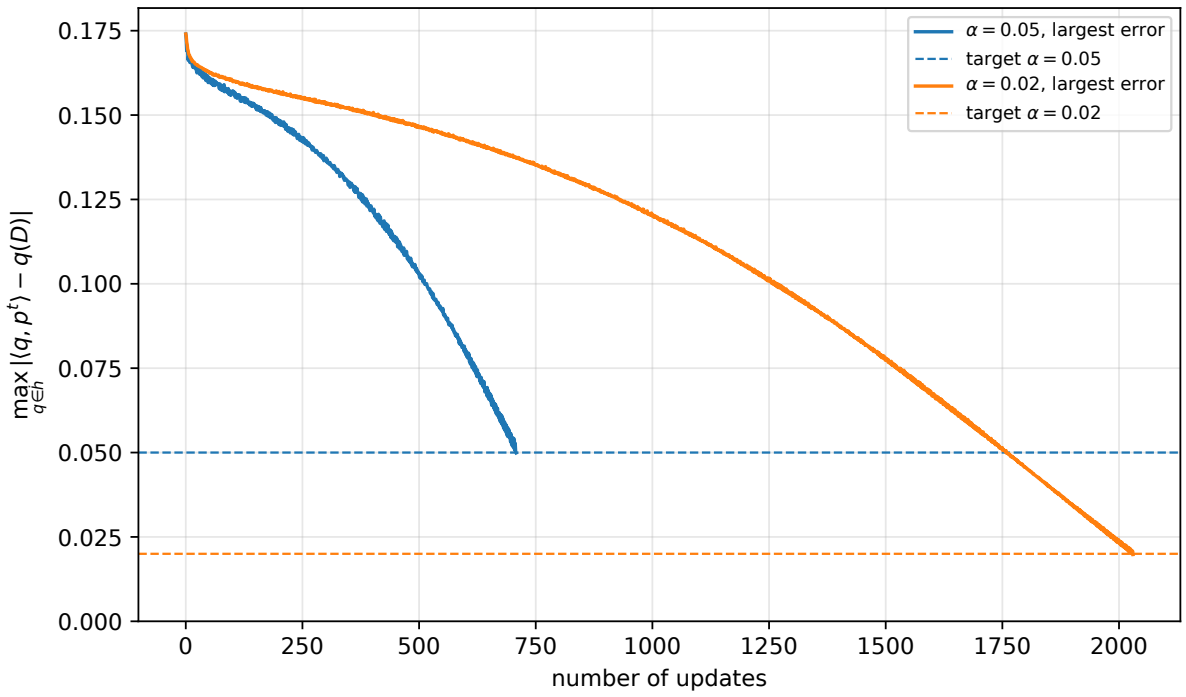


Figure 8.1.: Multiplicative weights for queries on the universe $\{0,1\}^8$, with the 128 one-way and two-way marginal queries $\mathbf{1}[x_j = a]$ and $\mathbf{1}[x_j = a, x_k = b]$. The dataset has 2,000 records whose eight features copy a hidden fair bit with probability 0.8. Each round picks the query with the largest error. The algorithm stops after about 700 updates for $\alpha = 0.05$ and about 2,000 for $\alpha = 0.02$, against the bounds $4 \ln 256/\alpha^2$ of 8,872 and 55,452.

8.4. Private multiplicative weights

The algorithm of Definition 8.5 reads the data in two places: it picks a query with a large error, and it reads the sign of that error. Both have private versions. The exponential mechanism picks a query with nearly the largest error, and the Laplace mechanism can measure the error itself, which has sensitivity $1/n$.

The exponential mechanism takes a finite set $\mathcal{R}$ of candidates and a score $u(D, r)$ whose sensitivity in D is at most Δ for each r. It outputs r with probability proportional to $\exp(\varepsilon u(D, r)/(2\Delta))$. It is ε-DP, and for every $t \geq 0$,

$$\Pr\left[u(D, \hat{r}) \leq \max_{r \in \mathcal{R}} u(D, r) - \frac{2\Delta}{\varepsilon}(\ln|\mathcal{R}| + t)\right] \leq e^{-t}, \tag{8.16}$$

where $\hat{r}$ is its output.

The private algorithm replaces the two steps of Definition 8.5 that read the data by these two mechanisms. It also fixes the number of rounds T in advance, so that the privacy budget can be split among them.

Definition 8.6 (Private multiplicative weights)**.** The inputs are a dataset $D \in \mathcal{X}^n$, a finite set $\mathcal{Q}$ of linear queries, an accuracy $\alpha \in (0, 1]$ and privacy parameters $\varepsilon \in (0, 1]$ and $\delta \in (0, 1/2]$. Set

$$T = \left\lceil \frac{64 \ln|\mathcal{X}|}{\alpha^2} \right\rceil, \qquad \eta = \frac{\alpha}{8}, \qquad \varepsilon_0 = \frac{\varepsilon}{4\sqrt{T \ln(1/\delta)}}, \tag{8.17}$$

and let p^1 be uniform on $\mathcal{X}$. In round $t = 1, \ldots, T$:

1. Choose $q^t \in \mathcal{Q}$ with the exponential mechanism, with privacy parameter ε_0 and score $u_t(D, q) = |\langle q, p^t\rangle - q(D)|$, whose sensitivity is $1/n$.
2. Measure $y_t = \langle q^t, p^t\rangle - q^t(D) + Z_t$, with $Z_t \sim \mathrm{Lap}(1/(\varepsilon_0 n))$.
3. If $|y_t| \leq \alpha/2$, output p^t and stop.
4. Otherwise let s_t be the sign of y_t and update $p^{t+1}(x) \propto p^t(x)\,(1 - \eta s_t q^t(x))$.

If no round stops, output p^{T+1}.

The score u_t has sensitivity $1/n$ because $\langle q, p^t\rangle$ does not depend on D, and $q(D)$ changes by at most $1/n$. The threshold $\alpha/2$ in step 3 sits between the errors that must cause an update and those small enough to stop on. It leaves room on both sides for the noise. Every access to the data goes through one of the two mechanisms, so privacy follows from composing them over the T rounds.

Theorem 8.3. *Private multiplicative weights is (ε, δ)-DP.*

Proof.

1. In round t, the distribution p^t is a function of the earlier outputs $q^1, y_1, \dots, q^{t-1}, y_{t-1}$ alone. So each round runs two mechanisms chosen from earlier outputs: the exponential mechanism, which is ε_0-DP, and the Laplace mechanism on a query of sensitivity $1/n$ with scale $1/(\varepsilon_0 n)$, which is also ε_0-DP.
2. A round after the algorithm stops can be treated as a mechanism with a fixed output, which is 0-DP. So the transcript of all T rounds is an adaptive composition of $k = 2T$ mechanisms, each ε_0-DP.
3. The value $\varepsilon_0 = \varepsilon/(4\sqrt{T\ln(1/\delta)})$ equals $\varepsilon/(2\sqrt{2k\ln(1/\delta)})$ for $k = 2T$. Lemma 8.1 makes the transcript (ε, δ)-DP.
4. The output is a function of the transcript, so it is (ε, δ)-DP by post-processing.

□

Basic composition would also work, and it would even give pure DP, but only with $\varepsilon_0 = \varepsilon/(2T)$. Then the condition on n in the next theorem would grow like T instead of $\sqrt{T}$, which means $1/\alpha^3$ instead of $1/\alpha^2$.

Accuracy needs enough records for the errors of both mechanisms to stay below $\alpha/4$ in every round. Then each update still makes progress, and the algorithm stops on a distribution that is accurate on every query.

Theorem 8.4 (Accuracy of private multiplicative weights)**.** *Let $|\mathcal{X}| \geq 2$ and $\beta \in (0,1)$, and let the other parameters be as in Definition 8.6. If*

$$n \geq \frac{32\sqrt{T\ln(1/\delta)}\,(\ln|\mathcal{Q}| + \ln(2T/\beta))}{\alpha\varepsilon}, \tag{8.18}$$

then with probability at least $1-\beta$ the output $\hat{p}$ satisfies $|\langle q, \hat{p}\rangle - q(D)| \leq \alpha$ for every $q \in \mathcal{Q}$.

Proof.

1. *The good event.* Let G be the event that in every round t that runs, the chosen query has $u_t(D, q^t) \geq \max_q u_t(D, q) - \alpha/4$ and the noise has $|Z_t| < \alpha/4$. Condition Equation 8.18 is the same as $\frac{2}{n\varepsilon_0}(\ln|\mathcal{Q}| + \ln(2T/\beta)) \leq \frac{\alpha}{4}$, by the value of ε_0. So Equation 8.16 with $t = \ln(2T/\beta)$ bounds the first failure in each round by $\beta/(2T)$. The same condition gives $\ln(2T/\beta)/(\varepsilon_0 n) \leq \alpha/4$, so the Laplace tail bounds the second failure by $\beta/(2T)$. A union bound over the two failures and the T rounds gives $\Pr[G] \geq 1-\beta$. The rest of the proof assumes G.
2. *Every update makes progress.* In a round that updates, $|y_t| > \alpha/2 > |Z_t|$. So the true error $\langle q^t, p^t\rangle - q^t(D) = y_t - Z_t$ has the sign s_t of y_t, and its absolute value exceeds $\alpha/2 - \alpha/4 = \alpha/4$.
3. *Few updates.* By step 2, the update is the one of Theorem 8.2, with $\eta = \alpha/8 \leq 1/2$. If U rounds update, each has error above $\alpha/4$, and the theorem gives

$$\frac{\alpha}{4}U < \frac{\alpha}{8}U + \frac{8\ln|\mathcal{X}|}{\alpha}, \qquad \text{that is,} \qquad U < \frac{64\ln|\mathcal{X}|}{\alpha^2} \leq T. \tag{8.19}$$

So the T rounds cannot all update, and the algorithm stops at step 3 in some round t. 4. *The output is accurate.* In that round, $|y_t| \leq \alpha/2$ and $|Z_t| < \alpha/4$, so $u_t(D, q^t) < 3\alpha/4$. On G, every query then has $u_t(D, q) \leq u_t(D, q^t) + \alpha/4 < \alpha$, and the output is $\hat{p} = p^t$. □

Since $\ln|\mathcal{X}|/\alpha^2 \geq \ln 2 > 1/2$, rounding up adds less than $2\ln|\mathcal{X}|/\alpha^2$ and $T \leq 66\ln|\mathcal{X}|/\alpha^2$. So condition Equation 8.18 asks for

$$n = O\left(\frac{\sqrt{\ln|\mathcal{X}|\,\ln(1/\delta)}\,(\ln|\mathcal{Q}| + \ln\frac{\ln|\mathcal{X}|}{\alpha\beta})}{\alpha^2\varepsilon}\right) \tag{8.20}$$

records, which is the last row of Table 8.1. It is logarithmic in both $|\mathcal{Q}|$ and $|\mathcal{X}|$, and the price is a worse dependence on α, $1/\alpha^2$ instead of $1/\alpha$. Each round evaluates every query on p^t and on D, so the running time is $O(T|\mathcal{Q}|(|\mathcal{X}| + n))$, linear in $|\mathcal{X}|$.

The output is a distribution $\hat{p}$ over the universe, a **synthetic dataset** that agrees with the original on every query in $\mathcal{Q}$ and that an analyst can use like the original. The logarithms pay off only when the number of queries is large, as a comparison of the three sufficient sample sizes shows.

Example 8.1. Take records with $d = 20$ binary features, so $|\mathcal{X}| = 2^{20}$, and $\alpha = 0.05$, $\varepsilon = 1$, $\delta = 10^{-6}$, $\beta = 0.05$. Private multiplicative weights runs for at most $T = 354{,}892$ rounds. For $|\mathcal{Q}| = 10^6$ queries, the sufficient sample sizes of Proposition 8.1, Proposition 8.2 and Theorem 8.4 are about 3.4×10^8, 3.5×10^6 and 4.3×10^7 records. For $|\mathcal{Q}| = 10^9$ they become 4.7×10^{11}, 1.6×10^8 and 5.3×10^7. Private multiplicative weights wins only for very many queries, because the constants in Equation 8.18 are large and its accuracy costs $1/\alpha^2$.

8.5. Summary

Table 8.2.: The chapter's results.

Result	Statement
Proposition 8.1	Laplace noise on every query: $n \geq \lvert\mathcal{Q}\rvert\ln(\lvert\mathcal{Q}\rvert/\beta)/(\alpha\varepsilon)$ under ε-DP
Lemma 8.1	k mechanisms at $\varepsilon/(2\sqrt{2k\ln(1/\delta)})$ each compose to (ε,δ)-DP
Proposition 8.2	Laplace with advanced composition: $n \geq 2\sqrt{2\lvert\mathcal{Q}\rvert\ln(1/\delta)}\ln(\lvert\mathcal{Q}\rvert/\beta)/(\alpha\varepsilon)$
Proposition 8.3	with a perfect expert, halving makes at most $\log_2 N$ mistakes
Proposition 8.4	weighted majority makes at most $2.41(\mathrm{OPT} + \log_2 N)$ mistakes
Theorem 8.1	multiplicative weights has regret $\eta T + \ln N/\eta$, or $2\sqrt{T\ln N}$
Theorem 8.2	with losses $s_t q^t$, the total query error is at most $2\sqrt{U\ln\lvert\mathcal{X}\rvert}$
Corollary 8.2	fewer than $4\ln\lvert\mathcal{X}\rvert/\alpha^2$ updates, then every query is within α
Theorem 8.3	private multiplicative weights is (ε,δ)-DP

Result	Statement
Theorem 8.4	accuracy α from $n = O(\sqrt{\log\|\mathcal{X}\| \log(1/\delta)} \log\|\mathcal{Q}\|/(\alpha^2\varepsilon))$ records, up to lower-order logarithmic terms

8.6. References

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9. Sparse vector

Queries may arrive sequentially, with each answer required before the next query is revealed. Adding independent noise to every answer can make error grow polynomially with the number of queries. Under differential privacy (DP), the sparse vector technique identifies queries above a threshold at a cost that depends on their number and logarithmically on the stream length. Its privacy proof requires noise on both the threshold and each query. The final section uses the technique to answer queries online with private multiplicative weights.

9.1. Answering queries online

Consider a stream $f_1, \dots, f_k \colon \mathcal{X}^n \to \mathbb{R}$ of queries, each with sensitivity at most 1. Neighbours $D \sim D'$ differ by one record, so this condition is $|f_i(D) - f_i(D')| \le 1$ for every neighbouring pair. Counting queries are a standard example. The algorithm must answer query f_i before f_{i+1} is revealed.

The Laplace mechanism adds $\mathrm{Lap}(b)$ noise with density $e^{-|z|/b}/(2b)$. For sensitivity 1, scale $b = 1/\varepsilon$ gives ε-DP, with tail probability $\Pr[|\mathrm{Lap}(b)| \ge tb] = e^{-t}$. Under pure DP, basic composition requires scale k/ε per query for total budget ε. Under approximate DP, the advanced composition theorem reduces this to order $\sqrt{k \ln(1/\delta)}/\varepsilon$. In the worst case, answering all queries still costs polynomially in k.

A cheaper task is to identify queries above a threshold. If all k queries are known in advance, the exponential mechanism selects one within $2(\ln k + t)/\varepsilon$ of the largest, except with probability e^{-t}. Repeating it c times costs order $c \ln k/\varepsilon$, or $\sqrt{c \ln(1/\delta)} \ln k/\varepsilon$ with advanced composition. This requires the full query list. In an online stream, the algorithm must compare each query with a fixed threshold.

> Given a public threshold τ, report the first c queries in the stream with $f_i(D) \ge \tau$, and stop.

The algorithm adds Laplace noise to each query and reports only whether the noisy value exceeds a noisy threshold $\hat{\tau}$. Noise on the threshold is essential; some published variants that omit it are not private.

9.2. AboveThreshold

The case $c = 1$ finds the first large query and halts. AboveThreshold draws one noisy threshold at the start, compares each noisy query with it, and answers with one of two symbols. The symbol $\top$ means “above the threshold” and $\bot$ means “below”.

Definition 9.1 (AboveThreshold)**.** The inputs are a dataset D, a stream of sensitivity-1 queries $f_1, f_2, ...$, a threshold $\tau \in \mathbb{R}$ and $\varepsilon > 0$.

1. Draw $\hat{\tau} = \tau + \mathrm{Lap}(2/\varepsilon)$.
2. For each query f_i in turn, draw $\nu_i \sim \mathrm{Lap}(4/\varepsilon)$. If $f_i(D) + \nu_i \geq \hat{\tau}$, output $\top$ and halt. Otherwise output $\perp$.

All noise draws are independent.

The algorithm and its privacy guarantee do not depend on the stream length, so the stream may be infinite. Accuracy does depend on length because more noise draws increase the maximum noise.

Queries may be fixed in advance or chosen adaptively from earlier answers. Before the algorithm halts, every answer is $\perp$. Conditional on the analyst's data-independent coins, each query is therefore fixed. The proof applies to each such stream; averaging over the analyst's coins preserves the guarantee.

Theorem 9.1. *AboveThreshold is ε-DP.*

Proof.

1. Fix neighbours $D \sim D'$ and a stream, and write Y and Y' for the outputs on D and D'. The outputs are $y = \perp^{t-1}\top$, which halts at query t, and the output that never halts: $\perp^k$ for a stream of finite length k, or $\perp^\infty$ for an infinite one. These form a countable set, so by the pointwise lemma for discrete outputs it suffices to show $\Pr[Y = y] \leq e^\varepsilon \Pr[Y' = y]$ for each of them.
2. *Fix the noise of the first $t-1$ queries.* Condition on $\nu_i = v_i$ for $i < t$, the same values on both datasets, and let

$$g(D) = \max_{i<t} \left(f_i(D) + v_i\right), \tag{9.1}$$

with $g = -\infty$ if $t = 1$. With the v_i fixed, g is a number, not a random variable. Each f_i has sensitivity 1, so $|g(D) - g(D')| \leq 1$. 3. The output is y exactly when every earlier query falls below $\hat{\tau}$ and query t does not, that is, when $g(D) < \hat{\tau} \leq f_t(D) + \nu_t$. Write ϕ_2 and ϕ_4 for the densities of $\hat{\tau}$ and ν_t, whose subscripts recall the scales $2/\varepsilon$ and $4/\varepsilon$. Then

$$\Pr[Y = y] = \iint \phi_2(u)\, \phi_4(v)\, \mathbf{1}\big[g(D) < u \leq f_t(D) + v\big]\, \mathrm{d}u\, \mathrm{d}v. \tag{9.2}$$

4. *Change variables to move from D to D'.* Substitute

$$u = u' + g(D) - g(D'), \qquad v = v' + g(D) - g(D') + f_t(D') - f_t(D). \tag{9.3}$$

This is a shift, so $\mathrm{d}u\, \mathrm{d}v = \mathrm{d}u'\, \mathrm{d}v'$. The condition $g(D) < u$ becomes $g(D') < u'$, and $u \leq f_t(D) + v$ becomes $u' \leq f_t(D') + v'$, so the indicator becomes $\mathbf{1}[g(D') < u' \leq f_t(D') + v']$. 5. *Pay for the shifts.* By step 2, $|u - u'| \leq 1$. By step 2 and the sensitivity of f_t, $|v - v'| \leq 2$. A Laplace density

of scale b changes by at most a factor $e^{s/b}$ when its argument moves by s, so $\phi_2(u) \le e^{\varepsilon/2}\phi_2(u')$ and $\phi_4(v) \le e^{2\varepsilon/4}\phi_4(v') = e^{\varepsilon/2}\phi_4(v')$. Hence

$$\Pr[Y = y] \le e^{\varepsilon} \iint \phi_2(u')\,\phi_4(v')\,\mathbf{1}\big[g(D') < u' \le f_t(D') + v'\big]\,\mathrm{d}u'\,\mathrm{d}v' = e^{\varepsilon}\Pr[Y' = y], \tag{9.4}$$

where both probabilities are conditional on the fixed $v_1, \dots, v_{t-1}$. 6. The v_i have the same distribution on both datasets, so averaging step 5 over them gives the unconditional inequality. 7. *The output that never halts.* It occurs exactly when $\hat{\tau} > \sup_i(f_i(D) + \nu_i)$, the supremum over all queries of the stream. Fixing all ν_i makes this supremum a number $g(D)$, possibly $+\infty$, with $|g(D) - g(D')| \le 1$ when it is finite, and the output has probability 0 on both datasets when it is not. Shifting u by at most 1 as in steps 4 and 5 gives the factor $e^{\varepsilon/2} \le e^{\varepsilon}$. □

Threshold noise accounts for all $\perp$ answers: the proof conditions on their query noise and shifts the threshold to absorb changes in their maximum. Query noise accounts for the single $\top$. Thus, privacy cost does not depend on the number of negative answers.

Because the output contains only threshold labels, accuracy means correct labels outside a slack interval around the threshold.

Definition 9.2 (Accuracy for threshold answers)**.** Let an algorithm answer a stream of queries $f_1, f_2, \dots$ with $\top$ or $\perp$, compared with a threshold τ, and write a_i for its answer to f_i. Let $\alpha \ge 0$ be a slack and $\beta \in (0, 1)$ a failure probability. The algorithm is (α, β)-**accurate** on D if, with probability at least $1 - \beta$, every answered query satisfies

$$a_i = \top \implies f_i(D) \ge \tau - \alpha \qquad \text{and} \qquad a_i = \perp \implies f_i(D) \le \tau + \alpha. \tag{9.5}$$

In words, no query more than α below the threshold gets a $\top$, and no query more than α above it gets a $\perp$. AboveThreshold meets this definition with a slack α that grows only with the logarithm of the number of queries. The theorem also asks that every query before the last be at least α below the threshold, so that the algorithm does not halt early.

Theorem 9.2 (Accuracy of AboveThreshold)**.** *Let $\beta \in (0, 1)$, and let $f_1, \dots, f_k$ be a stream with $f_i(D) \le \tau - \alpha$ for every $i < k$, where*

$$\alpha = \frac{8(\ln k + \ln(2/\beta))}{\varepsilon}. \tag{9.6}$$

Then AboveThreshold is (α, β)-accurate on D, and with probability at least $1 - \beta$ it does not halt before query k.

Proof.

1. *The good event.* Let G be the event that $|\hat{\tau} - \tau| < \alpha/2$ and $|\nu_i| < \alpha/2$ for every $i \le k$.
2. *Accuracy on G.* If $a_i = \top$, then $f_i(D) \ge \hat{\tau} - \nu_i > (\tau - \alpha/2) - \alpha/2 = \tau - \alpha$. If $a_i = \perp$, then $f_i(D) < \hat{\tau} - \nu_i < \tau + \alpha$.

3. *No early halt on G.* By step 2, a $\top$ at query i needs $f_i(D) > \tau - \alpha$, which the hypothesis rules out for $i < k$.
4. *G is likely.* The threshold noise has scale $2/\varepsilon$, so the tail formula gives $\Pr[|\hat{\tau} - \tau| \geq \alpha/2] = e^{-\varepsilon\alpha/4}$. Each ν_i has scale $4/\varepsilon$, so a union bound gives $\Pr[\max_{i \leq k} |\nu_i| \geq \alpha/2] \leq k e^{-\varepsilon\alpha/8}$.
5. With Equation 9.6, $k e^{-\varepsilon\alpha/8} = k \cdot \frac{1}{k} \cdot \frac{\beta}{2} = \beta/2$. Also $e^{-\varepsilon\alpha/4} = (\beta/(2k))^2 \leq \beta/2$. So $\Pr[G] \geq 1 - \beta$.

□

The error grows only logarithmically in k, like that of the exponential mechanism, but the queries can arrive online.

Example 9.1. Take $k = 10^6$ counting queries, $\varepsilon = 1$ and $\beta = 0.05$. Then Equation 9.6 gives $\alpha = 8(\ln 10^6 + \ln 40) = 8(13.82 + 3.69) = 140.0$. On a dataset of $n = 10^5$ records, a query is placed correctly unless its count lies within 140 of the threshold. Answering every query with the Laplace mechanism and basic composition would instead add noise of scale 10^6, ten times the size of the dataset.

9.3. Sparse vector

To find c large queries, restart AboveThreshold after each $\top$, with a fresh threshold, until c of them have been found.

Definition 9.3 (Sparse vector)**.** The inputs are those of AboveThreshold, a count $c \geq 1$ of $\top$ answers to find, and a parameter δ, with either $\delta = 0$ or $\delta \in (0, 1/2]$. Set the Laplace noise scale b to

$$b = \frac{2c}{\varepsilon} \text{ if } \delta = 0, \qquad b = \frac{\sqrt{32c \ln(1/\delta)}}{\varepsilon} \text{ if } \delta > 0. \tag{9.7}$$

1. Draw $\hat{\tau} = \tau + \mathrm{Lap}(b)$ and set count $= 0$.
2. For each query f_i in turn, draw $\nu_i \sim \mathrm{Lap}(2b)$. If $f_i(D) + \nu_i < \hat{\tau}$, output $\perp$. Otherwise output $\top$ and add 1 to count. If count $= c$, halt. If not, draw a fresh $\hat{\tau} = \tau + \mathrm{Lap}(b)$ and continue.

Between two $\top$ answers the algorithm runs AboveThreshold with parameter $\varepsilon' = 2/b$, whose noise scales are $2/\varepsilon' = b$ and $4/\varepsilon' = 2b$. The fresh threshold matters, because reusing $\hat{\tau}$ after a $\top$ would break this equivalence, and with it the proof. With the equivalence, sparse vector is a composition of c runs of AboveThreshold, and its privacy follows from the composition theorems.

Theorem 9.3. *Sparse vector with $\delta = 0$ is ε-DP for every $\varepsilon > 0$. With $\varepsilon \in (0, 1]$ and $\delta \in (0, 1/2]$ it is (ε, δ)-DP.*

Proof.

1. The output is the concatenation of at most c runs of AboveThreshold, each with parameter $\varepsilon' = 2/b$ and fresh noise. Each run takes the rest of the stream, which may depend on the earlier outputs, so the whole is an adaptive composition of c mechanisms. A run that never starts, because the algorithm halted, has a fixed output. Each run is ε'-DP by Theorem 9.1.
2. If $\delta = 0$, then $\varepsilon' = \varepsilon/c$, and basic composition gives $c\varepsilon' = \varepsilon$.
3. If $\delta > 0$, then $\varepsilon' = 2\varepsilon/\sqrt{32c\ln(1/\delta)} = \varepsilon/(2\sqrt{2c\ln(1/\delta)})$. The budget-splitting lemma says that m adaptively composed mechanisms, each $\varepsilon/(2\sqrt{2m\ln(1/\delta)})$-DP, are together (ε, δ)-DP when $\varepsilon \le 1$ and $\delta \le 1/2$. Apply it with $m = c$.

□

Accuracy is proved as for AboveThreshold. The only change is that the union bound now runs over every threshold drawn, one per run, as well as over the noise on each query.

Theorem 9.4 (Accuracy of sparse vector)**.** *Let $\beta \in (0, 1)$, let $f_1, \dots, f_k$ be a stream, and let*

$$\alpha = 4b\ln(2k/\beta). \tag{9.8}$$

With probability at least $1 - \beta$, every answered query satisfies the two implications of Definition 9.2. If moreover fewer than c of the queries have $f_i(D) > \tau - \alpha$, the algorithm answers all k queries. In terms of the parameters, the accuracy is

$$\alpha = \frac{8c(\ln k + \ln(2/\beta))}{\varepsilon} \quad \text{if } \delta = 0, \qquad \alpha = \frac{\sqrt{512\, c\ln(1/\delta)}\,(\ln k + \ln(2/\beta))}{\varepsilon} \quad \text{if } \delta > 0. \tag{9.9}$$

Proof.

1. Let G be the event that every threshold drawn and every ν_i with $i \le k$ lies within $\alpha/2$ of its centre. On G, steps 2 and 3 of the proof of Theorem 9.2 apply to each query, with whichever threshold is current. So the implications hold, and a $\top$ needs $f_i(D) > \tau - \alpha$.
2. If fewer than c queries have $f_i(D) > \tau - \alpha$, then on G there are fewer than c answers $\top$, and the algorithm never halts.
3. The query noise has scale $2b$, and $\alpha/(4b) = \ln(2k/\beta)$. A union bound over the k draws bounds its failure by $ke^{-\alpha/(4b)} = \beta/2$.
4. At most $k+1$ thresholds are drawn, each with scale b. Their failure is at most $(k+1)e^{-\alpha/(2b)} = (k+1)\beta^2/(4k^2) \le \beta/2$, since $k + 1 \le 2k$ and $\beta \le 1$.
5. Substituting the two values of b into $4b\ln(2k/\beta)$ gives Equation 9.9, because $4\sqrt{32} = \sqrt{512}$.

□

For $c = 1$ and $\delta = 0$, this recovers Theorem 9.2. Error grows polynomially with the number c of large queries and logarithmically with stream length k. Sparse vector therefore handles long streams when few queries exceed the threshold.

9.4. Numeric sparse vector

Sparse vector's privacy cost already scales with the number c of large queries. Releasing a noisy value for each adds c Laplace mechanisms at the same order of cost. Thus it can release the values with a constant-factor increase.

Definition 9.4 (Numeric sparse vector)**.** Run sparse vector with $\delta = 0$ and budget $\varepsilon/2$, so $b = 4c/\varepsilon$. Whenever it would output $\top$ for query f_i, output instead $a_i = f_i(D) + \mathrm{Lap}(2c/\varepsilon)$, with fresh noise.

Each answer is either $\perp$ or a numeric value. The guarantee combines Definition 9.2 with an error bound for each released value.

Theorem 9.5. *Numeric sparse vector is ε-DP. Let $f_1, \dots, f_k$ be a stream, $\beta \in (0,1)$ and $\alpha = 16c\ln(4k/\beta)/\varepsilon$. With probability at least $1-\beta$, every answered query satisfies the following, where $a_i \in \mathbb{R}$ means that the answer to f_i is a released number:*

$$a_i = \perp \implies f_i(D) \le \tau + \alpha, \qquad a_i \in \mathbb{R} \implies f_i(D) \ge \tau - \alpha \ \text{ and } \ |a_i - f_i(D)| \le \alpha, \tag{9.10}$$

and if fewer than c queries have $f_i(D) > \tau - \alpha$, all k are answered.

Proof.

1. *Privacy.* The algorithm is an adaptive composition of at most c runs of AboveThreshold, each with parameter $2/b = \varepsilon/(2c)$, and at most c Laplace mechanisms. Each Laplace mechanism measures one sensitivity-1 query, chosen from the earlier outputs, with scale $2c/\varepsilon$, so it is $\varepsilon/(2c)$-DP. Basic composition over the $2c$ mechanisms gives ε.
2. *Accuracy of the labels.* Theorem 9.4 with $b = 4c/\varepsilon$ and failure probability $\beta/2$ gives accuracy $4b\ln(4k/\beta) = \alpha$ for the labels, and no early halt, with probability at least $1 - \beta/2$.
3. *Accuracy of the values.* At most k values are released, one per query. By the tail formula and a union bound, some noise exceeds α with probability at most $k\,e^{-\alpha\varepsilon/(2c)} = k\,(\beta/(4k))^8 \le \beta/2$.

□

The released value must use fresh noise. Releasing $f_i(D) + \nu_i$, the noisy value used in the threshold test, is not private.

9.5. Online private multiplicative weights

The previous chapter develops private multiplicative weights for fixed linear queries $q\colon \mathcal{X} \to [0,1]$. This section uses four results: $\langle q, p\rangle = \sum_x q(x)p(x)$ is the expectation under p; the data histogram p_D satisfies $q(D) = \langle q, p_D\rangle$; multiplicative weights with losses $s_t q^t$ has cumulative error at most $\eta U + \ln|\mathcal{X}|/\eta$ after U updates; and the offline method uses the exponential mechanism to select a high-error query.

The exponential mechanism requires the full query set, so sparse vector replaces it online. The algorithm answers from a maintained distribution $\hat{p}$ while its error is small. When sparse vector detects a large error, it releases a noisy data answer and updates $\hat{p}$.

Here, query sensitivity is $1/n$. To handle sensitivity Δ, run sparse vector on f_i/Δ with threshold τ/Δ; this multiplies noise scales and the accuracy bound by Δ without changing privacy. The algorithm applies this rescaling to errors of $\hat{p}$ at threshold $\alpha/2$. Set c to the previous chapter's update bound and ε_0 to the budget for each of at most $2c$ mechanisms.

Definition 9.5 (Online private multiplicative weights)**.** The inputs are a dataset $D \in \mathcal{X}^n$, a stream of linear queries $f_1, f_2, \ldots$, an accuracy $\alpha \in (0,1]$, and $\varepsilon \in (0,1]$ and $\delta \in (0, 1/2]$. Set

$$c = \left\lceil \frac{64 \ln|\mathcal{X}|}{\alpha^2} \right\rceil, \qquad \eta = \frac{\alpha}{8}, \qquad \varepsilon_0 = \frac{\varepsilon}{4\sqrt{c \ln(1/\delta)}}, \tag{9.11}$$

let $\hat{p}$ be uniform on $\mathcal{X}$, set count $= 0$, and draw $\hat{\tau} = \alpha/2 + \mathrm{Lap}(2/(\varepsilon_0 n))$. For each query f_i in turn:

1. Let $e_i = |\langle f_i, \hat{p}\rangle - f_i(D)|$ and draw $\nu_i \sim \mathrm{Lap}(4/(\varepsilon_0 n))$.
2. If $e_i + \nu_i < \hat{\tau}$, answer $\langle f_i, \hat{p}\rangle$.
3. Otherwise answer $a_i = f_i(D) + Z_i$ with $Z_i \sim \mathrm{Lap}(1/(\varepsilon_0 n))$. Let s_i be the sign of $\langle f_i, \hat{p}\rangle - a_i$, update $\hat{p}(x) \propto \hat{p}(x)(1 - \eta s_i f_i(x))$, and add 1 to count. If count $= c$, halt. Otherwise draw a fresh $\hat{\tau} = \alpha/2 + \mathrm{Lap}(2/(\varepsilon_0 n))$.

Steps 1–3 apply numeric sparse vector to errors e_i at threshold $\alpha/2$, except that step 3 measures $f_i(D)$ rather than e_i. Step 2 uses only $\hat{p}$, which depends on earlier outputs, and incurs no additional privacy cost.

Theorem 9.6. *Online private multiplicative weights is (ε, δ)-DP.*

Proof.

1. Between two updates $\hat{p}$ is fixed, so the errors e_i form a stream of queries determined by the earlier outputs and the analyst. Each e_i has sensitivity $1/n$, since $\langle f_i, \hat{p}\rangle$ does not depend on D and $f_i(D)$ changes by at most $1/n$.
2. Each stretch between updates is therefore AboveThreshold with parameter ε_0, rescaled to sensitivity $1/n$, and so ε_0-DP by Theorem 9.1. Each measurement in step 3 is the Laplace mechanism on a query of sensitivity $1/n$ with scale $1/(\varepsilon_0 n)$, so it is also ε_0-DP. The answers of step 2 and the updates are post-processing.

3. There are at most c stretches and c measurements, so the algorithm is an adaptive composition of $2c$ mechanisms, each ε_0-DP. Since $\varepsilon_0 = \varepsilon/(2\sqrt{2 \cdot 2c \ln(1/\delta)})$, the budget-splitting lemma with $m = 2c$ mechanisms makes it (ε, δ)-DP.

□

The accuracy proof follows the offline analysis. If all noise terms are small, answers from $\hat{p}$ and noisy data answers are accurate, and each update makes progress. The update count therefore remains below c.

Theorem 9.7 (Accuracy of online private multiplicative weights)**.** *Let $|\mathcal{X}| \geq 2$, $\beta \in (0,1)$ and a stream of k queries, and let the other parameters be as in Definition 9.5. If*

$$n \geq \frac{128\sqrt{c \ln(1/\delta)}\, \ln\left((2c+k)/\beta\right)}{\alpha\varepsilon}, \tag{9.12}$$

then with probability at least $1-\beta$ the algorithm answers all k queries, each within α of $f_i(D)$.

Proof.

1. *The good event.* Let G be the event that every threshold noise, every ν_i and every Z_i is less than $\alpha/8$ in absolute value. These are at most $2c + k$ draws, because there are at most c thresholds and c measurements. The widest scale is $4/(\varepsilon_0 n)$, so each draw fails with probability at most $e^{-\alpha\varepsilon_0 n/32}$. Condition Equation 9.12 is the same as $\alpha\varepsilon_0 n/32 \geq \ln((2c+k)/\beta)$, by the value of ε_0. A union bound gives $\Pr[G] \geq 1-\beta$, and the rest assumes G.
2. *Answers from $\hat{p}$.* In step 2, $e_i < \hat{\tau} - \nu_i < \alpha/2 + \alpha/8 + \alpha/8 = 3\alpha/4$, so the answer is within α.
3. *Measured answers.* In step 3, $|a_i - f_i(D)| = |Z_i| < \alpha/8$. Also $e_i \geq \hat{\tau} - \nu_i > \alpha/2 - \alpha/4 = \alpha/4$. The difference $\langle f_i, \hat{p}\rangle - a_i$ is the true error minus Z_i, and $|Z_i| < \alpha/8 < e_i$, so s_i is the sign of the true error.
4. *Few updates.* By step 3, the updates are those of multiplicative weights for queries with $\eta = \alpha/8$, and each has error above $\alpha/4$. If there are U of them, the regret bound gives

$$\frac{\alpha}{4}U < \frac{\alpha}{8}U + \frac{8\ln|\mathcal{X}|}{\alpha}, \qquad \text{that is,} \qquad U < \frac{64\ln|\mathcal{X}|}{\alpha^2} \leq c. \tag{9.13}$$

So the count never reaches c, the algorithm never halts, and all k queries are answered. □

As in the previous chapter, $c \leq 66\ln|\mathcal{X}|/\alpha^2$. So condition Equation 9.12 asks for

$$n = O\left(\frac{\sqrt{\ln|\mathcal{X}|\, \ln(1/\delta)}\, \ln\frac{k+\ln|\mathcal{X}|/\alpha^2}{\beta}}{\alpha^2\varepsilon}\right) \tag{9.14}$$

records. This matches the offline algorithm, with the length k of the stream in place of the number $|\mathcal{Q}|$ of queries.

9.6. Summary

Table 9.1.: The chapter's results.

Result	Statement
Theorem 9.1	AboveThreshold, with noise on the threshold and on each query, is ε-DP
Theorem 9.2	it finds the first large query with error $8(\ln k + \ln(2/\beta))/\varepsilon$
Theorem 9.3	sparse vector, c restarts with fresh thresholds, is ε-DP or (ε, δ)-DP
Theorem 9.4	error $8c(\ln k + \ln(2/\beta))/\varepsilon$, or $\sqrt{512c\ln(1/\delta)}(\ln k + \ln(2/\beta))/\varepsilon$
Theorem 9.5	releasing the large values with fresh noise costs a constant factor
Theorem 9.6	online private multiplicative weights is (ε, δ)-DP
Theorem 9.7	it answers k online queries to accuracy α from $n = O(\sqrt{\log\lvert\mathcal{X}\rvert \log(1/\delta)}\log k/(\alpha^2\varepsilon))$ records, up to lower-order logarithmic terms

9.7. References

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10. Beyond global sensitivity

The Laplace mechanism scales noise to a function's global sensitivity: the largest change one record can cause between neighbouring datasets. This worst-case value may be much larger than the sensitivity near the dataset of interest, or may be infinite. This chapter develops methods that use dataset-dependent sensitivity while preserving differential privacy (DP).

Directly scaling noise to local sensitivity is not private because the scale depends on the data. Four techniques avoid this leak. Propose-test-release privately checks that the dataset is far from any dataset with high local sensitivity. A stability argument finds the most frequent element and all histogram counts, regardless of domain size. The final methods privately bound or smooth the sensitivity before using it.

10.1. Local sensitivity

Throughout, $f\colon \mathcal{X}^n \to \mathbb{R}$ is a real-valued function of the dataset. Its **global sensitivity** is the largest change caused by replacing one record, over all neighbouring datasets $D \sim D'$:

$$\Delta f = \max_{D \sim D'} |f(D) - f(D')|, \tag{10.1}$$

The Laplace mechanism adds $\mathrm{Lap}(\Delta f/\varepsilon)$ noise to $f(D)$ and is ε-DP. The maximum includes datasets unlike the one held. Fixing D gives a local measure.

Definition 10.1 (Local sensitivity)**.** The **local sensitivity** of f at a dataset $D \in \mathcal{X}^n$ is

$$\mathrm{LS}_f(D) = \max_{D' \sim D} |f(D) - f(D')|. \tag{10.2}$$

Global sensitivity is the maximum local sensitivity, $\Delta f = \max_D \mathrm{LS}_f(D)$, and may greatly exceed its value at a particular dataset. A natural mechanism releases $f(D) + \mathrm{Lap}(\mathrm{LS}_f(D)/\varepsilon)$, but its data-dependent noise scale can reveal information. The closest-pair distance illustrates how local sensitivity can change between neighbours.

Example 10.1. Let $\mathcal{X} = \mathbb{R}$, $n = 3$, and let $f(D)$ be the distance between the two closest records. On the real line f has infinite global sensitivity. For example, $(0, c, 2c)$ and its neighbour $(0, 0, 2c)$ have f equal to c and 0, for every $c > 0$, so the Laplace mechanism does not apply. Take $D = (0, 0, 0)$. Replacing any one record leaves two records at 0, so f stays 0 and $\mathrm{LS}_f(D) = 0$. Now take the neighbour $D' = (0, 0, 5)$. It also has $f(D') = 0$, but replacing one of its zeros by 10 gives $(0, 10, 5)$ with closest distance 5, so $\mathrm{LS}_f(D') \geq 5$.

For these neighbours, the naive mechanism adds no noise on one dataset and positive noise on the other. The output distributions are therefore distinguishable.

Proposition 10.1. *The mechanism $M(D) = f(D) + \mathrm{Lap}(\mathrm{LS}_f(D)/\varepsilon)$, where* $\mathrm{Lap}(0)$ *means no noise, is not ε-DP for any finite ε when f is the closest-pair distance of Example 10.1.*

Proof.

1. On $D = (0, 0, 0)$ the local sensitivity is 0, so $M(D) = 0$ with probability 1.
2. On the neighbour $D' = (0, 0, 5)$ the local sensitivity is positive, so $M(D')$ has a continuous distribution and $\Pr[M(D') = 0] = 0$.
3. So the event $\{0\}$ has probability 1 on D and 0 on D', and no factor e^{ε} with finite ε can bound the first by the second.

□

The local sensitivity is low at D and high at a neighbour, so the noise scale distinguishes them. The scale itself is a data-dependent statistic. The methods below account for sensitivity at nearby datasets, not only at D.

10.2. Propose-test-release

Propose-test-release chooses a bound b and privately tests whether it holds at D and nearby datasets. If the test passes, it releases $f(D)$ with noise scaled to b. The test measures the number of record changes needed to violate the bound. The **Hamming distance** $d(D, D')$ counts differing records.

Definition 10.2 (Distance to instability)**.** For a proposed bound $b \geq 0$ on the local sensitivity LS_f, the **distance to instability** of D is

$$\gamma_b(D) = \min\{\, d(D, D') : D' \in \mathcal{X}^n,\ \mathrm{LS}_f(D') > b \,\}, \tag{10.3}$$

the fewest records that must change to reach a dataset whose local sensitivity exceeds b, where d is the Hamming distance. If no dataset has local sensitivity above b, set $\gamma_b(D) = \infty$ for every D.

Thus, $\gamma_b(D) = 0$ exactly when $\mathrm{LS}_f(D) > b$. Since it counts record changes, γ_b has sensitivity 1, independent of f.

Lemma 10.1. *If γ_b is finite, then $|\gamma_b(D) - \gamma_b(D')| \leq 1$ for all neighbours $D \sim D'$.*

Proof.

1. If some dataset W has $\mathrm{LS}_f(W) > b$, every dataset is within distance n of W, so γ_b is finite everywhere.

2. Let W attain the minimum for D', so $d(D', W) = \gamma_b(D')$ and $\mathrm{LS}_f(W) > b$. The Hamming distance satisfies the triangle inequality, so $d(D, W) \le d(D, D') + d(D', W) = 1 + \gamma_b(D')$. Hence $\gamma_b(D) \le \gamma_b(D') + 1$.
3. Swapping the roles of D and D' gives the other inequality.

□

The Laplace mechanism releases this distance with privacy. The threshold ensures that a dataset with local sensitivity above b passes the test with probability at most $\delta/2$.

Definition 10.3 (Propose-test-release)**.** The mechanism takes a proposed bound $b > 0$ and privacy parameters $\varepsilon > 0$ and $\delta \in (0, 1)$. On input D, it does three steps.

1. compute $\hat{\gamma} = \gamma_b(D) + Z_1$ with $Z_1 \sim \mathrm{Lap}(1/\varepsilon)$;
2. if $\hat{\gamma} \le \ln(1/\delta)/\varepsilon$, output $\perp$, meaning "no answer";
3. otherwise output $f(D) + Z_2$ with $Z_2 \sim \mathrm{Lap}(b/\varepsilon)$, independent of Z_1.

The test and release each spend ε. The test may pass on a dataset with local sensitivity above b; δ accounts for this event. The release uses fixed scale b/ε, and its activation depends on the data only through the private estimate $\hat{\gamma}$.

Theorem 10.1. *Propose-test-release is $(2\varepsilon, \delta)$-DP.*

Proof. Fix neighbours $D \sim D'$ and an event $S \subseteq \mathbb{R} \cup \{\perp\}$. Write $\tau = \ln(1/\delta)/\varepsilon$ for the threshold. If γ_b is infinite at every dataset, the test always passes and $b \ge \Delta f$, so the output is the Laplace mechanism and is ε-DP. Assume from now on that γ_b is finite.

1. *The test is ε-DP.* By Lemma 10.1, γ_b has sensitivity 1, so $\hat{\gamma}$ is the Laplace mechanism and is ε-DP. The events $\{\hat{\gamma} \le \tau\}$ and $\{\hat{\gamma} > \tau\}$ are post-processing of $\hat{\gamma}$, so each has probability on D at most e^{ε} times its probability on D'.
2. *Case* $\mathrm{LS}_f(D) > b$. Then $\gamma_b(D) = 0$, and the test passes only if $Z_1 > \tau$, which has probability $\frac{1}{2}e^{-\varepsilon\tau} = \delta/2$. Splitting on whether the output is $\perp$ and using step 1,

$$\Pr[M(D) \in S] \le \Pr[M(D) = \perp]\,\mathbf{1}[\perp \in S] + \Pr[M(D) \ne \perp] \le e^{\varepsilon}\Pr[M(D') \in S] + \tfrac{\delta}{2}. \tag{10.4}$$

3. *Case* $\mathrm{LS}_f(D) \le b$. Then $|f(D) - f(D')| \le \mathrm{LS}_f(D) \le b$, so the release step is the Laplace mechanism for a pair at distance at most b, with scale b/ε. Its density at any y changes by a factor at most e^{ε} between D and D'. Write $S_{\mathbb{R}} = S \cap \mathbb{R}$. The two steps use independent noise, so

$$\Pr[M(D) \in S] = \Pr[\hat{\gamma} \le \tau]\,\mathbf{1}[\perp \in S] + \Pr[\hat{\gamma} > \tau]\,\Pr[f(D) + Z_2 \in S_{\mathbb{R}}]. \tag{10.5}$$

Each probability on the right is at most e^{ε} times the same probability on D', by step 1 and by the Laplace mechanism. So $\Pr[M(D) \in S] \le e^{2\varepsilon}\Pr[M(D') \in S]$.

4. In both cases $\Pr[M(D) \in S] \leq e^{2\varepsilon} \Pr[M(D') \in S] + \delta$. The pair was arbitrary, so this holds for every ordered pair of neighbours.

□

The test passes with high probability on datasets far from instability, because if $\gamma_b(D) \geq 2\ln(1/\delta)/\varepsilon$, then the output is $\perp$ only if $Z_1 \leq -\ln(1/\delta)/\varepsilon$, which has probability $\delta/2$. The price is computation, because γ_b is a minimum over all $|\mathcal{X}|^n$ datasets. So the distance to instability has to be worked out by hand for each function, and the median is the standard case where this is efficient.

10.3. Stability: the mode and the histogram

Propose-test-release is most striking with the bound $b = 0$. Then the test checks that the answer does not change at all near D, and the release step adds no noise. The mode, the most frequent element, is the standard example. If all n records equal the same value v, the mode stays v until about $n/2$ records change, so the mode is far from instability.

Let $\mathcal{X}$ be any set with a fixed total order, used only to break ties. Write $c_x(D)$ for the number of records of D equal to x, and $c_{(1)}(D) \geq c_{(2)}(D)$ for the largest and second-largest of these counts, with $c_{(2)}(D) = 0$ if all records are equal. The mode $\mathrm{mode}(D)$ is the element with the largest count, the least one in the order among ties. To change the mode, records must move from the most frequent element to the runner-up until it overtakes. So half the gap between the two largest counts is, up to 1, the distance to instability, and it is computed in one pass over the data whatever the domain.

Lemma 10.2. *Let $g(D) = (c_{(1)}(D) - c_{(2)}(D))/2$. For all neighbours $D \sim D'$:*

1. $|g(D) - g(D')| \leq 1$;
2. *if $g(D) > 1$, then* $\mathrm{mode}(D') = \mathrm{mode}(D)$.

Proof.

1. Replacing one record lowers one count by 1 and raises another by 1, so every count changes by at most 1.
2. The j-th largest count is the maximum, over sets S of j elements, of $\min_{x \in S} c_x$. If every count rises by at most 1, each such minimum rises by at most 1, and so does the maximum. The same holds for decreases, so $c_{(1)}$ and $c_{(2)}$ each change by at most 1, and g changes by at most $(1 + 1)/2 = 1$. This is claim 1.
3. For claim 2, let $a = \mathrm{mode}(D)$. In D', the count of a is at least $c_{(1)}(D) - 1$, and every other count is at most $c_{(2)}(D) + 1$. The condition $g(D) > 1$ means $c_{(1)}(D) - 1 > c_{(2)}(D) + 1$, so a has the strictly largest count in D'.

□

The stable mode is propose-test-release with $b = 0$, with the gap g in place of the distance to instability. Its threshold has an extra 1 because claim 2 protects the mode only when $g(D) > 1$.

Definition 10.4 (Stable mode)**.** Given $\varepsilon > 0$ and $\delta \in (0, 1)$, on input D, compute $\hat{g} = g(D) + Z$ with $Z \sim \mathrm{Lap}(1/\varepsilon)$, where $g(D)$ is half the gap between the two largest counts. If $\hat{g} \le 1 + \ln(1/\delta)/\varepsilon$, output $\perp$, and otherwise output $\mathrm{mode}(D)$ exactly.

The guarantee has two parts. The test costs δ in privacy, and it fails, by outputting $\perp$, only on datasets whose two largest counts are close.

Theorem 10.2. *The stable mode is (ε, δ)-DP. On every dataset whose two largest counts differ by at least $2 + 4\ln(1/\delta)/\varepsilon$, it outputs the mode with probability at least $1 - \delta/2$.*

Proof. Fix neighbours $D \sim D'$ and an event S, and write $\tau = 1 + \ln(1/\delta)/\varepsilon$.

1. *The gap is ε-DP.* By Lemma 10.2, claim 1, $\hat{g}$ is the Laplace mechanism for a function of sensitivity 1, so it is ε-DP.
2. *Case* $\mathrm{mode}(D') = \mathrm{mode}(D)$. The output on both datasets is the same function of $\hat{g}$, namely $\perp$ below the threshold and the common mode above it. Post-processing gives $\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S]$.
3. *Case* $\mathrm{mode}(D') \neq \mathrm{mode}(D)$. By Lemma 10.2, claim 2, $g(D) \le 1$. The mode is released only if $Z > \tau - g(D) \ge \ln(1/\delta)/\varepsilon$, which has probability at most $\delta/2$. As in the proof of Theorem 10.1, $\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S] + \delta/2$.
4. *Accuracy.* If $c_{(1)} - c_{(2)} \ge 2 + 4\ln(1/\delta)/\varepsilon$, then $g(D) \ge \tau + \ln(1/\delta)/\varepsilon$. The output is $\perp$ only if $Z \le -\ln(1/\delta)/\varepsilon$, which has probability $\delta/2$.

□

For typical parameters, the theorem requires a gap of a few dozen records.

Example 10.2. For $\varepsilon = 1$ and $\delta = 10^{-6}$, the threshold is $1 + \ln(10^6) = 14.82$. The required gap is $2 + 4 \times 13.82 = 57.3$, so a gap of 58 records suffices. The guarantee does not depend on domain size, even for an infinite domain.

The stable mode costs ε, not 2ε, because releasing the mode after the test uses no additional noise. Accuracy depends on the count gap, not the domain size. Pure DP cannot provide this guarantee: the packing chapter shows that an ε-DP mode algorithm over k elements needs at least $\ln(k/2)/\varepsilon$ records, even when all records are equal. Approximate DP replaces the domain cost $\ln k$ with $\ln(1/\delta)$.

The same method estimates all counts at once. It adds noise only to observed elements and reports those whose noisy count exceeds a threshold. A newly appearing element has count one; the threshold suppresses it except with probability $\delta/2$.

Definition 10.5 (Stability-based histogram)**.** Given $\varepsilon > 0$ and $\delta \in (0, 1)$, let $\tau = 1 + 2\ln(1/\delta)/\varepsilon$. On input D, for each element x whose count $c_x(D)$ of records is positive, draw an independent $Z_x \sim \mathrm{Lap}(2/\varepsilon)$ and set $\hat{c}_x = c_x(D) + Z_x$. Output the list of pairs $(x, \hat{c}_x)$ with $\hat{c}_x > \tau$. An element not on the list has estimate $\hat{c}_x = 0$.

The noise scale is $2/\varepsilon$ because replacement can change two counts. The threshold accounts for the additive privacy term δ. An unreported count has noisy value at most τ, so thresholding adds at most τ to its error.

Theorem 10.3. *The stability-based histogram is (ε, δ)-DP. For every $\beta \in (0, 1)$, with probability at least $1 - \beta$, every element $x \in \mathcal{X}$ has*

$$|\hat{c}_x - c_x(D)| \le \tau + \frac{2}{\varepsilon} \ln \frac{n}{\beta}. \tag{10.6}$$

Proof. Privacy. Let D' be D with one record equal to a replaced by $a' \neq a$. Let C be the set of elements that appear in both datasets. Any element that appears only in D is a, with count 1, and any element that appears only in D' is a', with count 1. Write $O(D) = O_C(D) \cup O_A(D)$ for the output list on D, where O_C is the part of the list from C and O_A the part from elements only in D, which can only be a. Define $O_C(D')$ and $O_{A'}(D')$ in the same way, where $O_{A'}$ can only contain a'. Fix an event S and let $p_C = \Pr[O_C(D') \in S]$.

1. *The extra element of D.* An element with count 1 is listed only if its noise exceeds $\tau - 1 = 2\ln(1/\delta)/\varepsilon$, which has probability $\frac{1}{2}e^{-\ln(1/\delta)} = \delta/2$. When $O_A(D)$ is empty, $O(D) = O_C(D)$, so $\Pr[O(D) \in S] \le \Pr[O_C(D) \in S] + \delta/2$.
2. *The common elements.* On C, the counts of D and D' differ by at most 1 at a and at a', so in ℓ_1 by at most 2. Laplace noise of scale $2/\varepsilon$ makes the noisy counts on C ε-DP, and the list is post-processing of them. So $\Pr[O_C(D) \in S] \le \min(1, e^{\varepsilon} p_C)$.
3. *The extra element of D'.* The parts $O_C(D')$ and $O_{A'}(D')$ are independent, and $O_{A'}(D')$ is empty with probability at least $1 - \delta/2$ by the computation of step 1. When it is empty, $O(D') = O_C(D')$. So $\Pr[O(D') \in S] \ge (1 - \delta/2)\, p_C$.
4. *Combining.* If $e^{\varepsilon} p_C \le 1$, then $\min(1, e^{\varepsilon} p_C) - e^{\varepsilon}(1 - \delta/2) p_C = e^{\varepsilon} p_C\, \delta/2 \le \delta/2$. Otherwise $1 - e^{\varepsilon}(1 - \delta/2) p_C < 1 - (1 - \delta/2) = \delta/2$. In both cases, steps 1 to 3 give

$$\Pr[O(D) \in S] \le e^{\varepsilon}(1 - \delta/2)\, p_C + \tfrac{\delta}{2} + \tfrac{\delta}{2} \le e^{\varepsilon} \Pr[O(D') \in S] + \delta. \tag{10.7}$$

Accuracy.

5. At most n elements appear in D. A Laplace variable of scale $2/\varepsilon$ exceeds s in absolute value with probability $e^{-\varepsilon s/2}$, so by a union bound all of them satisfy $|Z_x| \le (2/\varepsilon)\ln(n/\beta)$ with probability at least $1 - \beta$.
6. On that event, take any element x. If it is listed, its error is $|Z_x|$. If it appears in D but is not listed, then $c_x + Z_x \le \tau$, so its error is $c_x \le \tau + |Z_x|$. If it does not appear, both its count and its estimate are 0. In every case the error is at most Equation 10.6.

□

When $\delta \le 1/n$, Equation 10.6 is at most $1 + (2/\varepsilon)(2\ln(1/\delta) + \ln(1/\beta))$. Thus, every count has error of order $\ln(1/\delta)/\varepsilon$, independent of the domain size. A pure-DP Laplace histogram instead pays $\ln|\mathcal{X}|$, which the packing chapter shows is unavoidable.

10.4. Privately bounding the local sensitivity

Propose-test-release checks a proposed bound. A second method privately estimates local sensitivity and uses the estimate as the noise scale. It applies when local sensitivity itself has bounded global sensitivity.

Definition 10.6 (Noisy local sensitivity)**.** Suppose that the local sensitivity changes by at most $\lambda > 0$ between neighbours, that is, $|\mathrm{LS}_f(D) - \mathrm{LS}_f(D')| \le \lambda$ for all $D \sim D'$. Given $\varepsilon > 0$ and $\delta \in (0, 1)$, on input D:

1. compute $\hat{s} = \mathrm{LS}_f(D) + Z_1 + \lambda \ln(1/\delta)/\varepsilon$ with $Z_1 \sim \mathrm{Lap}(\lambda/\varepsilon)$;
2. if $\hat{s} \le 0$, output $\perp$, and otherwise output $f(D) + Z_2$ with $Z_2 \sim \mathrm{Lap}(\hat{s}/\varepsilon)$.

The shift $\lambda \ln(1/\delta)/\varepsilon$ makes $\hat{s}$ at least $\mathrm{LS}_f(D)$ except with probability $\delta/2$. On this event, the release noise is sufficient. The estimate and release each use budget ε, giving the 2ε guarantee.

Theorem 10.4. *Let $\lambda > 0$ satisfy $|\mathrm{LS}_f(D) - \mathrm{LS}_f(D')| \le \lambda$ for all neighbours $D \sim D'$, and let $\varepsilon > 0$ and $\delta \in (0, 1)$. Then noisy local sensitivity with these parameters is $(2\varepsilon, \delta)$-DP.*

Proof. Fix neighbours $D \sim D'$. It suffices to prove the guarantee for the pair $(\hat{s}, \text{output})$, since the output alone is post-processing of it.

1. *The good event.* Let $G = \{\hat{s} \ge \mathrm{LS}_f(D)\}$, computed with the data D. It fails only if $Z_1 < -\lambda \ln(1/\delta)/\varepsilon$, which has probability $\delta/2$.
2. *Density of the scale.* The function LS_f has sensitivity λ, so the density of $\hat{s}$ on D is at most e^{ε} times its density on D', at every value s.
3. *Density of the release.* Fix a value s with $s \ge \mathrm{LS}_f(D)$ and $s > 0$. Then $|f(D) - f(D')| \le \mathrm{LS}_f(D) \le s$, so the Laplace density of scale s/ε at any y changes by a factor at most $e^{\varepsilon |f(D) - f(D')|/s} \le e^{\varepsilon}$ between D and D'.
4. *Combining.* On values of $\hat{s}$ in G, the joint density of $(\hat{s}, \text{output})$ on D is at most $e^{2\varepsilon}$ times that on D', by steps 2 and 3. For an event S, split on G:

$$\Pr[(\hat{s}, M(D)) \in S] \le e^{2\varepsilon} \Pr[(\hat{s}, M(D')) \in S] + \Pr[\text{not } G] \le e^{2\varepsilon} \Pr[(\hat{s}, M(D')) \in S] + \tfrac{\delta}{2}. \quad (10.8)$$

□

For each function, one must derive the local sensitivity and a bound λ on its change. Direct computation may require $n(|\mathcal{X}| - 1)$ evaluations of f. Some functions, such as the mode, admit faster formulas based on the two largest counts. This method is simpler than propose-test-release but applies only when local sensitivity changes slowly everywhere.

10.5. Smooth sensitivity

Smooth sensitivity also accounts for nearby datasets, discounting sensitivity by distance from D. Consider the closest-pair function of Example 10.1 on $\mathcal{X} = [0, 10]$, where global sensitivity is finite. At $D = (0, 0, 0)$ local sensitivity is 0, while at its neighbour $(0, 0, 5)$ it is at least 5. A safe noise scale at D must account for this neighbour, with a distance-dependent discount.

Definition 10.7 (Smooth sensitivity)**.** For a smoothing parameter $s > 0$, the **smooth sensitivity** of f at D discounts the local sensitivity of every dataset D' by a factor e^{-s} for each record in which D' differs from D:

$$S_{f,s}(D) = \sup_{D' \in \mathcal{X}^n} \mathrm{LS}_f(D')\, e^{-s\, d(D,D')}, \tag{10.9}$$

where d is the Hamming distance.

The supremum covers all datasets. Since any two datasets are at distance at most n, $e^{-sn}\Delta f \leq S_{f,s}(D) \leq \Delta f$. Smooth sensitivity is finite exactly when global sensitivity is finite; it may be smaller by a factor of e^{sn}. The following two properties suffice for the privacy proof.

Lemma 10.3. *For every D, $S_{f,s}(D) \geq \mathrm{LS}_f(D)$. For all neighbours $D \sim D'$, $S_{f,s}(D) \leq e^s S_{f,s}(D')$.*

Proof.

1. The term $D' = D$ in the supremum is $\mathrm{LS}_f(D)$, so the supremum is at least that.
2. For any dataset W, the triangle inequality gives $d(D, W) \geq d(D', W) - 1$, so $\mathrm{LS}_f(W)e^{-s\, d(D,W)} \leq e^s\, \mathrm{LS}_f(W)e^{-s\, d(D',W)} \leq e^s S_{f,s}(D')$. Taking the supremum over W gives the second claim.

□

Laplace noise scaled to smooth sensitivity does not guarantee pure DP. The scale may change by a factor of e^s between neighbours, while a Laplace density changes more in the tail under a scale change. Polynomial-tail noise is less sensitive to rescaling. The standard Cauchy distribution is one such noise, with density

$$h(z) = \frac{1}{\pi(1 + z^2)}. \tag{10.10}$$

Theorem 10.5 (Smooth sensitivity with Cauchy noise)**.** *Let f be non-constant with finite global sensitivity Δf, let $\varepsilon > 0$, and write $S^* = S_{f,\varepsilon/2}$. The mechanism*

$$M(D) = f(D) + \frac{2\, S^*(D)}{\varepsilon}\, Z, \qquad Z \sim \textit{standard Cauchy}, \tag{10.11}$$

is ε-DP.

Proof.

1. *The scale is positive and finite.* Some dataset W_0 has $\mathrm{LS}_f(W_0) > 0$ because f is not constant, and every D is within distance n of it, so $S^*(D) \geq \mathrm{LS}_f(W_0)e^{-\varepsilon n/2} > 0$. Every term of the supremum is at most Δf, so $S^*(D) \leq \Delta f < \infty$.
2. *Two facts about h.* The derivative of $\ln(1+z^2)$ is $2z/(1+z^2)$, which lies in $[-1,1]$, so $h(z)/h(z+u) \leq e^{|u|}$ for all z and u. For every real κ and v, $e^{-\kappa}h(v)/h(e^{\kappa}v) = e^{-\kappa}(1+e^{2\kappa}v^2)/(1+v^2) \leq e^{-\kappa}\max(1, e^{2\kappa}) = e^{|\kappa|}$.
3. *The ratio of densities.* Fix neighbours $D \sim D'$ and write $a = \varepsilon/2$. The density of $M(D)$ at y is $(a/S^*(D))\, h(a(y-f(D))/S^*(D))$. Set $u = a(y-f(D))/S^*(D)$, $w = a(f(D)-f(D'))/S^*(D)$ and $e^{\kappa} = S^*(D)/S^*(D')$. Then $a(y-f(D'))/S^*(D') = e^{\kappa}(u+w)$, and the ratio of the densities on D and D' is

$$e^{-\kappa}\,\frac{h(u)}{h(e^{\kappa}(u+w))} = \frac{h(u)}{h(u+w)} \cdot \frac{e^{-\kappa}\,h(u+w)}{h(e^{\kappa}(u+w))} \leq e^{|w|}\,e^{|\kappa|}, \tag{10.12}$$

by the two facts of step 2.

4. *Bounding the exponents.* By Lemma 10.3, $|f(D)-f(D')| \leq \mathrm{LS}_f(D) \leq S^*(D)$, so $|w| \leq a = \varepsilon/2$. By the same lemma applied in both directions, $|\kappa| \leq \varepsilon/2$. The densities therefore differ by a factor at most e^{ε}, and integrating over any event gives ε-DP.

□

The Cauchy distribution has no mean. Its tail is $\Pr[|Z| > z] = \frac{2}{\pi}\arctan(1/z) \leq 2/(\pi z)$, so the error exceeds $4S^*(D)/(\pi\varepsilon\beta)$ with probability at most β. This bound scales as $1/\beta$, compared with $\ln(1/\beta)$ for Laplace or Gaussian noise. Those distributions, scaled to smooth sensitivity with a smaller smoothing parameter, give approximate DP and lighter tails. Computing $S_{f,s}$ requires maximizing over all datasets and is generally difficult; for the median, it takes polynomial time.

10.6. Summary

Table 10.1.: The chapter's results.

Result	Statement
Proposition 10.1	Laplace noise scaled to the local sensitivity is not private
Lemma 10.1	the distance to instability has sensitivity 1
Theorem 10.1	propose-test-release is $(2\varepsilon, \delta)$-DP
Theorem 10.2	the stable mode is (ε, δ)-DP and exact when the top two counts differ by $2 + 4\ln(1/\delta)/\varepsilon$
Theorem 10.3	the stability-based histogram is (ε, δ)-DP with every count within $\tau + (2/\varepsilon)\ln(n/\beta)$

Result	Statement
Theorem 10.4	a noisy overestimate of a Lipschitz local sensitivity gives $(2\varepsilon, \delta)$-DP
Lemma 10.3	smooth sensitivity bounds the local sensitivity and changes by at most e^s between neighbours
Theorem 10.5	Cauchy noise of scale $2S_{f,\varepsilon/2}(D)/\varepsilon$ gives ε-DP when $0 < \Delta f < \infty$

10.7. References

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11. Packing lower bounds

Previous chapters gave algorithms and upper bounds on their sample sizes. This chapter asks whether better algorithms can use less data. Under pure differential privacy (DP), a packing argument gives lower bounds: accuracy requires different output regions for many datasets, while privacy limits how much output probabilities can vary between them. Both requirements force a minimum dataset size.

These lower bounds apply to every ε-DP algorithm. For one-way marginals and histograms, they show that Laplace mechanisms are optimal up to constant or logarithmic factors. They also show that pure DP requires more data than approximate DP for these problems.

11.1. Two problems and their upper bounds

Both problems release many counting queries as fractions of the dataset. In the first, each record contains d binary answers, and the mechanism estimates the fraction of yes answers to each question.

Definition 11.1 (One-way marginals)**.** Let $\mathcal{X} = \{0,1\}^d$. For a record x_i, write x_{ij} for its coordinate j. The **one-way marginals** of a dataset $D \in \mathcal{X}^n$ are the d averages

$$f_j(D) = \frac{1}{n}\sum_{i=1}^{n} x_{ij}, \qquad j = 1, \dots, d. \tag{11.1}$$

In the second, each record is a symbol from a finite set, and the mechanism estimates the fraction of records equal to each symbol.

Definition 11.2 (Histogram)**.** Let $\mathcal{X} = [k] = \{1, \dots, k\}$. The **normalized histogram** of a dataset $D \in [k]^n$ is the vector $h(D) \in [0,1]^k$ with coordinates

$$h_j(D) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[x_i = j], \qquad j = 1, \dots, k, \tag{11.2}$$

where $\mathbf{1}[x_i = j]$ is 1 if record i equals j and 0 otherwise.

For both problems, error is the largest coordinate error, $\ell_\infty = \max_j |y_j - f_j(D)|$. Error at most α therefore bounds every query simultaneously.

The upper bounds use the Laplace mechanism. If answer vector f has ℓ_1 sensitivity $\Delta_1 f$, meaning $\sum_j |f_j(D) - f_j(D')| \le \Delta_1 f$ for neighbours that differ in one record, then adding independent

Laplace noise of scale $\Delta_1 f/\varepsilon$ to each coordinate gives ε-DP. Its analysis uses the tail probability for $Z \sim \mathrm{Lap}(b)$:

$$\Pr[|Z| \geq s] = e^{-s/b}. \tag{11.3}$$

The next proposition computes sensitivities and applies this tail bound to each coordinate.

Proposition 11.1. *Let $\varepsilon > 0$, $\alpha > 0$ and $\beta \in (0,1)$. With probability at least $1-\beta$:*

1. *the Laplace mechanism for the d one-way marginals, with scale $d/(n\varepsilon)$, has error at most $d\ln(d/\beta)/(n\varepsilon)$;*
2. *the Laplace mechanism for the histogram over $[k]$, with scale $2/(n\varepsilon)$, has error at most $2\ln(k/\beta)/(n\varepsilon)$.*

Both are ε-DP. So error at most α with probability at least $1/2$ needs only

$$n \geq \frac{d\ln(2d)}{\alpha\varepsilon} \quad \textit{for marginals}, \qquad n \geq \frac{2\ln(2k)}{\alpha\varepsilon} \quad \textit{for histograms}. \tag{11.4}$$

Proof.

1. *Sensitivity of the marginals.* Replacing one record changes each average f_j by at most $1/n$, so the ℓ_1 sensitivity of $(f_1, \ldots, f_d)$ is at most d/n. The Laplace mechanism with scale $d/(n\varepsilon)$ is therefore ε-DP.
2. *Sensitivity of the histogram.* Replacing a record equal to a by a record equal to $a' \neq a$ lowers h_a by $1/n$ and raises $h_{a'}$ by $1/n$, so the ℓ_1 sensitivity of h is $2/n$. The scale $2/(n\varepsilon)$ gives ε-DP.
3. *Union bound.* For m Laplace variables with scale b, the chance that some $|Z_j|$ is at least s is at most $me^{-s/b}$. Setting this equal to β gives $s = b\ln(m/\beta)$. With $m = d$ and $b = d/(n\varepsilon)$ this is claim 1, and with $m = k$ and $b = 2/(n\varepsilon)$ it is claim 2.
4. *Sample size.* Each error bound is at most α exactly when n satisfies Equation 11.4 with $\beta = 1/2$.

□

Approximate DP improves both bounds. For marginals, the Gaussian mechanism in the approximate DP chapter uses ℓ_2 sensitivity $\sqrt{d}/n$ rather than ℓ_1 sensitivity d/n, reducing the required n to order $\sqrt{d\ln(1/\delta)\ln d}/(\alpha\varepsilon)$. For histograms, the stability-based histogram needs n of order $\ln(1/\delta)/(\alpha\varepsilon)$ when $\delta \leq 1/n$, independent of k. The rest of this chapter shows that pure DP cannot achieve these improvements.

11.2. The packing argument

The argument resembles packing disjoint balls into a box. Here each dataset has a disjoint accuracy region in the output space. Accuracy places the output in its dataset's region with high probability; privacy requires other datasets to assign that region nonzero probability. Since probabilities over disjoint regions sum to at most one, only a limited number of datasets can satisfy both conditions.

Group privacy connects datasets that differ in multiple records; it was proved in the Laplace mechanism chapter. The **Hamming distance** between datasets in $\mathcal{X}^n$ counts differing records. An ε-DP mechanism is $t\varepsilon$-DP for datasets at distance at most t. Thus, for such D, D' and every event S,

$$\Pr[M(D) \in S] \le e^{t\varepsilon} \Pr[M(D') \in S]. \tag{11.5}$$

11.2.1. A first example: the mode

Consider estimating the mode, the most frequent symbol in $[k]$. For each j, let D_j contain n copies of j. The mode is j, and every pair of these datasets is at Hamming distance n. An accurate mechanism outputs j on D_j with probability at least one half. Group privacy gives a lower bound on the probability of each answer under every other dataset. The probabilities can sum to at most one only if n is sufficiently large.

Proposition 11.2 (Lower bound for the mode)**.** *Let $k \ge 2$ and $\varepsilon > 0$, and for each $j \in [k]$ let D_j be the dataset of n copies of the symbol j. Let $M\colon [k]^n \to [k]$ be ε-DP, and suppose that $\Pr[M(D_j) = j] \ge 1/2$ for every $j \in [k]$. Then*

$$n \ge \frac{\ln(k/2)}{\varepsilon}. \tag{11.6}$$

Proof.

1. *Pigeonhole.* Fix any j. The k probabilities $\Pr[M(D_j) = \ell]$, for $\ell \in [k]$, sum to 1. So at least one symbol ℓ has $\Pr[M(D_j) = \ell] \le 1/k$, since otherwise the sum would exceed 1.
2. *Accuracy.* The mechanism is correct on D_ℓ with probability at least $1/2$, so $\Pr[M(D_\ell) = \ell] \ge 1/2$.
3. *Group privacy.* The datasets D_ℓ and D_j are at Hamming distance at most n. By Equation 11.5 with the event $\{\ell\}$,

$$\frac{1}{2} \le \Pr[M(D_\ell) = \ell] \le e^{n\varepsilon} \Pr[M(D_j) = \ell] \le \frac{e^{n\varepsilon}}{k}. \tag{11.7}$$

4. *Rearranging.* The chain gives $e^{n\varepsilon} \ge k/2$, and taking logarithms gives $n\varepsilon \ge \ln(k/2)$.

□

Thus, a private mode requires order $\ln k/\varepsilon$ records. For $k = 1{,}000$ and $\varepsilon = 1$, the bound is $\ln 500 = 6.21$; six records cannot suffice for a reliable 1-DP mode estimate.

11.2.2. The packing theorem

The proof used three things: many datasets, disjoint sets of correct answers, and group privacy across a bounded distance. The theorem below keeps exactly these. It also lets the datasets be close to one common centre D, at distance t that may be much smaller than n, which the histogram bound will need.

Theorem 11.1 (Packing)**.** *Let $D_1, \dots, D_m \in \mathcal{X}^n$ be datasets, each at Hamming distance at most t from a fixed dataset $D \in \mathcal{X}^n$. Let $S_1, \dots, S_m$ be pairwise disjoint subsets of the output space $\mathcal{Y}$. If $M \colon \mathcal{X}^n \to \mathcal{Y}$ is ε-DP and, for some $p > 0$,*

$$\Pr[M(D_\ell) \in S_\ell] \geq p \qquad \text{for every } \ell \in [m], \tag{11.8}$$

then

$$\frac{1}{m} \geq p\, e^{-t\varepsilon}. \tag{11.9}$$

Proof.

1. *Accuracy and group privacy.* Fix ℓ. The datasets D_ℓ and D are at Hamming distance at most t, so Equation 11.5 gives $p \leq \Pr[M(D_\ell) \in S_\ell] \leq e^{t\varepsilon} \Pr[M(D) \in S_\ell]$. Hence $\Pr[M(D) \in S_\ell] \geq pe^{-t\varepsilon}$.
2. *Disjointness.* The sets S_ℓ are disjoint, so the probability of their union is the sum of their probabilities:

$$1 \geq \Pr\left[M(D) \in \bigcup_{\ell=1}^{m} S_\ell\right] = \sum_{\ell=1}^{m} \Pr[M(D) \in S_\ell] \geq m\, p\, e^{-t\varepsilon}. \tag{11.10}$$

3. *Rearranging.* Dividing by m gives Equation 11.9.

□

Taking logarithms gives $t\varepsilon \geq \ln(mp)$. With $t = n$ and $p = 1/2$, this becomes $n \geq \ln(m/2)/\varepsilon$, recovering Proposition 11.2 for $m = k$. Stronger lower bounds require many datasets, disjoint accuracy regions, and distances t small relative to n.

11.3. One-way marginals

For one-way marginals, take datasets that repeat a single record. For each $w \in \{0,1\}^d$, let D_w contain n copies of w. Its marginals equal the corner w of $[0,1]^d$. The 2^d corners differ by one in at least one coordinate, so error below $1/2$ identifies a unique corner. The packing theorem applies to these 2^d datasets.

Theorem 11.2 (Lower bound for one-way marginals)**.** *Let $\varepsilon > 0$. Let $M\colon (\{0,1\}^d)^n \to \mathbb{R}^d$ be ε-DP, and suppose that on every dataset D, with probability at least $1/2$, it answers every one-way marginal with error less than $1/2$:*

$$\Pr\left[\max_j |M(D)_j - f_j(D)| < \tfrac{1}{2}\right] \geq \tfrac{1}{2}. \tag{11.11}$$

Then

$$n \geq \frac{(d-1)\ln 2}{\varepsilon}. \tag{11.12}$$

Proof.

1. *The packing.* Use the $m = 2^d$ datasets D_w. For each w, let $S_w = \{y \in \mathbb{R}^d : |y_j - w_j| < 1/2 \text{ for all } j\}$ be the open ℓ_∞ ball of radius $1/2$ around the corner w. Since $f(D_w) = w$, the hypothesis says exactly that $\Pr[M(D_w) \in S_w] \geq 1/2$.
2. *Disjointness.* Let $w \neq w'$ differ in coordinate j, so $|w_j - w'_j| = 1$. For any y, the triangle inequality gives $|y_j - w_j| + |y_j - w'_j| \geq 1$, so the two distances cannot both be less than $1/2$. Hence S_w and $S_{w'}$ are disjoint.
3. *Distance.* Any two datasets in $\mathcal{X}^n$ differ in at most n records, so every D_w is within Hamming distance $t = n$ of $D = D_0$, the all-zero dataset.
4. *Packing.* Theorem 11.1 with $m = 2^d$, $p = 1/2$ and $t = n$ gives $2^{-d} \geq \frac{1}{2}e^{-n\varepsilon}$. Taking logarithms, $n\varepsilon \geq d\ln 2 - \ln 2$.

□

Under ε-DP, error below $1/2$ requires $n = \Omega(d/\varepsilon)$, compared with the upper bound $d\ln(2d)/(\alpha\varepsilon)$ in Proposition 11.1. Approximate DP needs only order $\sqrt{d}$ records, a factor of about $\sqrt{d}$ fewer. A matching approximate-DP lower bound also holds, but packing cannot prove it: group privacy degrades too quickly with t, so the proof uses fingerprinting codes.

This bound is independent of α because the cube's corners are far apart. For an α-dependent bound, use datasets that differ in only a few records, whose marginals are closer.

Corollary 11.1. *Let $\varepsilon > 0$ and $\alpha \in (0, 1/2)$. If an ε-DP mechanism $M\colon (\{0,1\}^d)^n \to \mathbb{R}^d$ answers every one-way marginal with error at most α, with probability at least $1/2$ on every dataset, then*

$$n \geq \frac{(d-1)\ln 2 - \varepsilon}{2\alpha\varepsilon}. \tag{11.13}$$

Proof.

1. *The packing.* Let $t = \lfloor 2\alpha n \rfloor + 1$, so that $t/n > 2\alpha$, and $t \leq n$ because $2\alpha < 1$. For each $w \in \{0,1\}^d$, let D_w consist of t copies of w and $n - t$ copies of the all-zero record. Its marginals are $f(D_w) = (t/n)\, w$.

2. *Disjointness.* Let S_w be the closed ℓ_∞ ball of radius α around $(t/n)\,w$, so that $\Pr[M(D_w) \in S_w] \geq 1/2$ by hypothesis. If $w \neq w'$ differ in coordinate j, the centres differ by $t/n > 2\alpha$ in that coordinate, and two closed intervals of half-width α whose centres are more than 2α apart do not meet. So the sets S_w are pairwise disjoint.
3. *Distance.* Each D_w differs from the all-zero dataset in at most t records.
4. *Packing.* Theorem 11.1 with $m = 2^d$ and $p = 1/2$ gives $\ln(2^{d-1}) \leq t\varepsilon \leq (2\alpha n + 1)\varepsilon$. Rearranging gives the claim.

□

For these parameters, the lower and Laplace upper bounds in Proposition 11.1 differ by a moderate factor.

Example 11.1. With $d = 1{,}000$ questions, $\alpha = 0.1$ and $\varepsilon = 1$, Corollary 11.1 says that $n \geq 3{,}458$ is necessary, and Equation 11.4 says that $n \geq 76{,}010$ suffices. The ratio, 22.0, is the factor $\ln(2d) = 7.60$ times constants, and it grows only logarithmically with d.

11.4. Histograms

For histograms, the corner construction gives only the mode bound $\ln(k/2)/\varepsilon$, with no α. The factor $1/\alpha$ comes from datasets that are all close to one centre, as in Corollary 11.1. Take the centre D_1 to be n copies of the symbol 1, and move a small block of records to another symbol.

Theorem 11.3 (Lower bound for histograms)**.** *Let $k \geq 2$, $\varepsilon > 0$ and $\alpha \in (0, 1/2)$. If an ε-DP mechanism $M \colon [k]^n \to \mathbb{R}^k$ estimates the normalized histogram with error at most α,*

$$\Pr\Big[\max_j |M(D)_j - h_j(D)| \leq \alpha\Big] \geq \tfrac{1}{2} \qquad \text{for every } D \in [k]^n, \tag{11.14}$$

then

$$n \geq \frac{\ln(k/2) - \varepsilon}{2\alpha\varepsilon}. \tag{11.15}$$

Proof.

1. *The packing.* Let $t = \lfloor 2\alpha n \rfloor + 1$, so that $t/n > 2\alpha$, and $t \leq n$ because $2\alpha < 1$. Let D_1 be n copies of the symbol 1, so its histogram puts all its mass on the symbol 1. For $\ell = 2, \ldots, k$, let D_ℓ consist of t copies of ℓ and $n - t$ copies of 1. Its histogram $h(D_\ell)$ puts mass t/n on the symbol ℓ, mass $1 - t/n$ on the symbol 1, and nothing elsewhere.
2. *Accuracy regions.* Let S_ℓ be the closed ℓ_∞ ball of radius α around $h(D_\ell)$. The hypothesis applied to the dataset D_ℓ says that $\Pr[M(D_\ell) \in S_\ell] \geq 1/2$.
3. *Disjointness.* Take $\ell \neq \ell'$. At least one of them is not 1, say $\ell \neq 1$. In coordinate ℓ, the centre $h(D_\ell)$ has value t/n and the centre $h(D_{\ell'})$ has value 0, because $D_{\ell'}$ contains only the symbols 1 and ℓ'. These differ by more than 2α, so S_ℓ and $S_{\ell'}$ are disjoint.
4. *Distance.* Each D_ℓ differs from D_1 in at most t records, the ones not equal to 1.

5. *Packing.* Theorem 11.1 with $m = k$, $p = 1/2$ and centre D_1 gives $1/k \geq \frac{1}{2}e^{-t\varepsilon}$, that is, $\ln(k/2) \leq t\varepsilon$. Since $t \leq 2\alpha n + 1$, this gives $\ln(k/2) \leq (2\alpha n + 1)\varepsilon$, which rearranges to Equation 11.15.

□

Figure 11.1 shows the construction for $k = 5$, $n = 20$ and $\alpha = 0.1$, where $t = 5$ and each moved block holds a quarter of the records. In coordinate ℓ, the band of acceptable answers for D_ℓ starts at 0.15, above the band of every other dataset, which ends at 0.1.

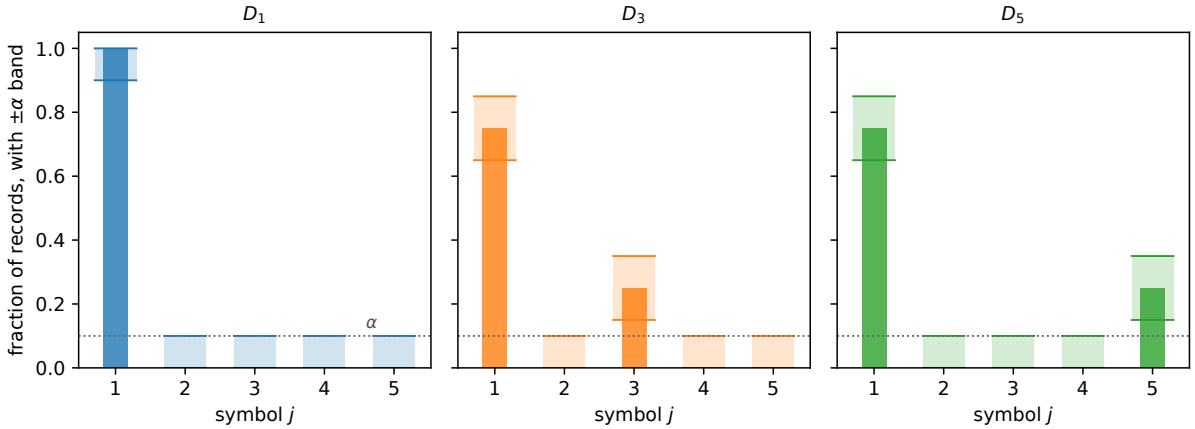


Figure 11.1.: The packing behind Theorem 11.3, for $k = 5$, $n = 20$ and $\alpha = 0.1$. Each panel shows the normalized histogram of one dataset as bars and the band of answers within α of each count as a shaded box. Three of the five datasets are shown. Any two datasets have non-overlapping bands in at least one coordinate, so no answer is acceptable for both.

The choice of t produces the factor $1/\alpha$. Moving the whole dataset, $t = n$, would also give disjoint regions, but then the packing theorem would bound only $n\varepsilon$, as in Proposition 11.2. A block of about $2\alpha n$ records is just enough to separate the answers, and group privacy pays $e^{t\varepsilon}$ for those t records only.

Example 11.2. With $k = 10^6$ symbols, $\alpha = 0.01$ and $\varepsilon = 1$, Equation 11.15 says that $n \geq 607$ is necessary, and the Laplace histogram of Proposition 11.1 succeeds once $n \geq 2{,}902$. The ratio is 4.8, and as k grows it falls towards 4, because the upper bound is $2\ln(2k)/(\alpha\varepsilon)$ and the lower bound is about $\ln k/(2\alpha\varepsilon)$. The union bound in Proposition 11.1 is nearly tight. In simulation with $k = 10^4$ and $n = 1{,}981$, the value Equation 11.4 gives for that k, the Laplace histogram met error α in 217 of 400 runs, 54%, against the 50% that the bound guarantees.

So under pure DP the Laplace histogram is optimal up to a constant factor, and the $\ln k$ in its sample size cannot be removed. The stability-based histogram, which replaces $\ln k$ by $\ln(1/\delta)$, shows that this cost belongs to pure DP and not to the problem.

11.5. Summary

Table 11.1.: The chapter's results. All lower bounds hold for every ε-DP mechanism that meets the accuracy requirement with probability at least $1/2$.

Result	Statement
Proposition 11.1	Laplace noise answers d marginals with $n \geq d \ln(2d)/(\alpha\varepsilon)$ and a k-bin histogram with $n \geq 2\ln(2k)/(\alpha\varepsilon)$
Proposition 11.2	an ε-DP mode that is right half the time needs $n \geq \ln(k/2)/\varepsilon$
Theorem 11.1	m datasets within distance t of a centre, with disjoint accuracy regions hit with probability p, force $1/m \geq pe^{-t\varepsilon}$
Theorem 11.2	error below $1/2$ on all one-way marginals needs $n \geq (d-1)\ln 2/\varepsilon$
Corollary 11.1	error at most α on all one-way marginals needs $n \geq ((d-1)\ln 2 - \varepsilon)/(2\alpha\varepsilon)$
Theorem 11.3	error at most α on a k-bin histogram needs $n \geq (\ln(k/2) - \varepsilon)/(2\alpha\varepsilon)$

11.6. References

- Moritz Hardt and Kunal Talwar, "On the Geometry of Differential Privacy", 2010. Packing lower bounds for linear queries under pure DP, stated geometrically.
- Amos Beimel, Hai Brenner, Shiva Prasad Kasiviswanathan and Kobbi Nissim, "Bounds on the Sample Complexity for Private Learning and Private Data Release", 2014. Packing arguments for private learning and for the release of point functions, which is the histogram problem.
- Mark Bun, Jonathan Ullman and Salil Vadhan, "Fingerprinting Codes and the Price of Approximate Differential Privacy", 2014. The lower bound of order $\sqrt{d}$ for one-way marginals under approximate DP.
- Thomas Steinke and Jonathan Ullman, "Between Pure and Approximate Differential Privacy", 2016. Tight bounds for one-way marginals across pure and approximate DP.
- Salil Vadhan, "The Complexity of Differential Privacy", 2017. Section 5 presents packing lower bounds in the generality used here.

Part IV.

Private learning and statistics

The earlier parts released statistics of fixed datasets under differential privacy (DP). This part studies model fitting and population inference. It distinguishes prediction from memorization: a model can predict a person's sensitive label without using that person's record. The next chapters cover private training of convex models and neural networks, then bound the sample size for private mean estimation. The final chapter shows how DP lets one dataset answer adaptively chosen questions without overfitting.

Table 11.2.: The chapters of this part and the questions they answer.

Chapter	Question it answers
What differential privacy protects	Which inferences from a trained model are privacy violations, and which are learning at work?
Private empirical risk minimization	How can a convex model be fitted privately, and at what cost in accuracy?
Private deep learning	How are neural networks trained privately, and how tightly can their privacy be accounted?
Private mean estimation	How many samples does a private estimate of a population mean need?
Adaptive data analysis	How can one dataset answer adaptively chosen questions without overfitting?

12. What differential privacy protects

Machine learning on sensitive data can produce invasive results, but not every such result violates privacy. This chapter distinguishes inferences that differential privacy (DP) must prevent from those it permits. A private learner can predict a sensitive label from population patterns, but should not memorize a secret found only in one person's records.

12.1. What the definition promises

Recall the definition from the randomized response chapter. Neighbours $D \sim D'$ differ in one record. A mechanism M is ε-DP if, for every such pair and output event S,

$$\Pr[M(D) \in S] \leq e^{\varepsilon} \Pr[M(D') \in S]. \tag{12.1}$$

The output distribution changes little when one record changes. Two properties from the Laplace mechanism chapter translate this guarantee to attackers. **Post-processing** preserves ε-DP when a function uses only the mechanism's output and independent randomness. **Group privacy** extends the bound to datasets differing in k entries:

$$\Pr[M(D) \in S] \leq e^{k\varepsilon} \Pr[M(D') \in S]. \tag{12.2}$$

Write $D^{(i \leftarrow z)}$ for D with record i replaced by z. An attacker is any possibly randomized function $\mathcal{A}$ of the output that returns 1 when it detects a specified harm. It may use all other records and any side information; these are fixed inputs to $\mathcal{A}$. The next proposition bounds the attacker's success when one record is replaced.

Proposition 12.1 (What an attacker gains from one record)**.** *Let M be ε-DP, and let $\mathcal{A}$ be a randomized function of the output whose coins are independent of the coins of M. Then for every dataset D, every index i and every record z,*

$$\Pr\left[\mathcal{A}(M(D)) = 1\right] \leq e^{\varepsilon} \Pr\left[\mathcal{A}(M(D^{(i \leftarrow z)})) = 1\right]. \tag{12.3}$$

Proof.

1. The composed algorithm $\mathcal{A} \circ M$ is ε-DP by post-processing.
2. The datasets D and $D^{(i \leftarrow z)}$ differ in at most one entry, so they are neighbours or equal. The definition, applied to the event $\{1\}$, gives the inequality.

□

Thus, replacing a person's record with a placeholder changes the probability of an output-caused harm by at most the DP factor. DP does not prevent harms that remain likely after replacement. To assess an alleged violation, ask:

1. Is the result substantially different if one person's record is changed or left out?
2. Is the harmful inference the same thing as the accuracy of the learned model?
3. If the inference still feels wrong, at what point, and by whom, was the information given away?

The second question is central. If an inference is exactly what an accurate model predicts, preventing it requires releasing an inaccurate model, such as one that ignores the data. Such a model is uninformative.

12.2. Learning a rule is not a violation

A learning algorithm takes training pairs (x_i, y_i), where x_i is a **feature vector** and y_i is a **label**. It outputs a **classifier**, which maps feature vectors to predicted labels. Under DP, the training set is private and the classifier is the protected output; test points used to measure accuracy are not protected.

Consider real-valued training pairs

$$(0,1),\ (2,3),\ (5,6),\ (-6,-5),\ (0.5,1.5),\ (\pi,\pi+1),\ \ldots \tag{12.4}$$

Every label is one more than its feature. A learner can infer the rule $y = x + 1$, predict 5 at $x = 4$, and predict $\pi + 1$ at $x = \pi$. The latter prediction matches a training label, but it is not a privacy violation: the rule holds across the population, so the prediction does not depend on whether $(\pi, \pi + 1)$ was collected.

A private algorithm makes this distinction explicit. Each record determines an offset $y_i - x_i$, and a rule $y = x + b$ specifies an offset. The stability-based histogram from the chapter on beyond global sensitivity can release frequently occurring offsets even though the domain is continuous. Its guarantee is approximate DP, defined in the approximate DP chapter by adding slack δ:

$$\Pr[M(D) \in S] \le e^{\varepsilon} \Pr[M(D') \in S] + \delta \quad \text{for all } D \sim D' \text{ and all } S. \tag{12.5}$$

The algorithm adds Laplace noise to each observed offset's count and releases offsets whose noisy counts exceed a threshold.

Definition 12.1 (Stable histogram of offsets)**.** Let $\varepsilon > 0$ and $\delta \in (0, 1)$, and set the threshold

$$\tau = 1 + \frac{2\ln(1/\delta)}{\varepsilon}. \tag{12.6}$$

On input $D = ((x_1, y_1), \ldots, (x_n, y_n))$, a list of n records with real entries, let $c_b = |\{i : y_i - x_i = b\}|$ be the number of records whose offset is b. For each b with $c_b > 0$, draw an independent Z_b from the Laplace distribution $\mathrm{Lap}(2/\varepsilon)$ with scale $2/\varepsilon$, and output the set

$$M(D) = \{\, b : c_b > 0 \text{ and } c_b + Z_b > \tau \,\}. \tag{12.7}$$

This applies the stability-based histogram to offsets $y_i - x_i$, then removes the noisy counts from the output. Zero-count bins are not considered, and threshold τ suppresses a bin supported by one record except with small probability. The resulting mechanism retains the histogram's privacy guarantee.

Theorem 12.1. *The stable histogram of offsets is (ε, δ)-DP.*

Proof.

1. Replacing one record changes at most one offset, so when $D \sim D'$ the two lists of offsets are neighbours or equal.
2. The stability-based histogram is (ε, δ)-DP on datasets of offsets, by the theorem of that chapter.
3. The output $M(D)$ is the set of elements on the histogram's list, a function of that output alone. So it is (ε, δ)-DP by post-processing, which holds for approximate DP as well.

□

The privacy cost does not depend on how many records follow the rule. Accuracy requires enough records for the offset count to exceed the threshold.

Proposition 12.2. *Suppose every record satisfies $y_i = x_i + 1$, let τ be the threshold of Definition 12.1, and let $\beta \in (0, 1/2)$ be a failure probability. If*

$$n \geq \tau + \frac{2}{\varepsilon} \ln \frac{1}{2\beta}, \tag{12.8}$$

then with probability at least $1 - \beta$ the output is exactly $\{1\}$, and the released rule predicts $y = x + 1$ at every feature value.

Proof.

1. Only the offset 1 has a positive count, $c_1 = n$, because all n records have that offset. So the output is either $\{1\}$ or empty.
2. The output is empty when the Laplace noise Z_1 on that count satisfies $n + Z_1 \leq \tau$. A $\mathrm{Lap}(2/\varepsilon)$ variable Z has $\Pr[Z \leq -s] = \frac{1}{2} e^{-\varepsilon s/2}$ for every $s \geq 0$. With $s = n - \tau$, which is not negative because $n \geq \tau$, this gives

$$\Pr[Z_1 \leq -(n - \tau)] = \tfrac{1}{2} e^{-\varepsilon (n - \tau)/2}. \tag{12.9}$$

3. The assumed bound on n makes this at most β.

□

Example 12.1. Take $\varepsilon = 1$, $\delta = 10^{-6}$ and $\beta = 0.01$. With these values the threshold is $\tau = 1 + 2\ln(10^6) = 1 + 2 \times 13.82 = 28.63$. The bound on n adds $2\ln 50 = 7.82$ to it, which gives 36.45. So 37 records that follow the rule are enough for the private algorithm to output it with probability at least 0.99.

Thus, Proposition 12.1 protects record $(\pi, \pi + 1)$ even though the rule predicts its label exactly. The prediction follows from a population-level rule. The relevant disclosure occurred when the feature value π became known; anyone who knows the rule can then calculate the label.

12.3. Inferences from what people reveal

The same distinction applies to a study of smoking and cancer. A smoker may fear higher insurance premiums if the study publishes an association. Here, "smokes" is the feature and "likely to develop cancer" is the label. The study would reach the same conclusion without this person. Preventing the resulting inference would require suppressing the population-level finding; the person's smoking status was disclosed separately.

12.4. What a violation looks like

A privacy violation is an inference that depends on the person's sensitive record being in the dataset.

The Netflix Prize. Netflix released dated movie ratings under pseudonymous identifiers. Narayanan and Shmatikov linked them to public reviews on the Internet Movie Database and identified ratings that matched users had not made public. This reidentification depends on the users' records. By contrast, inferring that someone watched the second film in a trilogy after seeing their public reviews of the first and third uses only information they chose to publish.

Memorized secrets. Carlini and coauthors trained language models on text containing secrets, such as a nine-digit identification number entered several times by one person. The model assigned the secret unusually high likelihood, enabling score-guided extraction. This is a violation because language patterns alone do not favour one nine-digit number; the preference comes from the person's records. DP and group privacy bound this risk, as the next proposition shows.

Proposition 12.3 (Secrets are not memorized)**.** *Let $\mathcal{S}$ be a finite set of N possible secrets. For each $s \in \mathcal{S}$, let $D(s)$ be a training set in which k entries depend on s and the other entries do not. Let M be ε-DP, and let G be any guessing rule, with coins independent of M, that maps the trained model to an element of $\mathcal{S}$. If the secret s is drawn uniformly from $\mathcal{S}$, then*

$$\Pr\left[G(M(D(s))) = s\right] \leq \frac{e^{k\varepsilon}}{N}. \tag{12.10}$$

Proof.

1. Let D_0 be the training set with the k entries that depend on the secret replaced by fixed placeholder records. For every s, the datasets $D(s)$ and D_0 differ in at most k entries.

2. By post-processing and group privacy, $\Pr[G(M(D(s))) = s] \le e^{k\varepsilon} \Pr[G(M(D_0)) = s]$ for every s.
3. The guess on D_0 does not depend on s, and it equals at most one secret, so $\sum_{s \in \mathcal{S}} \Pr[G(M(D_0)) = s] \le 1$.
4. The average of step 2 over the uniform secret, with step 3, gives

$$\Pr\left[G(M(D(s))) = s\right] = \frac{1}{N} \sum_{s \in \mathcal{S}} \Pr\left[G(M(D(s))) = s\right] \le \frac{e^{k\varepsilon}}{N}. \tag{12.11}$$

□

For a nine-digit number, $N = 10^9$. If $\varepsilon = 1$ and the secret appears $k = 4$ times, the attacker's success probability is at most $e^4/10^9 = 5.5 \times 10^{-8}$, compared with 10^{-9} for a blind guess. The bound grows exponentially with k, so repeated contributions require a user-level rather than record-level guarantee.

12.5. Summary

Table 12.1.: The chapter's results.

Result	Statement
Proposition 12.1	an attacker succeeds about as often when a person's record is replaced by any other
Theorem 12.1	releasing offsets whose noisy count exceeds τ is (ε, δ)-DP
Proposition 12.2	with $n \ge \tau + \frac{2}{\varepsilon} \ln \frac{1}{2\beta}$ records, the private algorithm finds $y = x + 1$
Proposition 12.3	a secret in k records of one person is guessed with probability at most $e^{k\varepsilon}/N$

12.6. References

- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014, whose Section 2.3 discusses what the definition promises and the smoking example.
- Arvind Narayanan and Vitaly Shmatikov, "Robust De-anonymization of Large Sparse Datasets", 2008, the linkage attack on the Netflix Prize data.
- Nicholas Carlini, Chang Liu, Úlfar Erlingsson, Jernej Kos and Dawn Song, "The Secret Sharer: Evaluating and Testing Unintended Memorization in Neural Networks", 2019, which extracts secrets that appear in few training records.
- Abhradeep Thakurta and Adam Smith, "Differentially Private Feature Selection via Stability Arguments, and the Robustness of the Lasso", 2013, and Salil Vadhan, "The Complexity of Differential Privacy", 2017, Section 3, on stability-based algorithms such as the histogram behind Theorem 12.1.

13. Private empirical risk minimization

Machine learning often fits a model by minimizing its average training loss. This chapter analyses three methods under differential privacy (DP) for convex losses: perturb the solver's output, perturb its objective, or perturb each gradient step. It proves privacy guarantees for all three and an accuracy guarantee for gradient descent, which the next chapter extends to neural networks.

13.1. Empirical risk minimization

Each record $z = (x, y)$ contains a feature vector x and label y; dataset $D = (z_1, \dots, z_n)$ contains n records. A parameter vector $\theta \in \mathbb{R}^d$ specifies the model. The **loss function** $\ell(\theta; z)$ measures its error on record z. Empirical risk minimization chooses an allowed parameter with low total loss. The definition also measures algorithms that approximate the minimum.

Definition 13.1 (Empirical risk minimization)**.** Let $\Theta \subseteq \mathbb{R}^d$ be a nonempty, closed, convex set of allowed parameters. The empirical risk of θ on D is the total loss

$$\mathcal{L}(\theta; D) = \sum_{i=1}^{n} \ell(\theta; z_i), \tag{13.1}$$

and $\theta^* \in \Theta$ denotes a minimizer of $\mathcal{L}(\cdot\,; D)$ over Θ. An algorithm that outputs $\tilde{\theta} \in \Theta$ has **excess empirical risk** $\mathbb{E}[\mathcal{L}(\tilde{\theta}; D)] - \mathcal{L}(\theta^*; D)$, where the expectation is over the algorithm's coins only. The dataset is fixed.

This problem is **empirical risk minimization (ERM)**. This chapter analyses empirical loss; when uniform convergence applies, low empirical loss also implies low loss on new data.

The analysis assumes that the parameter set is bounded, the loss is convex, and the loss is Lipschitz. Boundedness limits the parameter range; convexity makes every local minimum global; Lipschitz continuity bounds how quickly the loss changes. A **subgradient** extends the gradient to points where a convex function has a kink.

Definition 13.2 (Diameter, convexity, Lipschitz continuity)**.**

- The **diameter** of Θ is $\|\Theta\|_2 = \sup_{\theta, \theta' \in \Theta} \|\theta - \theta'\|_2$.
- A function f on a convex set K is **convex** if $f(t\theta + (1-t)\theta') \le t f(\theta) + (1-t) f(\theta')$ for all $\theta, \theta' \in K$ and $t \in [0, 1]$.
- A function f on K is **L-Lipschitz** if $|f(\theta) - f(\theta')| \le L\|\theta - \theta'\|_2$ for all $\theta, \theta' \in K$.
- A vector g is a **subgradient** of a convex f at θ if $f(\theta') \ge f(\theta) + \langle g, \theta' - \theta \rangle$ for all θ'. Where f is differentiable, its gradient $\nabla f(\theta)$, the vector of partial derivatives, is the only subgradient.

Assume Θ has finite diameter and, for every record z, $\ell(\cdot; z)$ is convex and L-Lipschitz on $\mathbb{R}^d$. Such a finite convex function has a subgradient everywhere. Write $\nabla\ell(\theta; z)$ for its gradient or, at nondifferentiable points, a subgradient. Lipschitz continuity bounds these vectors and therefore limits each record's effect on a gradient step.

Lemma 13.1. *If a convex function f is L-Lipschitz on $\mathbb{R}^d$, then every subgradient g of f satisfies $\|g\|_2 \le L$.*

Proof.

1. Suppose $g \neq 0$, take any point θ, and set $u = g/\|g\|_2$. The definition of a subgradient at $\theta' = \theta + u$ gives $f(\theta + u) - f(\theta) \ge \langle g, u\rangle = \|g\|_2$.
2. The Lipschitz condition gives $f(\theta + u) - f(\theta) \le L\|u\|_2 = L$. The two bounds together give $\|g\|_2 \le L$.

□

Common losses are convex; bounded features often also make them Lipschitz.

Example 13.1 (Common losses)**.** Take $x, \theta \in \mathbb{R}^d$.

- **Linear regression**, $y \in \mathbb{R}$: the squared loss $\ell(\theta; x, y) = (\langle x, \theta\rangle - y)^2$. It is convex, and Lipschitz only on bounded sets.
- **Logistic regression**, $y \in \{-1, 1\}$: $\ell(\theta; x, y) = \ln\left(1 + e^{-y\langle x,\theta\rangle}\right)$. Its gradient is $-y\,x/(1 + e^{y\langle x,\theta\rangle})$, of norm at most $\|x\|_2$, so it is $\|x\|_2$-Lipschitz.
- **Geometric median**, no label: $\ell(\theta; x) = \|\theta - x\|_2$. By the triangle inequality it is 1-Lipschitz. In one dimension it is $|\theta - x|$, and its minimizer is the median.
- **Support vector machine (SVM)**, $y \in \{-1, 1\}$: the hinge loss $\ell(\theta; x, y) = \max(0, 1 - y\langle x, \theta\rangle)$. It is a maximum of two functions that are linear in θ, so it is convex and $\|x\|_2$-Lipschitz.

Strong convexity and smoothness can improve rates but are not needed here. Before Section 13.5, treat the solver as a black box.

13.2. Why the minimizer is not private

An exact minimizer can reveal a record. The median gives a direct example.

Proposition 13.1. *Let $d = 1$, $\ell(\theta; x) = |\theta - x|$, records in $[-R, R]$ with $R > 0$, $\Theta = [-R, R]$, and n odd. Then the empirical risk has a unique minimizer, the median $x_{(m)}$ with $m = (n + 1)/2$, where $x_{(1)} \le \ldots \le x_{(n)}$ are the sorted records. The algorithm that outputs it is not (ε, δ)-DP for any ε and any $\delta < 1$.*

Proof.

1. Away from the records, the function $\theta \mapsto \sum_i |\theta - x_i|$ has slope equal to the number of records below θ minus the number above. For $\theta < x_{(m)}$ at least m records lie above, so the slope is at most $(m-1) - m = -1$. For $\theta > x_{(m)}$ the slope is at least 1 in the same way. The function is continuous, so $x_{(m)}$ is its unique minimizer.
2. On the dataset with every record equal to $-R$ the output is $-R$, and on the dataset with every record equal to R it is R. A path of n replacements joins the two datasets, so some neighbours $D \sim D'$ along it have different outputs $v \neq v'$.
3. The algorithm is deterministic. For the event $S = \{v\}$ this gives $\Pr[M(D) \in S] = 1$ and $\Pr[M(D') \in S] = 0$, and $1 \leq e^{\varepsilon} \cdot 0 + \delta$ fails for every $\delta < 1$.

□

Privatizing each record before using a non-private algorithm gives local DP, which needs more data for the same accuracy. The methods below instead add noise after pooling the data.

13.3. Output perturbation

Output perturbation adds noise calibrated to the minimizer's sensitivity. As Proposition 13.1 shows, that sensitivity can be large; regularization reduces it.

Polynomial regression illustrates the problem. For records $(x_i, y_i) \in \mathbb{R}^2$ and squared loss $\ell(\theta; x, y) = (\sum_{j=0}^{p} \theta_j x^j - y)^2$, the model has degree p. If $p = n-1$ and the features are distinct, the minimizer interpolates every point. It has zero excess risk on the training data but can fit new data poorly and change sharply when one point moves. The left panel of Figure 13.1 shows this instability.

Regularization penalizes model complexity. For a parameter-only function N, called the **regularizer**, and weight $\lambda > 0$, define the regularized objective as

$$F_D(\theta) = \mathcal{L}(\theta; D) + \lambda N(\theta). \tag{13.2}$$

At $\lambda = 0$ there is no regularization; as $\lambda \to \infty$, the penalty dominates the data. A common choice is $N(\theta) = \frac{1}{2}\|\theta\|_2^2$. Its effect on sensitivity follows from strong convexity, which requires a uniform lower bound on curvature. For the regularized objective, that lower bound is the weight λ.

Definition 13.3 (Strong convexity)**.** For $\lambda > 0$, a function f on a convex set K is **λ-strongly convex** if, for all $\theta, \theta' \in K$ and $t \in [0, 1]$,

$$f(t\theta + (1-t)\theta') \leq t f(\theta) + (1-t) f(\theta') - \frac{\lambda}{2} t(1-t) \|\theta - \theta'\|_2^2. \tag{13.3}$$

The function $\frac{1}{2}\|\theta\|_2^2$ is 1-strongly convex, with equality, by the identity $t\|a\|^2 + (1-t)\|b\|^2 - \|ta + (1-t)b\|^2 = t(1-t)\|a-b\|^2$. A λ-strongly convex function plus a convex function is λ-strongly convex. The next lemma shows that such a function grows at least quadratically away from its minimizer.

Lemma 13.2. *Let f be λ-strongly convex on a convex set K, with a minimizer $\hat{\theta}$ over K. Then for every $\theta \in K$,*

$$f(\theta) - f(\hat{\theta}) \geq \frac{\lambda}{2}\|\theta - \hat{\theta}\|_2^2. \tag{13.4}$$

In particular, the minimizer is unique.

Proof.

1. For $t \in (0,1)$, the point $\hat{\theta} + t(\theta - \hat{\theta})$ lies in K, so its value is at least $f(\hat{\theta})$.
2. Strong convexity, applied with weight t on θ, bounds that value by $tf(\theta) + (1-t)f(\hat{\theta}) - \frac{\lambda}{2}t(1-t)\|\theta - \hat{\theta}\|_2^2$.
3. The two bounds together, divided by t, give $f(\theta) - f(\hat{\theta}) \geq \frac{\lambda}{2}(1-t)\|\theta - \hat{\theta}\|_2^2$. Letting $t \to 0$ gives the claim.
4. A second minimizer θ would have $f(\theta) - f(\hat{\theta}) = 0$, which forces $\theta = \hat{\theta}$.

□

This growth keeps the regularized minimizer stable. When one record changes, the two regularized objectives differ by a Lipschitz function. Each objective grows quadratically around its own minimizer, so the two minimizers cannot be far apart.

Theorem 13.1 (Sensitivity of the regularized minimizer)**.** *Let Θ be nonempty, closed, convex and bounded. Suppose that for every record z the loss $\ell(\cdot\,; z)$ is convex and L-Lipschitz on Θ, and that N is continuous and 1-strongly convex. For each dataset D, let θ_D be the minimizer of $F_D = \mathcal{L}(\cdot\,; D) + \lambda N$ over Θ. Then for all neighbours $D \sim D'$, which differ in one replaced record,*

$$\|\theta_D - \theta_{D'}\|_2 \leq \frac{2L}{\lambda}. \tag{13.5}$$

Proof.

1. F_D is continuous on the closed bounded set Θ, so it has a minimizer, and it is λ-strongly convex, so the minimizer is unique by Lemma 13.2. The same holds for $F_{D'}$.
2. Write $\theta = \theta_D$ and $\theta' = \theta_{D'}$. Lemma 13.2 applied to each function gives

$$F_D(\theta') - F_D(\theta) \geq \frac{\lambda}{2}\|\theta - \theta'\|_2^2, \qquad F_{D'}(\theta) - F_{D'}(\theta') \geq \frac{\lambda}{2}\|\theta - \theta'\|_2^2. \tag{13.6}$$

3. Add the two inequalities. The left side becomes $h(\theta') - h(\theta)$, where $h = F_D - F_{D'}$, and the right side becomes $\lambda\|\theta - \theta'\|_2^2$.
4. If D and D' differ in record j, then $h(\cdot) = \ell(\cdot\,; z_j) - \ell(\cdot\,; z_j')$ is a difference of two L-Lipschitz functions, so $h(\theta') - h(\theta) \leq 2L\|\theta - \theta'\|_2$.
5. So $\lambda\|\theta - \theta'\|_2^2 \leq 2L\|\theta - \theta'\|_2$, and dividing by $\lambda\|\theta - \theta'\|_2$ gives the claim. If $\theta = \theta'$, there is nothing to prove.

□

The bound depends on neither n nor the dimension, and the Gaussian mechanism of the approximate DP chapter does the rest. Take a function f whose ℓ_2 sensitivity, the largest change in Euclidean norm between neighbours, is $\Delta_2 f$. The mechanism adds Gaussian noise $\mathcal{N}(0, \sigma^2 I_d)$ with standard deviation σ in each coordinate. For every $\varepsilon > 0$ and $\delta \in (0, 1)$ it is (ε, δ)-DP, the relaxation that allows an additive slack δ, when

$$\sigma = \frac{\Delta_2 f}{\varepsilon} \sqrt{2 \ln(1/\delta) + \varepsilon}. \tag{13.7}$$

The mechanism applied to the regularized minimizer $f(D) = \theta_D$, with $\Delta_2 f = 2L/\lambda$, gives the first private algorithm.

Theorem 13.2 (Output perturbation)**.** *Under the assumptions of Theorem 13.1, let $\varepsilon > 0$ and $\delta \in (0, 1)$. The algorithm that outputs $\tilde{\theta} = \theta_D + b$, where θ_D is the regularized minimizer, with $b \sim \mathcal{N}(0, \sigma^2 I_d)$ and*

$$\sigma = \frac{2L}{\lambda} \cdot \frac{\sqrt{2 \ln(1/\delta) + \varepsilon}}{\varepsilon}, \tag{13.8}$$

is (ε, δ)-DP. So is its projection onto Θ.

Proof. The map $D \mapsto \theta_D$ has ℓ_2 sensitivity at most $2L/\lambda$ by Theorem 13.1, so the Gaussian mechanism gives the guarantee. The projection is post-processing. □

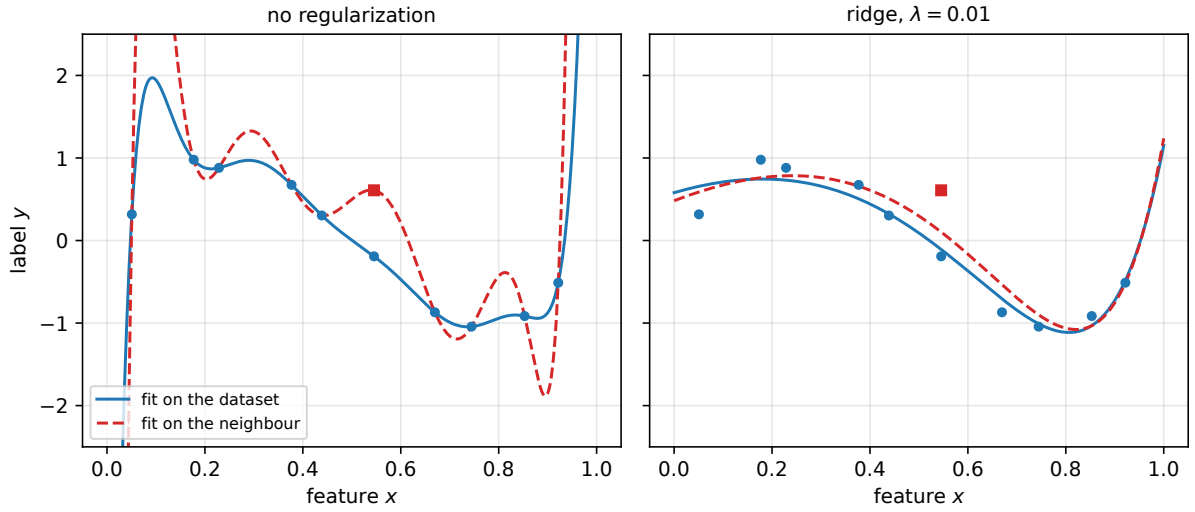


Figure 13.1.: Polynomials of degree 9 fitted by least squares to 10 points, on a dataset and on a neighbour with one label raised (the square). Without regularization, both fits pass through every point, and one moved point changes the whole curve. With the penalty $\frac{\lambda}{2}\|\theta\|_2^2$, the two fits stay close.

Output perturbation works with any exact solver, but privacy may require enough regularization to reduce accuracy.

13.4. Objective perturbation

Objective perturbation adds noise to the objective and releases its minimizer:

$$J_b(\theta) = \mathcal{L}(\theta; D) + \frac{\lambda}{2}\|\theta\|_2^2 + \langle b, \theta\rangle \tag{13.9}$$

for random vector b. The linear term randomly changes the objective's slope in each direction.

The proof covers **generalized linear** losses, including logistic regression and smoothed hinge loss. Such a loss has form $\ell(\theta; x, y) = \phi(\langle x, \theta\rangle, y)$, where for each y, $\phi(\cdot, y)$ is convex and twice continuously differentiable, $|\phi'| \le L$, and $0 \le \phi'' \le \beta$. Assume $\|x\|_2 \le 1$. Then $\nabla\ell = \phi'(\langle x, \theta\rangle, y)x$ has norm at most L, and the Hessian $\nabla^2\ell = \phi''(\langle x, \theta\rangle, y)xx^\top$ has rank at most one and eigenvalues in $[0, \beta]$.

The proof compares output densities on neighbouring datasets. The approximate-DP chapter defines the needed privacy-loss criterion: if p_D is the output density on D, the **privacy loss** at θ is $\ln(p_D(\theta)/p_{D'}(\theta))$. A mechanism is (ε, δ)-DP if, for every ordered neighbouring pair, this loss exceeds ε with probability at most δ when the output is drawn on D.

Theorem 13.3 (Objective perturbation)**.** *Let $\Theta = \mathbb{R}^d$, and let ℓ be a generalized linear loss as above, with $|\phi'| \le L$, $0 \le \phi'' \le \beta$ and features of norm at most 1. Let $\varepsilon \in (0, 4]$ and $\delta \in (0, 1)$, and set*

$$\lambda \ge \frac{\beta}{e^{\varepsilon/2} - 1}, \qquad \sigma = \frac{4L(1 + \sqrt{2\ln(1/\delta)})}{\varepsilon}. \tag{13.10}$$

Then the algorithm that draws $b \sim \mathcal{N}(0, \sigma^2 I_d)$ and outputs the exact minimizer $\tilde\theta$ of J_b is (ε, δ)-DP.

Proof.

1. *The output as a function of the noise.* For each b, the function J_b is continuous and λ-strongly convex on $\mathbb{R}^d$, so it has a unique minimizer, where its gradient vanishes. So $\tilde\theta = \theta$ exactly when $b = g_D(\theta)$, where $g_D(\theta) = -\nabla\mathcal{L}(\theta; D) - \lambda\theta$. Every θ is the minimizer for the noise $g_D(\theta)$, and each noise has one minimizer, so g_D is a bijection of $\mathbb{R}^d$ and $\tilde\theta = g_D^{-1}(b)$.
2. *The density of the output.* The map g_D is continuously differentiable with Jacobian $-(H_D(\theta) + \lambda I)$, where $H_D(\theta) = \sum_i \nabla^2\ell(\theta; z_i)$ is positive semidefinite. By the change of variables formula, $\tilde\theta$ has density

$$p_D(\theta) = \varphi_\sigma(g_D(\theta))\, \det\left(H_D(\theta) + \lambda I\right), \tag{13.11}$$

   where φ_σ is the density of $\mathcal{N}(0, \sigma^2 I_d)$.
3. *The determinants.* Let D and D' differ in the last record, $z = (x, y)$ in D and $z' = (x', y')$ in D'. Fix θ and write $A = \sum_{i<n} \nabla^2\ell(\theta; z_i) + \lambda I$, which satisfies $A \succeq \lambda I$. Then $H_D + \lambda I = A + a\, xx^\top$ with $a = \phi''(\langle x, \theta\rangle, y) \in [0, \beta]$. The matrix determinant lemma gives

$$\det(A + a\, xx^\top) = \det(A)\,(1 + a\, x^\top A^{-1} x) \in \left[\det A,\ (1 + \beta/\lambda)\det A\right], \tag{13.12}$$

   because $0 \le x^\top A^{-1} x \le \|x\|_2^2/\lambda \le 1/\lambda$. The same holds for D'. So the ratio of the two determinants is at most $1 + \beta/\lambda \le e^{\varepsilon/2}$, by the choice of λ.

4. *The Gaussian factors.* The two noise vectors that lead to θ differ by $u = g_{D'}(\theta) - g_D(\theta) = \nabla\ell(\theta; z) - \nabla\ell(\theta; z')$. This vector has norm at most $2L$, and it lies in the subspace V spanned by x and x', which does not depend on θ. With $b = g_D(\theta)$ and P_V the orthogonal projection onto V,
$$\ln \frac{\varphi_\sigma(b)}{\varphi_\sigma(b+u)} = \frac{2\langle b, u\rangle + \|u\|_2^2}{2\sigma^2} \le \frac{2L\,\|P_V b\|_2}{\sigma^2} + \frac{2L^2}{\sigma^2}, \tag{13.13}$$
by the Cauchy–Schwarz inequality applied to $\langle b, u\rangle = \langle P_V b, u\rangle$.
5. *The privacy loss.* When the output is drawn from $M(D)$, the vector $g_D(\tilde\theta)$ is the noise b itself. Steps 2 to 4 bound the privacy loss at the output by
$$\ln \frac{p_D(\tilde\theta)}{p_{D'}(\tilde\theta)} \le \frac{\varepsilon}{2} + \frac{2L\,\|P_V b\|_2}{\sigma^2} + \frac{2L^2}{\sigma^2}. \tag{13.14}$$
6. *The tail.* The subspace V has dimension $d_V \le 2$, so $\|P_V b\|_2^2/\sigma^2$ is chi-squared with d_V degrees of freedom. Its tail is at most that of two degrees of freedom, which is $e^{-r/2}$ beyond r. Write $s = \sqrt{2\ln(1/\delta)}$ for the cut-off. Then
$$\Pr\left[\|P_V b\|_2 > \sigma s\right] \le e^{-s^2/2} = \delta. \tag{13.15}$$
7. Outside that event, substituting $\sigma = 4L(1+s)/\varepsilon$ into step 5 gives
$$\ln \frac{p_D(\tilde\theta)}{p_{D'}(\tilde\theta)} \le \frac{\varepsilon}{2} + \frac{\varepsilon}{2}\left(\frac{s}{1+s} + \frac{\varepsilon}{4(1+s)^2}\right) \le \frac{\varepsilon}{2} + \frac{\varepsilon}{2}\cdot\frac{s+1}{1+s} = \varepsilon, \tag{13.16}$$
where the second step uses $\varepsilon \le 4$ and $1/(1+s)^2 \le 1/(1+s)$.
8. The privacy loss criterion turns this bound into (ε, δ)-DP.

□

The theorem requires the exact minimizer, which iterative solvers such as gradient descent only approximate. Floating-point arithmetic adds another concern: a naive floating-point Laplace mechanism can leak data through low-order output bits (Mironov). Later objective-perturbation methods allow approximate minimizers.

13.5. Gradient perturbation

The third method adds noise to each gradient-descent step. Define $\Pi_\Theta(u) = \arg\min_{\theta\in\Theta} \|\theta - u\|_2$, the **projection** onto Θ. Since Θ is closed and convex, this closest point exists and is unique.

Allowing random step directions G_t lets one analysis cover full gradients, sampled-record gradients, and noisy gradients.

Definition 13.4 (Projected stochastic gradient descent)**.** Fix a start $\theta_0 \in \Theta$, a number of steps T and a learning rate $\eta > 0$, also called the step size. For $t = 1, \dots, T$, draw a random vector G_t and set
$$\theta_t = \Pi_\Theta(\theta_{t-1} - \eta\, G_t). \tag{13.17}$$

Output the average $\bar{\theta} = \frac{1}{T}\sum_{t=0}^{T-1}\theta_t$. With $G_t = \nabla\mathcal{L}(\theta_{t-1}; D)$ this is projected **gradient descent**. With $G_t = n\,\nabla\ell(\theta_{t-1}; z_{I_t})$, for an index I_t drawn uniformly from $\{1, \dots, n\}$, it is projected stochastic gradient descent (SGD).

A step moves against the gradient, the direction of steepest increase. The projection brings a step that leaves Θ back to the nearest allowed point. The convergence proof compares each iterate with the minimizer, and for that it uses the fact that projection never moves a point farther from any point of Θ.

Lemma 13.3. *For every $u \in \mathbb{R}^d$ and every $\theta \in \Theta$, $\|\Pi_\Theta(u) - \theta\|_2 \le \|u - \theta\|_2$.*

Proof.

1. Let $v = \Pi_\Theta(u)$. For $t \in (0, 1]$ the point $v + t(\theta - v)$ is in Θ, so it is no closer to u than v: $\|u - v - t(\theta - v)\|^2 \ge \|u - v\|^2$. Expanding, dividing by t and letting $t \to 0$ gives $\langle u - v, \theta - v\rangle \le 0$.
2. Then $\|u - \theta\|^2 = \|u - v\|^2 + \|v - \theta\|^2 + 2\langle u - v, v - \theta\rangle \ge \|v - \theta\|^2$, because the last term is nonnegative by step 1.

□

The convergence theorem asks only that each G_t is right on average and not too large.

Theorem 13.4 (Convergence of projected SGD)**.** *Let F be convex on $\mathbb{R}^d$, and let θ^* minimize F over Θ. Suppose that, given $\theta_0, \dots, \theta_{t-1}$, the conditional mean $\mathbb{E}[G_t \mid \theta_0, \dots, \theta_{t-1}]$ is a subgradient of F at θ_{t-1}, and that $\mathbb{E}\|G_t\|_2^2 \le G^2$ for every t. With learning rate $\eta = \|\Theta\|_2/(G\sqrt{T})$,*

$$\mathbb{E}[F(\bar{\theta})] - F(\theta^*) \le \frac{\|\Theta\|_2\, G}{\sqrt{T}}. \tag{13.18}$$

Proof.

1. By Lemma 13.3 with $\theta = \theta^*$,
$$\|\theta_t - \theta^*\|^2 \le \|\theta_{t-1} - \eta G_t - \theta^*\|^2 = \|\theta_{t-1} - \theta^*\|^2 - 2\eta\langle G_t, \theta_{t-1} - \theta^*\rangle + \eta^2\|G_t\|^2. \tag{13.19}$$
2. Condition on the past and let g be the conditional mean of G_t, a subgradient at θ_{t-1}. The definition of a subgradient gives $\langle g, \theta_{t-1} - \theta^*\rangle \ge F(\theta_{t-1}) - F(\theta^*)$. Taking expectations in step 1,
$$\mathbb{E}[F(\theta_{t-1})] - F(\theta^*) \le \frac{\mathbb{E}\|\theta_{t-1} - \theta^*\|^2 - \mathbb{E}\|\theta_t - \theta^*\|^2}{2\eta} + \frac{\eta G^2}{2}. \tag{13.20}$$
3. Summing over $t = 1, \dots, T$, the distances telescope. The first is at most $\|\Theta\|_2^2$ and the last is nonnegative, so
$$\sum_{t=0}^{T-1}\left(\mathbb{E}[F(\theta_t)] - F(\theta^*)\right) \le \frac{\|\Theta\|_2^2}{2\eta} + \frac{T\eta G^2}{2}. \tag{13.21}$$
4. The average $\bar{\theta}$ lies in Θ, and convexity gives $F(\bar{\theta}) \le \frac{1}{T}\sum_t F(\theta_t)$. Step 3 divided by T, with the value of η, makes both terms $\|\Theta\|_2 G/(2\sqrt{T})$.

□

For the empirical risk, the theorem gives the number of steps that each method needs. The bound G on the step comes from the bound L on every gradient.

Corollary 13.1. *Suppose that Θ has finite diameter $\|\Theta\|_2$ and that every loss $\ell(\cdot\,; z)$ is convex and L-Lipschitz on $\mathbb{R}^d$. Then gradient descent and SGD both reach excess empirical risk at most α after $T = \lceil n^2 L^2 \|\Theta\|_2^2 / \alpha^2 \rceil$ steps. Gradient descent evaluates nT gradients of ℓ, and SGD evaluates T.*

Proof.

1. For gradient descent, $G_t = \sum_i \nabla \ell(\theta_{t-1}; z_i)$ is a subgradient of $\mathcal{L}$, and by Lemma 13.1 its norm is at most nL.
2. For SGD, the conditional mean of $G_t = n\nabla \ell(\theta_{t-1}; z_{I_t})$ is $\frac{1}{n}\sum_i n\nabla \ell(\theta_{t-1}; z_i)$, the same subgradient. Its second moment is $\frac{1}{n}\sum_i n^2 \|\nabla \ell(\theta_{t-1}; z_i)\|^2 \le n^2 L^2$.
3. In both cases $G = nL$, and Theorem 13.4 gives excess risk at most $\|\Theta\|_2 nL/\sqrt{T} \le \alpha$.

□

For constant α, gradient descent uses about n^3 gradient evaluations, compared with about n^2 for SGD. SGD is easier to privatize because each step uses one record. Private SGD adds calibrated Gaussian noise to each sampled gradient and adjusts the learning rate for that noise.

Definition 13.5 (Private SGD)**.** Let $\varepsilon \in (0, 1]$ and $\delta \in (0, 1)$. Run projected SGD for $T = n^2$ steps with

$$G_t = n\,\nabla \ell(\theta_{t-1}; z_{I_t}) + b_t, \qquad b_t \sim \mathcal{N}(0, \sigma^2 I_d), \qquad \sigma = \frac{16\, nL \sqrt{\ln(3.3n/\delta)\, \ln(2/\delta)}}{\varepsilon}, \tag{13.22}$$

with fresh, independent indices I_t and noise b_t, and learning rate $\eta = \|\Theta\|_2 / (G\sqrt{T})$ where $G^2 = n^2 L^2 + d\sigma^2$.

The accuracy analysis uses the noise's zero mean and its contribution to the second moment.

Theorem 13.5. *With $\ell(\cdot\,; z)$ convex and L-Lipschitz on $\mathbb{R}^d$, private SGD satisfies*

$$\mathbb{E}[\mathcal{L}(\bar{\theta}; D)] - \mathcal{L}(\theta^*; D) \le \|\Theta\|_2 \sqrt{L^2 + \frac{d\sigma^2}{n^2}} \le \|\Theta\|_2\, L \left(1 + \frac{16\sqrt{d\, \ln(3.3n/\delta) \ln(2/\delta)}}{\varepsilon}\right). \tag{13.23}$$

Proof.

1. Given the past, b_t has mean zero and is independent of I_t, so the conditional mean of G_t is the subgradient of step 2 of Corollary 13.1.

2. Expanding the square,

$$\mathbb{E}\|G_t\|^2 = n^2\,\mathbb{E}\|\nabla \ell(\theta_{t-1}; z_{I_t})\|^2 + 2n\,\mathbb{E}\langle \nabla \ell(\theta_{t-1}; z_{I_t}), b_t\rangle + \mathbb{E}\|b_t\|^2 \le n^2 L^2 + 0 + d\sigma^2. \tag{13.24}$$

   The first term is bounded by Lemma 13.1. The middle term vanishes because b_t is independent of the gradient and has mean zero. The last is a sum of d squared Gaussians of variance σ^2.
3. Theorem 13.4 with $T = n^2$ gives $\|\Theta\|_2 G/n$, which is the first bound. The second uses $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$ and the value of σ.

□

Up to logarithmic factors, excess risk is $O(\|\Theta\|_2 L\sqrt{d\ln(1/\delta)}/\varepsilon)$, so average excess risk per record decreases as $1/n$. The $\sqrt{d}$ privacy cost is unavoidable in some settings (Bassily, Smith and Thakurta), but the constant 16 reflects loose bounds. For $n = 10{,}000$, $\varepsilon = 1$, and $\delta = 10^{-6}$, the noise is $\sigma = 299.9\,nL$. At $d = 100$, the bound per record is $\|\Theta\|_2 L(1 + 2999)/10{,}000 = 0.30\,\|\Theta\|_2 L$. The next chapter gives a tighter analysis.

The privacy proof uses three tools. The first is the Gaussian mechanism, restated above. The second is **amplification by subsampling**, which says that a mechanism run on one random record is more private than the same mechanism run on a fixed record.

Lemma 13.4 (Amplification by sampling one record)**.** *Let M_0 be a mechanism whose input is a single record and which satisfies $\Pr[M_0(z) \in S] \le e^{\varepsilon_0}\Pr[M_0(z') \in S] + \delta_0$ for all records z, z' and all events S. Let $M(D) = M_0(z_I)$ with I uniform on $\{1, \dots, n\}$. Then M is $(\varepsilon_1, \delta_0/n)$-DP with*

$$\varepsilon_1 = \ln\left(1 + \frac{e^{\varepsilon_0} - 1}{n}\right) \le \frac{e^{\varepsilon_0} - 1}{n}. \tag{13.25}$$

Proof.

1. Let D and D' differ in record j, fix an event S, and write $p_i = \Pr[M_0(z_i) \in S]$ and $p'_j = \Pr[M_0(z'_j) \in S]$. Let r be the average of p_i over $i \ne j$, and write $q = 1/n$ for the sampling rate. Then

$$P = \Pr[M(D) \in S] = (1-q)r + qp_j, \qquad P' = \Pr[M(D') \in S] = (1-q)r + qp'_j. \tag{13.26}$$

2. The hypothesis, applied to z_j against z'_j and against each z_i and then averaged, gives two bounds:

$$p_j - p'_j \le (e^{\varepsilon_0} - 1)p'_j + \delta_0, \qquad p_j - p'_j \le e^{\varepsilon_0} r - p'_j + \delta_0. \tag{13.27}$$

3. Let $c = e^{\varepsilon_0} - 1$ and take the convex combination of the two bounds with weights w and $1 - w$, where $1 - w = c(1-q)/e^{\varepsilon_0} \in [0, 1)$. The coefficient of r becomes $c(1-q)$, and that of p'_j becomes $we^{\varepsilon_0} - 1 = cq$. So

$$p_j - p'_j \le c((1-q)r + qp'_j) + \delta_0 = cP' + \delta_0. \tag{13.28}$$

4. Then $P = P' + q(p_j - p'_j) \le (1 + qc)P' + q\delta_0$, which is the claim. The inequality $\ln(1+x) \le x$ gives the simpler bound on ε_1.

□

For $\varepsilon_0 \le 1$, convexity of the exponential gives $e^{\varepsilon_0} - 1 \le (e-1)\varepsilon_0 \le 2\varepsilon_0$. So a sample of one record of n cuts the privacy parameter by a factor of about $n/2$.

The third tool is advanced composition, from the advanced composition chapter. Take k mechanisms in sequence, each $(\varepsilon_0, \delta_0)$-DP and each chosen adaptively from the outputs of the previous ones. Their sequence is $(\varepsilon, k\delta_0 + \delta')$-DP for every $\delta' \in (0,1)$, where

$$\varepsilon = \varepsilon_0 \sqrt{2k \ln(1/\delta')} + k\varepsilon_0 \frac{e^{\varepsilon_0} - 1}{e^{\varepsilon_0} + 1} \le \varepsilon_0 \sqrt{2k \ln(1/\delta')} + \frac{k\varepsilon_0^2}{2}, \tag{13.29}$$

and the inequality uses $(e^x - 1)/(e^x + 1) = \tanh(x/2) \le x/2$ for $x \ge 0$.

The proof uses the three tools in turn. The Gaussian noise protects the sampled record in one step, the random choice of that record makes the step more private, and advanced composition adds up the steps.

Theorem 13.6 (Privacy of private SGD)**.** *For $\varepsilon \in (0,1]$ and $\delta \in (0,1)$, private SGD is (ε, δ)-DP, even if it releases every iterate $\theta_1, \dots, \theta_T$.*

Proof.

1. *One step, fixed index.* Fix the previous iterate and the index i. The step releases a post-processing of $n\nabla\ell(\theta_{t-1}; z_i) + b_t$. As a function of the single record z_i, the vector $n\nabla\ell$ has ℓ_2 sensitivity at most $2nL$, by Lemma 13.1. Set $\varepsilon_0 = \varepsilon/(4\sqrt{2\ln(2/\delta)})$, which is below 1, and $\delta_0 = \delta/(2n)$. The Gaussian mechanism is $(\varepsilon_0, \delta_0)$-DP with respect to z_i at the noise level

$$\sigma_0 = \frac{2nL}{\varepsilon_0} \sqrt{2\ln(1/\delta_0) + \varepsilon_0} \le \frac{2nL}{\varepsilon_0} \sqrt{2 \ln \frac{2\sqrt{e}\, n}{\delta}} \le \frac{2nL}{\varepsilon_0} \sqrt{2 \ln \frac{3.3n}{\delta}} = \sigma, \tag{13.30}$$

   where the first inequality uses $\varepsilon_0 \le 1 = 2\ln\sqrt{e}$, and the last equality is the σ of Definition 13.5. Noise of variance $\sigma^2 \ge \sigma_0^2$ is noise of variance σ_0^2 plus an independent Gaussian of variance $\sigma^2 - \sigma_0^2$, a post-processing, so the step is $(\varepsilon_0, \delta_0)$-DP with respect to z_i.
2. *One step, random index.* By Lemma 13.4, each step is $(\varepsilon_1, \delta_0/n)$-DP with respect to the dataset, where $\varepsilon_1 \le (e^{\varepsilon_0} - 1)/n \le 2\varepsilon_0/n$.
3. *All steps.* Each step is chosen adaptively from the previous iterates. Advanced composition over $T = n^2$ steps with $\delta' = \delta/2$ gives total failure probability $n^2 \cdot \delta/(2n^2) + \delta/2 = \delta$, and privacy parameter

$$\varepsilon_1 \sqrt{2n^2 \ln(2/\delta)} + \frac{n^2 \varepsilon_1^2}{2} \le 2\varepsilon_0 \sqrt{2\ln(2/\delta)} + 2\varepsilon_0^2 = \frac{\varepsilon}{2} + 2\varepsilon_0^2. \tag{13.31}$$

4. Finally, the second term is small because $\varepsilon \le 1$ and $\ln(2/\delta) > \ln 2 > 1/8$:

$$2\varepsilon_0^2 = \frac{\varepsilon^2}{16\ln(2/\delta)} \le \frac{\varepsilon}{2}. \tag{13.32}$$

   The average $\bar{\theta}$ is a post-processing of the iterates.

□

The privacy cost per step is about ε/n, and composition over n^2 steps multiplies it by only $\sqrt{n^2} = n$. Each piece of the argument is generic, so the same recipe extends to models with no convergence theorem.

13.6. Summary

Table 13.1.: The chapter's results.

Result	Statement
Proposition 13.1	the exact median is not (ε, δ)-DP for any $\delta < 1$
Theorem 13.1	the minimizer of loss plus λ·(1-strongly convex) has sensitivity $2L/\lambda$
Theorem 13.2	adding Gaussian noise to that minimizer is (ε, δ)-DP
Theorem 13.3	the exact minimizer of a Gaussian-perturbed objective is (ε, δ)-DP
Theorem 13.4	projected SGD has excess risk $\|\Theta\|_2 G/\sqrt{T}$
Lemma 13.4	sampling one record of n turns ε_0 into $\ln(1 + (e^{\varepsilon_0} - 1)/n)$
Theorem 13.6	private SGD with n^2 steps is (ε, δ)-DP
Theorem 13.5	its excess risk is $O(\|\Theta\|_2 L\sqrt{d \ln(1/\delta)}/\varepsilon)$ up to logarithmic factors

13.7. References

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14. Private deep learning

Private stochastic gradient descent (SGD) in the previous chapter assumes a convex, Lipschitz loss and has loose privacy constants. Neural-network losses generally satisfy neither assumption. This chapter adapts SGD to differential privacy (DP) by clipping each gradient, gives a tighter privacy accountant based on privacy-loss moments, and privatizes arbitrary learners by voting among models trained on disjoint data.

14.1. Neural networks

As in the previous chapter, each record contains a feature vector and label, and training minimizes total loss over parameter vector θ:

$$\mathcal{L}(\theta; D) = \sum_{i=1}^{n} \ell(\theta; x_i, y_i). \tag{14.1}$$

Here a feature vector x lies in $\mathbb{R}^{m_0}$, and a label is one of K classes, $y \in \{1, \dots, K\}$. A neural network maps a feature vector to a score for each class. Training adjusts its parameters θ. A multilayer perceptron stacks linear maps and fixed nonlinear functions.

Definition 14.1 (Multilayer perceptron)**.** A network with J layers sets $v^{(0)} = x$ and, for $j = 1, \dots, J$,

$$v^{(j)} = \phi\big(W^{(j)} v^{(j-1)} + c^{(j)}\big), \tag{14.2}$$

where $W^{(j)}$ is an $m_j \times m_{j-1}$ matrix of weights, $c^{(j)} \in \mathbb{R}^{m_j}$ is a vector of offsets, and the **activation function** $\phi \colon \mathbb{R} \to \mathbb{R}$ is applied to each coordinate. The last layer has no activation and outputs $o \in \mathbb{R}^K$, one score per class. The parameter vector θ collects all weights and offsets.

Without activations, the network is a single linear map. Common activations include the rectified linear unit $\phi(u) = \max(0, u)$ and bounded $\tanh(u) = (e^u - e^{-u})/(e^u + e^{-u})$.

The **softmax** maps scores to class probabilities, $\hat{y}_k = e^{o_k} / \sum_{j=1}^{K} e^{o_j}$. The predicted class maximizes $\hat{y}_k$. The **cross-entropy** loss is the negative log probability assigned to the true class:

$$\ell(\theta; x, y) = -\ln \hat{y}_y = -\sum_{k=1}^{K} \mathbf{1}[k = y] \, \ln \hat{y}_k, \tag{14.3}$$

where $\hat{y}$ depends on θ through every layer. The previous chapter's analysis does not apply: this loss is not convex in θ, and its gradients are not bounded. In a two-layer network, the second-layer

weights multiply the first-layer gradient and can be arbitrarily large. Thus one gradient can have unbounded sensitivity.

14.2. Differentially private SGD

Two changes make noisy SGD practical. First, **clipping** limits each gradient's norm and directly bounds sensitivity. Second, each step uses a random batch, called a **lot**, formed by Poisson sampling.

Definition 14.2 (Clipping)**.** For a clipping norm C, a positive number, the clipping of a vector g is

$$\mathrm{clip}_C(g) = g \cdot \min\left(1, \frac{C}{\|g\|_2}\right). \tag{14.4}$$

Each clipped vector has norm at most C.

DP-SGD clips each gradient in the lot, adds Gaussian noise to their sum, and divides by the expected lot size qn. The **noise multiplier** σ is the noise standard deviation divided by the clipping norm C; unlike the previous chapter's σ, it is not itself a standard deviation.

Definition 14.3 (DP-SGD)**.** Let the parameter vector θ have d coordinates. Fix a sampling rate q in $(0, 1]$, a noise multiplier $\sigma > 0$, a clipping norm C, a number of steps T, learning rates η_t and a start θ_0. For $t = 1, \dots, T$:

1. Form the lot B_t by including each index $i \in \{1, \dots, n\}$ independently with probability q.
2. Compute the noisy average of clipped gradients,

$$\tilde{g}_t = \frac{1}{qn}\left(\sum_{i \in B_t} \mathrm{clip}_C(\nabla \ell(\theta_{t-1}; z_i)) + b_t\right), \qquad b_t \sim \mathcal{N}(0, \sigma^2 C^2 I_d). \tag{14.5}$$

3. Step: $\theta_t = \theta_{t-1} - \eta_t \, \tilde{g}_t$.

Output θ_T.

Unlike the previous chapter's private SGD, each step uses a lot with expected size qn. Clipping bounds each gradient by construction, and the noise is calibrated to C, the maximum contribution of one record.

Lemma 14.1. *Adding or removing one record changes the clipped sum $\sum_{i \in B} \mathrm{clip}_C(\nabla \ell(\theta; z_i))$ by a vector of norm at most C. Replacing one record changes it by at most $2C$.*

Proof. Adding or removing a record adds or removes one clipped vector, of norm at most C. A replacement does both, and the triangle inequality gives $2C$. □

Clipping biases the average gradient. Consequently, the earlier convergence theorem no longer applies even to convex losses, although clipped SGD works well in practice.

Privacy amplification depends on the sampling scheme. **Poisson sampling** includes each record independently with probability q, producing a random lot size. Even when $q = 1/n$, a lot may be empty or contain multiple records, unlike a single-record sample. Fixed-size sampling without replacement requires a different analysis. The results below use Poisson sampling and add/remove neighbours, which differ by one record; Lemma 14.1 gives sensitivity C for this relation.

Implementations often shuffle once and divide the data into fixed-size batches. These batches are dependent, so the Poisson-sampling analysis does not apply.

14.3. Privacy accounting

Without subsampling, each DP-SGD step is a Gaussian mechanism with noise multiplier σ, so the previous chapter's analysis applies. If a step is $(\varepsilon_0, \delta_0)$-DP, Poisson subsampling gives approximately $(2q\varepsilon_0, q\delta_0)$-DP. Advanced composition over T steps then gives

$$\Big(O(q\varepsilon_0\sqrt{T\ln(1/\delta)}),\ Tq\delta_0 + \delta\Big)\text{-DP}. \tag{14.6}$$

Abadi and coauthors derived a tighter bound for this algorithm, requiring less noise for the same guarantee.

Theorem 14.1 (Moments accountant, asymptotic form)**.** *There are constants c_1 and c_2 such that, for every sampling rate q and number of steps T, every $\varepsilon < c_1 q^2 T$ and every $\delta > 0$, DP-SGD is (ε, δ)-DP under add/remove neighbours if*

$$\sigma \geq c_2\,\frac{q\sqrt{T\ln(1/\delta)}}{\varepsilon}. \tag{14.7}$$

Abadi and coauthors (2016, Theorem 1) prove this result; the rest of the section derives its ingredients. Compared with composition, the bound reduces σ by a factor of order $\sqrt{\ln(1/\delta_0)}$ and avoids the accumulated failure probability $Tq\delta_0$, as Corollary 14.1 shows. These factors can determine whether a privacy guarantee is useful.

The accountant starts from the privacy loss. For a mechanism M whose output has density p_D on dataset D, the **privacy loss** at an output y is $\ln\left(p_D(y)/p_{D'}(y)\right)$. It is a random variable when y is drawn from $M(D)$. By the privacy loss criterion of the approximate DP chapter, (ε, δ)-DP follows if, for every ordered pair of neighbours, the loss exceeds ε with probability at most δ. But tail probabilities do not compose simply, and moments do, by addition. So the accountant takes, for an order $\lambda > 0$, the logarithm of the average λ-th power of the ratio $p_D(y)/p_{D'}(y)$, for the worst pair of neighbours.

Definition 14.4 (Log-moments of the privacy loss)**.** Let p_D be the density of the output $M(D)$. For $\lambda > 0$, the λ-th log-moment of M is

$$\kappa_M(\lambda) = \sup_{D \sim D'} \ln \mathbb{E}_{y \sim M(D)} \left[\left(\frac{p_D(y)}{p_{D'}(y)} \right)^{\lambda} \right]. \tag{14.8}$$

Up to a factor, it is the Rényi divergence of order $\alpha = \lambda + 1$ between the two output distributions. A bound on it for every pair of neighbours is Rényi differential privacy (RDP). By Markov's inequality, a bound on one log-moment already gives approximate DP.

Theorem 14.2 (From moments to approximate DP)**.** *For every $\lambda > 0$ and $\varepsilon \geq 0$, the mechanism M is (ε, δ)-DP with*

$$\delta = \exp\left(\kappa_M(\lambda) - \lambda\varepsilon\right). \tag{14.9}$$

Proof.

1. Fix neighbours $D \sim D'$ and write $L = \ln(p_D(Y)/p_{D'}(Y))$ for the privacy loss random variable, with $Y \sim M(D)$. The map $u \mapsto e^{\lambda u}$ is increasing, so $\Pr[L > \varepsilon] \leq \Pr[e^{\lambda L} \geq e^{\lambda\varepsilon}]$.
2. Markov's inequality bounds the right side by $\mathbb{E}[e^{\lambda L}]\, e^{-\lambda\varepsilon}$, and $\mathbb{E}[e^{\lambda L}] \leq e^{\kappa_M(\lambda)}$ by definition.
3. So the privacy loss exceeds ε with probability at most δ, and the privacy loss criterion gives (ε, δ)-DP.

□

The conversion is useful because log-moments are easy to track over many steps. The next lemma shows that they add up when mechanisms run in sequence, even when the outputs of the previous mechanisms choose the next one.

Lemma 14.2 (Composition of moments)**.** *Let M_1 have output y_1, and let $M_2(\cdot\,; y_1)$ be a mechanism chosen as a function of y_1 with $\kappa_{M_2(\cdot; y_1)}(\lambda) \leq \kappa_2$ for every y_1. The mechanism M that outputs (y_1, y_2) satisfies $\kappa_M(\lambda) \leq \kappa_{M_1}(\lambda) + \kappa_2$. By induction, T adaptive steps have log-moments at most the sum of theirs.*

Proof.

1. The density of (y_1, y_2) on D is $p_D(y_1)\, p_D(y_2 \mid y_1)$, so the ratio for neighbours factors as $r_1(y_1)\, r_2(y_2 \mid y_1)$. Here r_1 is the density ratio of y_1, and r_2 is the density ratio of y_2 given y_1.
2. The expectation over y_2 first, given y_1, gives

$$\mathbb{E}[r_1^{\lambda} r_2^{\lambda}] = \mathbb{E}_{y_1}\Big[r_1(y_1)^{\lambda}\, \mathbb{E}_{y_2}[r_2(y_2 \mid y_1)^{\lambda} \,|\, y_1] \Big] \leq e^{\kappa_2}\, \mathbb{E}_{y_1}[r_1(y_1)^{\lambda}] \leq e^{\kappa_2 + \kappa_{M_1}(\lambda)}. \tag{14.10}$$

3. The logarithm and the supremum over neighbours give the claim.

□

For the Gaussian mechanism the log-moments have a closed form.

Lemma 14.3. *Let $M(D) = f(D) + b$ with $b \sim \mathcal{N}(0, \sigma^2 I_d)$, where f has ℓ_2 sensitivity $\Delta_2 f$. Then for every $\lambda > 0$,*

$$\kappa_M(\lambda) = \frac{\lambda(\lambda+1)(\Delta_2 f)^2}{2\sigma^2}. \tag{14.11}$$

Proof.

1. Fix neighbours, write $\mu = f(D)$ and $\mu' = f(D')$, and let $r = \|\mu - \mu'\|_2 \le \Delta_2 f$. For $y = \mu + \sigma\xi$ with ξ standard normal, the privacy loss is
$$\frac{\|y-\mu'\|^2 - \|y-\mu\|^2}{2\sigma^2} = \frac{r^2}{2\sigma^2} + \frac{\langle \xi, \mu - \mu' \rangle}{\sigma}. \tag{14.12}$$
2. So the privacy loss is normal with mean $m = r^2/(2\sigma^2)$ and variance $r^2/\sigma^2 = 2m$.
3. A normal variable Z with mean m and variance s^2 has $\mathbb{E}[e^{\lambda Z}] = e^{\lambda m + \lambda^2 s^2/2}$. Here the exponent is $\lambda m + \lambda^2 m = \lambda(\lambda+1) r^2/(2\sigma^2)$.
4. This increases with r, and the supremum is reached at $r = \Delta_2 f$.

□

The three results give a guarantee for many Gaussian steps without sampling. The log-moments of the steps add up, and the conversion turns their sum into (ε, δ)-DP at the best order λ.

Corollary 14.1. *Let $\rho = (\Delta_2 f)^2/(2\sigma^2)$. Any adaptive sequence of T Gaussian mechanisms, each with sensitivity $\Delta_2 f$ and noise $\mathcal{N}(0, \sigma^2 I_d)$, is (ε, δ)-DP for every $\delta \in (0,1)$, with*

$$\varepsilon = T\rho + 2\sqrt{T\rho \ln(1/\delta)}. \tag{14.13}$$

Proof.

1. By Lemma 14.3 and Lemma 14.2, the sequence has $\kappa(\lambda) \le \lambda(\lambda+1)T\rho$.
2. By Theorem 14.2, it is (ε, δ)-DP whenever $\lambda(\lambda+1)T\rho - \lambda\varepsilon \le \ln \delta$, that is, whenever $\varepsilon \ge (\lambda+1)T\rho + \ln(1/\delta)/\lambda$.
3. The right side is smallest at $\lambda = \sqrt{\ln(1/\delta)/(T\rho)}$, where it equals $T\rho + 2\sqrt{T\rho \ln(1/\delta)}$.

□

For comparison, take the same T Gaussian steps with $\Delta_2 f = 1$, so that $T\rho = T/(2\sigma^2)$. Advanced composition first needs a guarantee per step. The Gaussian mechanism of the approximate DP chapter gives ε_0 with $\sigma = \sqrt{2\ln(1/\delta_0) + \varepsilon_0}/\varepsilon_0$, so ε_0 is about $\sqrt{2\ln(1/\delta_0)}/\sigma$. Then it gives $\varepsilon \approx \varepsilon_0 \sqrt{2T \ln(1/\delta)}$, which is $2\sqrt{T \ln(1/\delta_0) \ln(1/\delta)}/\sigma$. The corollary gives $\sqrt{2T \ln(1/\delta)}/\sigma$ plus a lower-order term. That is smaller by a factor of about $\sqrt{2\ln(1/\delta_0)}$, with no failure probability $T\delta_0$ to pay.

DP-SGD adds Poisson sampling to each step, and its log-moments come from a one-dimensional pair of distributions. By Lemma 14.1, after a division by C, a step compares the noise distribution

$\mathcal{N}(0,\sigma^2)$ with the mixture $(1-q)\,\mathcal{N}(0,\sigma^2)+q\,\mathcal{N}(1,\sigma^2)$. In the mixture, the extra record is sampled with probability q.

Theorem 14.3 (Log-moments of the sampled Gaussian)**.** *For every integer $\lambda \geq 1$, one step of DP-SGD with sampling rate q and noise multiplier σ has, under add/remove neighbours,*

$$\kappa(\lambda) \leq \ln A_{\lambda+1}, \qquad A_a = \sum_{k=0}^{a} \binom{a}{k} (1-q)^{a-k} q^k \exp\left(\frac{k^2-k}{2\sigma^2}\right). \tag{14.14}$$

Mironov, Talwar and Zhang (2019, Section 3.3) prove the result. The binomial theorem expands the ath power of the mixture's density ratio and yields A_a. Each term is a Gaussian integral, $\mathbb{E}[e^{k\xi/\sigma^2}] = e^{k^2/(2\sigma^2)}$ for $\xi \sim \mathcal{N}(0,\sigma^2)$. The main step shows that this ordered pair is worst among all neighbouring pairs and both directions. For $\sigma \geq 1$, small q, and moderate λ, the bound is $q^2\lambda(\lambda+1)/((1-q)\sigma^2) + O(q^3\lambda^3/\sigma^3)$, which implies Theorem 14.1.

These pieces together give the **moments accountant**. It computes the log-moment of one step for every λ, multiplies by T by Lemma 14.2, and converts with Theorem 14.2 at the best λ:

$$\varepsilon(\delta) = \min_{\lambda \geq 1} \frac{T \ln A_{\lambda+1} + \ln(1/\delta)}{\lambda}. \tag{14.15}$$

The accountant depends only on q, σ, T, and δ, not on the data. You can therefore set the noise or number of steps to meet a privacy target before training. Figure 14.1 compares it with composition for one setting.

Example 14.1. With $q = 0.01$, $\sigma = 4$ and $\delta = 10^{-5}$, Figure 14.1 shows the two accountings of the same algorithm. After 100 epochs, or $T = 10{,}000$ steps, the composition route certifies $\varepsilon = 22.08$ and the moments accountant certifies $\varepsilon = 1.26$. After 400 epochs the values are 61.41 and 2.58. The first is no guarantee at all, and the second is a reasonable one.

These analyses use add/remove neighbours, whereas this book generally uses replace-one neighbours. A replacement is a removal and an addition, so conversion between the relations costs about twice ε.

14.4. DP-SGD in practice

DP-SGD differs from ordinary SGD in accuracy, architecture, and tuning.

Accuracy. On MNIST, a well-tuned DP-SGD model reached 98.1% accuracy at $\varepsilon = 2.93$ and $\delta = 10^{-5}$, compared with about 99.8% without privacy (Papernot and coauthors, 2020). On CIFAR-10, it reached 66.2% at $\varepsilon = 7.53$, a weaker result in both accuracy and privacy.

Architecture. Bounded activations such as tanh can outperform the rectified linear unit under DP-SGD. Network size also matters: the noise vector has expected squared norm $d\sigma^2C^2$, while the clipped-gradient sum has norm at most qnC. Thus the signal-to-noise ratio is at most about $qn/(\sigma\sqrt{d})$ and decreases as the parameter count d grows. Use a network large enough for the task, but no larger.

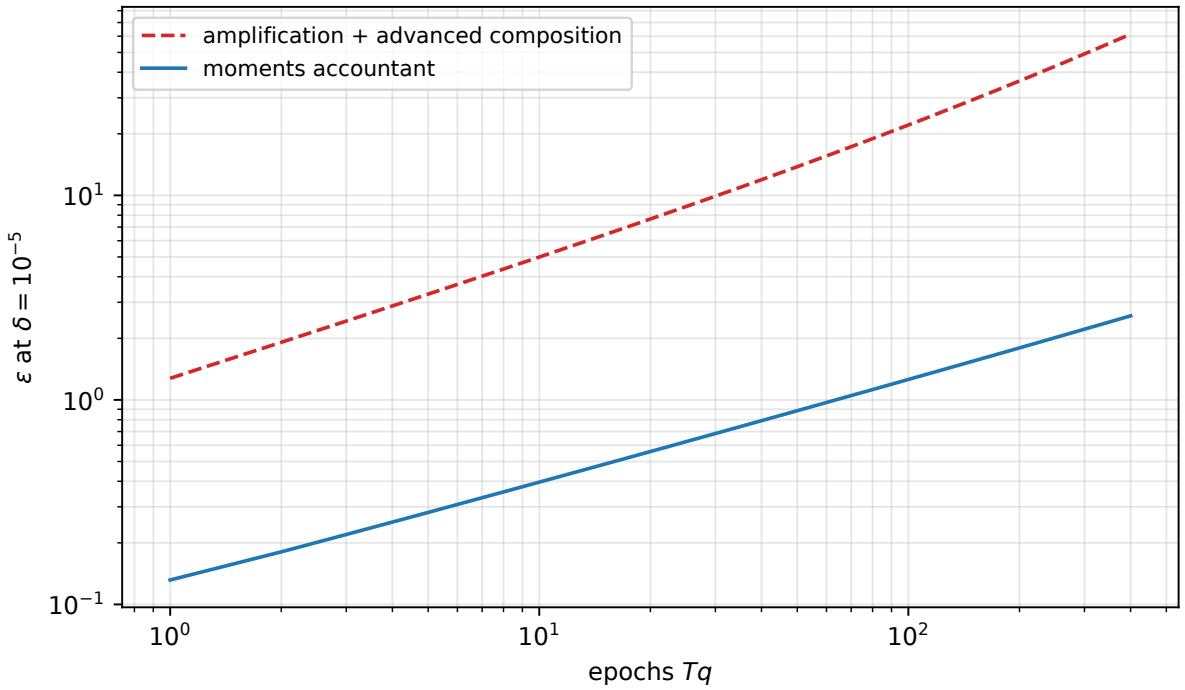


Figure 14.1.: Privacy parameter ε at $\delta = 10^{-5}$ of DP-SGD with sampling rate $q = 0.01$ and noise multiplier $\sigma = 4$, against the number of epochs of $1/q$ steps. Dashed: each step is $(\varepsilon_0, \delta_0)$-DP by Corollary 14.1 with $T = 1$, Poisson sampling makes it $(\ln(1 + q(e^{\varepsilon_0} - 1)), q\delta_0)$-DP, and advanced composition combines the steps, with δ split evenly. Solid: the moments accountant Equation 14.15.

Hyperparameters. Tuning the learning rate, lot size, clipping norm, or epoch count on private data spends privacy budget. Ignoring this cost overstates the final model's privacy. Private selection can choose among many private runs at a small constant factor in ε (Liu and Talwar). Public data can support tuning or pretraining, leaving fewer parameters to train privately.

14.5. Private aggregation of teacher ensembles

DP-SGD modifies one training algorithm. Private aggregation of teacher ensembles (PATE) treats the learner as a black box, so improvements to non-private learning also apply. PATE instead requires both a sensitive labelled training set and **public unlabelled** feature vectors.

PATE uses **sample and aggregate**: split the data into disjoint parts, train one non-private model on each part, then combine their outputs privately. Each record affects only one part, which limits the aggregation's sensitivity. PATE trains m classifiers, called **teachers**, and aggregates them by voting.

For a public feature vector x, each teacher votes for a label. Let c_k be the number of votes for label k. The **mode** a is the most-voted label, with ties resolved in favour of the smallest label. The **gap** $\Gamma = c_a - \max_{k \neq a} c_k$ measures the margin over the next label. The private mode releases a only when a noisy gap exceeds a threshold.

Definition 14.5 (Private mode of the teachers)**.** Let $\varepsilon > 0$ and $\delta \in (0, 1/2)$. Split the dataset into m fixed disjoint parts by position, train one teacher on each, and collect their votes on x. Let a be the mode of the votes and Γ its gap, as above. Draw $Z \sim \mathrm{Lap}(2/\varepsilon)$, Laplace noise with scale $2/\varepsilon$. If

$$\Gamma + Z > \tau, \qquad \tau = 2 + \frac{2}{\varepsilon} \ln \frac{1}{2\delta}, \tag{14.16}$$

output the mode a. Otherwise output a symbol $\perp$ for "no answer", or a uniformly random label.

When the votes agree, the mode is stable and can be released exactly. This is propose-test-release from the chapter on beyond global sensitivity: privately test whether the data is far from one that changes the answer, then release the answer without noise if the test passes. The threshold here is lower so the failure probability is δ, rather than $\delta/2$.

Theorem 14.4. *The private mode is (ε, δ)-DP. The teachers' training algorithm may be randomized, with coins independent of everything else.*

Proof.

1. Fix the teachers' coins; a guarantee for every fixed choice of coins implies one for random coins. Neighbours differ in one record, which lies in one part, so at most one teacher's vote changes. That moves one vote from one label to another, and changes each count by at most 1.

2. *The gap has sensitivity* 2. If the mode stays the same, both terms of Γ change by at most 1. If the mode changes from a on D to $a' \neq a$ on D', then $c_{a'} + 1 \geq c'_{a'} \geq c'_a \geq c_a - 1$, so $\Gamma \leq c_a - c_{a'} \leq 2$, and likewise $\Gamma' \leq 2$. In both cases $|\Gamma - \Gamma'| \leq 2$. So, by the Laplace mechanism, the indicator that the test passes is ε-DP.
3. *Same mode.* The output is a function of the test indicator and of the common mode, so it is ε-DP by post-processing.
4. *Different modes.* By step 2, $\Gamma \leq 2$. The test then passes on D with probability at most $\Pr[Z > \tau - 2] = \frac{1}{2} e^{-\varepsilon(\tau-2)/2} = \delta$. Outside that event the output is $\perp$, so for every set S of outputs,

$$\Pr[M(D) \in S] \leq \Pr[M(D) = \perp]\, \mathbf{1}[\perp \in S] + \delta \leq e^{\varepsilon} \Pr[M(D') = \perp]\, \mathbf{1}[\perp \in S] + \delta \leq e^{\varepsilon} \Pr[M(D') \in S] + \delta, \tag{14.17}$$

where the middle step is the ε-DP of the test from step 2. Replacing $\perp$ by a random label is post-processing.

□

Accuracy depends on teacher quality. For independent records and disjoint training subsets, teachers' votes on a fixed x are independent.

Proposition 14.1. *Let $\beta \in (0, 1)$ be a failure probability, and let τ be the threshold of Definition 14.5. Suppose each teacher, independently, votes for the correct label y of x with probability at least $3/4$. If*

$$m \geq \max\left(4\tau + \frac{8}{\varepsilon} \ln \frac{1}{\beta},\ 32 \ln \frac{2}{\beta}\right), \tag{14.18}$$

then the private mode outputs y with probability at least $1 - \beta$.

Proof.

1. Let V be the number of correct votes. Its mean is at least $3m/4$, and Hoeffding's inequality for independent indicators gives $\Pr[V \leq 5m/8] \leq e^{-2m(1/8)^2} = e^{-m/32} \leq \beta/2$.
2. If $V > 5m/8$, then y has a majority, so it is the mode, and every other label has at most $m - V$ votes. So $\Gamma \geq 2V - m > m/4$.
3. The test then fails with probability $\Pr[Z \leq \tau - \Gamma] \leq \frac{1}{2} e^{-\varepsilon(m/4-\tau)/2}$, because $m/4 > \tau$. The first lower bound on m makes this at most $\beta/2$.
4. A union bound over the two failure events gives the claim.

□

Example 14.2. With $\varepsilon = 1$, $\delta = 10^{-5}$ and $\beta = 0.01$, the threshold is $\tau = 2 + 2\ln(50{,}000) = 23.64$. The two lower bounds are $94.56 + 8 \ln 100 = 131.4$ and $32 \ln 200 = 169.5$, so 170 teachers suffice. If a teacher needs n_0 records to be right three times in four, the private prediction needs $170\, n_0$.

The required number of teachers grows as $\ln(1/\delta)/\varepsilon$. This sample-and-aggregate cost is about a factor $1/\varepsilon$ more data than the non-private algorithm requires, often more than a task-specific private method.

Answering k queries with separate private-mode runs costs about $\varepsilon\sqrt{2k\ln(1/\delta')}$ by advanced composition. Sparse vector is more efficient when most queries are stable: it pays only for answers above a threshold. Here, each query tests whether the next point's vote gap is small. The method therefore pays for unstable points, returning random labels for them, while releasing stable modes exactly. Its privacy cost grows with the square root of the number of unstable queries, rather than all k (Bassily, Thakkar and Thakurta).

The labelled pairs $(x_1, \hat{y}_1), \ldots, (x_k, \hat{y}_k)$ use public features and private labels. PATE trains a non-private **student** on them. If labeling is (ε, δ)-DP and the features are independent of private data, the student inherits that guarantee by post-processing and can answer further queries without additional cost. PATE thus converts a bounded number of private predictions into a private model.

14.6. Summary

Table 14.1.: The chapter's results.

Result	Statement
Lemma 14.1	one record changes the clipped sum by at most C, or $2C$ under replacement
Theorem 14.1	DP-SGD is (ε, δ)-DP with $\sigma = O(q\sqrt{T\ln(1/\delta)}/\varepsilon)$
Theorem 14.2	a log-moment bound $\kappa(\lambda)$ gives $(\varepsilon, e^{\kappa(\lambda)-\lambda\varepsilon})$-DP
Lemma 14.2	log-moments add up under adaptive composition
Lemma 14.3	the Gaussian mechanism has $\kappa(\lambda) = \lambda(\lambda+1)(\Delta_2 f)^2/(2\sigma^2)$
	T Gaussian steps give $\varepsilon = T\rho + 2\sqrt{T\rho\ln(1/\delta)}$
Corollary 14.1	
Theorem 14.3	a sampled Gaussian step has $\kappa(\lambda) \le \ln A_{\lambda+1}$
Equation 14.15	the moments accountant for T steps of DP-SGD
Theorem 14.4	releasing the teachers' mode when its noisy gap exceeds τ is (ε, δ)-DP
Proposition 14.1	$O(\ln(1/\delta)/\varepsilon + \ln(1/\beta)/\varepsilon)$ accurate teachers suffice

14.7. References

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15. Private mean estimation

Earlier chapters estimated properties of fixed datasets. Statistical estimation instead treats the dataset as a sample and targets a property of its generating distribution. This chapter bounds the sample size needed for an estimate under differential privacy (DP) of a distribution's mean within error α, and compares it with the non-private requirement.

For bounded data, privacy adds a lower-order sample cost. For unbounded data, assume prior knowledge such as a bound R on the mean. Direct clipping to $[-R, R]$ costs samples proportional to R; first locating the data reduces this dependence to logarithmic in R. Heavy tails require biased clipping and increase the privacy cost.

15.1. The mean of a dataset and of a distribution

Two earlier results are used throughout. A Laplace variable $Z \sim \mathrm{Lap}(b)$ with scale b has density proportional to $e^{-|z|/b}$ and tail

$$\Pr[|Z| > t] = e^{-t/b} \qquad (t \geq 0). \tag{15.1}$$

The Laplace mechanism privatizes a function f of the dataset with values in $\mathbb{R}^d$. Its **sensitivity** $\Delta_1 f$ is the largest change that the replacement of one record can cause,

$$\Delta_1 f = \max_{D \sim D'} \|f(D) - f(D')\|_1. \tag{15.2}$$

If $\Delta_1 f$ is finite, adding independent $\mathrm{Lap}(\Delta_1 f/\varepsilon)$ noise to each coordinate of $f(D)$ gives an ε-DP release.

For bits $X_i \in \{0, 1\}$, the empirical mean is $\tilde{p} = \frac{1}{n}\sum_i X_i$. Replacing one bit changes it by at most $1/n$, so the Laplace mechanism releases

$$\hat{p} = \tilde{p} + \mathrm{Lap}\left(\frac{1}{\varepsilon n}\right). \tag{15.3}$$

Now suppose the bits are a sample, $X_1, \ldots, X_n$ independent draws from $\mathrm{Bern}(p)$, and the target is the unknown p.

Theorem 15.1 (Private Bernoulli mean)**.** *The estimator Equation 15.3 is ε-DP. If $X_1, \ldots, X_n$ are independent* $\mathrm{Bern}(p)$ *variables, then for every $\beta \in (0, 1)$, with probability at least $1 - \beta$,*

$$|\hat{p} - p| \leq \sqrt{\frac{\ln(4/\beta)}{2n}} + \frac{\ln(2/\beta)}{\varepsilon n}. \tag{15.4}$$

Proof.

1. Privacy is the Laplace mechanism with sensitivity $1/n$.
2. By the triangle inequality, $|\hat{p} - p| \le |\tilde{p} - p| + |\hat{p} - \tilde{p}|$.
3. The bits are independent, take values in $[0, 1]$, and have mean p. Hoeffding's inequality gives $\Pr[|\tilde{p} - p| > t] \le 2e^{-2nt^2}$, which equals $\beta/2$ at $t = \sqrt{\ln(4/\beta)/(2n)}$.
4. The noise $\hat{p} - \tilde{p}$ is $\mathrm{Lap}(1/(\varepsilon n))$, so by the Laplace tail it exceeds $\ln(2/\beta)/(\varepsilon n)$ with probability $\beta/2$.
5. A union bound over the two failures gives the claim.

□

Both terms are at most $\alpha/2$ if

$$n \ge \frac{2\ln(4/\beta)}{\alpha^2} + \frac{2\ln(2/\beta)}{\alpha\varepsilon}, \tag{15.5}$$

Thus, for fixed β, $n = O(1/\alpha^2 + 1/(\alpha\varepsilon))$ samples suffice. The first term is the non-private sample complexity; the second is the **cost of privacy**. For constant ε and small α, the latter is smaller by a factor of α. The proof separates **sampling error** $\tilde{p} - p$ from the added **noise error** $\hat{p} - \tilde{p}$; this decomposition recurs below.

15.2. Unbounded data

The Bernoulli mean has bounded sensitivity because records lie in $[0, 1]$. For real-valued records such as heights, replacing 160 cm with 10^5 cm shifts the empirical mean by about $10^5/n$. The mean therefore has infinite sensitivity. Group privacy shows why no private algorithm can estimate it accurately on every dataset: if an ε-DP mechanism's inputs differ in k records, then for every event S,

$$\Pr[M(D) \in S] \le e^{k\varepsilon} \Pr[M(D') \in S], \tag{15.6}$$

This follows by applying the DP definition along a path of k neighbouring datasets. The next proposition uses this bound to show that every ε-DP algorithm fails on some real-valued dataset, for any finite error tolerance α.

Proposition 15.1. *Let $M : \mathbb{R}^n \to \mathbb{R}$ be ε-DP and $\alpha > 0$. Some dataset $D \in \mathbb{R}^n$ has $\Pr[|M(D) - \bar{X}| \le \alpha] < 1/2$, where $\bar{X}$ is its empirical mean.*

Proof.

1. Suppose instead that $\Pr[|M(D) - \bar{X}| \le \alpha] \ge 1/2$ for every D. Take an integer $m > 2e^{\varepsilon n}$ and, for $j = 0, \dots, m-1$, the dataset $D^{(j)}$ whose n entries all equal $3\alpha j$, so that its empirical mean is $3\alpha j$.
2. The intervals $I_j = [3\alpha j - \alpha, 3\alpha j + \alpha]$ are disjoint, and $\Pr[M(D^{(j)}) \in I_j] \ge 1/2$ by assumption.

3. The datasets $D^{(0)}$ and $D^{(j)}$ differ in n entries, so group privacy gives $\Pr[M(D^{(0)}) \in I_j] \geq e^{-\varepsilon n}/2$.
4. The sum over the disjoint intervals gives $1 \geq me^{-\varepsilon n}/2 > 1$, a contradiction.

□

The remedy is prior knowledge independent of the data. For example, an algorithm can clip heights to the known range $[0, 300]$ cm.

Definition 15.1 (Clipping)**.** For $u \leq v$, the **clipping** map onto $[u, v]$ is $g_{[u,v]}(x) = \min(\max(x, u), v)$. It leaves points of $[u, v]$ unchanged and moves every other point to the nearer endpoint.

Clipping to a fixed range bounds each record's effect on the mean, allowing the Laplace mechanism to privatize it.

Lemma 15.1. *Let $u < v$ be fixed before seeing the data, and let $g_{[u,v]}$ be the clipping map of Definition 15.1. The estimator*

$$M(D) = \frac{1}{n}\sum_{i=1}^{n} g_{[u,v]}(X_i) + \mathrm{Lap}\Big(\frac{v-u}{\varepsilon n}\Big) \tag{15.7}$$

is ε-DP. If every X_i lies in $[u, v]$, then $|M(D) - \bar{X}| \leq (v-u)\ln(1/\beta)/(\varepsilon n)$ with probability at least $1 - \beta$.

Proof.

1. Each clipped value lies in $[u, v]$, so a replaced record moves the clipped mean by at most $(v - u)/n$. The Laplace mechanism gives privacy.
2. If every X_i lies in $[u, v]$, clipping changes nothing, and $M(D) - \bar{X}$ is the Laplace noise, whose tail gives the bound.

□

For heights in $[0, 300]$, $n = 1{,}000$, and $\varepsilon = 1$, the noise scale is $300/1{,}000 = 0.3$ cm. Choose the range before inspecting the data. A range selected afterward to exclude an outlier depends on that record and invalidates the guarantee.

15.3. Estimating a parameter of a distribution

From this point, the dataset is a sample from an unknown distribution and the target is its mean. The estimator must be private on every dataset and accurate on samples from the assumed distribution.

Definition 15.2 (Private estimation)**.** Let $\mathcal{P}$ be a class of distributions on $\mathbb{R}$, each with a mean $\mu(P)$. An algorithm $M \colon \mathbb{R}^n \to \mathbb{R}$ is an ε**-DP** (α, β)**-estimator** for $\mathcal{P}$ if

1. M is ε-DP, as a map on all datasets in $\mathbb{R}^n$, and
2. for every $P \in \mathcal{P}$, if $X_1, \dots, X_n$ are independent draws from P, then $\Pr[|M(D) - \mu(P)| \le \alpha] \ge 1 - \beta$.

The smallest such n is the **sample complexity** of the class.

Privacy is worst-case: it covers datasets with probability zero under every $P \in \mathcal{P}$. Thus privacy does not depend on whether the analyst's distributional model is correct. Accuracy is measured over samples from P and requires assumptions even without privacy. For example, a Cauchy distribution has no mean, and its sample mean does not converge.

The main example is the class of unit-variance Gaussians with bounded means:

$$\mathcal{G}_R = \{\, \mathcal{N}(\mu, 1) : |\mu| \le R \,\}. \tag{15.8}$$

A bound on the mean plays the role of the range of heights. The argument of Proposition 15.1 shows that some bound is necessary.

Proposition 15.2. *Let $\alpha > 0$, $\beta \in (0, 1)$, and $m = \lfloor 2R/(3\alpha) \rfloor + 1$. Every ε-DP (α, β)-estimator for $\mathcal{G}_R$ uses*

$$n \ge \frac{\ln\big((1-\beta)m\big)}{\varepsilon} \tag{15.9}$$

samples. Without a bound on $|\mu|$ ($R = \infty$), no finite n suffices.

Proof.

1. The means $\mu_j = -R + 3\alpha j$, for $j = 0, \dots, m-1$, lie in $[-R, R]$, and the intervals $I_j = [\mu_j - \alpha, \mu_j + \alpha]$ are disjoint.
2. Let D be a sample from $\mathcal{N}(\mu_0, 1)$ and set $D^{(j)} = D + (\mu_j - \mu_0)$, which shifts every entry. Then $D^{(j)}$ is a sample from $\mathcal{N}(\mu_j, 1)$, and it differs from D in all n entries.
3. For each fixed D, group privacy gives $\Pr_M[M(D) \in I_j] \ge e^{-\varepsilon n} \Pr_M[M(D^{(j)}) \in I_j]$. The average over D, with accuracy for $\mathcal{N}(\mu_j, 1)$, gives $\Pr[M(D) \in I_j] \ge e^{-\varepsilon n}(1 - \beta)$.
4. The intervals are disjoint, so the sum over j gives $1 \ge m e^{-\varepsilon n}(1 - \beta)$. This inequality, solved for n, gives the bound, and $m = \infty$ when $R = \infty$.

□

So a term of order $\ln(R/\alpha)/\varepsilon$ is unavoidable, and the rest of the chapter finds algorithms that pay little more than that.

15.4. Clipping the Gaussian

A bound on the mean is not a bound on the records. But with the light tails of the Gaussian, it gives one with high probability. The lemma below adapts part 3 of Lemma 5.3 in the approximate DP chapter, which says that a standard normal W has $\Pr[W > u] \le e^{-u^2/2}$ for every $u \ge 0$, to both tails.

Lemma 15.2 (Gaussian tail)**.** *If $Y \sim \mathcal{N}(0,1)$ and $t \ge 0$, then $\Pr[|Y| \ge t] \le 2e^{-t^2/2}$.*

Proof.

1. Part 3 of Lemma 5.3 with $u = t$ gives $\Pr[Y \ge t] = \Pr[Y > t] \le e^{-t^2/2}$, because Y has a density.
2. The lower tail is the same by symmetry, and the two add.

□

Fix $\beta \in (0,1)$ and write

$$\tau = \sqrt{2\ln(2n/\beta)}. \tag{15.10}$$

For a sample from $\mathcal{N}(\mu,1)$, each point leaves $[\mu-\tau, \mu+\tau]$ with probability at most $2e^{-\tau^2/2} = \beta/n$. So by a union bound all n points stay inside with probability at least $1-\beta$, and the width τ grows only like $\sqrt{\ln n}$.

The first estimator clips to an interval known to contain the mean, widened by τ. It takes any public centre c and radius r, because the later algorithms call it with a centre that they computed privately.

Theorem 15.2 (Clipped Laplace estimator)**.** *Let $c \in \mathbb{R}$ and $r \ge 0$ be public, $\beta \in (0,1)$, and τ as in Equation 15.10. With $g_{[u,v]}$ the clipping map of Definition 15.1, the estimator*

$$M_{c,r}(D) = \frac{1}{n}\sum_{i=1}^{n} g_{[c-r-\tau,\, c+r+\tau]}(X_i) + Z, \qquad Z \sim \mathrm{Lap}\left(\frac{2(r+\tau)}{\varepsilon n}\right), \tag{15.11}$$

is ε-DP. Let $X_1, \dots, X_n$ be independent $\mathcal{N}(\mu,1)$ draws. With probability at least $1-2\beta$ over the sample, an event E holds on which every $M_{c,r}$ with $|\mu - c| \le r$ satisfies

$$|M_{c,r}(D) - \mu| \le \sqrt{\frac{2\ln(2/\beta)}{n}} + |Z|. \tag{15.12}$$

In particular, if $|\mu - c| \le r$, then with probability at least $1-3\beta$,

$$|M_{c,r}(D) - \mu| \le s_\varepsilon(r) = \sqrt{\frac{2\ln(2/\beta)}{n}} + \frac{2(r+\tau)\ln(1/\beta)}{\varepsilon n}. \tag{15.13}$$

Proof.

1. *Privacy.* The clipping interval has length $2(r+\tau)$, so the clipped mean has sensitivity $2(r+\tau)/n$, and the Laplace mechanism applies.
2. *The event.* Let G be the event that every X_i lies in $[\mu-\tau, \mu+\tau]$, with probability at least $1-\beta$ by the union bound above. Let S be the event that the empirical mean $\tilde{\mu} = \frac{1}{n}\sum_i X_i$ satisfies $|\tilde{\mu}-\mu| \le \sqrt{2\ln(2/\beta)/n}$. Since $\tilde{\mu}-\mu \sim \mathcal{N}(0, 1/n)$, Lemma 15.2 with $t = \sqrt{2\ln(2/\beta)}$ gives $\Pr[S] \ge 1-\beta$. Set $E = G \cap S$.
3. *No clipping on E.* If $|\mu - c| \le r$, the interval $[\mu-\tau, \mu+\tau]$ lies inside $[c-r-\tau, c+r+\tau]$. So on G clipping changes no point, and $M_{c,r}(D) = \tilde{\mu} + Z$.
4. *Error on E.* By the triangle inequality, $|M_{c,r}(D) - \mu| \le |\tilde{\mu} - \mu| + |Z|$, and S bounds the first term.
5. *Noise.* The Laplace tail gives $|Z| \le 2(r+\tau)\ln(1/\beta)/(\varepsilon n)$ with probability $1-\beta$. A union bound with E gives Equation 15.13.

□

Decompose the estimation error into three terms. For clipping map g, let $\bar{g} = \frac{1}{n}\sum_i g(X_i)$ be the clipped mean, $\tilde{\mu}$ the empirical mean, and Z the added noise. Then

$$\hat{\mu} - \mu = \underbrace{(\tilde{\mu} - \mu)}_{\text{sampling}} + \underbrace{(\bar{g} - \tilde{\mu})}_{\text{clipping bias}} + \underbrace{Z}_{\text{noise}}. \tag{15.14}$$

Here the interval is wide enough that clipping bias vanishes with high probability. In Section 15.8, the interval is narrower and the bias-noise trade-off matters.

With $c = 0$ and $r = R$, each term of Equation 15.13 is at most $\alpha/2$ when

$$n \ge \frac{8\ln(2/\beta)}{\alpha^2} + \frac{4(R+\tau)\ln(1/\beta)}{\alpha\varepsilon}, \tag{15.15}$$

that is, $n = \tilde{O}(1/\alpha^2 + R/(\alpha\varepsilon))$, where $\tilde{O}$ hides factors logarithmic in n and $1/\beta$. When R is a small constant, this matches the Bernoulli case. But R measures how little the analyst knows, and for an unfamiliar quantity R may have to be 10^6, so that the noise swamps the signal. The next two sections replace R by $\ln R$.

15.5. A coarse estimate from a histogram

The first improvement locates the data before it takes the average. Cut the range into bins of width 2τ. The whole sample fits in an interval of that width, so one or two bins hold all the points. A private histogram can find one of them at a cost logarithmic in the number of bins.

Lemma 15.3 (Coarse estimate)**.** *Let τ be as in Equation 15.10, $m = \lfloor R/\tau \rfloor + 2$, and $a_j = -R - \tau + 2\tau j$ for $j = 0, \dots, m-1$. Let h_j count the records in the bin $B_j = [a_j, a_j + 2\tau)$. The algorithm outputs $c = a_{\hat{\jmath}} + \tau$, where $\hat{\jmath}$ maximizes $h_j + Z_j$ over j, with independent $Z_j \sim \mathrm{Lap}(2/\varepsilon)$. It is ε-DP. If $X_1, \dots, X_n$ are independent $\mathcal{N}(\mu, 1)$ draws with $|\mu| \le R$ and*

$$n > \frac{8}{\varepsilon} \ln \frac{m}{\beta}, \tag{15.16}$$

then $|\mu - c| \le 2\tau$ with probability at least $1 - 2\beta$.

Proof.

1. *Privacy.* A replaced record moves out of at most one bin and into at most one bin, so the vector of counts has sensitivity 2. Adding $\mathrm{Lap}(2/\varepsilon)$ to each count is ε-DP by the Laplace mechanism, and the argmax is post-processing.
2. *The bins catch the sample.* On the event G of Theorem 15.2, every record lies in $[\mu - \tau, \mu + \tau] \subseteq [-R - \tau, R + \tau]$. The bins cover $[-R - \tau, -R - \tau + 2\tau m)$, and $2\tau m > 2R + 2\tau$, so every record is counted.
3. *One or two bins hold everything.* The interval $[\mu - \tau, \mu + \tau]$ has length 2τ, so it meets at most two bins. Hence some bin has $h_j \ge n/2$, and every bin that misses the interval has $h_j = 0$.
4. *The noise is small.* Set $t = (2/\varepsilon) \ln(m/\beta)$. Each $|Z_j| > t$ with probability β/m, so with probability at least $1 - \beta$ all $|Z_j| \le t$. Then the fullest bin has noisy count at least $n/2 - t$, and an empty bin at most t. Since $n > 4t$, the winning bin $\hat{\jmath}$ is not empty.
5. *The winner is near the mean.* On G, the bin $B_{\hat{\jmath}}$ holds a record x with $|x - \mu| \le \tau$ and $a_{\hat{\jmath}} \le x < a_{\hat{\jmath}} + 2\tau$. So $\mu \in [a_{\hat{\jmath}} - \tau, a_{\hat{\jmath}} + 3\tau] = [c - 2\tau, c + 2\tau]$. The two failures have total probability at most 2β.

□

After the coarse step, the mean is known to within $2\tau = O(\sqrt{\ln(n/\beta)})$ instead of R. The fine step is Theorem 15.2 on that short interval. Basic composition, from the Laplace mechanism chapter, says that an ε_1-DP algorithm followed by an ε_2-DP algorithm that may depend on the first output is $(\varepsilon_1 + \varepsilon_2)$-DP. So each step gets half the budget.

Theorem 15.3 (Histogram, then clipping)**.** *Let m be the number of bins in Lemma 15.3. Run that algorithm with $\varepsilon/2$ (noise $\mathrm{Lap}(4/\varepsilon)$) to get c, then output $\hat{\mu} = M_{c,2\tau}(D)$ from Theorem 15.2 with $\varepsilon/2$. The result is ε-DP. If $X_1, \dots, X_n$ are independent $\mathcal{N}(\mu, 1)$ draws with $|\mu| \le R$ and $n > (16/\varepsilon) \ln(m/\beta)$, then with probability at least $1 - 4\beta$,*

$$|\hat{\mu} - \mu| \le \sqrt{\frac{2 \ln(2/\beta)}{n}} + \frac{12\,\tau \ln(1/\beta)}{\varepsilon n}. \tag{15.17}$$

Proof.

1. *Privacy.* The two steps are $\varepsilon/2$-DP each, and basic composition adds them.

2. *Coarse step.* Lemma 15.3 with $\varepsilon/2$ needs $n > (16/\varepsilon)\ln(m/\beta)$. Its two failure events are G and the event that some $|Z_j|$ is large, of total probability at most 2β. Outside them, $|\mu - c| \le 2\tau$.
3. *Fine step.* On the event $E = G \cap S$ of Theorem 15.2, the fine estimate satisfies $|\hat{\mu} - \mu| \le \sqrt{2\ln(2/\beta)/n} + |Z|$, since $|\mu - c| \le 2\tau$. The event S adds probability β of failure.
4. The noise Z is $\mathrm{Lap}(2(2\tau + \tau)/((\varepsilon/2)n)) = \mathrm{Lap}(12\tau/(\varepsilon n))$, whatever c is, and it exceeds $12\tau\ln(1/\beta)/(\varepsilon n)$ with probability β. The union bound over G, the histogram noise, S and Z gives $1 - 4\beta$.

□

The error is at most α once $n \ge 8\ln(2/\beta)/\alpha^2 + 24\tau\ln(1/\beta)/(\alpha\varepsilon)$, and the coarse step needs $n > (16/\varepsilon)\ln(m/\beta)$ with $m \le R/\tau + 2$. So

$$n = \tilde{O}\Big(\frac{1}{\alpha^2} + \frac{1}{\alpha\varepsilon} + \frac{\ln R}{\varepsilon}\Big). \tag{15.18}$$

This reduces dependence on R from linear to logarithmic. The first term is necessary without privacy; Proposition 15.2 shows that the last is necessary, and the middle term is also necessary up to logarithmic factors.

15.6. Shrinking confidence intervals

The histogram locates the data in one step, but a histogram over a region in $\mathbb{R}^d$ needs exponentially many bins. An alternative repeatedly applies the clipped Laplace estimator. Each call returns a shorter confidence interval that still contains μ with high probability.

Given interval $[c - r, c + r]$ and privacy budget ε', one step computes $c' = M_{c,r}(D)$ and returns $[c' - s, c' + s]$, where $s = s_{\varepsilon'}(r)$ from Equation 15.13. This radius depends on r and public parameters, not the data.

Lemma 15.4 (One step)**.** *Let τ be as in Equation 15.10, $n \ge 16$ and $n \ge 16\ln(1/\beta)/\varepsilon'$. Consider one step, as described above, from the interval $[c - r, c + r]$ to the interval $[c' - s, c' + s]$. Then*

$$s_{\varepsilon'}(r) \le \tfrac{1}{2}\max(r, \tau). \tag{15.19}$$

Moreover, on the event E of Theorem 15.2, if $|\mu - c| \le r$, then $\mu \in [c' - s, c' + s]$ with probability at least $1 - \beta$ over the step's noise.

Proof.

1. Since $2\ln(2/\beta) \le 2\ln(2n/\beta) = \tau^2$ and $n \ge 16$, the first term of $s_{\varepsilon'}(r)$ is at most $\tau/\sqrt{n} \le \tau/4$.
2. Since $n \ge 16\ln(1/\beta)/\varepsilon'$, the second term is at most $2(r + \tau)/16 = (r + \tau)/8 \le \max(r, \tau)/4$.
3. Adding the two terms gives the first claim, because $\tau/4 \le \max(r, \tau)/4$.
4. On E, Theorem 15.2 gives $|c' - \mu| \le \sqrt{2\ln(2/\beta)/n} + |Z|$, where Z is the step's Laplace noise. That noise is at most $2(r + \tau)\ln(1/\beta)/(\varepsilon' n)$ with probability $1 - \beta$, and then $|c' - \mu| \le s$.

□

The first claim is deterministic, and the second needs the data to be Gaussian. Each step halves the radius until it reaches the floor τ, where the tails of the data stop the clipping from doing better.

Theorem 15.4 (Shrinking intervals, then clipping)**.** *Let $T = \lceil \log_2(R/\tau) \rceil$ (or $T = 0$ if $R \le \tau$), $\varepsilon' = \varepsilon/(2T)$, $r_0 = R$, $c_0 = 0$, and $r_t = s_{\varepsilon'}(r_{t-1})$. For $t = 1, \dots, T$, let $c_t = M_{c_{t-1}, r_{t-1}}(D)$ computed with ε'. Output $\hat{\mu} = M_{c_T, r_T}(D)$ computed with $\varepsilon/2$. The result is ε-DP. Suppose $X_1, \dots, X_n$ are independent $\mathcal{N}(\mu, 1)$ draws with $|\mu| \le R$,*

$$n \ge 16 \qquad \text{and} \qquad n \ge \frac{32\, T \ln(1/\beta)}{\varepsilon}. \tag{15.20}$$

Then with probability at least $1 - (T+3)\beta$,

$$|\hat{\mu} - \mu| \le \sqrt{\frac{2\ln(2/\beta)}{n}} + \frac{4(r_T + \tau)\ln(1/\beta)}{\varepsilon n} \le \sqrt{\frac{2\ln(2/\beta)}{n}} + \frac{8\,\tau \ln(1/\beta)}{\varepsilon n}. \tag{15.21}$$

Proof.

1. *Privacy.* The radii r_t depend only on n, ε, β and R, so each step is an instance of Theorem 15.2 with public parameters. There are T steps of $\varepsilon/(2T)$ and one of $\varepsilon/2$, each using the outputs before it, and basic composition gives ε.
2. *The radii shrink.* The condition on n is the one of Lemma 15.4 with $\varepsilon' = \varepsilon/(2T)$, so $r_t \le \max(r_{t-1}, \tau)/2$. By induction, $r_t \le \max(R/2^t, \tau/2)$. Since $2^T \ge R/\tau$, this gives $r_T \le \tau$.
3. *The intervals cover μ.* The event E holds with probability at least $1 - 2\beta$, and it concerns only the sample, so it serves every step. On E, by induction on t, Lemma 15.4 keeps $|\mu - c_t| \le r_t$ unless the noise of step t fails, which has probability β.
4. *Final step.* On E and with $|\mu - c_T| \le r_T$, Theorem 15.2 with $\varepsilon/2$ bounds the error by the first expression, unless its noise fails, with probability β. Step 2 gives $r_T \le \tau$, hence the second expression.
5. The failures are E (2β), the T steps ($T\beta$) and the final noise (β).

□

The steps need $n = O(\ln(R)\ln(1/\beta)/\varepsilon)$, and the final step needs $n = \tilde{O}(1/\alpha^2 + 1/(\alpha\varepsilon))$ for error α. So shrinking intervals reach the same rate Equation 15.18 as the histogram, with only the Laplace mechanism. The example compares the three bounds when the prior bound R is large.

Example 15.1. Take $n = 4{,}000$, $\varepsilon = 1$, $R = 10^6$ and $\beta = 0.002$, so that $\tau = 5.51$. Then $T = \lceil \log_2(10^6/5.51) \rceil = 18$ steps, which need $n \ge 32 \cdot 18 \ln 500 = 3{,}580$, and all $T + 3 = 21$ events hold with probability at least $1 - 21 \cdot 0.002 = 0.958$. The radius reached by the recursion gives the error bound 0.098. The coarse step of Theorem 15.3 uses $m = 181{,}360$ bins, needs $n > 293$, and gives the bound 0.162. The clipped estimator on $[-R, R]$ alone has the bound 3,107.

The bounds are conservative, and Figure 15.1 compares the simulated errors.

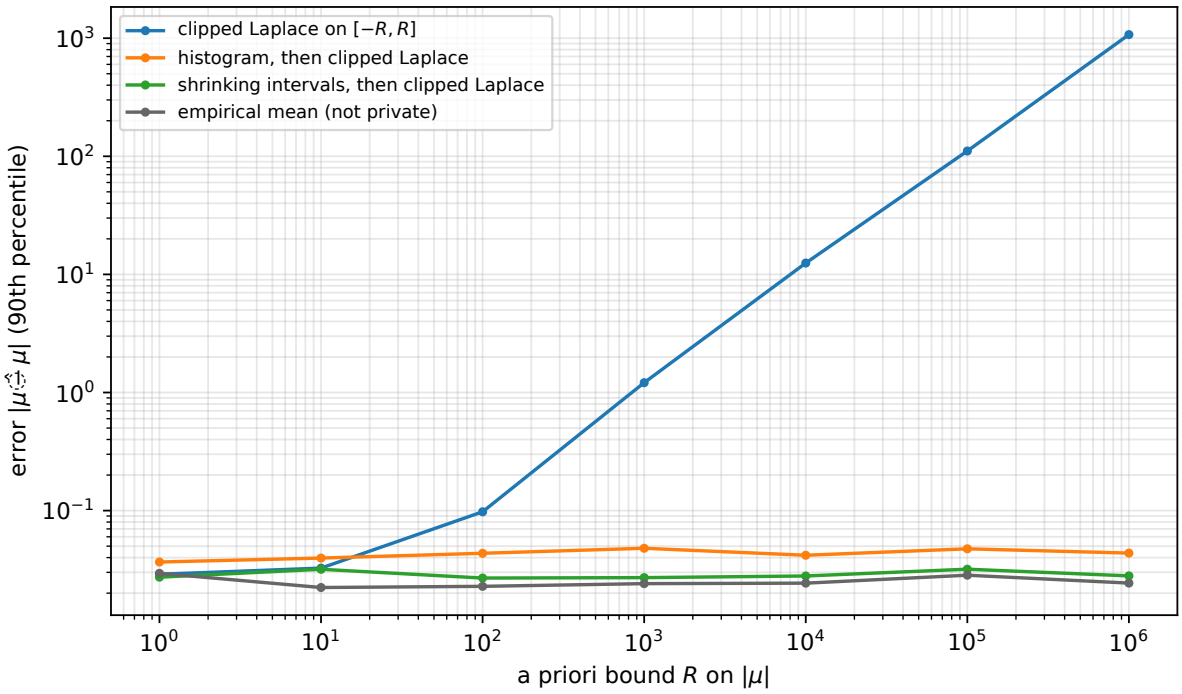


Figure 15.1.: Error of three ε-DP estimators of the mean of $\mathcal{N}(\mu, 1)$ with $\mu = R/3$, as the a priori bound R grows, for $n = 4{,}000$, $\varepsilon = 1$ and $\beta = 0.002$. Each point is the 90th percentile of $|\hat{\mu} - \mu|$ over 100 runs. The clipped estimator on $[-R, R]$ degrades linearly in R. The two estimators that first locate the data stay within a factor of about two of the non-private empirical mean for every R up to 10^6. At $R = 10^6$, over 300 runs, the 90th percentiles are 0.0250 without privacy, 0.0461 for the histogram and 0.0295 for shrinking intervals.

15.7. Many dimensions

The shrinking argument has an analogue in many dimensions. It keeps a confidence set for the parameter, starting from the prior knowledge, and repeats three moves.

1. Clip the data to the confidence set, widened by a tail bound for the class of distributions.
2. Compute the empirical estimate of the parameter, and add noise scaled to its sensitivity, which the clipping bounds.
3. Replace the confidence set by one centred at the noisy estimate, whose size accounts for the sampling error and the noise.

For a Gaussian $\mathcal{N}(\mu, I_d)$ in $\mathbb{R}^d$ with $\|\mu\|_2 \le R$, the confidence set is a Euclidean ball, and the noise is Gaussian, because clipping to a ball bounds the ℓ_2 sensitivity. Two facts replace those used above. First, the Gaussian mechanism adds $\mathcal{N}(0, \sigma^2 I_d)$ to a function f whose ℓ_2 sensitivity is $\Delta_2 f$. For every $\varepsilon > 0$ and $\delta \in (0,1)$ it is (ε, δ)-DP with $\sigma = \Delta_2 f \sqrt{2\ln(1/\delta) + \varepsilon}/\varepsilon$. Approximate DP also composes by adding the ε's and the δ's. Second, a standard Gaussian vector $Y \sim \mathcal{N}(0, I_d)$ has norm about $\sqrt{d}$, with

$$\Pr\left[\|Y\|_2 \ge \sqrt{d} + t\right] \le e^{-t^2/2} \qquad (t \ge 0). \tag{15.22}$$

This follows from the Gaussian concentration inequality for 1-Lipschitz functions (Boucheron, Lugosi and Massart, 2013, Theorem 5.6), applied to the norm, with $\mathbb{E}\|Y\|_2 \le \sqrt{\mathbb{E}\|Y\|_2^2} = \sqrt{d}$.

Clipping to the ball of centre c and radius ρ maps x to $c + (x-c)\min(1, \rho/\|x-c\|_2)$, which moves no point already inside the ball. Two radii play the role of τ in one dimension,

$$\tau_d = \sqrt{d} + \sqrt{2\ln(n/\beta)}, \qquad L = \sqrt{d} + \sqrt{2\ln(1/\beta)}. \tag{15.23}$$

The first widens each ball before clipping. By Equation 15.22 and a union bound, every record lies within τ_d of μ with probability at least $1 - \beta$. The second bounds the norm of a single standard Gaussian vector with probability at least $1 - \beta$, and so it bounds both the sampling error and the noise.

Theorem 15.5 (Multivariate Gaussian mean)**.** *Let $\varepsilon > 0$ and $\delta \in (0,1)$. A step on the ball $B(c, r)$ with budget (ε', δ') clips every record to $B(c, r + \tau_d)$, averages, and adds $\mathcal{N}(0, \sigma^2 I_d)$ with $\sigma = 2(r + \tau_d)\sqrt{2\ln(1/\delta') + \varepsilon'}/(\varepsilon' n)$. Its new radius is*

$$s(r) = \frac{L}{\sqrt{n}} + \sigma L. \tag{15.24}$$

Run $T = \lceil \log_2(R/\tau_d) \rceil$ steps with $(\varepsilon/(2T), \delta/(2T))$ from $B(0, R)$, then one more with $(\varepsilon/2, \delta/2)$, and output its noisy mean $\hat{\mu}$. This is (ε, δ)-DP. If $X_1, \dots, X_n$ are independent $\mathcal{N}(\mu, I_d)$ draws with $\|\mu\|_2 \le R$, and

$$n \ge 16, \qquad n \ge \frac{32\, T L \sqrt{2\ln(2T/\delta) + \varepsilon}}{\varepsilon}, \tag{15.25}$$

then with probability at least $1-(T+3)\beta$,

$$\|\hat{\mu}-\mu\|_2 \le \frac{L}{\sqrt{n}} + \frac{8\,\tau_d L\sqrt{2\ln(2/\delta)+\varepsilon}}{\varepsilon n}. \tag{15.26}$$

Proof. The proof is that of Theorem 15.4, with these changes.

1. *Privacy.* The clipped mean has ℓ_2 sensitivity $2(r+\tau_d)/n$, so each step is the Gaussian mechanism. The radii are public, and composition adds the budgets to (ε,δ).
2. *No clipping.* By Equation 15.22 with $t=\sqrt{2\ln(n/\beta)}$ and a union bound, every $\|X_i-\mu\|_2 \le \tau_d$ with probability at least $1-\beta$. If $\|\mu-c\|_2 \le r$, the ball $B(c, r+\tau_d)$ then contains every record.
3. *Sampling.* The empirical mean satisfies $\tilde{\mu}-\mu = Y/\sqrt{n}$ with $Y\sim\mathcal{N}(0,I_d)$, so Equation 15.22 gives $\|\tilde{\mu}-\mu\|_2 \le L/\sqrt{n}$ with probability $1-\beta$.
4. *Noise.* The noise is $\sigma Y'$ with $Y'\sim\mathcal{N}(0,I_d)$, so its norm is at most σL with probability $1-\beta$. With steps 2 and 3, the new ball has radius $s(r)$ and contains μ.
5. *Halving.* Since $L\le\tau_d$ and $n\ge 16$, the first term of $s(r)$ is at most $\tau_d/4$. With $\varepsilon'=\varepsilon/(2T)$ and $\delta'=\delta/(2T)$, $2\ln(1/\delta')+\varepsilon' \le 2\ln(2T/\delta)+\varepsilon$, so $\sigma L \le 4T(r+\tau_d)L\sqrt{2\ln(2T/\delta)+\varepsilon}/(\varepsilon n)$, and the condition on n makes this at most $(r+\tau_d)/8$. So $s(r)\le \max(r,\tau_d)/2$, and after T steps the radius is at most τ_d.
6. *Final step.* With $r_T \le \tau_d$, the final noise, with budget $(\varepsilon/2,\delta/2)$, has $\sigma = 4(r_T+\tau_d)\sqrt{2\ln(2/\delta)+\varepsilon/2}/(\varepsilon n) \le 8\tau_d\sqrt{2\ln(2/\delta)+\varepsilon}/(\varepsilon n)$, and steps 3 and 4 give the bound. The failures add up as before.

□

Because L and τ_d scale as $\sqrt{d}$ up to logarithmic factors, achieving error α for $\varepsilon\le 1$ requires

$$n = \tilde{O}\left(\frac{d}{\alpha^2} + \frac{d\sqrt{\ln(1/\delta)}}{\alpha\varepsilon} + \frac{\sqrt{d}\,\ln R\,\sqrt{\ln(1/\delta)}}{\varepsilon}\right). \tag{15.27}$$

The first two terms are d times the one-dimensional ones. The cost of locating the data grows only like $\sqrt{d}$, because the noise of each step is compared with a radius that already includes the $\sqrt{d}$ spread of the data.

15.8. Heavy-tailed data

The preceding bounds require light tails and therefore apply to **sub-Gaussian** distributions, whose tails decay at least as fast as Gaussian tails. The parameter τ then scales with the tail width. For heavier-tailed data, only a few moments may be bounded. Consider distributions on $\mathbb{R}$ satisfying

$$\mu = \mathbb{E}[X]\in[-1,1], \qquad \operatorname{Var}(X)\le 1. \tag{15.28}$$

The problem is to estimate μ to a precision α well below 1.

The previous strategy clips only outside the range occupied by the sample. Chebyshev's inequality gives $\Pr[|X-\mu| \ge \sqrt{n/\beta}] \le \beta/n$, so clipping to $[-1-C, 1+C]$ with $C = \sqrt{n/\beta}$ leaves every sample point unchanged with probability at least $1-\beta$. The resulting noise scale, $2(1+C)/(\varepsilon n)$, exceeds $2/(\varepsilon\sqrt{\beta n})$. Its tail is at most α only if

$$n \ge \frac{4\ln^2(1/\beta)}{\beta\,\alpha^2\varepsilon^2}, \tag{15.29}$$

which is worse than the Gaussian rate in both α and ε. Instead, clip more tightly and control the resulting bias, one of the terms in Equation 15.14.

Lemma 15.5 (Clipping bias)**.** *Let X have mean μ and variance at most σ_X^2, and let $[u,v] \supseteq [\mu-C, \mu+C]$ with $C>0$. Then $g = g_{[u,v]}$ satisfies*

$$\bigl|\mathbb{E}[g(X)] - \mu\bigr| \le \frac{\sigma_X^2}{C}, \qquad \mathrm{Var}(g(X)) \le \sigma_X^2. \tag{15.30}$$

Proof.

1. Since $\mu = \mathbb{E}[X]$, $|\mathbb{E}[g(X)] - \mu| = |\mathbb{E}[g(X) - X]| \le \mathbb{E}|g(X) - X|$.
2. If $|X-\mu| \le C$, then $X \in [u,v]$ and $g(X) = X$. Otherwise X moves towards μ and does not pass it, because $\mu \in [u,v]$, so $|g(X)-X| \le |X-\mu|$.
3. So $|g(X)-X| \le |X-\mu|\,\mathbf{1}[|X-\mu| > C] \le (X-\mu)^2/C$, and the expectation gives the first bound.
4. Clipping is 1-Lipschitz. With X' an independent copy of X, $\mathrm{Var}(g(X)) = \frac{1}{2}\mathbb{E}[(g(X) - g(X'))^2] \le \frac{1}{2}\mathbb{E}[(X-X')^2] = \mathrm{Var}(X)$.

□

Bounding the bias removes the need to retain every sample point. The next estimator clips at distance C beyond the possible mean range; the three terms in Equation 15.14 determine the choice of C.

Theorem 15.6 (Heavy-tailed mean)**.** *For $C>0$, the estimator*

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} g_{[-1-C,\,1+C]}(X_i) + \mathrm{Lap}\!\left(\frac{2(1+C)}{\varepsilon n}\right) \tag{15.31}$$

is ε-DP. If $X_1, \dots, X_n$ are independent draws from a distribution with mean $\mu \in [-1,1]$ and variance at most 1, then with probability at least $1-2\beta$,

$$|\hat{\mu} - \mu| \le \frac{1}{C} + \frac{1}{\sqrt{\beta n}} + \frac{2(1+C)\ln(1/\beta)}{\varepsilon n}. \tag{15.32}$$

Proof.

1. Privacy is Lemma 15.1 with an interval of length $2(1+C)$.

2. Since $|\mu| \le 1$, the interval $[-1-C, 1+C]$ contains $[\mu - C, \mu + C]$. Write $\nu = \mathbb{E}[g(X_1)]$ and $\bar{g}$ for the clipped mean. Then $\hat{\mu} - \mu = (\nu - \mu) + (\bar{g} - \nu) + Z$, with Z the noise.
3. *Bias.* Lemma 15.5 with $\sigma_X^2 = 1$ gives $|\nu - \mu| \le 1/C$.
4. *Sampling.* By Lemma 15.5, $\mathrm{Var}(\bar{g}) \le 1/n$, so Chebyshev's inequality gives $|\bar{g} - \nu| \le 1/\sqrt{\beta n}$ with probability at least $1 - \beta$.
5. *Noise.* The Laplace tail gives $|Z| \le 2(1 + C)\ln(1/\beta)/(\varepsilon n)$ with probability $1 - \beta$. A union bound finishes the proof.

□

Increasing C reduces bias but increases noise, so an optimal clipping norm balances them. If each term in Equation 15.32 is at most $\alpha/3$, the sample complexity follows directly.

Corollary 15.1. *Let $\alpha \in (0, 1]$, and let $\hat{\mu}$ be the estimator of Theorem 15.6 with $C = 3/\alpha$. If*

$$n \ge \frac{9}{\beta\alpha^2} \qquad \text{and} \qquad n \ge \frac{24\ln(1/\beta)}{\alpha^2\varepsilon}, \tag{15.33}$$

then $|\hat{\mu} - \mu| \le \alpha$ with probability at least $1 - 2\beta$.

Proof.

1. The bias term is $1/C = \alpha/3$, and the first condition makes the sampling term at most $\alpha/3$.
2. Since $\alpha \le 1$, $1 + C \le 4/\alpha$, so the noise term is at most $8\ln(1/\beta)/(\alpha\varepsilon n)$, which the second condition makes at most $\alpha/3$.

□

Thus C is of order $1/\alpha$, and for fixed β and $\varepsilon \le 1$, $n = O(1/(\alpha^2\varepsilon))$ suffices. This exceeds the Gaussian rate $O(1/\alpha^2 + 1/(\alpha\varepsilon))$ because the terms multiply rather than add. The increase is necessary: with only bounded variance, every ε-DP estimator has error at least order $1/\sqrt{\varepsilon n}$ (Barber and Duchi, 2014). For fixed n, the clipping norm minimizing $1/C + 2C\ln(1/\beta)/(\varepsilon n)$ is

$$C^* = \sqrt{\frac{\varepsilon n}{2\ln(1/\beta)}}, \tag{15.34}$$

and with $n = 4{,}000$, $\varepsilon = 1$ and $\beta = 0.05$ it is $C^* = 25.8$, where Equation 15.32 equals 0.150. Figure 15.2 shows the trade-off.

15.9. Summary

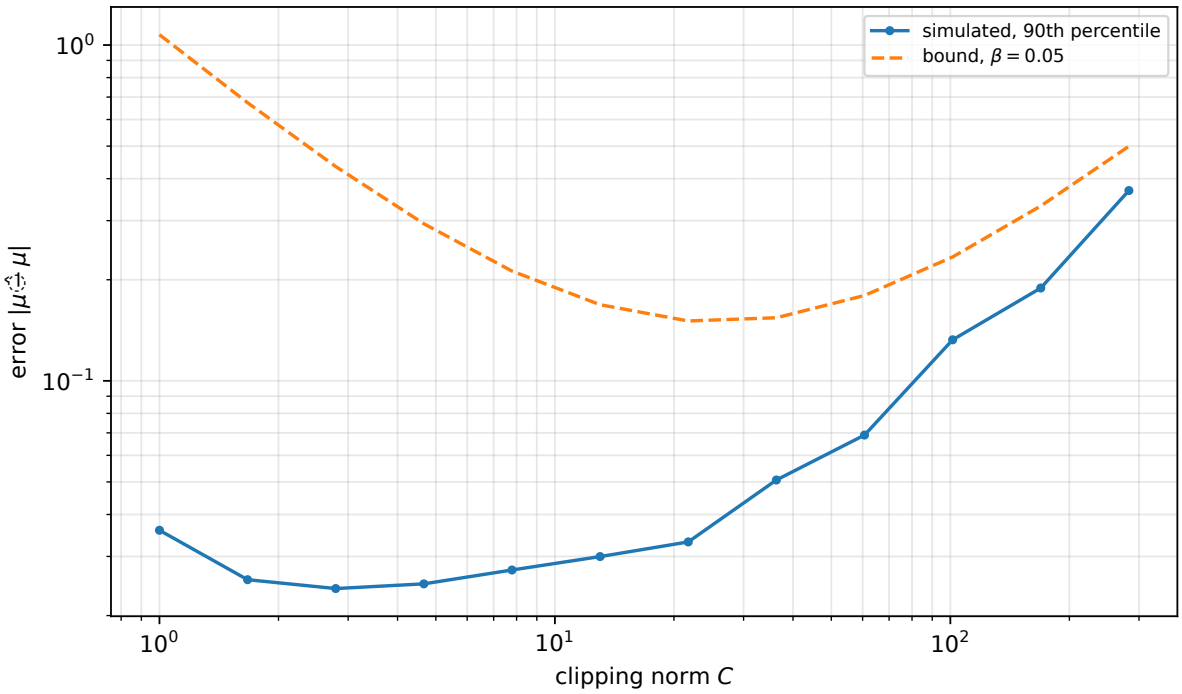


Figure 15.2.: Error of the clipped estimator of Theorem 15.6 on a Student t distribution with 3 degrees of freedom, scaled to variance 1, with $\mu = 0.5$, $n = 4{,}000$ and $\varepsilon = 1$, as the clipping norm C varies. The solid line is the 90th percentile of $|\hat{\mu} - \mu|$ over 300 runs, and the dashed line is the bound Equation 15.32 with $\beta = 0.05$. Both rise steeply at large C, where the noise dominates. The simulated error is smallest at a smaller C than the bound, because the clipping bias of this distribution is far below the worst case $1/C$.

Table 15.1.: The chapter's results.

Result	Statement
Theorem 15.1	Bernoulli mean with error $\sqrt{\ln(4/\beta)/(2n)} + \ln(2/\beta)/(\varepsilon n)$
Proposition 15.1	no ε-DP algorithm estimates the mean of every real dataset to finite accuracy
Lemma 15.1	clipping to a public $[u, v]$ gives sensitivity $(v-u)/n$
Proposition 15.2	estimating a Gaussian mean with $\lvert\mu\rvert \le R$ needs $n \ge \ln((1-\beta)m)/\varepsilon$, with m about $2R/(3\alpha)$
Theorem 15.2	clipping to $[-R-\tau, R+\tau]$ needs $n = \tilde{O}(1/\alpha^2 + R/(\alpha\varepsilon))$
Lemma 15.3	a noisy histogram locates μ within 2τ once $n > (8/\varepsilon)\ln(m/\beta)$
Theorem 15.3	histogram, then clipping, needs $n = \tilde{O}(1/\alpha^2 + 1/(\alpha\varepsilon) + \ln R/\varepsilon)$
Lemma 15.4	one clipped step halves the confidence interval down to the floor τ
Theorem 15.4	$T = \lceil \log_2(R/\tau) \rceil$ shrinking steps, then clipping, reach the same rate
Theorem 15.5	in $\mathbb{R}^d$, $n = \tilde{O}(d/\alpha^2 + d/(\alpha\varepsilon) + \sqrt{d}\ln R/\varepsilon)$ under (ε, δ)-DP, up to factors in $\ln(1/\delta)$
Lemma 15.5	clipping at distance C from the mean biases it by at most σ_X^2/C
Corollary 15.1	with variance 1 and $C = 3/\alpha$, $n = O(1/(\alpha^2\varepsilon))$ samples suffice

15.10. References

- Cynthia Dwork, Frank McSherry, Kobbi Nissim and Adam Smith, "Calibrating Noise to Sensitivity in Private Data Analysis", 2006, which introduces the Laplace mechanism and the sensitivity of the mean of bounded data.
- Vishesh Karwa and Salil Vadhan, "Finite Sample Differentially Private Confidence Intervals", 2018, on the histogram approach to the Gaussian mean, with lower bounds that make the rate Equation 15.18 optimal up to logarithmic factors.
- Cynthia Dwork and Aaron Roth, *The Algorithmic Foundations of Differential Privacy*, 2014, on the Laplace and Gaussian mechanisms and the composition theorems of this chapter.
- Stéphane Boucheron, Gábor Lugosi and Pascal Massart, *Concentration Inequalities*, 2013, whose Theorem 5.6 is the Gaussian concentration inequality behind Equation 15.22.

- Rina Foygel Barber and John Duchi, "Privacy and Statistical Risk: Formalisms and Minimax Bounds", 2014, whose Proposition 2 gives the $1/\sqrt{\varepsilon n}$ lower bound for bounded variance.

16. Adaptive data analysis

A laboratory may test successive hypotheses on one dataset, choosing each question from earlier results, until one is significant. Fresh data often fails to reproduce that finding. The individual tests may be valid, but reusing the data to select and answer questions makes the results fit the sample rather than the population.

This chapter formalizes the failure for averages of predicates, called statistical queries, and shows how differential privacy (DP) limits it. Queries fixed in advance require sample size logarithmic in their number k. Adaptive queries can overfit unless the sample size is of order k. A DP answering mechanism needs only about $\sqrt{k}$ samples, because privacy limits its ability to select queries where sample and population disagree.

16.1. Statistical queries

The dataset is sampled from an unknown distribution $\mathcal{D}$ on universe $\mathcal{X}$. Each query asks for the population fraction with a binary property and has both a distribution value and a sample value.

Definition 16.1 (Statistical query)**.** A **statistical query** is a function $q\colon \mathcal{X} \to \{0,1\}$. Its value on the distribution and its empirical value on a dataset $D = (X_1, \dots, X_n)$ are

$$q(\mathcal{D}) = \mathbb{E}_{x\sim\mathcal{D}}[q(x)], \qquad q(D) = \frac{1}{n}\sum_{j=1}^{n} q(X_j). \tag{16.1}$$

Thus, $q(\mathcal{D})$ is the population fraction with property q, and $q(D)$ is the sample fraction. These are the earlier chapters' counting queries divided by n. For independent samples, $q(D)$ is unbiased for $q(\mathcal{D})$. A fixed list of queries requires sample size logarithmic in its length.

Proposition 16.1. *Let $\alpha > 0$ be the allowed error and $\beta \in (0,1)$ the allowed failure probability. Let $q_1, \dots, q_k$ be statistical queries fixed before the sample is drawn, and let $X_1, \dots, X_n$ be independent draws from $\mathcal{D}$. If*

$$n \geq \frac{\ln(2k/\beta)}{2\alpha^2}, \tag{16.2}$$

then some query has error at least α with probability at most β,

$$\Pr\left[\exists i\colon |q_i(D) - q_i(\mathcal{D})| \geq \alpha\right] \leq \beta. \tag{16.3}$$

Proof.

1. For a fixed i, the values $q_i(X_1), \dots, q_i(X_n)$ are independent, lie in $[0,1]$, and have mean $q_i(\mathcal{D})$. Hoeffding's inequality gives $\Pr[|q_i(D) - q_i(\mathcal{D})| \geq \alpha] \leq 2e^{-2\alpha^2 n}$.
2. A union bound over the k queries gives failure probability at most $2ke^{-2\alpha^2 n}$, which the condition on n makes at most β.

□

A sample of size n therefore supports about $e^{2\alpha^2 n}$ fixed queries. The proof requires each query to be independent of the sample; adaptivity violates that condition.

16.2. Adaptivity breaks the bound

In an adaptive analysis, each query may depend on previous answers. The definition specifies this interaction and the corresponding privacy guarantee.

Definition 16.2 (Adaptive analysis)**.** An **analyst** $\mathcal{A}$ and a **mechanism** M interact for k rounds. In round i, the analyst chooses a statistical query q_i as a function of the earlier answers $a_1, \dots, a_{i-1}$ and its own random bits, called its coins. The mechanism returns an answer a_i, computed from the dataset D, its own coins and the history. The mechanism is (ε, δ)**-DP against adaptive analysts** if for every analyst $\mathcal{A}$ the map from D to the transcript $(q_1, a_1, \dots, q_k, a_k)$ is (ε, δ)-DP.

The empirical mechanism $a_i = q_i(D)$ answers each query exactly on the sample, but an adaptive analyst can use it to overfit.

Proposition 16.2. *Let $\mathcal{X} = \{1, \dots, N\}$, let $\mathcal{D}$ be uniform on $\mathcal{X}$, and let the mechanism answer $a_i = q_i(D)$. Some analyst with $k = N + 1$ queries ends with a query q_k such that, for every sample,*

$$q_k(D) - q_k(\mathcal{D}) \geq 1 - \frac{n}{k-1}. \tag{16.4}$$

Proof.

1. For $i = 1, \dots, N$, the analyst asks $q_i(x) = \mathbf{1}[x = i]$. The answer a_i is the fraction of records equal to i, so after N rounds the analyst knows the set $S = \{X_1, \dots, X_n\}$ of values in the sample.
2. The last query is $q_k(x) = \mathbf{1}[x \in S]$. Every record lies in S, so $q_k(D) = 1$.
3. The set S has at most n elements, so $q_k(\mathcal{D}) = |S|/N \leq n/N = n/(k-1)$.

□

Each of the first N queries could be fixed in advance. The analyst constructs the final query from their answers; its sample and population values differ by nearly 1 unless n is of order k. Adaptivity therefore reduces the supported query count from exponential to linear in sample size.

Sample splitting assigns fresh data to each query. Queries remain adaptive, but each answer uses records unseen by earlier answers.

Proposition 16.3 (Sample splitting)**.** *Let k divide n. Split D into k disjoint parts $D^{(1)}, \dots, D^{(k)}$ of n/k records each, and answer $a_i = q_i(D^{(i)})$. If the records are independent draws from $\mathcal{D}$ and*

$$n \geq \frac{k \ln(2k/\beta)}{2\alpha^2}, \tag{16.5}$$

then for every analyst $\Pr\left[\exists i \colon |a_i - q_i(\mathcal{D})| \geq \alpha\right] \leq \beta$.

Proof.

1. The query q_i depends only on the coins of the analyst and on $a_1, \dots, a_{i-1}$, which depend only on the parts $D^{(1)}, \dots, D^{(i-1)}$. So q_i is independent of $D^{(i)}$.
2. Conditioned on q_i, the part $D^{(i)}$ is still n/k independent draws from $\mathcal{D}$. Step 1 of the proof of Proposition 16.1 gives $\Pr[|a_i - q_i(\mathcal{D})| \geq \alpha] \leq 2e^{-2\alpha^2 n/k}$.
3. A union bound over i gives at most $2ke^{-2\alpha^2 n/k}$, which the condition makes at most β.

□

Sample splitting is valid but needs about k times the nonadaptive sample size. Differential privacy can reduce this cost. With Gaussian noise, sample size

$$n = O\left(\frac{\sqrt{k}\,\ln^{3/2}(k/(\alpha\beta))}{\alpha^2}\right) \tag{16.6}$$

suffices for error α (Corollary 16.1). Under a standard cryptographic assumption, no computationally efficient mechanism can generally improve the $\sqrt{k}$ dependence.

16.3. Privacy implies generalization

The overfitting analyst in Proposition 16.2 learned information about individual records. If every answer passes through a DP mechanism, each answer reveals little about any one record. A **transfer theorem** formalizes how sample accuracy and privacy imply population accuracy.

Theorem 16.1 (Transfer theorem)**.** *Let $\alpha, \beta \in (0, 0.1)$, and let M answer k adaptively chosen statistical queries. Suppose that*

1. *the mechanism is $(\alpha/64, \alpha\beta/32)$-DP against adaptive analysts, and*
2. *the mechanism is accurate on the sample, that is, for every analyst and every dataset D,* $\Pr\left[\exists i \colon |a_i - q_i(D)| > \alpha/8\right] \leq \alpha\beta/16$*, with probability over the coins of M and of the analyst.*

Then for every analyst and every distribution $\mathcal{D}$, if the n records are independent draws from $\mathcal{D}$,

$$\Pr\left[\exists i \colon |a_i - q_i(\mathcal{D})| > \alpha\right] \leq \beta. \tag{16.7}$$

This is Theorem 3.4 of Bassily, Nissim, Smith, Steinke, Stemmer and Ullman (2016), for queries of sensitivity $1/n$. The sample failure probability must be order $\alpha\beta$ to obtain population failure probability β. The proof runs about $1/\beta$ independent interactions, each on a separate sample. A proof-only monitor, which knows $\mathcal{D}$, selects the query with the largest population error. This selection post-processes DP outputs and remains private. The multi-sample version of Lemma 16.1 below shows that the monitor cannot find a large population error on average.

Two results from earlier chapters apply the theorem. The Gaussian mechanism answers a query f with Gaussian noise, and the sensitivity Δf sets the standard deviation. With $\sigma = \Delta f \sqrt{2\ln(1/\delta_0) + \varepsilon_0}/\varepsilon_0$, the mechanism is $(\varepsilon_0, \delta_0)$-DP for every $\varepsilon_0 > 0$ and $\delta_0 \in (0,1)$. A corollary of advanced composition says that, for $\varepsilon \in (0,1]$ and $\delta' \in (0, 1/2]$, k adaptively chosen mechanisms that are each $(\varepsilon_0, \delta_0)$-DP with

$$\varepsilon_0 = \frac{\varepsilon}{2\sqrt{2k\ln(1/\delta')}} \tag{16.8}$$

are, together, $(\varepsilon, k\delta_0 + \delta')$-DP. A statistical query has sensitivity $1/n$, because a replaced record changes one term of the average by at most 1. The corollary adds Gaussian noise to each empirical answer, at a scale that meets both hypotheses of Theorem 16.1, and gives the sample size that this needs.

Corollary 16.1 (Gaussian answers generalize)**.** *Let $\alpha, \beta \in (0, 0.1)$. Answer each query by $a_i = q_i(D) + Z_i$ with independent $Z_i \sim \mathcal{N}(0, \sigma^2)$, where*

$$\varepsilon_0 = \frac{\alpha}{128\sqrt{2k\ln(64/(\alpha\beta))}}, \qquad \sigma = \frac{\sqrt{2\ln(64k/(\alpha\beta)) + \varepsilon_0}}{\varepsilon_0\, n}. \tag{16.9}$$

If

$$n \geq \frac{2048\sqrt{3}}{\alpha^2}\sqrt{k\ln\frac{64}{\alpha\beta}}\ \ln\frac{64k}{\alpha\beta}, \tag{16.10}$$

then for every analyst, $\Pr\left[\exists i \colon |a_i - q_i(\mathcal{D})| > \alpha\right] \leq \beta$.

Proof.

1. *Each answer is private.* With $\delta_0 = \alpha\beta/(64k)$, the noise is the Gaussian mechanism's for sensitivity $1/n$ and budget $(\varepsilon_0, \delta_0)$. For a fixed analyst, the i-th answer is this mechanism applied to a query fixed by the earlier answers and the coins of the analyst.
2. *All answers are $(\alpha/64, \alpha\beta/32)$-DP.* The corollary of advanced composition applies with $\varepsilon = \alpha/64$ and $\delta' = \alpha\beta/64$. The total δ is $k\delta_0 + \delta' = \alpha\beta/32$. The queries in the transcript are a post-processing of the answers and the coins of the analyst, so the transcript has the same guarantee.
3. *Accuracy on the sample.* The error $a_i - q_i(D)$ is the noise Z_i, and a Gaussian variable exceeds t standard deviations in absolute value with probability at most $2e^{-t^2/2}$. With $t = \sqrt{2\ln(32k/(\alpha\beta))}$, each $|Z_i|$ exceeds $t\sigma$ with probability at most $\alpha\beta/(16k)$, so by a union bound all k answers are within $t\sigma$ of the sample values with probability at least $1 - \alpha\beta/16$.

4. *The threshold.* Write $\ell = \ln(64k/(\alpha\beta))$. Since $\varepsilon_0 < 1 < \ell$, we have $2\ell + \varepsilon_0 \le 3\ell$, and $t \le \sqrt{2\ell}$. So

$$t\sigma \le \frac{\sqrt{6}\,\ell}{\varepsilon_0 n} = \frac{256\sqrt{3}}{\alpha n}\sqrt{k \ln \frac{64}{\alpha\beta}}\;\ell, \tag{16.11}$$

which Equation 16.10 makes at most $\alpha/8$. Steps 2 to 4 are the hypotheses of Theorem 16.1, which gives the claim. □

The $\sqrt{k}$ in Equation 16.10 is the $\sqrt{k}$ of advanced composition. The constant is large because the constants of the transfer theorem are. But the number of queries that a sample supports grows from linear in n, for sample splitting, to quadratic in n. For a finite universe and a very large k, private multiplicative weights does better still. For the average error over the queries, it needs a sample only logarithmic in k (Bassily et al., 2016, Corollary 6.3).

16.4. An easier transfer theorem

A weaker transfer theorem, with accuracy in expectation instead of with high probability, has a short complete proof. It turns a bound on the expected worst error on the sample into one on the population.

Theorem 16.2 (Transfer in expectation)**.** *Let M be (ε, δ)-DP against adaptive analysts. Suppose that for every analyst, if the records are independent draws from $\mathcal{D}$, then*

$$\mathbb{E}\Big[\max_{i \le k} |a_i - q_i(D)|\Big] \le \alpha. \tag{16.12}$$

Then for every analyst,

$$\mathbb{E}\Big[\max_{i \le k} |a_i - q_i(\mathcal{D})|\Big] \le \alpha + e^{\varepsilon} - 1 + \delta. \tag{16.13}$$

For small ε, the excess is about $\varepsilon + \delta$. Most of the work is in a lemma about one query. It says that a query chosen by a DP algorithm has on average about the same value on the sample as on the population. So a DP algorithm cannot produce the last query of Proposition 16.2, on which the two disagree.

Lemma 16.1 (A private query generalizes)**.** *Let W be an (ε, δ)-DP algorithm that maps a dataset in $\mathcal{X}^n$ to a statistical query. Let $X_1, \ldots, X_n$ be independent draws from $\mathcal{D}$ and $q = W(D)$. Then*

$$\big|\mathbb{E}[q(D)] - \mathbb{E}[q(\mathcal{D})]\big| \le e^{\varepsilon} - 1 + \delta, \tag{16.14}$$

with expectations over the sample and the coins of W.

Proof.

1. *Linearity.* Since q takes values in $\{0,1\}$,

$$\mathbb{E}[q(D)] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[q(X_i)] = \frac{1}{n}\sum_{i=1}^{n}\Pr[q(X_i)=1]. \tag{16.15}$$

2. *A fresh record.* Fix i, draw X_i' from $\mathcal{D}$ independently of everything else, and let $D^{(i)}$ be D with entry i replaced by X_i'. The datasets D and $D^{(i)}$ are neighbours, $D \sim D^{(i)}$, since they differ in one replaced record.
3. *Privacy.* Write $W(D)(X_i)$ for the value of the query $W(D)$ on the record X_i. For fixed values of D and X_i', the queries q' with $q'(X_i)=1$ form an event about the output of W. DP for the neighbours D and $D^{(i)}$ gives

$$\Pr_W\left[W(D)(X_i)=1\right] \le e^{\varepsilon}\Pr_W\left[W(D^{(i)})(X_i)=1\right] + \delta. \tag{16.16}$$

The average over D and X_i' keeps the inequality. 4. *Swapping.* The pair $(D^{(i)}, X_i)$ has the same distribution as the pair (D, X_i'), since each is n independent draws from $\mathcal{D}$ plus one more independent draw. Hence $\Pr[W(D^{(i)})(X_i)=1] = \Pr[q(X_i')=1]$. The record X_i' is independent of $q = W(D)$, so this probability is $\mathbb{E}[q(\mathcal{D})]$. 5. *One direction.* Steps 1, 3 and 4 give $\mathbb{E}[q(D)] \le e^{\varepsilon}\,\mathbb{E}[q(\mathcal{D})] + \delta$. Since $\mathbb{E}[q(\mathcal{D})] \le 1$,

$$\mathbb{E}[q(D)] - \mathbb{E}[q(\mathcal{D})] \le (e^{\varepsilon}-1)\,\mathbb{E}[q(\mathcal{D})] + \delta \le e^{\varepsilon} - 1 + \delta. \tag{16.17}$$

6. *Other direction.* DP applied from $D^{(i)}$ to D gives $\Pr[W(D^{(i)})(X_i)=1] \le e^{\varepsilon}\Pr[W(D)(X_i)=1] + \delta$. With steps 1 and 4, this is $\mathbb{E}[q(\mathcal{D})] \le e^{\varepsilon}\,\mathbb{E}[q(D)] + \delta$, and the same argument as in step 5 bounds $\mathbb{E}[q(\mathcal{D})] - \mathbb{E}[q(D)]$ by $e^{\varepsilon} - 1 + \delta$.

□

The proof of Theorem 16.2 lets a private algorithm pick out the query with the worst error, and two tricks make that query fit the lemma.

Proof.

1. *Removing the absolute value.* Whenever the analyst asks q_i and receives a_i, add the query $1 - q_i$ with the answer $1 - a_i$. The extended transcript is a post-processing of the original, so it is still (ε, δ)-DP. The errors on the sample are unchanged, because $|(1-a_i) - (1-q_i)(D)| = |a_i - q_i(D)|$. Since $(1-q_i)(\mathcal{D}) - (1-a_i) = -(q_i(\mathcal{D}) - a_i)$, the $2k$ queries q_j satisfy

$$\max_{i \le k}|q_i(\mathcal{D}) - a_i| = \max_{j \le 2k}(q_j(\mathcal{D}) - a_j). \tag{16.18}$$

2. *The selector.* Let W run the interaction on D and output the query $q^* = q_{j^*}$, with answer $a^* = a_{j^*}$, where j^* maximizes $q_j(\mathcal{D}) - a_j$. The selector uses $\mathcal{D}$, which is fixed and is not the data, so W is a post-processing of the transcript and is (ε, δ)-DP. It is a device of the proof and is never run.
3. *Splitting the error.* By linearity of expectation,

$$\mathbb{E}\Big[\max_i |q_i(\mathcal{D}) - a_i|\Big] = \mathbb{E}[q^*(\mathcal{D}) - q^*(D)] + \mathbb{E}[q^*(D) - a^*]. \tag{16.19}$$

4. *The two terms.* Lemma 16.1, applied to W, bounds the first term by $e^{\varepsilon} - 1 + \delta$. The second is at most $\mathbb{E}[\max_j |q_j(D) - a_j|] \le \alpha$, by the accuracy hypothesis and step 1.

□

The Gaussian mechanism applies this theorem too, with a proof that cites nothing. The only change from Corollary 16.1 is that the error on the sample is bounded in expectation.

Corollary 16.2. *Let $\alpha \in (0, 1]$. Answer each query by $a_i = q_i(D) + Z_i$ with independent $Z_i \sim \mathcal{N}(0, \sigma^2)$, where*

$$\varepsilon_0 = \frac{\alpha}{2\sqrt{2k \ln(2/\alpha)}}, \qquad \sigma = \frac{\sqrt{2\ln(2k/\alpha) + \varepsilon_0}}{\varepsilon_0\, n}. \tag{16.20}$$

If

$$n \ge \frac{4\sqrt{3}}{\alpha^2}\sqrt{k \ln \frac{2}{\alpha}}\ \ln \frac{2k}{\alpha}, \tag{16.21}$$

then for every analyst, $\mathbb{E}[\max_i |a_i - q_i(\mathcal{D})|] \le 4\alpha$.

Proof.

1. *Privacy.* As in steps 1 and 2 of the proof of Corollary 16.1, each answer is the Gaussian mechanism with budget $(\varepsilon_0, \alpha/(2k))$, and the corollary of advanced composition with $\varepsilon = \alpha$ and $\delta' = \alpha/2$ makes the mechanism (α, α)-DP against adaptive analysts.
2. *Expected worst noise.* For $\lambda > 0$, Jensen's inequality and $e^{\lambda \max_i |Z_i|} \le \sum_i e^{\lambda |Z_i|}$ give

$$e^{\lambda\, \mathbb{E}[\max_i |Z_i|]} \le \sum_{i=1}^{k} \mathbb{E}[e^{\lambda |Z_i|}] \le 2k\, e^{\lambda^2 \sigma^2 / 2}, \tag{16.22}$$

because $\mathbb{E}[e^{\lambda|Z|}] \le \mathbb{E}[e^{\lambda Z}] + \mathbb{E}[e^{-\lambda Z}] = 2e^{\lambda^2\sigma^2/2}$. The logarithm, with $\lambda = \sqrt{2\ln(2k)}/\sigma$, gives $\mathbb{E}[\max_i |Z_i|] \le \sigma\sqrt{2\ln(2k)}$. 3. *The threshold.* Write $\ell = \ln(2k/\alpha) \ge \ln 2$. Since $\varepsilon_0 \le 1/(2\sqrt{2\ln 2}) < \ln 2 \le \ell$, we have $2\ell + \varepsilon_0 \le 3\ell$, and $\ln(2k) \le \ell$. So

$$\sigma\sqrt{2\ln(2k)} \le \frac{\sqrt{6}\,\ell}{\varepsilon_0 n} = \frac{4\sqrt{3}}{\alpha n}\sqrt{k \ln \frac{2}{\alpha}}\ \ell, \tag{16.23}$$

which the condition on n makes at most α. Hence the expected worst error on the sample is at most α. 4. Theorem 16.2 with $\varepsilon = \delta = \alpha$ gives at most $\alpha + (e^{\alpha} - 1) + \alpha \le 4\alpha$, since $e^{\alpha} - 1 \le 2\alpha$ for $\alpha \le 1$. □

16.5. Summary

Table 16.1.: The chapter's results.

Result	Statement
Proposition 16.1	k fixed queries generalize once $n \geq \ln(2k/\beta)/(2\alpha^2)$
Proposition 16.2	exact answers to k adaptive queries can be off by $1 - n/(k-1)$ on the population
Proposition 16.3	sample splitting generalizes once $n \geq k\ln(2k/\beta)/(2\alpha^2)$
Theorem 16.1	an $(\alpha/64, \alpha\beta/32)$-DP mechanism with error $\alpha/8$ on the sample, except with probability $\alpha\beta/16$, has error α on the population, except with probability β
Corollary 16.1	Gaussian answers generalize with n of order $\sqrt{k}$ up to logarithmic factors
Lemma 16.1	a query chosen by an (ε, δ)-DP algorithm has $\lvert\mathbb{E}[q(D)] - \mathbb{E}[q(\mathcal{D})]\rvert \leq e^{\varepsilon} - 1 + \delta$
Theorem 16.2	expected worst error on the population is at most $\alpha + e^{\varepsilon} - 1 + \delta$
Corollary 16.2	Gaussian answers give expected worst error 4α with n of order $\sqrt{k}$

16.6. References

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- Cynthia Dwork, Guy N. Rothblum and Salil Vadhan, "Boosting and Differential Privacy", 2010, which proves advanced composition.

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Part V.

Deployments

The earlier parts developed differential privacy (DP) with a trusted curator who holds the data and releases noisy answers. This part studies deployments with many users, repeated data collection, and external analysts. The first chapter removes the curator: each person randomizes their own data, and the chapter quantifies the resulting error. The second introduces the shuffle model, then examines selection, composition, and budget allocation in central deployments.

Table 16.2.: The chapters of this part.

Chapter	Question it answers
Local differential privacy	What does it cost to trust no one, and how do large deployments live with that cost?
The shuffle model and central deployments	How much does an anonymizing shuffler recover, and how do central deployments select, compose and budget?

17. Local differential privacy

Previous mechanisms assumed a trusted curator with access to raw data. The local model removes that assumption: each person randomizes their record before sending it. This chapter defines the model and shows that, for a basic estimation task, accuracy costs a factor of about $\sqrt{n}$. No noninteractive protocol avoids this penalty.

The chapter then analyses three deployments that address limitations of the basic protocol. Google's RAPPOR protects repeated answers, Apple's protocol discovers previously unknown values, and Microsoft's protocol encodes numeric data in one bit and handles changing values.

17.1. The local model

In the central model from the first chapter on the definition, people send raw data to a curator, and privacy constrains the curator's output. In the local model, each person randomizes their data before transmission. An adversary may observe every outgoing message; the curator only analyses those messages.

Definition 17.1 (Local randomizer)**.** Let $\varepsilon \geq 0$. A randomized algorithm $R\colon \mathcal{X} \to \mathcal{Y}$ is an **ε-local randomizer** if for all inputs $x, x' \in \mathcal{X}$ and all events $S \subseteq \mathcal{Y}$,

$$\Pr[R(x) \in S] \leq e^{\varepsilon} \Pr[R(x') \in S]. \tag{17.1}$$

This is differential privacy (DP) applied to a dataset with one record. The simplest protocol has each person send one message simultaneously.

Definition 17.2 (Noninteractive local protocol)**.** A **noninteractive ε-local protocol** gives each person i an ε-local randomizer R_i. Person i sends $Y_i = R_i(X_i)$, using randomness independent of everyone else's, and the analyst outputs a function of the messages $(Y_1, \dots, Y_n)$.

Local privacy implies central privacy: the analyst sees all messages, but only one message depends on any given record.

Proposition 17.1. *The messages $(Y_1, \dots, Y_n)$ of a noninteractive ε-local protocol, viewed as the output of a mechanism on D, are ε-DP. So is any function of them that the analyst computes.*

Proof.

1. Let $D \sim D'$ be neighbours, which differ by replacing entry j, and take an event S. Write y_{-j} for a value of all the messages except the j-th, and $S_{y_{-j}} = \{y_j : (y_j, y_{-j}) \in S\}$ for the slice of S through it.
2. The other messages have the same distribution under D and D', because they depend only on the other records and on independent randomness. Conditioning on them gives

$$\Pr_{D}[Y \in S] = \mathbb{E}_{Y_{-j}}\left[\Pr[R_j(X_j) \in S_{Y_{-j}}]\right] \le e^{\varepsilon}\, \mathbb{E}_{Y_{-j}}\left[\Pr[R_j(X'_j) \in S_{Y_{-j}}]\right] = e^{\varepsilon} \Pr_{D'}[Y \in S], \tag{17.2}$$

where the inequality applies Definition 17.1 to each slice. 3. The analyst's output is a post-processing of the messages, so it is ε-DP by the post-processing property of the Laplace mechanism chapter. □

Randomized response reports a true bit with probability $e^{\varepsilon}/(1+e^{\varepsilon})$ and is therefore an ε-local randomizer. The book's first private algorithm was a local protocol.

Protocols may also be interactive. In a **sequentially interactive** protocol, the analyst contacts people in sequence; each person's randomizer may depend on earlier messages, but each person speaks once. In a **fully interactive** protocol, a person may answer repeatedly, and the entire message sequence must be an ε-local randomizer. This chapter proves bounds only for noninteractive protocols.

17.2. Counting bits, central and local

The cost of this trust change appears in bit-mean estimation. Each person holds $X_i \in \{0,1\}$, and the analyst wants $\mu = \frac{1}{n}\sum_i X_i$. Write $\mathrm{Lap}(b)$ for the Laplace distribution with density $\frac{1}{2b}e^{-|z|/b}$. The Laplace mechanism adds $\mathrm{Lap}(\Delta f/\varepsilon)$ noise to a query with sensitivity Δf. A central curator can therefore privatize the mean directly.

Proposition 17.2. *For every $\varepsilon > 0$ and $\beta \in (0,1)$, the central estimate $\hat{\mu} = \mu + \mathrm{Lap}(1/(\varepsilon n))$ is ε-DP, and with probability at least $1-\beta$,*

$$|\hat{\mu} - \mu| \le \frac{\ln(1/\beta)}{\varepsilon n}. \tag{17.3}$$

Proof.

1. Changing one bit changes μ by at most $1/n$, so the Laplace mechanism with $\Delta f = 1/n$ is ε-DP.
2. The Laplace tail is $\Pr[|\mathrm{Lap}(b)| > t] = 2\int_t^{\infty} \frac{1}{2b}e^{-z/b}\,dz = e^{-t/b}$. Setting $b = 1/(\varepsilon n)$ and $t = b\ln(1/\beta)$ makes the tail equal to β.

□

In the local model, each person applies randomized response and sends Y_i: it equals X_i with probability $e^\varepsilon/(1+e^\varepsilon)$ and $1-X_i$ otherwise. Since $\Pr[Y_i=1]=(1+(e^\varepsilon-1)X_i)/(e^\varepsilon+1)$ is linear in X_i, inverting this relation gives the unbiased estimate

$$\hat{\mu}=\frac{1}{n}\sum_{i=1}^{n}\frac{(e^\varepsilon+1)\,Y_i-1}{e^\varepsilon-1}. \tag{17.4}$$

This is the debiased estimator from the randomized response chapter, expressed using ε through $2\kappa=(e^\varepsilon-1)/(e^\varepsilon+1)$. That chapter proves that, with probability at least $1-\beta$,

$$|\hat{\mu}-\mu|\le \coth\!\left(\frac{\varepsilon}{2}\right)\sqrt{\frac{\ln(2/\beta)}{2n}},\qquad \coth\!\left(\frac{\varepsilon}{2}\right)\ge\frac{2}{\varepsilon}. \tag{17.5}$$

The central bound guarantees error α when $n\ge \ln(1/\beta)/(\varepsilon\alpha)$. The local bound requires $n\ge \coth^2(\varepsilon/2)\ln(2/\beta)/(2\alpha^2)$, at least $2\ln(2/\beta)/(\varepsilon^2\alpha^2)$. Thus, as Figure 17.1 shows, the local estimate is quadratically more costly in both $1/\alpha$ and $1/\varepsilon$.

Example 17.1. Take $\varepsilon=1$, $\alpha=0.01$ and $\beta=0.05$. The central estimate needs $n\ge \ln 20/0.01$, that is, 300 people. The local one needs $n\ge \coth^2(0.5)\ln 40/(2\times 10^{-4})$, that is, 86,370 people, about 288 times as many.

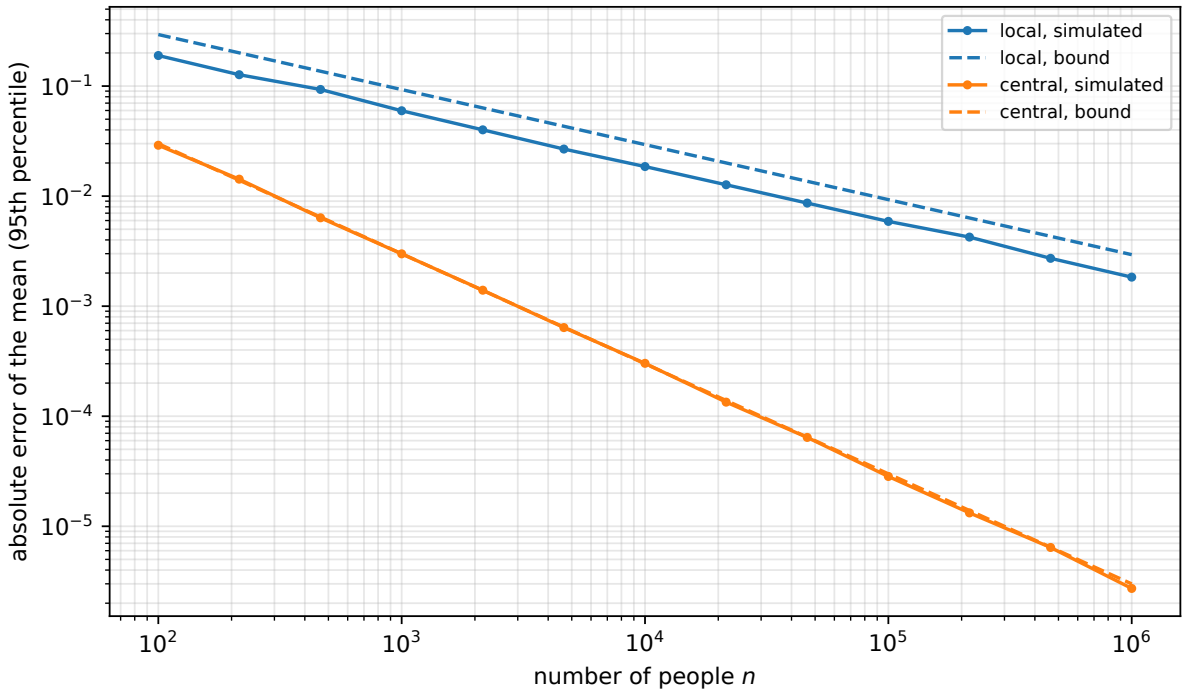


Figure 17.1.: Error of the mean of n bits at $\varepsilon=1$. Solid lines are the 95th percentile of the error over 2,000 simulated runs with $\mu=0.3$, dashed lines the bounds of Proposition 17.2 and Equation 17.5 with $\beta=0.05$. The central error falls as $1/n$ and the local one as $1/\sqrt{n}$, so the gap widens with n.

This gap helps explain why local privacy has been deployed mainly by companies with hundreds of millions of users. With only thousands of users, the estimate can be dominated by noise.

17.3. The price is unavoidable

This error is not specific to randomized response: every noninteractive local protocol has error at least order $1/(\varepsilon\sqrt{n})$. Local randomization contracts differences between input distributions by a factor determined by ε, and n independent messages cannot recover the lost information.

Two distances between distributions P and Q on a countable set measure this. The **total variation distance** is $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_y |P(y)-Q(y)|$, the largest gap $P(S)-Q(S)$ over events S. The **chi-squared divergence** is

$$\chi^2(P\,\|\,Q) = \sum_y \frac{(P(y)-Q(y))^2}{Q(y)}, \tag{17.6}$$

where a term in which both $P(y)$ and $Q(y)$ are zero is dropped, and a term in which only $Q(y)$ is zero makes the divergence infinite. The lower bound needs two facts about these two distances. Total variation is at most a constant times the square root of the chi-squared divergence, and the chi-squared divergence behaves simply on products, which is where the n independent messages enter.

Lemma 17.1. *For distributions on a countable set,*

1. $\mathrm{TV}(P,Q) \le \frac{1}{2}\sqrt{\chi^2(P\,\|\,Q)}$;
2. *for product distributions,* $1+\chi^2(P_1\times P_2\,\|\,Q_1\times Q_2) = (1+\chi^2(P_1\,\|\,Q_1))(1+\chi^2(P_2\,\|\,Q_2))$.

Proof.

1. Write $|P(y)-Q(y)| = \frac{|P(y)-Q(y)|}{\sqrt{Q(y)}}\sqrt{Q(y)}$ and apply the Cauchy–Schwarz inequality to the sum. Since $\sum_y Q(y) = 1$, this gives $\sum_y |P(y)-Q(y)| \le \sqrt{\chi^2(P\,\|\,Q)}$.
2. Expanding the square shows $1+\chi^2(P\,\|\,Q) = \sum_y P(y)^2/Q(y)$. For a product, this sum factorizes as $\sum_{y_1} P_1(y_1)^2/Q_1(y_1)\cdot\sum_{y_2} P_2(y_2)^2/Q_2(y_2)$.

□

The key step is a contraction. Feed a local randomizer an input drawn from P_0 or from P_1, and compare the two resulting distributions of its message.

Lemma 17.2. *Let R be an ε-local randomizer with a countable message set, and let P_0, P_1 be distributions on $\mathcal{X}$. Write W_0 and W_1 for the distributions of the message $R(x)$ when x is drawn from P_0 and from P_1. Then*

$$\chi^2(W_0\,\|\,W_1) \le (e^\varepsilon-1)^2\,\mathrm{TV}(P_0,P_1)^2. \tag{17.7}$$

Proof.

1. Fix a message y and write $Q(y \mid x) = \Pr[R(x) = y]$ and $c_y = \inf_x Q(y \mid x)$. By Definition 17.1, every $Q(y \mid x)$ lies in $[c_y, e^{\varepsilon} c_y]$. If $c_y = 0$, then $Q(y \mid x) = 0$ for every x, and y contributes nothing.
2. Since $\sum_x (P_0(x) - P_1(x)) = 0$, subtracting the midpoint $m_y = (1 + e^{\varepsilon})c_y/2$ changes nothing:

$$W_0(y) - W_1(y) = \sum_x \bigl(Q(y \mid x) - m_y\bigr)\bigl(P_0(x) - P_1(x)\bigr). \tag{17.8}$$

3. Each factor $|Q(y \mid x) - m_y|$ is at most $(e^{\varepsilon} - 1)c_y/2$, and $\sum_x |P_0(x) - P_1(x)| = 2\,\mathrm{TV}(P_0, P_1)$. So $|W_0(y) - W_1(y)| \le (e^{\varepsilon} - 1)c_y\,\mathrm{TV}(P_0, P_1)$.
4. The denominator satisfies $W_1(y) \ge c_y$. Hence each term of χ^2 is at most $(e^{\varepsilon} - 1)^2 c_y\,\mathrm{TV}(P_0, P_1)^2$.
5. Summing over y and using $\sum_y c_y \le \sum_y Q(y \mid x) = 1$ for any fixed x gives the bound.

□

Now let the data be random. If each X_i is drawn independently from $\mathrm{Bern}(p)$, the Bernoulli distribution with mean p, then an accurate estimate of the mean must tell $p = \frac{1}{2} - \alpha$ from $p = \frac{1}{2} + \alpha$.

Theorem 17.1 (Lower bound for local counting)**.** *Let $\alpha \in (0, \frac{1}{2}]$, and let $X_1, \dots, X_n$ be independent draws from $\mathrm{Bern}(\frac{1}{2} + \alpha)$ or from $\mathrm{Bern}(\frac{1}{2} - \alpha)$. Write P_+ and P_- for the two distributions of the messages of a noninteractive ε-local protocol with countable message sets. Then*

$$\mathrm{TV}(P_+, P_-) \le \frac{1}{2}\sqrt{\exp\bigl(4n(e^{\varepsilon} - 1)^2\alpha^2\bigr) - 1}. \tag{17.9}$$

Consequently, if an estimate $\tilde{\mu}$ computed from the messages satisfies $|\tilde{\mu} - p| < \alpha$ with probability at least $3/4$ in both cases, then

$$n \ge \frac{\ln 2}{4(e^{\varepsilon} - 1)^2\alpha^2}, \qquad \text{and for } \varepsilon \le 1, \quad n \ge \frac{1}{18\,\varepsilon^2\alpha^2}. \tag{17.10}$$

Proof.

1. The two input distributions have $\mathrm{TV}(\mathrm{Bern}(\frac{1}{2} + \alpha), \mathrm{Bern}(\frac{1}{2} - \alpha)) = 2\alpha$. By Lemma 17.2, each person's message satisfies $\chi^2 \le 4(e^{\varepsilon} - 1)^2\alpha^2$ between the two cases.
2. The messages are independent, because the records and the coins are. By the product rule of Lemma 17.1 and $1 + z \le e^z$, we get $1 + \chi^2(P_+ \,\|\, P_-) \le \exp(4n(e^{\varepsilon} - 1)^2\alpha^2)$. The first part of the same lemma gives the bound on TV.
3. Take the test that answers "$p = \frac{1}{2} + \alpha$" when $\tilde{\mu} \ge \frac{1}{2}$. In each case it errs only if $|\tilde{\mu} - p| \ge \alpha$, so each error probability is at most $\frac{1}{4}$ and the two add up to at most $\frac{1}{2}$. For any test, the error probabilities add up to at least $1 - \mathrm{TV}(P_+, P_-)$, because the chance of answering "+" differs between the cases by at most TV. So $\mathrm{TV}(P_+, P_-) \ge \frac{1}{2}$.
4. Combining steps 2 and 3 gives $\exp(4n(e^{\varepsilon} - 1)^2\alpha^2) \ge 2$, which is the first bound on n.
5. For $\varepsilon \le 1$, convexity of e^{ε} gives $e^{\varepsilon} - 1 \le (e - 1)\varepsilon$, and $\ln 2/(4(e - 1)^2) = 0.0587 > 1/18$.

□

The theorem is about data drawn at random, but it also constrains estimates of the empirical mean on fixed data. Suppose a protocol is within $\alpha/2$ of the empirical mean with probability at least $7/8$ on every dataset. When $n \geq 2\ln(16)/\alpha^2$, Hoeffding's inequality puts the empirical mean strictly within $\alpha/2$ of p with probability at least $7/8$. So the protocol is strictly within α of p with probability at least $3/4$, and the theorem applies to it. This reduction says nothing about smaller n, and Chan, Shi and Song (2012) give a direct argument for fixed data.

For $\varepsilon \leq 1$, comparing the theorem with Equation 17.5 shows that randomized response is optimal up to a constant factor. The central estimate, which needs only about $1/(\varepsilon\alpha)$ people, is out of reach of every noninteractive local protocol.

17.4. Repeated questions: RAPPOR

Google's first large deployment, RAPPOR, collected statistics from the Chrome browser. It met a problem that a one-off survey never has. The same question goes to the same person every day, and fresh randomness each day leaks the answer.

Example 17.2 (Averaging attack)**.** Take $\varepsilon = \ln 4$, so that randomized response tells the truth with probability $4/5$. A single "yes" from a person whose answer is "no" happens with probability $1/5$, which leaves good deniability. Now ask the same question on 50 days with fresh coins, and suppose 40 reports say "yes". If the answer is "no", this happens with probability $\binom{50}{40}(1/5)^{40}(4/5)^{10} = 1.21 \times 10^{-19}$. If it is "yes", it happens with probability 0.140. The likelihood ratio is $4^{30} \approx 1.15 \times 10^{18}$, and the deniability is gone.

The attack is composition at work. Basic composition, from the Laplace mechanism chapter, says that k answers with fresh coins are $k\varepsilon$-DP, which here is $50 \ln 4 = 69.3$. The attack meets this bound exactly, because the all-"yes" outcome has likelihood ratio 4^{50}. So the fix is to randomize once and repeat.

Definition 17.3 (Memoization)**.** A person who is asked a question for the first time runs their randomizer and stores the result. When asked the same question again, they send the stored result.

Memoization keeps the privacy cost of a single answer, however often the question comes.

Proposition 17.3. *If the stored result comes from an ε-local randomizer, then any number of memoized answers to the same question is ε-locally private.*

Proof. The sequence of answers is a function of the single stored result, so the claim follows from post-processing. □

Memoization protects honest repetition, not repetition by an adversarial analyst. For example, an analyst can ask for "your PIN" and then for "your PIN reversed", two questions with the same information that each get a fresh randomization.

RAPPOR handles values from any finite domain, not just bits, so it first encodes a value as a bit vector.

Definition 17.4 (Bloom filter)**.** Fix a length K and h hash functions $H_1, \dots, H_h : \mathcal{X} \to \{1, \dots, K\}$. The Bloom filter of a value v is the vector $B \in \{0,1\}^K$ with $B_j = 1$ exactly when $H_l(v) = j$ for some l.

With $h = 1$ and no collisions the Bloom filter is a one-hot encoding of v. In general, B has at most h ones. RAPPOR then applies two layers of randomized response, bit by bit.

Definition 17.5 (RAPPOR)**.** Fix $\nu \in (0,1)$, the noise level of the permanent layer, and probabilities $p_0 < p_1$ in $(0,1)$, the chances of reporting a one in the instantaneous layer. A person with value v and Bloom filter B of length K, from Definition 17.4, does the following.

1. **Permanent randomized response.** Once, they draw $B' \in \{0,1\}^K$ with independent coordinates, where B'_j is 1 with probability $\nu/2$, 0 with probability $\nu/2$, and B_j otherwise. They memoize B'.
2. **Instantaneous randomized response.** Each time they report, they send a fresh $Y \in \{0,1\}^K$ with independent coordinates, where $Y_j = 1$ with probability p_1 if $B'_j = 1$ and with probability p_0 if $B'_j = 0$.

The permanent layer carries the long-term guarantee. It holds against an analyst who sees every report the person ever sends, because all of them come from the one memoized vector.

Theorem 17.2. *Run RAPPOR with h hash functions and permanent noise level ν. As a function of the value v, the memoized vector B' is an ε_∞-local randomizer with*

$$\varepsilon_\infty = 2h \ln \frac{1 - \nu/2}{\nu/2}. \tag{17.11}$$

So is the whole sequence of reports Y, however long.

Proof.

1. Let v, v' have Bloom filters $B, \tilde{B}$. Each has at most h ones, so they differ in at most $2h$ coordinates.
2. Fix an outcome $b \in \{0,1\}^K$. Coordinates are independent, so $\Pr[B' = b]$ is a product over j. The factors at coordinates where $B_j = \tilde{B}_j$ are equal for v and v' and cancel in the ratio.
3. Each remaining factor is $1 - \nu/2$ or $\nu/2$, so each ratio is at most $(1 - \nu/2)/(\nu/2)$, and there are at most $2h$ of them. By the pointwise lemma of the randomized response chapter, this bound on single outcomes gives ε_∞-privacy.
4. The reports depend on v only through B', so they are a post-processing of it.

□

The instantaneous layer gives a stronger guarantee for an analyst who sees one report. With both layers, a reported bit is 1 with probability $p_1^* = (1 - \frac{\nu}{2})p_1 + \frac{\nu}{2}p_0$ when $B_j = 1$ and $p_0^* = \frac{\nu}{2}p_1 + (1 - \frac{\nu}{2})p_0$ when $B_j = 0$. Then $p_1^* > p_0^*$, because $p_1^* - p_0^* = (1 - \nu)(p_1 - p_0)$.

Theorem 17.3. *Run RAPPOR with h hash functions, and let p_1^* and p_0^* be the combined probabilities above. A single report Y, as a function of v, is an ε_1-local randomizer with*

$$\varepsilon_1 = h \ln \frac{p_1^*(1 - p_0^*)}{p_0^*(1 - p_1^*)}. \tag{17.12}$$

Proof.

1. Given B, the coordinates of Y are independent, with $\Pr[Y_j = 1] = p_1^*$ or p_0^* according to B_j. As before, coordinates where $B_j = \tilde{B}_j$ cancel.
2. Let A be the set of coordinates with $B_j = 1$ and $\tilde{B}_j = 0$, and A' the set with the reverse. Both have at most h elements, since each lies inside the ones of one filter.
3. On A the ratio is $p_1^*/p_0^* > 1$ if $y_j = 1$ and $(1 - p_1^*)/(1 - p_0^*) < 1$ if $y_j = 0$, so it is at most p_1^*/p_0^*. On A' it is at most $(1 - p_0^*)/(1 - p_1^*)$ by the same reasoning.
4. Multiplying, the ratio is at most $\left(p_1^*(1-p_0^*)/(p_0^*(1-p_1^*))\right)^h$, and the pointwise lemma finishes the proof.

□

Typical parameters show how far apart the two guarantees can be.

Example 17.3. With $h = 2$, $\nu = 1/2$, $p_0 = 1/2$ and $p_1 = 3/4$, the long-term guarantee is $\varepsilon_\infty = 4 \ln 3 = 4.39$. The combined probabilities are $p_1^* = 0.6875$ and $p_0^* = 0.5625$, so a single report has $\varepsilon_1 = 1.07$.

The second layer has one more purpose, which local privacy does not capture. A memoized B' of a few hundred bits is almost surely unique, so two reports that both reveal it link to the same person, as a cookie would. Fresh noise on each report breaks that link for an analyst who sees few reports. An analyst who sees many reports can average the noise away and recover B', but then still faces the ε_∞ guarantee of Theorem 17.2.

The analyst decodes the reports by debiasing the bit counts and fitting candidate values against them, in the spirit of Equation 17.4. Memoization fails when the value drifts. For example, a person's age in days changes daily, each new value needs a new B', and k distinct values cost $k\,\varepsilon_\infty$ by composition.

17.5. Values no one knew: Apple

Apple deployed local privacy on its phones and computers. Around the randomizers it put measures outside the model of DP, for example it discards network addresses and limits the number of reports per day. Independent researchers later measured from the software of macOS that each report had ε of 1 or 2, but that a day could spend up to 16.

Most of Apple's protocols build on a private count mean sketch, a hashed table that estimates frequencies. A device hashes its value into a short table with one of several hash functions, and randomizes the table before it sends it. A variant with the Hadamard transform lets each device send a single bit. That matters on a phone, where a report must cost a few bytes at most.

The new problem Apple solved is **discovery**. A hash maps a value to its encoding, but the encoding cannot be inverted. So to estimate how often a word is typed, the analyst must already have the word in its dictionary and test its encoding against the reports. Slang and new words are exactly the values that no dictionary holds, and a search of all strings is hopeless, since there are $26^8 \approx 2.1 \times 10^{11}$ strings of eight letters.

The **sequence fragment puzzle** breaks a word into pieces short enough to enumerate. Take, for illustration, eight-letter words and fragments of two letters, and fix an eight-bit hash function H from words to $\{0, \dots, 255\}$. A person with an eight-letter word w picks one of the starting positions $j \in \{1, 3, 5, 7\}$ uniformly at random. They send j together with a private report of the *puzzle piece* $(w_j w_{j+1}, H(w))$, the two letters at position j and the hash of the whole word. The pieces come from a set of $26^2 \times 256 = 173{,}056$ candidates, small enough for the analyst to estimate the frequency of every one at each position.

The position needs no protection, because the person chooses it at random, independently of the word, as the next result shows.

Proposition 17.4. *If the private report of the pair is an ε-local randomizer, then the whole message (j, report) is ε-locally private.*

Proof. The position j is drawn independently of w. Conditional on j, the message is an ε-local randomizer of w. For an event S, write S_j for the set of reports r with $(j, r) \in S$. Then $\Pr[(j, \text{report}) \in S \mid w] = \sum_j \frac{1}{4} \Pr[\text{report} \in S_j \mid w, j]$ is at most e^{ε} times the same sum for any other word. □

An example shows how the analyst puts the pieces back together.

Example 17.4. Suppose many people type "snowfall", and $H(\text{snowfall}) = 117$. Their messages are spread over $(1, \text{sn}, 117)$, $(3, \text{ow}, 117)$, $(5, \text{fa}, 117)$ and $(7, \text{ll}, 117)$. Each of the four pairs is frequent at its position. Pieces that share the hash 117 are joined in order of position, giving the candidate "snowfall". A second round of ordinary frequency estimation over the discovered words then counts it, and it removes candidates that a hash collision assembled from different words.

With this method Apple found abbreviations and slang that no dictionary held, and measured the frequency of each emoji, which differed between English and French keyboards.

17.6. Numbers in one bit: Microsoft

Microsoft collects usage telemetry from Windows, such as how long people spend in each application. These values are numbers in a range $[0, m]$, and the analyst wants their mean μ. The Laplace mechanism works locally. Person i sends $Y_i = X_i + \mathrm{Lap}(m/\varepsilon)$, which is ε-local because the identity map on $[0, m]$ changes by at most m. The mean of the reports then has standard deviation exactly $\sqrt{2}\, m/(\varepsilon\sqrt{n})$, which is optimal for local protocols by Theorem 17.1 applied to values in $\{0, m\}$. But each report is a real number of size about m/ε, so it costs many bits, while a single bit sent with a probability that grows linearly with the value suffices.

Definition 17.6 (One-bit mean protocol)**.** Person i, with $X_i \in [0, m]$, sends a bit Y_i with

$$\Pr[Y_i = 1] = \frac{1}{e^\varepsilon + 1} + \frac{X_i}{m} \cdot \frac{e^\varepsilon - 1}{e^\varepsilon + 1}. \tag{17.13}$$

The analyst outputs

$$\hat{\mu} = \frac{m}{n} \sum_{i=1}^{n} \frac{(e^\varepsilon + 1)\, Y_i - 1}{e^\varepsilon - 1}. \tag{17.14}$$

At $X_i = 0$ and $X_i = m$ this is randomized response, and between them it interpolates linearly. The estimate of the analyst inverts that linear relation, as Equation 17.4 did for bits, and it keeps the privacy and the accuracy of randomized response.

Theorem 17.4. *The one-bit protocol is ε-locally private, and $\hat{\mu}$ is unbiased. With probability at least $1 - \beta$,*

$$|\hat{\mu} - \mu| \le m \coth\Big(\frac{\varepsilon}{2}\Big) \sqrt{\frac{\ln(2/\beta)}{2n}}, \tag{17.15}$$

and the standard deviation of $\hat{\mu}$ is at most $m \coth(\varepsilon/2)/(2\sqrt{n})$.

Proof.

1. *Privacy.* For every $x \in [0, m]$, $\Pr[Y = 1]$ lies between $1/(e^\varepsilon + 1)$ and $e^\varepsilon/(e^\varepsilon + 1)$, and so does $\Pr[Y = 0] = 1 - \Pr[Y = 1]$. The ratio of two numbers in that interval is at most e^ε, and the pointwise lemma applies.
2. *Unbiasedness.* Substituting $\mathbb{E}[Y_i] = \Pr[Y_i = 1]$ gives $(e^\varepsilon + 1)\mathbb{E}[Y_i] - 1 = (X_i/m)(e^\varepsilon - 1)$, so each term of the sum has mean X_i/m, and $\mathbb{E}[\hat{\mu}] = \mu$.
3. *Error.* Each term $Z_i = m((e^\varepsilon + 1)Y_i - 1)/(e^\varepsilon - 1)$ takes the two values $-m/(e^\varepsilon - 1)$ and $me^\varepsilon/(e^\varepsilon - 1)$, an interval of length $m(e^\varepsilon + 1)/(e^\varepsilon - 1) = m \coth(\varepsilon/2)$. The Z_i are independent, and Hoeffding's inequality for their average gives the bound, as in the randomized response chapter.
4. *Variance.* A bit has variance at most $1/4$, so $\mathrm{Var}(Z_i) \le m^2 \coth^2(\varepsilon/2)/4$, and the average of n independent terms has variance at most $m^2 \coth^2(\varepsilon/2)/(4n)$.

□

Since $\coth(\varepsilon/2)/2 \approx 1/\varepsilon$ for small ε, one bit loses nothing against the $\sqrt{2}/\varepsilon$ of the Laplace protocol. At $\varepsilon = 1$ the one-bit standard deviation is at most 0.765 times the Laplace one.

17.6.1. Memoization with drifting values

Daily usage is never exactly the same two days running, so the memoization of RAPPOR does not apply. And rounding to bins before memoizing poses a dilemma. Narrow bins change often and force new randomizations, while wide bins make each report wrong by up to half a bin. Randomized rounding solves the second problem. It rounds each value to 0 or to m at random, with probabilities chosen so that the rounded value is right on average, so two bins suffice without bias.

Definition 17.7 (Memoized one-bit protocol)**.** Before any data arrive, person i does three things, independently of their data and of each other.

1. They run the one-bit protocol on the input 0 and store the bit $A_i(0)$.
2. They run it on the input m and store the bit $A_i(m)$.
3. They draw an offset u_i uniformly from $[0, m)$.

On a day when their value is $x \in [0, m]$, they round it to $r_i(x) = m$ if $x + u_i \geq m$ and to $r_i(x) = 0$ otherwise, and they send $A_i(r_i(x))$.

The random offset is what makes the rounding right on average.

Lemma 17.3. *For every $x \in [0, m]$, $\Pr[r_i(x) = m] = x/m$, so $\mathbb{E}[r_i(x)] = x$.*

Proof. The offset is uniform on $[0, m)$, so $\Pr[u_i \geq m - x] = (m - (m - x))/m = x/m$. □

Unbiased rounding, together with stored bits that are themselves outputs of the one-bit protocol, makes the bit of each day look like a fresh run of that protocol.

Theorem 17.5. *On each day, the bit sent by the memoized protocol has exactly the distribution of the one-bit protocol run on that day's value x. So each day's bit is ε-locally private, and the estimator of Definition 17.6, applied to one day's bits, is unbiased with the error bound of Theorem 17.4.*

Proof. The offset and the two stored bits are independent. By Lemma 17.3,

$$\Pr[\text{bit} = 1] = \frac{x}{m} \cdot \frac{e^{\varepsilon}}{e^{\varepsilon} + 1} + \left(1 - \frac{x}{m}\right) \frac{1}{e^{\varepsilon} + 1} = \frac{1}{e^{\varepsilon} + 1} + \frac{x}{m} \cdot \frac{e^{\varepsilon} - 1}{e^{\varepsilon} + 1}, \tag{17.16}$$

which is the probability in Definition 17.6. Different people use independent coins, so the bits of one day are independent, as the error bound requires. □

Over many days, the bits depend on the data only through the sequence of rounded values. While the value of a person stays on one side of their threshold $m - u_i$, they repeat one stored bit, and the analyst learns nothing new. A crossing shows in the reports only if the two stored bits differ, which happens with probability $(1 + e^{2\varepsilon})/(1 + e^{\varepsilon})^2$, or 0.607 at $\varepsilon = 1$. Then the analyst learns the day on which the value crossed a threshold drawn uniformly at random. No ε-local guarantee for arbitrary sequences of values covers this leak, so the deployed protocol adds a further layer of randomness to the sent bits to blur the crossing points.

17.7. Summary

Table 17.1.: The chapter's results.

Result	Statement
Proposition 17.1	a noninteractive ε-local protocol is ε-DP in the central sense
Proposition 17.2	central mean of bits: error $\ln(1/\beta)/(\varepsilon n)$
Equation 17.5	local mean of bits by randomized response: error $\coth(\varepsilon/2)\sqrt{\ln(2/\beta)/(2n)}$
Lemma 17.2	an ε-local randomizer shrinks χ^2 to at most $(e^{\varepsilon} - 1)^2\,\mathrm{TV}^2$
Theorem 17.1	every noninteractive local protocol needs $n \geq \ln 2/(4(e^{\varepsilon} - 1)^2\alpha^2)$
Proposition 17.3	memoized answers to one question cost ε once
Theorem 17.2	RAPPOR's memoized vector is $2h\ln((1 - \nu/2)/(\nu/2))$-locally private
Theorem 17.3	one RAPPOR report is $h\ln\left(p_1^*(1 - p_0^*)/(p_0^*(1 - p_1^*))\right)$-locally private
Proposition 17.4	a fragment report with a public random position keeps its ε
Theorem 17.4	one bit per person: unbiased, ε-local, error $m\coth(\varepsilon/2)\sqrt{\ln(2/\beta)/(2n)}$
Theorem 17.5	memoizing two bits with a random offset reproduces the one-bit protocol each day

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18. The shuffle model and central deployments

The previous chapter showed that local privacy can incur $\sqrt{n}$ times the error of the central model. This chapter studies the shuffle model, where an intermediary mixes messages to provide differential privacy (DP) with less local noise, then examines central deployments: analyst query services and public data releases.

18.1. Encode, shuffle, analyse

Google's Prochlo system addressed a limitation of local privacy. Its designers wanted to count calls from a thousand applications to a hundred system interfaces, or 10^5 pairs. They estimated that a local protocol such as RAPPOR would require about a trillion reports for useful counts.

The architecture has three stages: **encode, shuffle, analyse**. Each person encodes data on their device, for example with a local randomizer. A *shuffler* collects messages, removes metadata such as arrival times and network addresses, and sends them in uniform random order. The *analyst* sees only the shuffled batch. The shuffler's purpose is to hide the sender-message mapping.

Users must trust the shuffler to hide the order. Prochlo uses an Intel SGX trusted-hardware enclave, placing trust in the hardware vendor; side-channel attacks such as Spectre show that this hardware can leak. Alternatively, several servers can each shuffle and re-encrypt the batch. The sender mapping remains hidden unless all servers collude.

18.2. The shuffle model

Prochlo motivated a formal model between the local and central models. The adversary sees the shuffled messages but not their senders.

Definition 18.1 (Shuffle model)**.** A protocol in the **shuffle model** gives each person the same randomizer $R\colon \mathcal{X} \to \mathcal{Y}$. Person i sends $Y_i = R(X_i)$, using randomness independent of everyone else's, and the **shuffler** outputs $(Y_{\pi(1)}, \dots, Y_{\pi(n)})$ for a permutation π drawn uniformly at random. The protocol is (ε, δ)**-DP in the shuffle model** if this shuffled output, as a mechanism on the dataset D, is (ε, δ)-DP in the central sense.

The uniform permutation adds no information beyond the message multiset. A local protocol is also a shuffle protocol with the same guarantee, since shuffling is post-processing. The shuffle model is useful when individual messages are not private but their shuffled collection is.

Use bit counting as a benchmark. In the central model, adding Laplace noise of scale $1/\varepsilon$ gives error of order $1/\varepsilon$. In the local model, the n independent noises yield standard deviation of order

$\sqrt{n}/\varepsilon$, and the previous chapter's lower bound shows that no local protocol does better. The goal is to approach the central error in the shuffle model.

18.3. Amplification by shuffling

Shuffling amplifies the privacy of any local randomizer, as subsampling does in an earlier chapter. The first bound shows that a randomizer with privacy parameter ε_0 yields a shuffled batch with parameter smaller by order $\sqrt{n}$.

Theorem 18.1 (Amplification by shuffling)**.** *Let $n \geq 1000$, $\delta \in (0, \frac{1}{100})$, and let R be an ε_0-local randomizer with $0 < \varepsilon_0 < \frac{1}{2}$. Then the shuffled output of $R(X_1), \ldots, R(X_n)$ is (ε, δ)-DP with*

$$\varepsilon = 12\,\varepsilon_0 \sqrt{\frac{\ln(1/\delta)}{n}}. \tag{18.1}$$

Erlingsson and coauthors (2019, Theorem 7 and Corollary 9) prove this result. Because everyone uses the same randomizer, shuffling records before randomization preserves the output distribution. Their proof places the differing record first, randomizes its position, and bounds how much the remaining messages reveal about that position.

This theorem requires an already-private randomizer, with $\varepsilon_0 \leq 1/2$, a regime that limits local accuracy. Feldman, McMillan and Talwar (2021) analyse the more useful regime ε_0 of order $\ln n$, where the shuffled guarantee can remain constant. For $\varepsilon_0 \leq \ln(n/(16\ln(2/\delta)))$, they obtain $\varepsilon = O((1 - e^{-\varepsilon_0})\sqrt{e^{\varepsilon_0}\ln(1/\delta)/n})$. Their proof treats each other person's message as a possible "clone" of the differing person's message; the number of clones hides that person's contribution. The next section proves this idea for a simple randomizer.

18.4. Counting bits in the shuffle model

For bit sums, a direct shuffle protocol achieves error of order $\sqrt{\ln(1/\delta)}/\varepsilon$, independent of n. It uses randomized response with a low probability of replacing the true bit by a fair coin.

Definition 18.2 (Shuffled randomized response)**.** Fix a coin probability $\nu \in (0, 1]$. Person i, with bit X_i, sends $Y_i = X_i$ with probability $1 - \nu$, and otherwise sends a fresh fair coin. The analyst sees the shuffled reports, counts their ones as $Z = \sum_{i=1}^{n} Y_i$, and estimates the count $c = \sum_{i=1}^{n} X_i$ by

$$\tilde{c} = \frac{Z - n\nu/2}{1 - \nu}. \tag{18.2}$$

In the local model, ν would be constant. Here ν is order $\ln(1/\delta)/(\varepsilon^2 n)$, so about $\ln(1/\delta)/\varepsilon^2$ people send coins. These coins hide each person's contribution, analogous to the clones above. The proof uses two lemmas and $\tau = \tanh(\varepsilon/2)$, for which $(1 + \tau)/(1 - \tau) = e^{\varepsilon}$. The first lemma shows that, for large s, the count of s fair coins conceals whether one additional bit is included.

Lemma 18.1. *Let $\varepsilon > 0$, let $\tau = \tanh(\varepsilon/2)$, and let $s \geq 2/\tau$ be an integer. Let N be binomial with s trials and success probability $\frac{1}{2}$, and let B and B' be Bernoulli variables independent of N, with any parameters. Then for every set S of integers,*

$$\Pr[N + B \in S] \leq e^{\varepsilon} \Pr[N + B' \in S] + 2e^{-s\tau^2/8}. \tag{18.3}$$

Proof.

1. Write $P(k) = \Pr[N = k] = \binom{s}{k} 2^{-s}$. For $1 \leq k \leq s$, the ratio of consecutive probabilities is $P(k-1)/P(k) = k/(s+1-k)$.
2. Let K be the set of integers k with $|k - \frac{s+1}{2}| \leq \frac{s+1}{2}\tau$. Writing $k = \frac{s+1}{2}(1+v)$ with $|v| \leq \tau$, the ratio becomes $(1+v)/(1-v)$, which lies in $[e^{-\varepsilon}, e^{\varepsilon}]$. Since $\tau < 1$, every $k \in K$ satisfies $1 \leq k \leq s$.
3. For $k \in K$ and any Bernoulli parameter b, $\Pr[N + B = k] = (1-b)P(k) + bP(k-1)$ is at most $\max(P(k), P(k-1))$. By step 2 this is at most $e^{\varepsilon} \min(P(k), P(k-1))$, which is at most $e^{\varepsilon} \Pr[N + B' = k]$.
4. Outside K, use $|N + B - \frac{s+1}{2}| \leq |N - \frac{s}{2}| + \frac{1}{2}$. So $N + B \notin K$ requires $|N - \frac{s}{2}| > \frac{(s+1)\tau - 1}{2}$, and this threshold is at least $s\tau/4$ because $s\tau \geq 2$. Hoeffding's inequality for s fair coins gives $\Pr[N + B \notin K] \leq 2\exp(-2(s\tau/4)^2/s) = 2e^{-s\tau^2/8}$.
5. Split S into $S \cap K$, bounded by step 3, and the rest, bounded by step 4.

□

The second lemma bounds the probability that too few people send coins. For small means, it gives a sharper binomial lower-tail bound than Hoeffding's inequality.

Lemma 18.2. *Let N be binomial with mean μ. Then*

$$\Pr\left[N \leq \frac{\mu}{2}\right] \leq \exp\left(-\frac{(1 - \ln 2)\,\mu}{2}\right). \tag{18.4}$$

Proof.

1. Let N have m trials with success probability p, so $\mu = mp$. For $\lambda > 0$, Markov's inequality applied to $e^{-\lambda N}$ gives $\Pr[N \leq a] \leq e^{\lambda a}\, \mathbb{E}[e^{-\lambda N}]$.
2. The trials are independent, so $\mathbb{E}[e^{-\lambda N}] = (1 - p + pe^{-\lambda})^m$. Using $1 + z \leq e^z$, this is at most $\exp\left(\mu(e^{-\lambda} - 1)\right)$.
3. Take $a = \mu/2$ and $\lambda = \ln 2$. The exponent becomes $\mu \ln 2/2 - \mu/2$.

□

The privacy theorem chooses ν so that, by Lemma 18.2, enough people send coins with high probability. By Lemma 18.1, those coins then hide the one person in question.

Theorem 18.2 (Privacy of shuffled randomized response)**.** *Let $\varepsilon > 0$, $\delta \in (0,1)$, $\tau = \tanh(\varepsilon/2)$ and $n \geq 1 + 16\ln(4/\delta)/\tau^2$, and set*

$$\nu = \frac{16\ln(4/\delta)}{\tau^2(n-1)}. \tag{18.5}$$

Then shuffled randomized response is (ε, δ)-DP in the shuffle model.

Proof.

1. *Reduce to the count.* Given the count Z, every arrangement of the shuffled bits with Z ones is equally likely. So the shuffled output and Z are functions of each other, up to independent randomness, and it suffices to prove that Z is (ε, δ)-DP.
2. *Isolate one person.* The count does not depend on the order of the people, so we may take neighbours $D \sim D'$ that differ by replacing the record of person 1. Let J be the set of persons $i \geq 2$ who send coins. Its law does not depend on the data, and given J,

$$Z = C_J + N + Y_1, \qquad C_J = \sum_{i \geq 2,\ i \notin J} X_i, \tag{18.6}$$

where N counts the ones among the $|J|$ fair coins. Both C_J and the law of N are the same under D and D', and only the Bernoulli variable Y_1 changes. 3. *Enough coins.* Let $s_0 = 8\ln(4/\delta)/\tau^2$. The size $|J|$ is binomial with mean $(n-1)\nu = 2s_0$. By Lemma 18.2, $\Pr[|J| < s_0] \leq \exp(-(1 - \ln 2)\, 8\ln(4/\delta)/\tau^2)$. Since $8(1-\ln 2) > 2$ and $\tau < 1$, this is at most $(\delta/4)^2 \leq \delta/4$. 4. *Each large J is private.* If $|J| \geq s_0$, then $|J|\tau \geq 8\ln 4/\tau > 2$, so Lemma 18.1 applies. Its error term is $2e^{-|J|\tau^2/8} \leq 2e^{-\ln(4/\delta)} = \delta/2$. Shifting by the constant C_J preserves the inequality, so for every event S, $\Pr_D[Z \in S \mid J] \leq e^{\varepsilon} \Pr_{D'}[Z \in S \mid J] + \delta/2$. 5. *Average over J.* Summing over sets with $|J| \geq s_0$ and adding the probability of the others,

$$\Pr_D[Z \in S] \leq e^{\varepsilon} \Pr_{D'}[Z \in S] + \frac{\delta}{2} + \frac{\delta}{4} \leq e^{\varepsilon} \Pr_{D'}[Z \in S] + \delta. \tag{18.7}$$

□

The coins also add error to the count. But only about $n\nu$ people send them, so the error depends on $n\nu$, which Equation 18.5 keeps independent of n.

Theorem 18.3 (Accuracy of shuffled randomized response)**.** *For every dataset, $\tilde{c}$ is an unbiased estimate of c with variance at most $n\nu/(2(1-\nu)^2)$. With ν as in Equation 18.5 and $\beta \in (0,1)$, with probability at least $1-\beta$,*

$$|\tilde{c} - c| \leq \frac{1}{1-\nu}\sqrt{\frac{n\nu}{2\beta}} \leq \frac{4}{(1-\nu)\,\tau}\sqrt{\frac{\ln(4/\delta)}{\beta}}. \tag{18.8}$$

For $\varepsilon \leq 1$ and $\nu \leq \frac{1}{2}$, the right-hand side is at most $18\sqrt{\ln(4/\delta)/\beta}\,/\varepsilon$.

Proof.

1. Each report is 1 with probability $(1-\nu)X_i + \nu/2$, so $\mathbb{E}[Z] = (1-\nu)c + n\nu/2$, and $\tilde{c}$ is unbiased.
2. Each report is a Bernoulli variable with parameter $\nu/2$ or $1-\nu/2$, so its variance is $\frac{\nu}{2}(1-\frac{\nu}{2}) \le \frac{\nu}{2}$. The reports are independent, so $\mathrm{Var}(Z) \le n\nu/2$ and $\mathrm{Var}(\tilde{c}) \le n\nu/(2(1-\nu)^2)$.
3. Chebyshev's inequality gives the first bound. For the second, $n\nu = 16\ln(4/\delta)\, n/(\tau^2(n-1)) \le 32\ln(4/\delta)/\tau^2$ when $n \ge 2$.
4. The function tanh is concave on $[0,\infty)$, so $\tau \ge \tanh(\frac{1}{2})\,\varepsilon = 0.462\,\varepsilon$ for $\varepsilon \le 1$. With $1/(1-\nu) \le 2$, the constant is $8/0.462 < 18$.

□

So the shuffled count has error $O(\sqrt{\ln(1/\delta)}/\varepsilon)$, which does not grow with n. That is within a factor $\sqrt{\ln(1/\delta)}$ of the central model and far from the local model's $\sqrt{n}/\varepsilon$, as Figure 18.1 shows over n and the example below shows at one setting.

Example 18.1. Take $\varepsilon = 1$, $\delta = 10^{-6}$ and $n = 10^6$. Then $\tau = 0.4621$, and Equation 18.5 gives $\nu = 0.001139$, so about 1,139 people send coins. The standard deviation of $\tilde{c}$ is at most 23.9. Local randomized response at the same ε has standard deviation $\sqrt{n}\,e^{1/2}/(e-1) = 959.5$, and central Laplace noise has $\sqrt{2} = 1.414$.

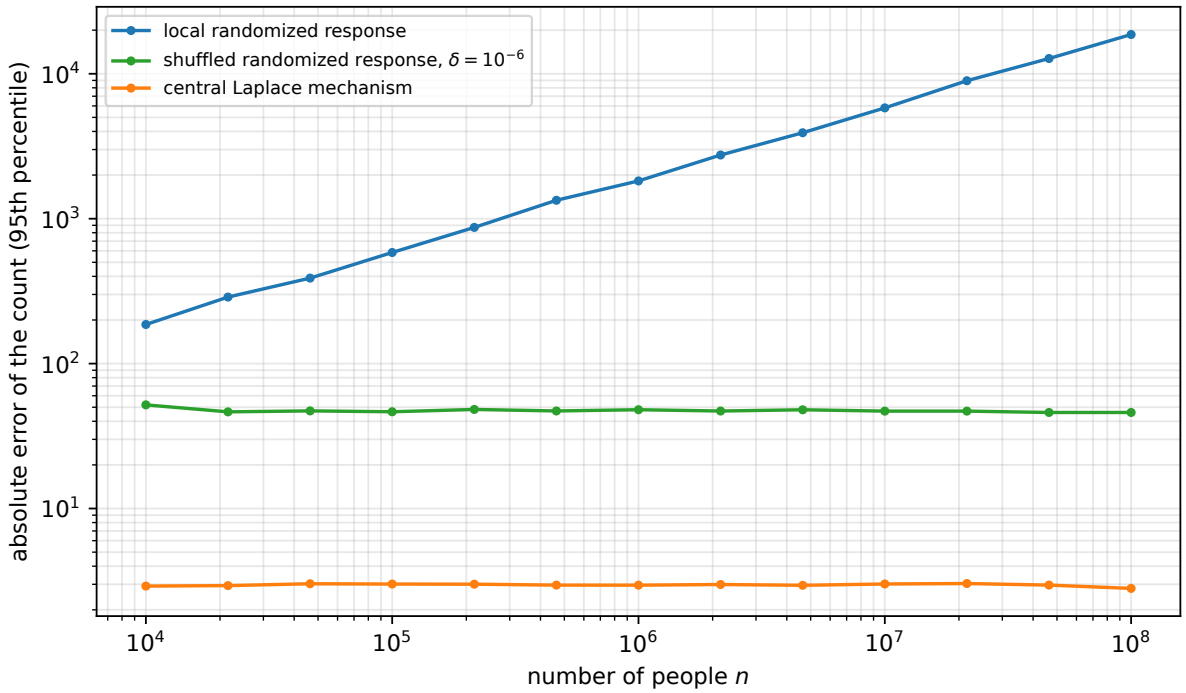


Figure 18.1.: Error of three private counts of n bits at $\varepsilon = 1$, as the 95th percentile of the absolute error over 2,000 simulated runs on data with half the bits equal to one. The local error grows as $\sqrt{n}$. The shuffled error, with ν from Equation 18.5 and $\delta = 10^{-6}$, stays near 47, and the central Laplace error stays near 3.

The price is trust in the shuffler. Each message on its own is randomized response that keeps the true bit with probability $1-\nu/2$, so it is only ε_0-locally private with $\varepsilon_0 = \ln((2-\nu)/\nu) = 7.47$ in

the example. So a shuffler that leaked the order would expose almost every bit. This ε_0 is below $\ln\left(n/(16\ln(2/\delta))\right) = 8.37$, so the protocol lies in the range that the clone analysis of the previous section covers.

18.5. A query service: LinkedIn

LinkedIn's audience engagements service lets advertisers ask how members in a target audience engage with content, such as which articles or topics they click on most. It runs in the central model, so LinkedIn holds the data and answers each query with a private mechanism. This section covers two results of the research for it, a sharper analysis of the exponential mechanism and a fast way to run it.

18.5.1. Bounded range

Recall the exponential mechanism. It takes a score $u(D, r)$ for each candidate r in a countable set, whose value changes by at most Δu between neighbours. It outputs r with probability proportional to $\exp(\varepsilon u(D, r)/(2\Delta u))$, and it is ε-DP. Write

$$L(y) = \ln \frac{\Pr[M(D) = y]}{\Pr[M(D') = y]} \tag{18.9}$$

for the privacy loss of output y between neighbours D and D'. Pure ε-DP says that $|L(y)| \le \varepsilon$, so the losses lie in an interval of width 2ε. The exponential mechanism satisfies a stronger condition.

Definition 18.3 (Bounded range)**.** A mechanism M with a countable output set is **ε-bounded range** if for all neighbours $D \sim D'$ and all outputs y, y',

$$\Pr[M(D) = y]\,\Pr[M(D') = y'] \le e^{\varepsilon}\,\Pr[M(D') = y]\,\Pr[M(D) = y']. \tag{18.10}$$

When the outputs have positive probability, the definition says that $L(y) - L(y') \le \varepsilon$. So the privacy losses lie in an interval of width ε, which need not be centred at zero. The next result shows that the two notions are within a factor of two of each other.

Proposition 18.1. *An ε-bounded range mechanism is ε-DP. An ε-DP mechanism with a countable output set is 2ε-bounded range.*

Proof.

1. *Same support.* Let M be ε-bounded range. Suppose some output y has $\Pr[M(D) = y] > 0$ but $\Pr[M(D') = y] = 0$. With this y, the right-hand side of the definition is zero for every y', and the left-hand side is $\Pr[M(D) = y]$ times $\Pr[M(D') = y']$. So the definition forces $\Pr[M(D') = y'] = 0$ for every y', which is impossible. So $M(D)$ and $M(D')$ give positive probability to the same outputs, and L is finite on them.

2. *Both signs occur.* Both distributions sum to one over the common support, so some output y_+ has $L(y_+) \geq 0$ and some output y_- has $L(y_-) \leq 0$.
3. *Width ε around zero.* For every y, bounded range gives $L(y) \leq L(y_-) + \varepsilon \leq \varepsilon$ and $L(y) \geq L(y_+) - \varepsilon \geq -\varepsilon$. The pointwise lemma of the randomized response chapter turns $|L| \leq \varepsilon$ into ε-DP.
4. *Converse.* If $|L| \leq \varepsilon$ everywhere, then $L(y) - L(y') \leq 2\varepsilon$.

□

The general conversion loses a factor of two, but the exponential mechanism does not, because its privacy losses already lie in an interval of width ε.

Proposition 18.2. *The exponential mechanism with parameter ε is ε-bounded range.*

Proof.

1. Write $Z_D = \sum_r \exp(\varepsilon u(D, r)/(2\Delta u))$ for the normalizing constant. Then $L(r) = \frac{\varepsilon}{2\Delta u}(u(D, r) - u(D', r)) + \ln(Z_{D'}/Z_D)$.
2. The constant cancels in a difference: $L(r) - L(r') = \frac{\varepsilon}{2\Delta u}\big[(u(D, r) - u(D', r)) - (u(D, r') - u(D', r'))\big]$.
3. Each bracketed difference is at most Δu in absolute value, so $L(r) - L(r') \leq \varepsilon$.

□

The factor of two matters when many selections are composed, and two lemmas make the composition precise. The first turns a tail bound on the privacy loss into a privacy guarantee. It holds for any output space, with L the logarithm of the ratio of mass functions or of densities.

Lemma 18.3. *Let $\varepsilon' \geq 0$ and $\delta \in [0, 1]$. If for all neighbours $D \sim D'$ the loss satisfies $\Pr_{y \sim M(D)}[L(y) > \varepsilon'] \leq \delta$, then M is (ε', δ)-DP.*

Proof. Fix an event S and let $G = \{y : L(y) \leq \varepsilon'\}$. On $S \cap G$ the probability under D is at most $e^{\varepsilon'}$ times that under D', so $\Pr[M(D) \in S \cap G] \leq e^{\varepsilon'} \Pr[M(D') \in S]$. The rest of S has probability at most $\Pr[M(D) \notin G] \leq \delta$ under D. □

The second lemma bounds the mean loss. It uses Hoeffding's lemma, the key step in the proof of Hoeffding's inequality: a random variable W in an interval of length ℓ satisfies $\mathbb{E}[e^{\lambda(W - \mathbb{E}W)}] \leq e^{\lambda^2 \ell^2/8}$ for every real λ.

Lemma 18.4. *If M is ε-bounded range and $D \sim D'$, then $\mathbb{E}_{y \sim M(D)}[L(y)] \leq \varepsilon^2/8$.*

Proof.

1. Under $y \sim M(D)$, the variable $W = -L(y)$ lies in an interval of length ε, and $\mathbb{E}[e^W] = \sum_y \Pr[M(D) = y] \frac{\Pr[M(D') = y]}{\Pr[M(D) = y]} = 1$.
2. Hoeffding's lemma with $\lambda = 1$ gives $1 = \mathbb{E}[e^W] \leq e^{\mathbb{E}W + \varepsilon^2/8}$, so $\mathbb{E}[L] = -\mathbb{E}[W] \leq \varepsilon^2/8$.

□

The two lemmas combine into a composition theorem. The total privacy loss of k steps is a sum of k losses, each with mean at most $\varepsilon^2/8$ and a range of width ε. So the sum concentrates around $k\varepsilon^2/8$, and Lemma 18.3 turns that concentration into a guarantee.

Theorem 18.4 (Composition of bounded-range mechanisms)**.** *Run k mechanisms in sequence on the same dataset, where the j-th may be chosen as a function of the outputs of the first $j-1$, and each is ε-bounded range whatever those outputs are. For every $\delta \in (0,1)$, the sequence of outputs is (ε', δ)-DP with*

$$\varepsilon' = \frac{k\varepsilon^2}{8} + \varepsilon\sqrt{\frac{k\ln(1/\delta)}{2}}. \tag{18.11}$$

Proof.

1. The probability of an output sequence is the product of the conditional probabilities of its terms, so the loss is a sum $L = \sum_{j=1}^{k} L_j$, where L_j is the loss of the j-th output given the earlier ones.
2. Given the earlier outputs, L_j lies in an interval of length ε, by bounded range, and its conditional mean is at most $\varepsilon^2/8$, by Lemma 18.4. Let V_j be L_j minus its conditional mean, so $L \le \sum_j V_j + k\varepsilon^2/8$.
3. For $\lambda > 0$, Hoeffding's lemma applied conditionally gives $\mathbb{E}[e^{\lambda V_j} \mid \text{earlier outputs}] \le e^{\lambda^2\varepsilon^2/8}$. Taking conditional expectations from the last term to the first gives $\mathbb{E}[e^{\lambda \sum_j V_j}] \le e^{k\lambda^2\varepsilon^2/8}$.
4. By Markov's inequality, $\Pr[\sum_j V_j \ge t] \le e^{-\lambda t + k\lambda^2\varepsilon^2/8}$. Choosing $\lambda = 4t/(k\varepsilon^2)$ makes this $e^{-2t^2/(k\varepsilon^2)}$, which equals δ at $t = \varepsilon\sqrt{k\ln(1/\delta)/2}$.
5. So $\Pr[L > \varepsilon'] \le \delta$ for every pair of neighbours, and Lemma 18.3 finishes the proof.

□

Advanced composition gives $\varepsilon\sqrt{2k\ln(1/\delta)} + k\varepsilon\tanh(\varepsilon/2)$ for k mechanisms that are known only to be ε-DP. The square-root term of Theorem 18.4 is half as large. Its linear term $k\varepsilon^2/8$ is also about a quarter of $k\varepsilon\tanh(\varepsilon/2) \approx k\varepsilon^2/2$ for small ε.

Example 18.2. Compose $k = 100$ exponential mechanisms, each with $\varepsilon = 0.1$, at $\delta = 10^{-6}$. Basic composition gives 10.00, advanced composition 5.76, and Theorem 18.4 gives 2.75.

18.5.2. Top-k selection with Gumbel noise

An advertiser usually wants the few most popular items in a histogram, such as the top topics among an audience, not every count. LinkedIn's algorithms depend on two things. The **domain** of items is known in advance or not. And a person's **contribution** is **m-restricted** when each person adds one to at most m counts, and unrestricted otherwise. For example, hires are 1-restricted, because a hire adds one to one employer, but skills are not, because one person can list many.

With m-restricted contributions, replacing one person changes the vector of counts by at most $2m$ in ℓ_1 norm. So Laplace noise of scale $2m/\varepsilon$ on every count, followed by taking the top k, is ε-DP, but its noise grows with m. For unrestricted contributions it is better to select with the exponential mechanism, with the count of an item as its score. A count changes by at most one between neighbours, whatever the number of counts that a person touches. **Peeling** selects the top item, removes it, and repeats k times. Each round is the exponential mechanism, so by Proposition 18.2 and Theorem 18.4 its cost depends on k and not on m.

Sampling from the exponential mechanism needs its normalizing constant, and peeling needs it k times. Adding noise of the right shape and taking a maximum avoids both, and the right shape is the Gumbel distribution.

Definition 18.4 (Gumbel distribution)**.** The **Gumbel distribution** with scale $b > 0$ has cumulative distribution function $\Pr[G \le g] = \exp(-e^{-g/b})$ for real g.

Adding independent Gumbel noise to each score and taking the largest gives exactly the distribution of the exponential mechanism, and taking the k largest gives exactly peeling.

Theorem 18.5 (Gumbel-max trick)**.** *Let $u_1, \dots, u_d$ be scores, let $G_1, \dots, G_d$ be independent Gumbel variables with scale b, and write $w_i = e^{u_i/b}$. Then*

1. *the index of the largest noisy score $u_i + G_i$ is i with probability $w_i / \sum_j w_j$;*
2. *the indices of the k largest noisy scores, in decreasing order, have the distribution of k rounds of peeling, where each round picks a remaining item i with probability proportional to w_i.*

Proof.

1. *Exponential race.* Let $E_i = e^{-(u_i+G_i)/b}$. Then $\Pr[E_i > t] = \Pr[G_i < -b \ln t - u_i] = \exp(-w_i t)$ for $t > 0$. So the E_i are independent exponential variables with rates w_i, and the noisy scores in decreasing order are the E_i in increasing order.
2. *The first finisher.* With $W = \sum_j w_j$, the chance that E_i is the smallest is

$$\int_0^\infty w_i e^{-w_i t} \prod_{j \ne i} e^{-w_j t}\, dt = \frac{w_i}{W}. \tag{18.12}$$

This proves the first claim. 3. *Memorylessness.* Condition on E_i being smallest, with value t. This is the event that $E_j > t$ for every $j \ne i$, and given it, the excesses $E_j - t$ are independent exponential variables with the original rates w_j, because $\Pr[E_j > t + s \mid E_j > t] = e^{-w_j s}$. 4. *Induction.* By step 3, the order of the remaining E_j is a fresh exponential race among the remaining items. Step 2 applied to it shows that the next item is chosen with probability proportional to its w_j, and repeating gives the second claim. □

Counts change by at most one between neighbours, so take $b = 2/\varepsilon_0$. Then $w_i = e^{\varepsilon_0 u_i/2}$, and the first claim is the exponential mechanism with parameter ε_0. By the second claim, adding Gumbel noise once and reporting the k largest noisy counts *is* peeling, so it has the guarantee of peeling from Theorem 18.4. It draws d noise values and selects the top k once, in time of order d plus the time of that selection. Peeling takes time of order kd.

18.5.3. Queries from outside

Here, unlike in the previous chapter, outside analysts choose the queries, and they can be careless or malicious. Each advertiser has a privacy budget for its own queries. Two advertisers with budget ε each can combine their answers, and then basic composition gives only 2ε about the members. Many advertisers together would erase the guarantee. The mechanisms cannot rule this out, so LinkedIn relies on vetting and on agreements not to share results. It also chooses ε and the monthly number of queries so that the attacks it models fail.

18.6. Data releases

The last two deployments do not answer queries. Each publishes a fixed dataset once, computed in the central model.

18.6.1. Mobility reports

During the Covid-19 pandemic, Google published community mobility reports from the location history of users who opted in. For each region and day, they gave the change in visits to places such as parks, workplaces and residences, relative to a baseline from before the pandemic. The counts of visits get Laplace noise after a bound on what each person adds. A person counts at most once per category and place, and in at most four such pairs per day, kept at random. So the contributions are m-restricted, as in the counts of LinkedIn. The reports drop metrics whose noisy count of people is below 100, and changes that the noise makes too uncertain.

18.6.2. A URL dataset with zero-concentrated privacy

Facebook and the academic organization Social Science One released a dataset on how users interact with web links shared on Facebook. It covers more than 38 million URLs, and researchers get access through a proposal process. The raw data are events, such as "a woman aged 35 to 44 in Germany shared this URL on this date". The release aggregates them by URL, month, country, age and gender, and publishes noisy counts of views, clicks, shares, likes and other reactions.

The noise is Gaussian, and the guarantee is zero-concentrated differential privacy, a variant of DP that suits Gaussian noise. It uses the Rényi divergence. For an order $\alpha > 1$ and two distributions P and Q, write p and q for their densities, or their mass functions on a countable set. The **Rényi divergence** of order α is

$$D_\alpha(P \,\|\, Q) = \frac{1}{\alpha - 1} \ln \int p(y)^\alpha q(y)^{1-\alpha} \, dy, \tag{18.13}$$

with a sum in place of the integral on a countable set. It is zero when $P = Q$ and grows as they move apart. Zero-concentrated privacy asks that the divergence between the outputs on neighbours grow at most linearly in the order α.

Definition 18.5 (Zero-concentrated differential privacy)**.** Let $\rho \geq 0$. A mechanism M is **ρ-zero-concentrated differentially private** if for all neighbours $D \sim D'$ and all $\alpha > 1$,

$$D_\alpha(M(D) \,\|\, M(D')) \leq \rho\,\alpha. \tag{18.14}$$

It has one parameter, like pure DP, but composes like advanced composition, as the next three results show. It lies between the two, because an ε-DP mechanism is $(\varepsilon^2/2)$-zero-concentrated, and the converse holds with a δ.

The first two results follow from private deep learning, where Definition 14.4 defines the log-moment $\kappa_M(\lambda)$. It is the worst case over neighbours of the logarithm of the average λ-th power of the density ratio p/q under p. For one pair of neighbours and $\alpha = \lambda + 1$, that logarithm is a multiple of the Rényi divergence of order α, by its definition:

$$\ln \int p^\alpha q^{1-\alpha}\, dy = (\alpha - 1)\, D_\alpha(M(D) \,\|\, M(D')). \tag{18.15}$$

So a mechanism is ρ-zero-concentrated exactly when $\kappa_M(\lambda) \leq \rho\lambda(\lambda+1)$ for every $\lambda > 0$.

Lemma 18.5. *Let M_1 be ρ_1-zero-concentrated, and let $M_2(\cdot\,; y_1)$ be a mechanism chosen as a function of the output y_1 of M_1, which is ρ_2-zero-concentrated for every y_1. Then the mechanism that outputs (y_1, y_2) is $(\rho_1 + \rho_2)$-zero-concentrated. In particular, this holds for two mechanisms run with independent coins.*

Proof. By the equivalence above, $\kappa_{M_1}(\lambda) \leq \rho_1\lambda(\lambda+1)$ and $\kappa_{M_2(\cdot;y_1)}(\lambda) \leq \rho_2\lambda(\lambda+1)$ for every y_1 and every $\lambda > 0$. Lemma 14.2 adds the two bounds, so $\kappa_M(\lambda) \leq (\rho_1 + \rho_2)\lambda(\lambda+1)$, which is the claim. □

The second result gives the guarantee of the Gaussian mechanism, which is why the release uses it.

Proposition 18.3 (Gaussian mechanism)**.** *Let $f\colon \mathcal{X}^n \to \mathbb{R}^d$ change by at most $\Delta_2 f$ in Euclidean norm between neighbours. Then $M(D) = f(D) + Z$, with Z a vector of d independent normal variables of mean 0 and variance σ^2, is $(\Delta_2 f)^2/(2\sigma^2)$-zero-concentrated.*

Proof. Lemma 14.3 gives $\kappa_M(\lambda) = \lambda(\lambda+1)(\Delta_2 f)^2/(2\sigma^2)$ for every $\lambda > 0$, which by the equivalence above is ρ-zero-concentrated privacy with $\rho = (\Delta_2 f)^2/(2\sigma^2)$. □

The third result converts a zero-concentrated guarantee into an (ε, δ) guarantee, the form in which a release usually states its budget.

Proposition 18.4. *A ρ-zero-concentrated mechanism is $(\rho + 2\sqrt{\rho \ln(1/\delta)}, \delta)$-DP for every $\delta \in (0, 1)$.*

Proof.

1. If $\rho = 0$, every Rényi divergence is 0, so $M(D)$ and $M(D')$ have the same distribution and the claim holds. Otherwise let $y \sim M(D)$, with density p, and let $L = \ln(p(y)/q(y))$, where q is the density of $M(D')$. For $\alpha > 1$, $\mathbb{E}[e^{(\alpha-1)L}] = \int p^\alpha q^{1-\alpha} = e^{(\alpha-1)D_\alpha} \le e^{(\alpha-1)\alpha\rho}$.
2. Markov's inequality gives $\Pr[L > t] \le e^{(\alpha-1)(\alpha\rho - t)}$.
3. Take $t = \rho + 2\sqrt{\rho \ln(1/\delta)}$ and $\alpha = 1 + \sqrt{\ln(1/\delta)/\rho}$. Writing $\ell = \ln(1/\delta)$, the exponent is $\sqrt{\ell/\rho}\,(\rho + \sqrt{\rho\ell} - \rho - 2\sqrt{\rho\ell}) = -\ell$, so $\Pr[L > t] \le \delta$.
4. Lemma 18.3 finishes the proof.

□

The budget of the release raises a question that recurs in deployments: what is a neighbour? **User-level** privacy hides everything one person did, and **action-level** privacy hides one action, such as one click. User-level privacy is the goal, but a few users act thousands of times, and hiding them would drown every count in noise. So the release protects each action strongly enough that a user with at most a high percentile of the activity distribution gets a stated user-level guarantee. Heavier users still have the action-level one.

Example 18.3. Suppose each action adds one to a single cell, and the actions of one user fall in different cells. Replacing one action changes two counts by one each, a Euclidean change of $\sqrt{2}$. Replacing all k actions of a user changes at most $2k$ counts, a change of at most $\sqrt{2k}$. Then by Proposition 18.3, k actions cost k times the action-level ρ. To give users with at most 20 actions $(0.5, 10^{-6})$-DP, solve $\rho + 2\sqrt{\rho \ln 10^6} = 0.5$ for $\rho = 0.00444$ at the user level. The action level is then $\rho = 0.000222$, which Gaussian noise of standard deviation $\sigma = \sqrt{2}/\sqrt{2\rho} = 67.1$ achieves. A single action alone is protected at $(0.111, 10^{-6})$-DP.

The largest deployment, the US Census Bureau's use of DP for the 2020 census, is complex enough to need a separate treatment.

18.7. Summary

Table 18.1.: The chapter's results.

Result	Statement
Definition 18.1	the adversary sees the messages in a uniformly random order, not who sent which
Theorem 18.1	shuffling turns ε_0-local into $(12\varepsilon_0\sqrt{\ln(1/\delta)/n}, \delta)$-DP, for $\varepsilon_0 < \frac{1}{2}$ and $n \ge 1000$
Theorem 18.2	randomized response with $\nu = 16\ln(4/\delta)/(\tau^2(n-1))$ is (ε, δ)-DP once shuffled
Theorem 18.3	its count has error $O(\sqrt{\ln(1/\delta)/\beta}/\varepsilon)$, independent of n
Proposition 18.1	ε-bounded range implies ε-DP, which implies 2ε-bounded range

Result	Statement
Proposition 18.2	the exponential mechanism is ε-bounded range
Theorem 18.4	k bounded-range steps cost $k\varepsilon^2/8 + \varepsilon\sqrt{k\ln(1/\delta)/2}$
Theorem 18.5	Gumbel noise and a maximum sample the exponential mechanism, and the top k is peeling
Proposition 18.3	Gaussian noise with sensitivity $\Delta_2 f$ is $(\Delta_2 f)^2/(2\sigma^2)$-zero-concentrated
Proposition 18.4	ρ-zero-concentrated implies $(\rho + 2\sqrt{\rho\ln(1/\delta)}, \delta)$-DP

18.8. References

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A. Glossary

Acronyms used in the book and the pages that use them.

API Application programming interface. Used in The shuffle model and central deployments.

DNA Deoxyribonucleic acid. Used in Attempts at data privacy.

DP Differential privacy. Used in The basics of differential privacy, Randomized response and the definition, The Laplace mechanism, Approximate differential privacy, Advanced composition, The exponential mechanism, Query release and its limits, Private multiplicative weights, Sparse vector, Beyond global sensitivity, Packing lower bounds, Private learning and statistics, What differential privacy protects, Private empirical risk minimization, Private deep learning, Private mean estimation, Adaptive data analysis, Deployments, Local differential privacy, The shuffle model and central deployments.

ERM Empirical risk minimization. Used in Private empirical risk minimization.

LP Linear programming. Used in Reconstruction attacks.

PAC Probably approximately correct. Used in The exponential mechanism.

PATE Private aggregation of teacher ensembles. Used in Private deep learning.

PIN Personal identification number. Used in Local differential privacy.

RDP Rényi differential privacy. Used in Private deep learning.

SGD Stochastic gradient descent. Used in Private empirical risk minimization, Private deep learning.

SVM Support vector machine. Used in Private empirical risk minimization.

ZIP Zone Improvement Plan code, the US postal code. Used in Attempts at data privacy.

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